

Studies on the Semantic Web 63

Cogan Shimizu, Annette Ten Teije, Heiko Paulheim,
Marta Sabou, Tassilo Pellegrini, Giray Havur,
Blerina Spahiu, Sahar Vahdati and Angelo Salatino (Eds.)

Bridging the Gap Between Curated and Induced Semantics

Proceedings of the 22nd International
Conference on Semantic Systems,
15-17 September 2026, Ghent, Belgium

BRIDGING THE GAP BETWEEN CURATED AND INDUCED SEMANTICS

Studies on the Semantic Web

Semantic Web has grown into a mature field of research. Its methods find innovative applications on and off the World Wide Web. Its underlying technologies have significant impact on adjacent fields of research and on industrial applications. This book series reports on the state of the art in foundations, methods, and applications of Semantic Web and its underlying technologies. It is a central forum for the communication of recent developments and comprises research monographs, textbooks and edited volumes on all topics related to the Semantic Web.

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Abstract. This volume presents the proceedings of SEMANTiCS 2026, the 22nd International Conference on Semantic Systems. SEMANTiCS is a premier forum for researchers, industry professionals, and technology leaders exploring the evolving landscape of semantic computing and artificial intelligence. At the conference participants delve into the transformative potential of semantically powered AI while critically examining their real-world challenges and limitations. The conference annually attracts a diverse community of information managers, IT architects, software engineers, and academics from across the globe, representing leading research institutions, public sector organizations, non-profits, and major enterprises.

Keywords. Semantic Systems, Knowledge Graphs, Artificial Intelligence, Semantic Web, Linked Data, Machine Learning, Knowledge Discovery

SEMANTiCS serves as a dynamic forum for the exchange of cutting-edge research and innovation in semantic systems. Beyond its core focus, the conference embraces emerging challenges across a wide range of disciplines, including knowledge engineering, artificial intelligence and machine learning, logic programming, content engineering, data science, social computing, and the Semantic Web. Now in its 22nd year, SEMANTiCS has matured into a globally recognized event that effectively bridges the gap between academic research and industrial application.

Participants benefit from direct engagement with leading researchers and seasoned practitioners, gaining critical insights into the latest trends, technologies, and application scenarios in semantic computing. The strength of the SEMANTiCS community lies in its diversity, bringing together professionals from fields such as artificial intelligence, data science, knowledge discovery and management, big data analytics, e-commerce,

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enterprise search, technical documentation, business intelligence, and enterprise vocabulary management.

In 2026, the conference embraced the theme **“Bridging the Gap Between Curated and Induced Semantics”** and particularly welcomed submissions pertaining to the following topics:

- Web Semantics & Linked (Open) Data
- Enterprise Knowledge Graphs, Graph Data Management
- Machine Learning Techniques for/using Knowledge Graphs (e.g. reinforcement learning, deep learning, data mining and knowledge discovery)
- Generative AI and Knowledge Graphs (e.g., Retrieval-Augmented Generation (RAG) with knowledge graph integration, generative model grounding)
- Reasoning, Rules, and Policies on RAG
- Knowledge Engineering and Management (e.g., knowledge acquisition, extraction, integration, and publication workflows)
- Terminology, Thesaurus & Ontology Management, Ontology Engineering Web Agents
- Natural Language Processing for/using Knowledge Graphs (e.g. entity linking and resolution using target knowledge such as Wikidata and DBpedia, foundation models)
- Crowdsourcing for/using Knowledge Graphs
- Data Quality Management and Assurance
- Mathematical and Logical Foundations of Knowledge-aware AI
- Multimodal Knowledge Graphs (e.g., text, image, audio fusion in graph structures)
- Semantic-Enhanced Data Science Pipelines and Processes
- Semantics in Blockchain environments (e.g., traceability, decentralized knowledge representation)
- Trust, Data Privacy, and Security with Semantic Technologies
- Internet of Things (IoT), Stream Processing, and Temporal Data Management (e.g., real-time semantic processing and predictive analytics)
- Conversational AI and Dialogue Systems powered by Knowledge Graphs
- Provenance and Data Change Tracking (e.g., semantic versioning, data updates in distributed settings)
- Semantic Interoperability (e.g., cross-domain standards, mapping frameworks, ontology alignment)
- Linked Data storage, triple stores, graph databases
- Robust, Scalable, and Fault-Tolerant Semantic Data Systems (e.g., distributed querying, optimization)
- User Interfaces and Usability of Semantic Technologies (e.g., visualizations, intelligent user interaction)
- Explainable and Interoperable AI
- Decentralised and Federated Knowledge Graphs (e.g., federated querying, link traversal)
- Neurosymbolic AI methods for reasoning and learning
- Neurosymbolic Architectures, including mechanisms for composition of neural and symbolic components
- Hybrid intelligence (human-machine intelligence, human-machine interaction)
- Hybrid Learning and Reasoning methods

Applied Semantic Technologies and AI in Real-World Scenarios, such as, but not limited to:

- Biomedicine and Health (e.g., Knowledge Graphs for biomedical applications, AI-driven diagnostics, personalized health)
- AI for Environmental and Climate Solutions (e.g., semantic modeling for environmental impact, biodiversity knowledge graphs)
- Scientific Knowledge Graphs and Open Science (e.g., FAIR data principles, enhanced scholarly communication)
- Semantic Technologies in GLAM (Galleries, Libraries, Archives, and Museums)
- Knowledge Graphs and Hybrid AI for Industry 4.0/5.0 and Predictive Maintenance
- Digital Humanities and Cultural Heritage Preservation
- Legal Technology, AI Ethics, and Regulatory Compliance (e.g., AI and legal frameworks, semantic-enabled compliance with the EU AI Act)
- Economics and Governance of Data Ecosystems (e.g., data marketplaces, semantic service interoperability, data policy)

The Research and Innovation track garnered significant attention with **72** (60 for the regular research track and 12 for the Blue Sky track) submissions after a call for papers was publicly announced. To ensure high-quality evaluations, an esteemed program committee comprising **130** members (109 for the regular research track and 21 for the Blue Sky track) collaborated to identify the papers of utmost impact and scientific merit. We applied a double-anonymous review process for the research track, wherein author identities and the reviewers were obscured to assure anonymity, and a single blind review for the Blue Sky track. A minimum of three independent reviews were conducted for each submission. Upon completion of all reviews, the chairing team carefully compared and deliberated on the evaluations, addressing any disparities or differing viewpoints with the reviewers. This comprehensive approach facilitated a meta-review, enabling the committee to recommend acceptance or rejection of each paper. Ultimately, we were pleased to accept **19** papers, resulting in an acceptance rate of **26%**.

In addition to the peer-reviewed work, the conference had four renowned keynotes from

- Juan Sequeda, Principal Data Strategist & Researcher at ServiceNow
- Alessandra Mileo, Associate Professor at Dublin City University
- Daniel Garijo, Associate Professor at Universidad Politécnica de Madrid
- Prof. Deborah McGuinness, Tetherless World Senior Constellation Chair, Professor of Computer, Cognitive, and Web Sciences, Rensselaer Polytechnic Institute

Additionally, the conference held a posters and demos session, a comprehensive set of workshops, as well as interactive sessions with leaders from industry and academia.

We thank all authors who submitted papers. We particularly thank the program committee members, who provided careful reviews in a quick turnaround time. Their service is essential for the quality of the conference.

Special thanks also go to our sponsors, without whom this event would not be possible:

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Sincerely yours,

The Editors

Ghent, September 2026

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Ontology Engineering & Domain Modeling

SEOntology: A Domain Ontology for Semantic Modeling of Search Engine Optimization Workflows

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Abstract. Search engine optimization (SEO) workflows remain difficult to automate and integrate because they span heterogeneous tools while relying on loosely structured data models that provide limited machine-interpretable representations of content, actions, and performance signals. This fragmentation hinders interoperability and makes provenance tracking across SEO systems brittle. We present SEOntology, a Schema.org-aligned OWL 2 ontology that models SEO workflows as knowledge graph structures to support structured querying and provenance-aware analysis. The ontology enables queries for content-quality inspection, entity-coverage analysis, and end-to-end tracing of analytical actions through their inputs, outputs, and generated signals. We assess SEOntology via structural ontology evaluation and competency-question validation, and report supporting evidence from production deployment. These results indicate logical consistency, coverage of the defined requirements, and practical utility as a semantic integration layer for representing and querying SEO workflows. SEOntology is designed as a reusable semantic resource for knowledge-graph-driven and agent-assisted SEO systems, and is extensible toward emerging generative and AI-mediated search environments.

Keywords. Search Engine Optimization, Ontology, Schema.org Extension, Semantic Web, Knowledge Graphs, Provenance, Agent-Assisted SEO

1. Introduction

Search engine optimization (SEO) plays a central role in the modern WWW ecosystem, yet its workflows remain fragmented and difficult to formalize. Practitioners operate across heterogeneous tools, proprietary platforms, and inconsistent terminology, which complicates data integration and limits opportunities for automation [27]. These conditions introduce substantial cognitive overhead and inefficiencies in optimization pro-

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cesses, sometimes described as “Delphic costs”, referring to the effort required to interpret opaque search engine behaviors and evolving ranking mechanisms [7].

At the same time, advances in semantic search and knowledge-based retrieval have transformed the optimization landscape. Search engines increasingly rely on entity-centric indexing, structured data, and contextual understanding rather than keyword matching alone, while AI-driven interfaces now generate synthesized answers from multiple sources instead of presenting ranked lists of links. These developments place a growing emphasis on machine-readable content, structured metadata, and interoperability across heterogeneous systems.

Despite these shifts, most SEO tools and workflows still rely on loosely structured data models and platform-specific heuristics. Core concepts such as search intent, content structure, internal linking relationships, and performance signals are rarely represented through shared semantic models, which limits reproducibility, automation, and systematic analysis of optimization processes.

In that context, ontologies provide a promising solution by introducing formal vocabularies and explicit relationships that enable machine-interpretable representations of complex workflows and reasoning over structured knowledge graphs (KGs) [19,18]. However, widely adopted vocabularies such as Schema.org primarily address general web metadata and do not explicitly model the operational concepts required for SEO analysis, provenance, and decision-making [17].

This paper introduces *SEOntology*, a domain ontology designed to formalize SEO concepts and workflows while remaining interoperable with existing Semantic Web standards. *SEOntology* extends Schema.org through a conservative OWL 2 design that models search queries and intent, content segmentation, linking structures, performance signals, and agent-driven analytical workflows.

The design and evaluation of *SEOntology* are guided by the following research questions:

- **RQ1:** How can core SEO concepts and workflows be formally represented in a domain ontology while maintaining interoperability with existing Semantic Web standards?
- **RQ2:** To what extent can competency-question-driven ontology design capture the operational information needs of SEO practitioners?
- **RQ3:** Can an ontology-driven representation of SEO workflows support practical automation and analysis in real-world production systems?

To address these questions, we present the design of *SEOntology*, validated through competency-question-based requirements engineering and deployment within a production SEO platform.

This paper makes four contributions. First, we introduce *SEOntology*, an OWL 2 ontology aligned with Schema.org that models SEO concepts spanning queries and intent, content segmentation, internal linking, performance measurement, and agent-driven workflows. Second, we present a requirements-driven ontology engineering process grounded in stakeholder elicitation and competency questions expressed as executable SPARQL queries. Third, we demonstrate the practical applicability of the ontology through two representative deployed applications: automated Schema.org markup generation and action-based modeling of computational SEO workflows. Fourth, we evaluate the ontology through competency question validation and operational deploy-

ment evidence, showing that it supports real-world SEO workflows while enabling consistent and traceable automation.

2. Related Work

Research on search engine optimization intersects with several established areas of Semantic Web and information retrieval, including semantic search, structured web metadata, ontology engineering, and emerging generative search systems. This section summarizes the most relevant foundations and positions the contribution of SEOntology.

2.1. Semantic Search and Structured Metadata

Web search has evolved from keyword-based retrieval toward semantic and entity-centric understanding. Early work emphasized the importance of semantic relationships and contextual information for improving retrieval quality [5]. Later research demonstrated that entities and their attributes provide more stable representations of information needs than lexical terms alone [6]. Advances in representation learning and semantic indexing, including models such as CBOW [22] and approaches such as S3D [11], further strengthened this shift and contributed to the adoption of KGs and entity-based ranking signals in modern search systems.

Structured metadata plays a central role in this ecosystem. Schema.org has become the dominant vocabulary for embedding machine-readable metadata in web pages [14], enabling enhanced search features such as rich results and knowledge panels. Linked data initiatives such as Open SESMO [9] and Wikidata-based KG applications in libraries and archives [10] further demonstrate how structured entity representations can improve discoverability across heterogeneous information systems.

2.2. Knowledge Graph and Ontology Approaches

Several studies have explored knowledge graphs (KGs) and ontologies for supporting search optimization and content discovery. Systems such as Biperpedia construct ontologies of concepts derived from user queries to expand long-tail content coverage [15], while other approaches combine inductive learning with structured KGs for ranking optimization and semantic reasoning [23]. KGs are also used to organize complex information ecosystems, for example in digital library management where heterogeneous metadata and entity relationships are integrated into graph-based semantic models [21].

Complementary work has investigated automated generation of structured metadata for SEO. Abbasi et al. [1] propose a framework for automatically generating Schema.org markup to improve search visibility. However, such systems primarily focus on markup deployment rather than formal semantic models capable of representing broader optimization workflows.

2.3. Ontology Engineering and Emerging Generative Search

Ontology engineering methodologies provide established practices for constructing reusable semantic models. Approaches such as NeOn emphasize requirements-driven de-

velopment, vocabulary reuse, and validation through competency questions [25]. These practices enable domain ontologies to remain interoperable with widely adopted standards while supporting automated reasoning over structured KGs.

Recent developments in generative search introduce additional modeling challenges. Generative engines synthesize answers from multiple sources rather than presenting ranked lists of documents, leading to the emergence of Generative Engine Optimization (GEO), which focuses on influencing citation and visibility within AI-generated responses [8,2]. Existing vocabularies provide limited support for representing prompts, outputs, and machine-mediated interactions between AI systems and web content.

2.4. Research Gap

Although prior work provides important foundations in semantic search, structured metadata, and ontology-based knowledge representation, these approaches typically address only isolated aspects of SEO-related information modeling. Research on semantic search and entity-centric retrieval focuses on improving document discovery and ranking signals, while vocabularies such as Schema.org primarily describe web resources through structured metadata. KG and ontology-based systems have also been applied to tasks such as entity extraction, metadata generation, and content organization.

However, these approaches do not provide a unified semantic representation of SEO workflows. Existing models rarely capture, within a single framework, search demand and intent, fine-grained content structures, internal linking relationships, performance signals, and the computational processes used to analyze and optimize them. In particular, provenance-aware representations of analytical actions and their associated agents and tools remain largely unsupported, especially in ways that maintain interoperability with widely adopted vocabularies such as Schema.org.

SEOntology addresses this gap by introducing a Schema.org-aligned ontology that integrates these elements into a unified semantic framework for representing SEO workflows. The ontology models queries, content units, links, performance signals, and analytical actions as interconnected KG entities with explicit provenance, enabling machine-interpretable analysis and semantic interoperability across heterogeneous SEO systems.

3. Methodology

SEOntology was developed as a domain ontology for representing search engine optimization workflows in a machine-interpretable and operationally useful form. The development followed a requirements-driven ontology engineering approach grounded in established Semantic Web practices, particularly competency question validation and iterative refinement [25]. The ontology was implemented in OWL 2 DL [3] and developed using Protégé 5.5.0 [12].

The methodological objective was to construct a semantically coherent model that supports executable validation, interoperability with web standards, and traceable representation of SEO analysis workflows. Figure 1 summarizes the ontology engineering cycle adopted in this work.

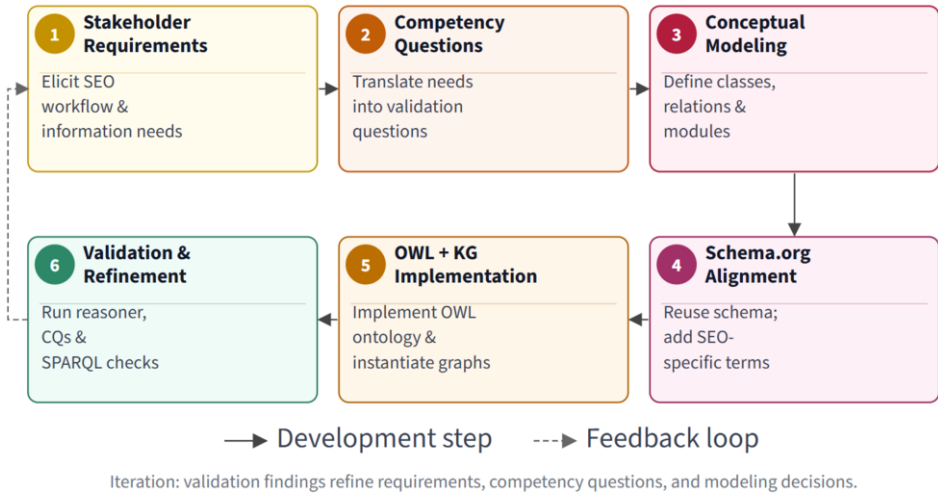


Figure 1. The SEOntology development cycle.

3.1. Design Principles

The design of SEOntology was guided by three principles: standards alignment, relationship-centric modeling, operational traceability.

Standards alignment: The ontology was designed as a Schema.org-aligned extension in order to preserve compatibility with existing structured data ecosystems. Domain specific concepts were introduced only when they were not adequately covered by the reused vocabulary.

Relationship centric modeling: Rather than relying on deep taxonomic hierarchies, the ontology emphasizes explicit relationships between entities such as queries, web-pages, chunks, links, agents, and analytical actions. This reflects the interconnected nature of SEO workflows.

Operational traceability: SEO analyses and optimization activities are represented as explicit actions linked to their agents, tools, targets, and outputs. This design supports provenance tracking and reproducibility across automated and human driven workflows.

3.2. Ontology Engineering Process

Ontology development followed a requirements-driven engineering process consisting of six connected stages: stakeholder requirement elicitation, competency-question formulation, conceptual modeling, Schema.org alignment, OWL implementation with KG instantiation, and validation-driven refinement. This sequence corresponds to the development cycle shown in Figure 1.

Stakeholder requirements: Domain requirements were gathered from $n = 18$ stakeholders, including SEO practitioners, business stakeholders, semantic web engineers, and developers of SEO tooling systems. Input was collected through semi-structured interviews and asynchronous reviews, focusing on recurring optimization tasks, decision points, and data integration challenges.

Competency-question formulation: The elicited requirements were translated into competency questions expressing the information needs that the ontology should support once instantiated with data. Examples include identifying the primary entity associated with a webpage, detecting semantic gaps between queries and content coverage, retrieving relationships between linked pages, and determining which agent executed a particular analysis. The elicitation process produced $m = 47$ competency questions, which were mapped to candidate ontology classes and properties and compiled into the SEOntologyCQ dataset together with metadata describing the corresponding stakeholder requirements. These questions provided the basis for scope definition, module derivation, and later SPARQL-based validation.

Conceptual modeling: The competency questions and stakeholder requirements were translated into candidate domain concepts and relationships. Particular attention was given to entities that recur across SEO workflows, including queries, intents, webpages, content segments, internal links, performance signals, agents, and analytical actions. The resulting model favors a relatively flat hierarchy combined with dense semantic relations.

Schema.org alignment: Candidate concepts were compared against existing Schema.org terms. Where compatible terms existed, they were reused directly. New classes and properties were introduced only when necessary to capture domain-specific constructs, such as chunk-level content representation or SEO-specific analytical actions.

OWL implementation and knowledge graph instantiation: The conceptual model was implemented in OWL 2 DL and instantiated as RDF KGs containing synthetic and production-derived examples. This implementation step made the ontology executable for reasoning, SPARQL querying, and competency-question validation.

Validation and refinement: Modeling decisions were iteratively revised against the competency-question set, reasoner checks, and SPARQL query results. This ensured that ontology development remained grounded in explicit informational needs while preserving logical consistency and operational usefulness.

3.3. Derivation of Ontology Modules

The modular structure of SEOntology emerged from analysis of the competency question set, rather than from a predefined architecture. Because competency questions capture recurring information needs, they provide a useful representation of the conceptual regions of the domain.

Competency questions were embedded and examined in a semantic space to identify groups of related requirements. The resulting distribution revealed stable clusters corresponding to recurring categories of SEO analysis and decision making tasks, as illustrated in Figure 2.

Interpretation of these clusters together with stakeholder requirements led to the identification of six ontology modules:

- **Query and Intent Module:** modeling search demand, intent, sentiment, and query driven information needs.
- **Content Module:** modeling webpages, chunks, topics, and entity oriented content structure.
- **Linking Module:** modeling internal links, anchor semantics, and site structure.

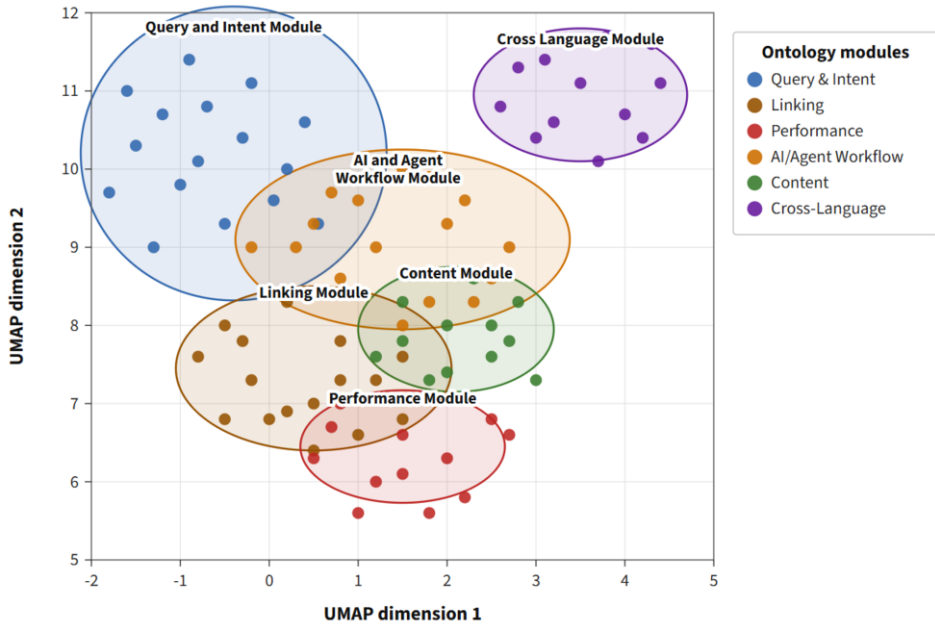


Figure 2. Competency-question distribution across the six SEOntology modules, mapping stakeholder information needs to the ontology structure

- **Performance Module:** modeling quality signals and measurable optimization outcomes.
- **Cross Language Module:** modeling multilingual alignment and persona driven content adaptation.
- **AI and Agent Workflow Module:** modeling agents, tools, prompts, analytical actions, and outputs.

These modules correspond to the dominant categories of SEO work identified during stakeholder elicitation and competency question analysis. The resulting ontology structure provides the conceptual foundation for the data model described in Section 4.

4. The SEOntology Data Model

SEOntology models SEO workflows through a relationship-centric data model extending Schema.org (Figure 3 shows the core WebPage class). The ontology currently contains 33 classes, 62 object properties, and 160 data properties, with OntoMetrics analysis indicating high relationship richness (0.91) and a shallow class hierarchy (average depth 1.21), reflecting the interconnected nature of SEO concepts.

A key feature is chunk-level content representation: the class `seovoc:Chunk` models semantically meaningful content units within webpages and links them to their parent pages via `seovoc:hasChunk` and `seovoc:isChunkOf`, enabling fine-grained semantic analysis and alignment between content segments and search intent. SEOntology also extends the Schema.org Action framework with classes such as `seovoc:AnalyzeAction`

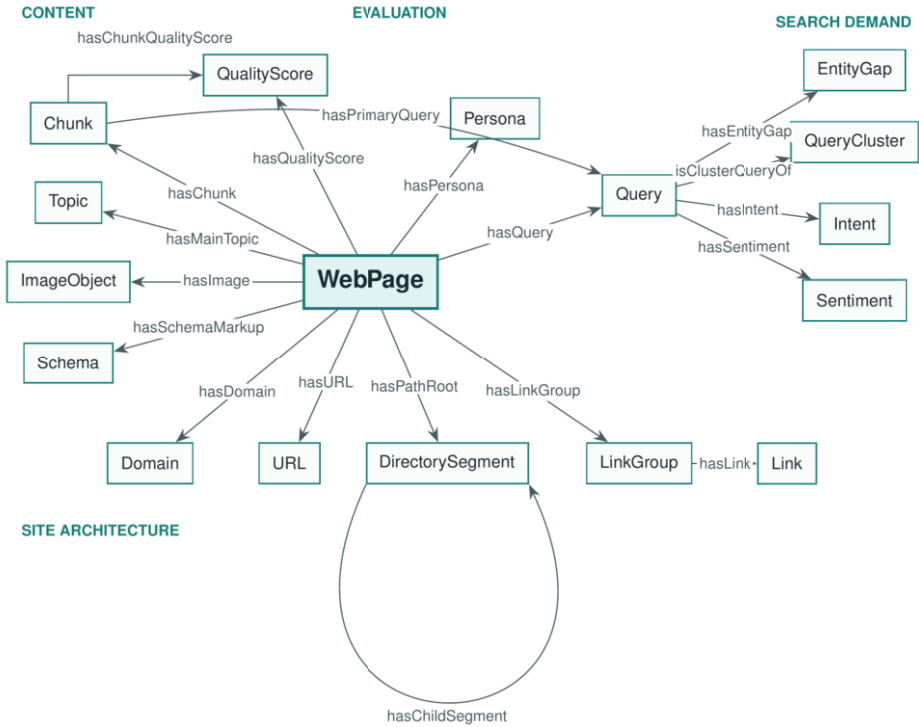


Figure 3. Conceptual overview of the SEOntology's core module **WebPage**. Full figure available on GitHub.

and `seovoc:ExtractAction` to represent computational SEO processes linked to agents, tools, targets, and outputs, enabling provenance-aware analysis. Performance signals are modeled through entities such as `seovoc:QualityScore`, which can be associated with webpages, chunks, or actions to monitor optimization outcomes. Together, these design choices allow SEO workflows to be represented as dynamic, traceable KG processes, rather than static metadata structures.

4.1. Schema.org Alignment and Data-Source Mapping

SEOntology follows a conservative Schema.org extension strategy. Existing Schema.org terms are reused where they provide sufficient semantic coverage for web-native resources, metadata, and actions. This includes page-level descriptions, entity descriptions, and action metadata, which are represented using Schema.org-compatible concepts such as `WebPage`, `Thing`, and `Action`.

SEOntology introduces new `seovoc:` classes and properties only where operational SEO workflows require more specific representation, such as chunk-level content units, query-intent relations, internal-link groupings, quality signals, and provenance-aware analytical actions. This design choice reflects the dual nature of SEO data. On the one hand, SEO workflows depend on web-facing structured data that should remain compatible with existing Schema.org and JSON-LD practices. On the other hand, operational SEO analysis requires concepts that are not directly covered by Schema.org, including con-

tent chunks, entity gaps, quality scores, workflow agents, prompts, and analysis targets. SEOntology therefore uses Schema.org as the interoperability layer and `seovoc:` as the domain-specific extension layer.

Table 1 summarizes how operational SEO data sources are lifted into SEOntology instances.

Table 1. Layered mapping from operational SEO data sources to SEOntology representations.

Mapping layer	Operational source	Extracted elements	Ontology representation
Web resource layer page identity	CMS and HTML pages; page metadata	URL, canonical URL, headline, body text, publication date	<code>schema:WebPage</code> , <code>seovoc:WebPage</code> , <code>seovoc:URL</code> , <code>schema:headline</code> , <code>schema:datePublished</code>
Content layer semantic units	Page body and editorial structure	Paragraphs, sections, meaningful content units	<code>seovoc:Chunk</code> , <code>seovoc:hasChunk</code> , <code>seovoc:isChunkOf</code> , <code>seovoc:chunkText</code>
Entity layer topical meaning	Named-entity extraction and entity linking	Main entity, secondary mentions, external identifiers	<code>schema:Thing</code> , <code>seovoc:about</code> , <code>seovoc:mentions</code> , <code>owl:sameAs</code> , <code>seovoc:thingLabel</code>
Site-structure layer links and hierarchy	Crawl data and internal-link analysis	Internal links, anchor text, domains, directory segments	<code>seovoc:Link</code> , <code>seovoc:LinkGroup</code> , <code>seovoc:AnchorText</code> , <code>seovoc:Domain</code> , <code>seovoc:DirectorySegment</code>
Performance layer signals and gaps	Search Console, SERP, and performance tools	Queries, rankings, CTR trends, entity gaps, quality indicators	<code>seovoc:Query</code> , <code>seovoc:Intent</code> , <code>seovoc:EntityGap</code> , <code>seovoc:QualityScore</code>
Workflow layer provenance	Agent and workflow execution logs	Agent, tool, prompt, target object, timestamp, generated output	<code>schema:Action</code> , <code>seovoc:AnalyzeAction</code> , <code>seovoc:ExtractAction</code> , <code>seovoc:Agent</code> , <code>seovoc:Tool</code> , <code>seovoc:Prompt</code>

This mapping clarifies how heterogeneous SEO workflow data become queryable semantic representations. For example, a page extracted from a CMS is represented as both a Schema.org-compatible web resource and a `seovoc:WebPage`; its body text is segmented into `seovoc:Chunk` instances; extracted entities are linked through `seovoc:about` and `seovoc:mentions`; crawl data adds linking and site-structure relations; and workflow logs instantiate provenance-bearing analytical actions. The resulting graph connects content, structure, performance signals, and computational provenance in a single representation.

5. Use Cases and Deployment

SEOntology can be operationalized through multiple applications which utilize its KG representation of SEO workflows. We present two representative applications that illustrate how the ontology supports production-level SEO processes. Rather than serving as isolated demonstrations, these applications show how SEOntology functions as a semantic integration layer connecting web content, analytical actions, agents, and performance signals within a unified KG.

5.1. Applications

5.1.1. Ontology-driven structured markup generation

The operational pipeline underlying this application is illustrated in Figure 4. Editorial web content is processed through an automated pipeline that extracts named entities, generates structured metadata, and maps the resulting data into ontology instances stored within a KG.

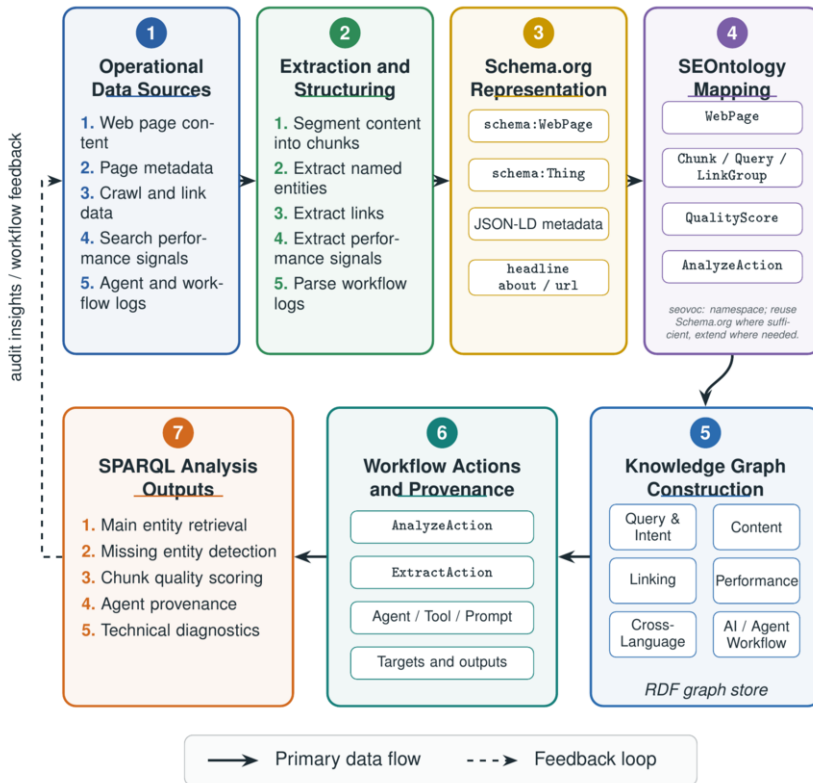


Figure 4. The SEOntology pipeline architecture.

Named entities are identified using spaCy [16], after which Schema.org JSON-LD markup is generated to capture the dominant entity of the content, secondary mentions, authorship information, publication timestamps, headlines, and canonical URLs.

The generated Schema.org artifacts are subsequently mapped to SEOntology classes and properties. Each page is represented as a `seovoc:WebPage` instance associated with a `seovoc:URL` object, while the primary subject of the content is linked through the `seovoc:about` relation. Secondary referenced entities are represented through `seovoc:mentions` relationships.

This semantic lifting step creates a bridge between web-native structured data and ontology-based knowledge representations suitable for reasoning and automated analysis. As a result, content structures become systematically auditable through SPARQL queries executed over the resulting KG.

5.1.2. Action-based modeling of SEO workflows

The second application models SEO analysis workflows using extensions of the Schema.org Action framework [24]. While Schema.org provides a general vocabulary for describing interactions, its taxonomy offers limited support for representing computational processes. SEOntology therefore introduces specialized subclasses, including `seovoc:AnalyzeAction` and `seovoc:ExtractAction`, to model automated SEO operations.

These classes inherit standard Schema.org properties such as `schema:agent`, `schema:object`, `schema:startTime`, and `schema:actionStatus`, while extending them with SEO-specific attributes describing the scope of analysis, tools used, and generated outputs. Actions are linked to responsible agents, prompts, analysis targets, and tools, enabling explicit modeling of automated SEO workflows and providing detailed provenance information for executed analyses.

Analysis outcomes are modeled as `seovoc:QualityScore` instances linked to webpages, chunks, or actions, enabling systematic monitoring of SEO signals and retrospective inspection of optimization results together with their computational provenance.

5.2. Query Examples

The ontology enables expressive reasoning and auditing through SPARQL queries executed over the resulting KG. The listings below illustrate two representative query patterns used during validation. Each query was executed against populated RDF graphs containing synthetic fixtures and production-derived instances. A query was considered successful when it returned the expected type of result using only ontology-defined classes and properties, without requiring ad-hoc schema terms. Additional competency-question queries, including missing-entity detection and technical SEO diagnostics, are provided in the accompanying repository.

Entity verification. The following query retrieves the primary entity associated with a webpage together with its semantic type and external identifiers.

```

1 PREFIX seovoc: <https://w3id.org/seovoc#>
2 PREFIX owl:  <http://www.w3.org/2002/07/owl#>
3
4 SELECT ?aboutName ?aboutType ?sameAs
5 WHERE {
6   ?page    a seovoc:WebPage ;
7           seovoc:hasURL [ seovoc:value "https://example.com/some-page" ] ;
8           seovoc:about ?about .
9   ?about a ?aboutType ;

```



```

10     seovoc:thingLabel ?aboutName .
11     OPTIONAL { ?about owl:sameAs ?sameAs }
12 }

```

Listing 1: Main entity retrieval for a given page.

Workflow provenance. Because analytical actions are represented explicitly, SEOntology supports queries that combine content evaluation results with agent provenance. The following query retrieves chunk-level quality scores together with the agent responsible for the analysis.

```

1 PREFIX seovoc: <https://w3id.org/seovoc#>
2
3 SELECT ?chunkText ?scoreValue ?scoreType ?agentName
4 WHERE {
5     ?analysis a seovoc:AnalyzeAction ;
6               seovoc:hasAnalysisTarget ?chunk ;
7               seovoc:hasAgent ?agent .
8     ?chunk    seovoc:isChunkOf ?page ;
9               seovoc:chunkText ?chunkText ;
10              seovoc:hasChunkQualityScore ?scoreObj .
11     ?scoreObj seovoc:score ?scoreValue ;
12              a ?scoreType .
13     ?agent    seovoc:name ?agentName .
14     ?page     seovoc:hasURL [ seovoc:value "https://example.com/page-audit" ] .
15 }

```

Listing 2: Retrieving chunk-level quality scores with agent provenance.

Such queries verify that entity extraction reflects the intended topical focus and links to authoritative knowledge bases such as Wikidata [26] or DBpedia [4].

6. Validation and Evaluation

SEOntology was evaluated along three complementary dimensions: (a) structural ontology quality, (b) requirements validation through competency questions, and (c) operational deployment in a production SEO system. The first two dimensions assess whether the ontology is logically coherent and expressive enough to answer the requirements defined during ontology engineering. The third dimension assesses whether the ontology-supported workflow representation is useful in operational SEO settings.

6.1. Ontology Structural Evaluation

The structural characteristics of SEOntology were assessed using the OntoMetrics framework [20]. Table 2 summarizes the main indicators.

The high relationship richness reflects the relational nature of SEO workflows, where interactions among queries, webpages, content segments, links, quality signals, and analytical actions dominate over deep class hierarchies. The shallow average hierarchy depth is consistent with this design choice: SEOntology is intended less as a taxonomic classification scheme and more as a relationship-centric model for connecting operational SEO entities. Logical consistency and satisfiability were verified with the

HermiT reasoner in Protégé, confirming suitability for reasoning and query execution in KG environments.

Table 2. Structural metrics of SEOntology.

Size		Expressivity & Structure	
Classes	33	DL Expressivity	ALUHF (D)
Object Properties	62	Relationship Richness	0.91
Data Properties	160	Average Hierarchy Depth	1.21
Logical Axioms	537		

6.2. Requirements Validation via Competency Questions

Validation used the SEOntologyCQ dataset of $m = 47$ competency questions spanning all six ontology modules. Each competency question was translated into an executable SPARQL query and evaluated against populated RDF test graphs containing synthetic fixtures and production-derived examples. Validation focused on three criteria: whether the information need could be expressed using SEOntology vocabulary, whether the query returned semantically interpretable results, and whether the returned results corresponded to the intent of the original competency question.

All competency questions were successfully operationalized without ad-hoc schema extensions, confirming sufficient representational coverage across search demand and intent, content segmentation, internal linking, performance signals, cross-language alignment, and agent-driven workflows. The full competency-question mapping, representative SPARQL queries, and minimal KG fixtures are provided in the accompanying repository to support inspection and reproducibility [13].

6.3. Operational Evaluation in Production

A six-month deployment within an industrial SEO automation platform (2024–H1 2025) yielded execution logs for 1,233 production tasks. No workflows were modified for evaluation purposes, so the evaluation reflects naturally occurring operational work rather than a controlled laboratory benchmark.

Tasks were grouped into five recurring activity clusters: content creation and editing, research and analysis, structured data and schema work, project management, and technical SEO and plugins. Savings were estimated at task-type level where manual-to-agent benchmarks were available, then aggregated to the corresponding cluster. For each eligible task type, saved hours were calculated as the difference between historically established manual execution time and observed ontology-driven execution time, multiplied by the number of observed tasks. The resulting percentages therefore represent estimated gross operational effort reduction, not direct effects on search rankings, business outcomes, or net return on investment.

Initial adoption costs, including ontology customization, data-source mapping, KG setup, and integration with existing SEO tools, are not included in these savings percentages, as they were not systematically logged in the retrospective production dataset. The reported values should therefore be interpreted as recurring workflow-efficiency estimates after deployment, while a full cost-benefit analysis would require organization-

specific implementation and maintenance costs. Table 3 reports task volume, effort, and estimated gross automation savings by activity cluster.

Table 3. Production SEO tasks and estimated gross savings by activity cluster.

Task Cluster	Tasks	Hours	Savings (%)
Content Creation & Editing	381	2,471.3	15.0
Research & Analysis	300	1,394.8	63.1
Structured Data & Schema	158	981.5	63.2
Project Management	342	2,123.6	7.1
Technical SEO & Plugins	52	209.1	59.8
Total	1,233	7,180.3	29.9

The largest gains occur in research-intensive and rule-based clusters, including SEO analysis, SERP monitoring, structured-data validation, and recurring diagnostic checks, where ontology-driven automation can formalize analytical procedures and reuse structured representations. Content creation and project coordination yield smaller savings, reflecting the importance of human judgment in creative and organizational work.

Table 4. Comparison of manual and ontology-driven execution for recurring SEO analysis tasks.

Use Case	Tasks	Per task (h)		Aggregate (h)		Savings (%)
		Manual	Auto	Manual	Saved	
Search Console Anomaly Analysis	118	5.20	0.08	613.6	603.8	98.4
SERP Competitive Analysis	62	4.91	0.17	304.4	294.1	96.6
Local SEO Audit (Multi-location)	24	4.02	0.17	96.5	92.5	95.9
SERP Feature Monitoring	11	4.18	0.05	46.0	45.5	98.9
Entity Gap Analysis	6	3.17	0.17	19.0	18.0	94.7
Total (selected)	221	–	–	1,079.5	1,053.9	97.6

Table 4 compares manual and ontology-driven execution times for selected recurring analytical tasks. These task types were selected because they are repeated often enough to permit comparison and because their inputs, outputs, and decision logic can be represented explicitly in the KG. For the selected recurring analytical tasks, ontology-driven execution reduces overhead by formalizing decision logic, reusing structured representations, and making quality signals queryable through KGs. These results should be interpreted as evidence of operational efficiency, consistency, and workflow traceability after deployment, rather than as evidence of improved rankings or net economic return.

7. Limitations

SEOntology focuses on six core modules that capture recurring SEO practices while maintaining interoperability with structured data standards. This scope provides conceptual stability while leaving room for future specialization, especially for Generative Engine Optimization (GEO), which remains an emerging field without a stable conceptual framework. Future extensions may model generative search visibility, prompt-driven discovery, and citation dynamics in AI-generated responses. At the implementation level, `seovoc:Chunk` provides a bridge to such extensions by representing semantically meaningful webpage segments with preserved page-level provenance and entity context through relations such as `seovoc:isChunkOf`, `seovoc:about`, and `seovoc:mentions`.

Empirical validation reflects the current deployment stage. SEOntology was evaluated through 47 expert-derived competency questions and a six-month industrial deployment, supporting its expressivity and practical applicability. Broader validation across additional datasets, industries, and longer observation periods remains necessary to assess generalizability.

8. Summary and Future Work

SEOntology was introduced as a Schema.org-aligned OWL 2 ontology for modeling SEO workflows as machine-interpretable KGs. It covers search demand and intent, content segmentation, internal linking, performance signals, and agent-driven analytical actions across six competency-question-derived modules, validated through ontology metrics, SPARQL queries, and production deployment evidence. Future work will extend SEOntology toward Generative Engine Optimization, AI-mediated discovery, prompt-driven retrieval, and citation dynamics, with further validation across multilingual datasets, industries, and longer deployment periods.

Data and Code Availability

The SEOntology ontology, documentation, representative SPARQL competency-question queries, and minimal KG fixtures are publicly available in the project repository [13]. The ontology namespace is available at <https://w3id.org/seovoc#>. Operational evaluation results were derived from internal task tracking and execution logs from a production SEO system. However, due to contractual and privacy constraints, the underlying production data and full execution logs cannot be publicly released.

Appendix A. Declaration on Generative AI

Large Language Models (LLMs) were used as auxiliary tools for coding assistance in prototype development, example query formulation, and minor language editing. All conceptual contributions, ontology design decisions, validation methodology, analyses, and conclusions were developed and verified by the authors, who take full responsibility for the content of the paper.

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Synthesising Causal Knowledge with Semantic Technologies

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Abstract.

MOTIVATION: AI systems relying solely on correlations often fail when training and production distributions differ, limiting their utility for reliable decision-making. While incorporating causal knowledge offers a more robust alternative, synthesizing such knowledge from heterogeneous sources remains a fragmented and cognitively demanding task. There is currently a lack of support for integrating, comparing, and resolving discrepancies between multiple, often conflicting, causal graphs.

OBJECTIVES: We present SoCK (Synthesis of Causal Knowledge), a semantic framework designed to integrate, analyse, and explore causal graphs derived from heterogeneous sources. Our goal is to provide a unified representation of causal knowledge that supports expert-in-the-loop exploration and transforms differing causal graphs into a queryable knowledge base. The framework aims to provide an accessible interface for multiple stakeholders, focusing on knowledge engineers, causal discovery researchers, and domain experts.

METHODS: Following a requirements-driven design, we developed three core contributions: (i) the *SoCK ontology* provides a unified formalization of causal graphs and their provenance; (ii) the *Causal Knowledge Graph* integrates multi-source inputs into a unified knowledge base; and (iii) an *Exploration Interface* operationalizes complex semantic queries through task-based interactions.

RESULTS: We evaluate SoCK using the CausalAssembly benchmark dataset. Results demonstrate that the ontology provides high representational coverage, successfully answering 14 out of 15 competency questions. Furthermore, scenario-based evaluations show that the framework effectively supports stakeholder tasks, such as algorithm selection and structural discrepancy analysis, by enabling systematic comparison and exploration of causal graphs through the interface.

CONCLUSION: SoCK addresses a critical challenge in causal discovery by providing the first semantic infrastructure for managing the heterogeneity and ambiguity inherent in multi-source causal discovery. By integrating fragmented causal knowledge and providing an interactive exploration interface to query complex metadata, the framework offers a systematic foundation for comparing causal models and resolving structural causal discrepancies.

Keywords. Ontology Engineering, Causal Discovery, Data Integration, Causal Knowledge

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1. Introduction

Current AI systems rely primarily on statistical correlations for their predictions, which lack generalizable explanatory power. Consequently, they often fail to support reliable decision-making when deployment and training environments differ [22]. Unlike correlations, causal relations can oftentimes more realistically be assumed to hold beyond the training environment. Incorporating such causal knowledge into AI systems has emerged as a promising direction to enable reliable predictions, even in the face of data shifts [24]. Moreover, causal reasoning is intuitive, offering predictions that are closer to human-understandable, transparent decision-making [15].

Causal knowledge about a system is typically encoded as a causal graph in which system components are connected through directed edges, representing direct and indirect causal relations. Fig. 1 illustrates a typical workflow for obtaining causal knowledge about a smart home system to understand the factors influencing energy use, and to use this knowledge in causality-based AI systems. Several complementary approaches exist for obtaining such causal graphs. (i) *Expert-curated causal graphs* are informed by physical and domain understanding. This yields high-quality causal graphs, but comes at a significant labour cost and lacks empirical validation (Graph 1, Fig. 1). (ii) *Causal discovery algorithms* derive graphs from observational (i.e., without intervening on the system) or experimental data. When restricted to observational data, the causal structure typically cannot be uniquely determined. These ambiguities are represented by ambiguous edges (e.g., $\circ\text{--}\circ$ in Graph 2, Fig. 1), indicating a relationship between Occupancy and Ventilation where the direction remains uncertain) [27]. (iii) *LLM-based causal discovery* derives causal knowledge by prompting Large Language Models to create a causal graph of a system based on its context and/or observational data [12,25,28].

While each approach provides insights on the causal structure of an existing system, they often produce conflicting causal knowledge. As shown in Fig. 1, graphs from different sources may look similar but differ in details: varying graph types, missing relations, or contradictory causal directions. For large-scale industrial systems, summarising and analysing these discrepancies manually is challenging. Despite a well-established research community in causal discovery, a gap remains in causal knowledge management. Current state-of-the-art research focuses predominantly on the statistical properties and performance of individual algorithms but neglects the integration and conflict resolution of their outputs in real data applications [3]. Specifically, there is no unified representation that accounts for the diverse graph types, underlying assumptions (e.g., faithfulness, causal sufficiency), and metadata (e.g., provenance, dataset versions) associated with different causal graphs. While semantic web technologies offer a potential solution to this heterogeneity, existing ontologies for causality remain too high-level and fail to capture the intricacies of causal knowledge representation, such as ambiguous edges ($\circ\text{--}\circ$) or algorithmic constraints, required for industrial-scale causal analysis [6,10,26]. To the best of our knowledge, there is currently no support for synthesizing such heterogeneous causal knowledge. We argue that integrating, synthesizing and interactively exploring causal graphs from multiple methods can yield more reliable causal graphs.

Various stakeholders can potentially benefit from integrated causal knowledge when building causality-based AI systems. *Knowledge Engineers (KE)* require reliable causal knowledge for AI system integration, also ensuring its provenance and traceability. *Causal Discovery Researchers (CDR)* need to compare different algorithmic assump-

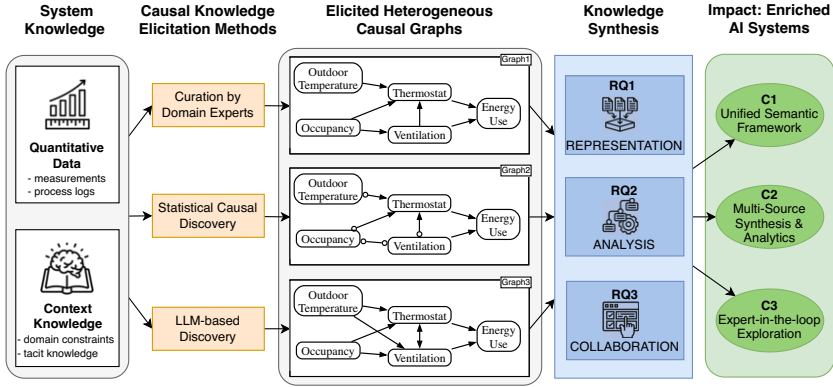


Figure 1. Illustration of the research context, showing typical settings for creating causality-based AI system—using diverse methods. This figure is using resources from flaticon.com.

tions, as well as their outputs to ensure the suitability of specific methods for a given use case. *Domain Experts (DE)* need to interpret and validate causal findings, to compare causal graphs and assess relations against their domain expertise. The practical utility of multi-source causal discovery is currently constrained by the absence of a unified framework to mediate between conflicting knowledge sources, as well as enable collaboration between multiple stakeholders. To facilitate the collaborative synthesis and validation of these structures, we investigate the following research questions:

RQ1 How can causal graphs, their provenance, and associated metadata be formally represented in a unified semantic model to enable the *integration of causal knowledge from heterogeneous sources*?

RQ2 How can a semantic framework leverage this unified semantic model (RQ1) to facilitate the *systematic management and comparative analysis of causal graphs* derived from different causal discovery settings?

RQ3 In what ways can this semantic framework (RQ2) *support experts in exploring the causal discovery space*?

We developed **SoCK (Synthesis of Causal Knowledge)**, a novel semantic framework designed to unify and synthesize heterogeneous causal knowledge, to address these research questions. While automated causal discovery is outside the scope of this work, **SoCK** provides a framework to manage, analyse, and explore the outputs of various discovery processes. Developed following a requirements-driven design cycle, **SoCK** makes three primary contributions. First, we propose a *Unified Semantic Framework (C1)* that leverages ontology-based modelling to represent causal graphs alongside their provenance, algorithmic assumptions, and underlying datasets. Second, we develop a *Multi-Source Synthesis and Analytics Engine (C2)*, which unifies multi-source causal graphs into a queryable knowledge base to allow complex analytics, ranging from provenance tracking to the identification of structural discrepancies across models. Third, we demonstrate *Expert-in-the-Loop Exploration (C3)* by providing an exploration interface for knowledge engineers, causal discovery researchers as well as domain experts to interactively investigate causal structures and discrepancies.

We evaluate **SoCK** through a proof-of-concept implementation applied to the CausalAssembly benchmark dataset [7]. This benchmark dataset represents a complex manufacturing setup with 98 sensors, distributed over five stations, providing a repre-

sentative industrial-scale use case for causal analysis. To reflect real-world applications, we apply multiple discovery algorithms to this dataset, generating a set of candidate causal graphs that serve as the foundation for our analysis. First, we assess the representational capabilities of the **SoCK** ontology through a competency question-based evaluation (RQ1), verifying its ability to capture and retrieve complex causal, contextual, and provenance information. Second, we evaluate the practical utility of the framework through scenario-based analysis (RQ2, RQ3). By grounding this analysis in previously defined stakeholder scenarios for Knowledge Engineers, Causal Discovery Researchers, and Domain Experts, we demonstrate how **SoCK** supports task-driven workflows by mitigating the complexities of causal synthesis compared to a baseline approach. Our results indicate that our approach not only identifies structural inconsistencies but also provides the necessary transparency to support robust, human-understandable causal reasoning.

This paper is structured as follows: Sec. 2 reviews causal discovery methods and the state of causal knowledge management. Sec. 3 outlines the framework requirements, while Sec. 4 presents the **SoCK** framework architecture. Sec. 5 details the evaluation on industrial benchmark data, and Sec. 6 concludes with a discussion and future work. Code and resources of our framework are available as supplemental material online.²

2. Background and Related Work

2.1. Preliminaries on Causal Graphs

Causal graphs represent cause-effect relationships among variables as graphs [15]. To account for the varying assumptions and constraints of different discovery algorithms, researchers have established a diverse range of graph formalisms. Understanding the semantic nuances of these representations is crucial for developing an ontology capable of capturing the full complexity of the causal discovery space.

Arguably the most common representation is a *directed acyclic graph* (DAG) $\mathcal{G} = (V, E)$, where nodes V correspond to variables and directed edges E encode direct causal influences (see Graph 1 in Fig. 1). The *Markov condition* then connects the graph structure to the data-generating distribution: Each variable is statistically independent of its non-descendants in the graph when conditioning on its parents in the graph [15,27]. While the Markov condition forms the basis for learning causal graphs from data, we also assume the Markov condition for causal graphs from domain knowledge or LLMs.

From observational data alone, a DAG is typically not uniquely identifiable: Multiple DAGs can encode the same conditional independencies and are said to belong to the same *Markov equivalence class* [5]. This class can be represented compactly as a *Completed Partially Directed Acyclic Graph* (CPDAG), which uses directed edges where the orientation (i.e., the roles of cause and effect) is unambiguous ($\rightarrow$) and undirected edges where it is ambiguous ($-$).

When unmeasured common causes, also referred to as *latent confounders*, are present, the appropriate identifiable object is a *Partial Ancestral Graph* (PAG) [20]. PAGs enrich the set of possible edges by using endpoint marks (Table 1) that encode varying degrees of uncertainty about causal direction and confounding. An *Acyclic Directed Mixed Graph* (ADMG) extends DAGs with bidirected edges ($\leftrightarrow$) representing shared la-

²<https://github.com/wu-semsys/SOCK-framework>

Mark	Notation	Interpretation at endpoint
Arrowhead	$\rightarrow$	The adjacent node is an effect
Tail	$-$	The adjacent node is not an effect
Ambiguous	$\circ\rightarrow$	Ambiguous (arrowhead or tail)

Table 1. PAG edge-mark vocabulary. Each edge carries one mark per endpoint; a full edge is, e.g., $\circ\rightarrow$.

tent causes (i.e., unmeasured variables influencing variables at each edge mark). ADMGs arise naturally via *latent projection*: when focusing on a subsystem O within a larger DAG $\mathcal{G}$, marginalising out the remaining variables yields an ADMG over O , where any variable outside the subsystem that is a common cause of two variables within it becomes a bidirected edge [20].

Latent projection becomes important when analysing causal relations in a smaller subsystem, as simply deleting causal edges outside the specified subsystem would remove causal knowledge about (now unobserved) latent variables outside of the subsystem. This is exactly how the ground-truth graph is constructed in our evaluation: The CausalAssembly ground-truth DAG covers the entire assembly line, but marginalizing to the sensor variables of a single station introduces bidirected edges that encode confounding from variables in other stations.

Causal discovery algorithms rely on three key assumptions: the *Markov assumption* (stated above), *faithfulness* (every conditional independence in the distribution also corresponds to a graphical separation in $\mathcal{G}$, ruling out accidental cancellations of causal effects) and *causal sufficiency* (all common causes of observed variables are themselves observed) [27]. Whether an algorithm requires all three, or only a subset, determines both its output type and the extent of its causal claims. Some algorithms additionally rely on the assumption that relationships between variables are linear or assumptions on the distribution of variables.

2.2. Causal Discovery Methods

In this paper, we consider three primary sources of causal graphs: data-driven causal discovery, domain expertise, and LLM-based causal discovery. Below, we detail prominent data-driven and LLM-based methodologies.

Constraint-based methods identify the causal graph by testing conditional independencies in the data. The PC (Peter–Clark) algorithm [27] returns a CPDAG under causal sufficiency and faithfulness, using pairwise conditional independence tests (typically partial correlation for continuous Gaussian data) at significance level α . The FCI (Fast Causal Inference) algorithm [27] relaxes causal sufficiency, returning a PAG that remains valid in the presence of latent confounders; the cost is greater edge-mark ambiguity. *Score-based methods* frame causal structure learning as an optimisation problem. NOTEARS [30] recasts DAG learning as a continuous optimisation problem via a differentiable acyclicity constraint, assuming linearity and causal sufficiency; sparsity is controlled via an ℓ_1 penalty λ . *Regression-based methods* exploit asymmetries between cause and effect. R2SORT [18,19] sorts variables by their scale-invariant R^2 regression statistics and applies a greedy regression to recover a DAG, also under linearity and causal sufficiency. Table 2 summarises the four methods used in our evaluation.

LLM-based causal discovery derives causal knowledge about a system by prompting LLMs to create a causal model of a system [12]. Some of these approaches leverage only

Algorithm	Output	Causal Sufficiency	Faithfulness	Linearity
PC	CPDAG	✓	✓	(✓)
FCI	PAG		✓	(✓)
NOTEARS	DAG	✓	✓	✓
R2SORT	DAG	✓	✓	✓

Table 2. Causal discovery algorithms used in the [SoCK](#) evaluation and their required assumptions. A (✓) indicates that PC and FCI, depending on the conditional independence test, may be implicitly assuming linearity.

variable names, while others include observational data, or context knowledge in their prompts [25,28]. LLM-based causal discovery can potentially create high-quality causal graphs, but their verification remains an open research area [29].

A fundamental challenge in working with causal knowledge is *heterogeneity*: Different algorithms operate under different assumptions and produce structurally incomparable outputs (CPDAGs, PAGs, or DAGs) even when applied to the same data. The richer edge-mark language of a PAG cannot be directly compared to the binary directed/undirected vocabulary of a CPDAG, and neither can it be directly merged with a DAG returned by a score-based method. This incompatibility motivates the need for a unifying representation of causal knowledge learned by various algorithms. To the best of our knowledge, no such unifying effort has been established.

2.3. Causal Knowledge Management

The semantic web community has investigated the representation of causal knowledge in knowledge graphs [10,11,26]. However, these approaches usually focus on representing unambiguous causality, disregarding that causal directions can not always be inferred, and represented differently by various sources. Further, there is a disconnect between such semantic models and the outputs of the discovery methods detailed in Sec. 2.1.

One approach is the [ODP \(Ontology Design Pattern\)](#) for causal events [26], which models causal relations between events as well as their associated context and provenance. This pattern differentiates between concrete events (fixed in time and space) and abstract events (general occurrences), allowing for a nuanced representation of how observed events interact and originate.

The Causality [ODP](#) [10] proposes a similar pattern, where causality between two events can be direct, or indirect (connected by a mediator). In addition to [26], [10] also introduced effect weights as an additional parameter to represent the effect size of a causal relation. While valuable for event modelling, these patterns lack the vocabulary to represent nuances in causal graphs and algorithmic assumptions (e.g., faithfulness, causal sufficiency) that are crucial for interpreting causal discovery results.

In the domain of Cyber-Physical Systems (CPS), the [SENSE](#) ontology [6] integrates causality with topology and observation knowledge to provide user-centered explanations for system anomalies. While [SENSE](#) supports causal reasoning over anomalies at runtime, its focus remains on the application of causal knowledge for explainability rather than the synthesis of conflicting graphs from heterogeneous discovery methods. A similar industrial focus is found in the [causalgraph](#) package [16], which embeds causal graphs within context knowledge to derive localized insights.

Despite these advancements in specific areas like CPS [6,16] or the biomedical domain [23], there remains no shared conceptualization that facilitates the integration of causal knowledge across different elicitation methods.

	ID	Competency Question
Knowledge Engineer	CQ1	How are <i>VariableA</i> and <i>VariableB</i> related to each other?
	CQ2	Does <i>VariableA</i> cause <i>VariableB</i> directly?
	CQ3	What data was <i>CausalGraphX</i> based on?
	CQ4	What algorithm was <i>CausalGraphX</i> produced by?
Causal Discovery Researcher	CQ5	What are differing assumptions between <i>AlgorithmA</i> and <i>AlgorithmB</i> ?
	CQ6	If I am not willing to <i>SetOfAssumptionsX</i> , what is the causal relation between <i>VariableA</i> and <i>VariableB</i> ?
	CQ7	Which algorithm assumes <i>SetOfAssumptionsX</i> ? (e.g. causal sufficiency and linearity)
	CQ8	What type of causal graph is created by <i>AlgorithmX</i> ?
	CQ9	Which conditional independence tests are applied by <i>AlgorithmRunX</i> , and what are their results?
	CQ10	What separations are implied by <i>CausalGraphX</i> ?
Domain Expert	CQ11	How similar are <i>CausalGraphA</i> and <i>CausalGraphB</i> in terms of their hamming distance?
	CQ12	How similar are <i>CausalGraphA</i> and <i>CausalGraphB</i> in terms of their separation distance?
	CQ13	What are all the causal paths from <i>VariableA</i> to <i>VariableB</i> ?
	CQ14	How does the output differ between <i>AlgorithmA</i> and <i>AlgorithmB</i> ?
	CQ15	How do <i>CausalGraphA</i> and <i>CausalGraphB</i> differ?

Table 3. Competency Questions based on each stakeholder’s professional needs in the context of causal discovery.

In summary, despite the extensive range of causal discovery methods, the field lacks a formal mechanism to consolidate the structurally diverse and often conflicting outputs of its algorithms. Conversely, existing semantic models remain too general to capture the technical intricacies (e.g., edge-mark semantics and statistical assumptions) required to integrate these diverse causal discovery results. This leaves a research gap in the synthesis of multi-source causal knowledge. SoCK addresses this by providing a novel semantic infrastructure that mediates between heterogeneous discovery methods and their distinct representations, enabling the systematic integration of causal knowledge.

3. Framework Requirements

The development of SoCK follows a requirements-driven design cycle aimed at bridging the gap between raw causal discovery outputs and expert decision-making. We operationalize the Task-Technology Fit [8] of the framework through a scenario-based design approach [21], utilizing user stories to capture stakeholder interactions. Following established ontology engineering practices [2,17], we define the functional scope of SoCK through CQs (competency questions) [9]. These natural-language questions specify the information needs of stakeholders. The CQs are derived from the limitations of current “baseline” manual workflows of our three primary stakeholders (KE, CDR, DE). Table 3 summarizes the resulting CQs.

KE: Metadata and Provenance Integration. Manual tracking of dataset versions, prompts (for LLMs), and algorithmic parameters is error-prone. The framework must formally link causal edges to their provenance and metadata (CQs 1–4). This supports the systematic integration of heterogeneous causal models and their associated context.

CDR: Method Selection and Filtering. CDRs currently need to manually inspect documentation to ensure algorithm assumptions (e.g., linearity, causal sufficiency) match the dataset. The framework must enable systematic filtering of algorithms based on queryable methodological properties (CQs 5–10). This allows CDRs to investigate methodological differences and ensure the validity of discovery runs.

DE: Structural Synthesis and Resolution. DEs currently rely on visual inspection of adjacency matrices to compare graphs, which can be cognitively demanding for large

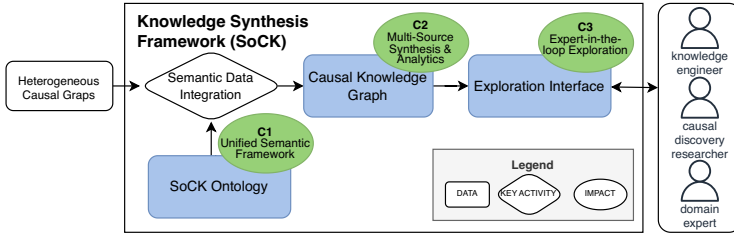


Figure 2. SoCK: a knowledge synthesis framework to enable enriched AI applications of causal knowledge from various sources of causal knowledge

systems. The framework must provide quantitative metrics (e.g., Hamming distance) and local representations to identify shared, conflicting, or unique edges (CQs 11–15). This enables DEs to interpret and validate causal relationships within their domain knowledge, facilitating informed decision-making.

A more detailed description of stakeholders, user stories, and task scenarios, is provided in the supplemental material. In Sec. 5, we evaluate SoCK via SPARQL queries that formally represent the CQs. We evaluate their correctness and utility in representative task scenarios to demonstrate support for practical causal knowledge synthesis.

4. SoCK: a Framework for Synthesising Causal Knowledge

We introduce the SoCK framework, a semantic infrastructure that enables the integration and exploration of causal knowledge from heterogeneous sources. As illustrated in Fig. 2, SoCK functions as a synthesis pipeline that bridges the gap between raw algorithmic outputs and domain-specific decision-making. The core innovation of SoCK lies in its three primary technical contributions:

The *SoCK Ontology* (C1, Sec. 4.1) provides the formal vocabulary required to map diverse causal graphs, their metrics, the causal discovery methods they are produced by, as well as their datasource and context. By ingesting multi-source inputs into a coherent *Causal Knowledge Graph* (CKG) (C2, Sec. 4.2), the framework creates a central “source of truth” for a specific (industrial) system that represents causal knowledge from different sources, including their structural and semantic discrepancies. The *Exploration Interface* (C3, Sec. 4.3) operationalizes the CKG, allowing stakeholders to interactively query, visualize, and explore causal knowledge about a system, making the CKG accessible to non-experts in semantic technologies.

Via its main contributions, the framework supports the full lifecycle of causal knowledge synthesis, including integration, representation, and exploration for its three stakeholder groups (KE, CDR, DE).

4.1. SoCK Ontology (C1)

We propose the SoCK ontology as the foundation of the knowledge synthesis framework. The ontology enables the representation of causal graphs within a unified knowledge graph that not only captures the graph structure itself, but also incorporates contextual information and provenance metadata, describing how each causal graph was generated. A visual overview of the ontology is provided in Fig. 3. To illustrate the ontology, a

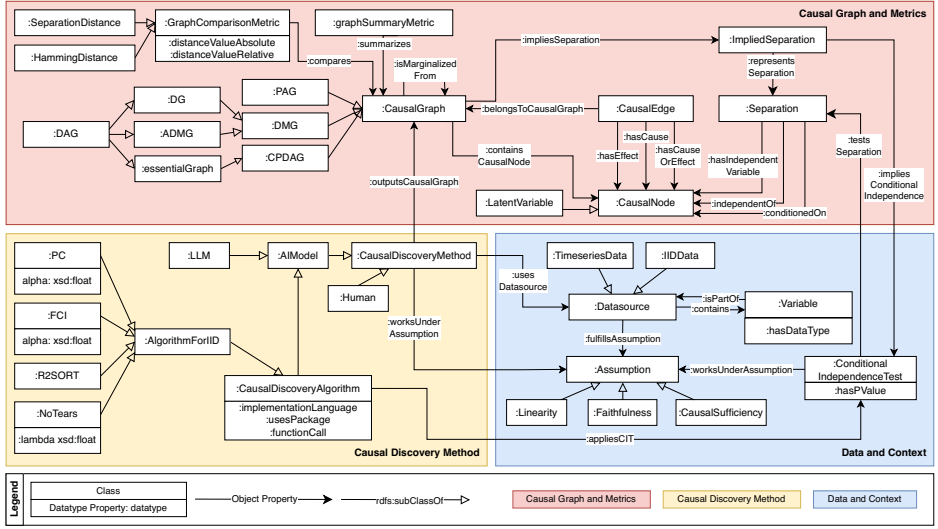


Figure 3. A visualization of the SoCK ontology in chowlk notation [4], representing *Causal Graphs and Metrics*, *Causal Discovery Method*, and *Data and Context* knowledge.

snippet of a potential representation of a use case is shown in Listing 4. It is based on the smart home and its causal relations from Fig. 1. The SoCK ontology is available as a permanent resource via w3id.³ There are three main modules of the ontology:

Causal Graph and Metrics focuses on the representation of causal graphs and associated structural and comparison metrics. A `:CausalDiscoveryMethod` `:outputsCausalGraph` a `:CausalGraph`, whose specific type depends on the applied method (see Sec. 2.1). Each `:CausalGraph` `:containsCausalNode` a set of `:CausalNodes` connected via `:CausalEdges`. Different types of `:CausalEdges` are encoded using different edge end-point types: `:hasEffect` ($\rightarrow$), `:hasCause` ($\leftarrow$), and `:hasCauseOrEffect` ($\leftrightarrow$) (see Table 1), allowing the representation of both directed and uncertain causal relations. Each causal graph `:impliesSeparation` a set of `:ImpliedSeparations` of `:CausalNodes`, which are determined by the set of `:CausalEdges` in a `:CausalGraph`. In addition, the ontology supports quantitative analysis through `:GraphComparisonMetrics`, which capture distance or similarity measures between pairs of causal graphs (e.g., Hamming distance), and `:GraphSummaryMetrics`, which describe structural properties of individual graphs (e.g., graph density or connectivity).

Causal Discovery Method focuses on the processes and methods used to generate a causal graph. A `:CausalDiscoveryMethod` can be a `:Human`, or an `:AIModel`. For `:CausalDiscoveryAlgorithms`, the ontology captures relevant implementation details, including the software package used (`:usesPackage`), `:implementationLanguage`, and algorithm-specific hyperparameter settings (i.e., `:alpha`, `:lambda`).

Data and Context captures the data sources underlying causal discovery and their embedding within a broader application context. A `:CausalDiscoveryMethod` `:usesData-source` `:Datasource` to base its discovery on.

³<http://w3id.org/explainability/SoCK#>

1	PREFIX : <http://w3id.org/explainability/SoCK#>	
2	:CausalDiscoveryRunA a :FCI ;	Discovery Method Details on the FCI run, R implementation, and datasource.
3	:implementationLanguage "R" ;	
4	:usesPackage "pcalg" ;	
5	:alpha 0.001 ;	
6	:outputsCausalGraph :Graph2;	
7	:usesDatasource :SmartHome1 .	Data & Context Defining the datasource and variable types.
8	:SmartHome1 a :IIDData ;	
9	:contains :Ventilation, :Thermostat ;	
10	:Ventilation a :Variable, :CausalNode ;	Graph Metrics Structural definition of the produced causal graph (PAG) and specific edges.
11	:hasDataType xsd:numeral .	
12	:Graph2 a :PAG ;	
13	:containsCausalNode :Ventilation, :Thermostat .	
14	:CausalEdge1 a :CausalEdge ;	
15	:hasCauseOrEffect :Ventilation ;	
16	:hasEffect :Thermostat .	

Figure 4. SoCK representation snippet (Graph 2 from Fig 1).

4.2. Causal Knowledge Graph (C2)

The Causal Knowledge Graph (CKG) is the instantiated result of the semantic data integration process for a specific system of interest (e.g., the smart home in Fig. 1). It serves as the framework’s central repository, structuring heterogeneous inputs, including causal graph structures, discovery metadata, and data sources, according to the [SoCK](#) ontology.

To populate the CKG, the framework utilizes RML mappings implemented via the `rdflib` Python package [13]. This mapping transforms raw outputs from diverse causal discovery methods into a coherent RDF representation. Currently, [SoCK](#) supports the integration of DAGs, CPDAGs, and PAGs, preserving the structural nuances of each (e.g., ambiguous edge marks of a PAG).

The CKG is designed not merely for storage, but for knowledge synthesis. Once instantiated, the framework computes graph comparison metrics (e.g., Hamming distance) between the integrated causal graphs. These metrics are stored in the CKG, enabling users to perform queries about causal graphs and relations across different discovery runs. By combining structural data with these derived analytics, the CKG provides the necessary foundation for the interactive exploration described in the next Section.

4.3. Exploration Interface (C3)

The *Exploration Interface* provides the mechanism for expert-in-the-loop exploration (C3), operationalizing the requirements defined in Sec. 3 by leveraging the CQs (Table 3) for user interaction. By abstracting the complexity of the underlying CKG, the interface enables stakeholders to perform sophisticated causal synthesis without direct exposure to SPARQL or RDF syntax.

Fig. 5 illustrates the overall workflow of the interface. Users select a query, such as identifying differences between two causal graphs (Fig. 5), and specify relevant parameters, such as the causal graphs they need to compare. The interface then translates these selections into SPARQL queries executed against the CKG, and returns structured answers based on the integrated knowledge base.

While some CQs can be answered through direct query results, others require more exploratory analysis. To support this, the interface provides additional visualization and interaction mechanisms that enable users to investigate complex relationships in greater depth. As shown in Fig. 5, differences between two causal graphs (CQ15) are presented through a colour-coded graph visualization. Edges are highlighted to indicate agreement

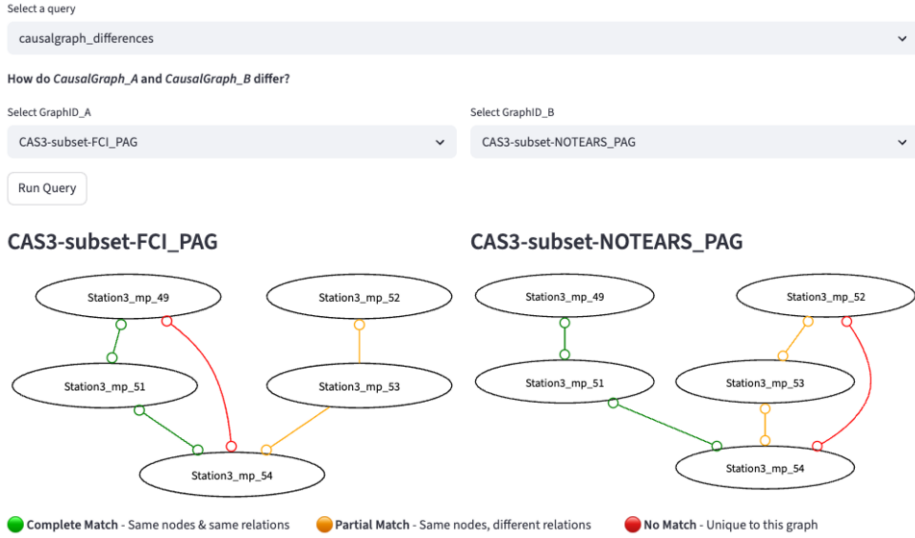


Figure 5. Screenshots of the exploration interface - example for CQ15 “How do *CausalGraphA* and *CausalGraphB* differ?”. The visualisation is reduced to 5 sensors for illustrative purposes.

(green), disagreement in edge type (yellow), and edges unique to one graph (red). The interface also provides this knowledge in tabular form for deeper analysis. A full demo of the tool, featuring the CausalAssembly dataset, can be accessed online.⁴ This combined representation allows users to move beyond isolated query results and explore global structural differences between causal graphs. This enables users to visually compare how relations between variables differ across methods, and to identify patterns of agreement or conflict at the graph level.

Ultimately, the interface serves as a collaborative bridge between stakeholders with different expertise for causal synthesis. It allows KEs to verify integration integrity, CDRs to conduct methodological sensitivity analyses, and DEs to interpret and reconcile conflicting causal claims.

5. Evaluation and Results

We evaluate the **SoCK** framework using a proof-of-concept implementation based on the CausalAssembly benchmark dataset [7]. Our evaluation follows a twofold strategy. First, we assess the representational capabilities of the ontology using a CQ-based evaluation (Sec. 5.2), focusing on its ability to capture and retrieve the required causal, contextual, and provenance information for each CQ (RQ1). Second, we evaluate the practical utility of the framework through a scenario-based analysis (Sec. 5.3), demonstrating how different stakeholders (KE, CDR, DE) can leverage the system to support decision-making tasks (RQ2, RQ3).

⁴<https://semsys.ai.wu.ac.at/sock-ui/>

5.1. Experimental Setup

We evaluate **SoCK** using the CausalAssembly benchmark dataset [7], a synthetic dataset generated from a causal graph and real data from an industrial assembly line. The dataset contains 98 sensor variables across five stations and contains 15,581 observations. The ground-truth causal graph is a DAG over all 98 variables without hidden confounders.

For our experiments we focus on Station 3, which contains 16 sensor variables. This station contains a moderate number of sensors and confounding sensors from previous stations influence causal relations at this station, which allows us to investigate complex scenarios. Restricting attention to this subsystem requires latent projection of the ground-truth DAG onto the Station 3 variables: Sensor variables from other stations are marginalised out. As a result, the ground-truth structure over Station 3 variables becomes an ADMG, with bidirected edges representing latent confounding introduced by inter-station dependencies (see Sec. 2.1). Because of those bidirected edges, the causal sufficiency assumption is violated. As we use only observational data for causal discovery, the identifiable learning target from Station 3 data is a PAG (since PAGs summarize Markov equivalence classes of ADMGs). This setup reflects a realistic scenario in which not all relevant variables are observed, making it suitable for evaluating the framework’s ability to integrate and compare causal graphs under differing assumptions.

We apply the four causal discovery algorithms introduced in Table 2 to the Station 3 dataset. PC and FCI are run via the `pcalg` R package using Gaussian partial correlation tests at significance level $\alpha = 0.001$; FCI is run via the `fciPlus` variant with selection bias disabled. NOTEARS is run via the `dagma` Python package with ℓ_1 penalty $\lambda = 0.02$. R2SORT is run via the `CausalDisco` Python package with default settings.

The DAG outputs of NOTEARS and R2SORT are converted to their corresponding CPDAGs and PAGs to enable comparison with the PAG produced by FCI and the ground-truth ADMG. This harmonization step ensures that all resulting causal graphs can be represented within a common formalism for subsequent analysis. All resulting graphs, along with their methodological and contextual metadata, are then integrated into the CKG, which is comprised of 67,117 triples to represent the CausalAssembly system, focusing on Station 3.

5.2. Ontology Evaluation (RQ1)

This section evaluates the representational capacities of the **SoCK** ontology by assessing its ability to answer the 15 CQs defined in Sec. 3. We operationalize this evaluation by translating each natural language CQ into one or more SPARQL queries executed against the CKG instantiated with the CausalAssembly data (Sec. 5.1) using the **SoCK** ontology. A CQ is considered satisfied if the query returns a complete and correct result.

The framework currently successfully answers 14 out of 15 CQs, demonstrating that the **SoCK** ontology provides sufficient expressiveness to successfully represent and integrate heterogeneous causal knowledge. The full set of SPARQL queries, along with the dataset and ontology specification, is provided in the supplemental material.

The only currently unsupported query, CQ13 (“What are all the causal paths from Variable A to Variable B?”), highlights a limitation of the query language rather than of the ontology. All required information is present in the knowledge graph, as causal relationships are fully represented via `:CausalNode` and `:CausalEdge` instances. How-

ever, although SPARQL supports property path queries of arbitrary length, it provides no built-in functionality to enumerate the (causal) paths between two nodes, so CQ13 cannot be expressed as a standard SPARQL query. Such path-finding is available only as extensions or functions in some triplestores (e.g., GraphDB [14]) and libraries (e.g., Apache Jena [1]), outside the standard SPARQL specification. We are exploring several workarounds, e.g., an implementation-specific SPARQL extension or temporarily translating reified relations into direct relations for path querying. As each carries its own drawbacks, we deliberately propose no specific solution here and instead highlight this as an area for future improvement.

5.3. Scenario-Based Evaluation (RQ2, RQ3)

To evaluate the practical utility of the SoCK framework, we conduct a scenario-based evaluation grounded in the previously defined stakeholders (KE, CDR, DE). This evaluation demonstrates how the framework supports users in performing realistic tasks related to causal knowledge synthesis and analysis. Each stakeholder is associated with distinct requirements and CQs, which can be answered by the knowledge base via predefined SPARQL queries. For each stakeholder, we defined two representative task scenarios, which they may face in their work. Due to space limitations, we present selected scenarios (S3 and S5) in detail, while the remaining scenarios are provided in the supplemental material. Each scenario is structured as a task-driven workflow, comparing a baseline approach (without the framework) with the supported workflow enabled by SoCK. This highlights how SoCK mitigates manual overhead and enhances analytical capabilities.

5.3.1. Scenario S3: Selection and Filtering (CDR)

A causal discovery researcher is presented with a new dataset from the CausalAssembly setup and they are instructed to analyse the causal relationships within a smaller subsystem. All variables in the data are continuous and, based on physical domain knowledge, can be assumed to exhibit linear relationships. However, since only a subsystem is considered, causal sufficiency may be violated. The task is to select an appropriate causal discovery algorithm that does not violate key assumptions to ensure trustworthy results.

Baseline. The researcher manually inspects documentation, code repositories, or academic publications for candidate algorithms (e.g., PC, FCI, NOTEARS). It involves identifying and comparing assumptions such as linearity, causal sufficiency, or distributional constraints. This approach is time-consuming and error-prone, as critical assumptions may be overlooked, potentially invalidating downstream analysis.

SoCK-supported workflow. Using the SoCK exploration interface, the researcher can systematically filter and compare algorithms in the causal knowledge graph based on their assumptions and expected outputs:

1. *Filtering by assumptions.* The user queries which algorithms satisfy or violate specific assumptions (e.g., causal sufficiency, linearity), narrowing down the candidate set (CQ6, CQ7).
2. *Inspecting output representations.* The user retrieves the type of causal graph produced by each algorithm (e.g., DAG, CPDAG, PAG), ensuring compatibility with the intended analysis (CQ5).
3. *Comparing methodological differences.* The user compares remaining candidates to identify differences in assumptions and methodological properties (CQ8-10).

SoCK enables a structured and transparent selection process, allowing informed decision-making based on explicit, queryable knowledge rather than manual inspection. This reduces the risk of invalid model assumptions and improves reproducibility of the causal discovery process.

5.3.2. Scenario S5: Structural Discrepancy Resolution (DE)

A domain expert analyses two causal graphs of an assembly process, one generated by FCI and another by NOTEARS. Two causal discovery models provide conflicting causal knowledge when analysing a spike in temperature at a specific sensor. The expert aims to identify where and why the models diverge.

Baseline. The expert compares the two causal graphs manually, often by visual inspection. This can involve analysing the adjacency matrix of each causal graph, tracing relationships across multiple variables and identifying differences in edge structure. For larger graphs, this process is cognitively demanding, error-prone, and does not provide quantitative measures of disagreement.

SoCK-supported workflow. The SoCK framework enables a systematic comparison of causal graphs through both quantitative and visual analysis:

1. *Quantifying global differences.* The user computes graph comparison metrics (e.g., Hamming distance, separation distance) to assess the overall level of disagreement between the two models (CQ11, CQ12).
2. *Identifying structural discrepancies.* The user retrieves a structured comparison of edges, highlighting which common relationships, as well as discrepancies (CQ15). In addition to a tabular comparison, the interface provides a colour-coded visualization of the two graphs, where shared edges are highlighted in green, conflicting edges in yellow, and unique edges in red. This allows the expert to immediately identify regions of structural disagreement.
3. *Tracing methodological differences.* Building on the identified differences, the user can further query how the outputs of the underlying algorithms differ (CQ14), linking structural discrepancies back to methodological assumptions and data characteristics.

SoCK transforms a manual and qualitative comparison task into a systematic and interpretable analysis. By combining quantitative metrics, structured queries, and visual exploration, it enables domain experts to identify and understand discrepancies between causal graphs more effectively.

Across the evaluated scenarios, the [SoCK](#) framework supports users in systematically exploring, comparing, and interpreting causal knowledge from multiple sources. By operationalizing competency questions through a unified knowledge graph and interactive interface, the framework reduces manual effort, increases transparency, and facilitates informed decision-making across different user roles.

6. Conclusion

In this work, we introduced [SoCK](#), a semantic framework for the integration, exploration, and comparison of causal knowledge from heterogeneous sources. By bridging the gap between raw algorithmic discovery and expert decision-making, [SoCK](#) addresses a critical bottleneck in causal workflows: the lack of structured support for reconciling and interpreting conflicting causal models.

The framework's impact is grounded in three primary contributions. First, the **SoCK** ontology (C1) provides a novel, unified formalization of causal graphs, their underlying methodological assumptions, and their provenance. Second, the instantiation of a Causal Knowledge Graph (C2) enables the synthesis of multi-source inputs into a single, queryable "source of truth." Third, the exploration interface (C3) provides a novel abstraction layer that operationalizes the CKG. By translating complex semantic queries into interactive elements, the interface reduces the technical expertise required to interact with the underlying knowledge, facilitating an expert-in-the-loop workflow that allows stakeholders to resolve discrepancies through interactive visual and semantic analysis.

Our evaluation demonstrated that the ontology provides high representational coverage, successfully satisfying 14 out of 15 competency questions. Furthermore, scenario-based analysis confirmed the framework's utility in reducing manual overhead for knowledge engineers, causal discovery researchers, and domain experts, particularly in complex tasks such as method selection and structural discrepancy analysis of causal graphs.

Despite these contributions, several limitations remain. While the ontology captures a wide range of causal knowledge and metadata, certain analytical tasks, such as causal path identification, require extensions beyond standard query capabilities. Further, the ontology is currently limited to causal graphs for static systems.

Future work will address these limitations by extending the framework with graph traversal capabilities, as well as integrating new causal discovery algorithms that support time series data. These algorithms come with additional peculiarities that will require an extension of the **SoCK** ontology to represent aspects that are specific to the time series case (e.g., causes are always temporally preceding effects). Furthermore, we plan to conduct user studies to systematically evaluate the usability and effectiveness of the exploration interface in real-world decision-making settings.

Overall, **SoCK** demonstrates how semantic technologies can move beyond isolated causal discovery results toward a systematic and interpretable process of causal knowledge synthesis. By operationalizing expert requirements through a unified knowledge graph, the framework provides the transparency necessary to explore conflicting causal models and support informed decision-making in complex domain applications.

Declaration on Generative AI

During the preparation of this work, the author(s) used Gemini 3, GPT 5 and Claude Sonnet 4.6 in order to: paraphrase and reword, improve writing style, abstract drafting. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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A Knowledge Graph–Based Framework for Harmonising and Comparing Social Surveys

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Abstract. A wide range of tools in the social sciences requires systematic methods for reuse, integration, and international comparison. To address this challenge, we present an integrated solution that combines ontological modelling and knowledge-extraction techniques to detect, import, and reuse questionnaires in a simple and intuitive way. The system integrates a user-friendly web interface, a questionnaire import module, and a semantic knowledge graph that formalises and interlinks core survey entities, such as questions, variables, and groups. This infrastructure supports both the construction of new questionnaires and meaningful comparison across heterogeneous studies and diverse contexts. Built upon semantic technologies and established ontology design patterns, the system’s architecture ensures an interoperable representation of survey components and their relationships to social variables. In general, the proposed system addresses a critical limitation of current research infrastructures in the social sciences: the difficulty of designing FAIR-by-design survey instruments capable of identifying, operationalising and systematically comparing relevant social variables across countries and over time.

Keywords. Ontology, Knowledge Graph, Social Sciences Data, Survey Data Integration, Data Reuse

1. Introduction

Researchers are exposed to an ever-growing amount of information which, within the traditional document-oriented workflow, presents structural limitations: many scientific outputs are published in forms that are insufficient to enable their reuse in further research or their full reproducibility. The adoption of knowledge graphs (KGs) within the social sciences domain represents a significant step towards the creation of FAIR by design surveys formally modelled surveys [1,2]. In fact, KGs provide a tool for building systems that bridge existing data and survey management. Additionally, by offering a formal representation of entities and

their semantic relationships, they enable the integration of heterogeneous datasets and support systematic information linking across disciplines and institutions. This capability is particularly relevant to the principle of reuse: KGs make it possible to discover, compare, and connect data originally produced for different research purposes, thus fostering cumulative, transparent, and collaborative forms of inquiry.

Therefore, this work relies on KGs to semantically represent and harmonise different studies, facilitating the comparison of research projects and the efficient identification of survey questions previously used to investigate specific variables. As a consequence, new studies can be designed as interoperable by design, drawing on a shared and structured repository of concepts, items, and metadata. Our evaluation shows that the system reduces the time required for survey design while providing an effective and usable instrumented environment for connecting, reusing, and extending existing studies. The system consists of three main components:

- a knowledge graph modelled on the DDI-RDF Discovery Vocabulary [3] and the Survey [4,5];
- an automated data-import module that extracts and formalises surveys from LimeSurvey;
- Survey Builder, a web interface that allows users to search, edit, and create questionnaires directly within the knowledge-graph ecosystem.

This work originates from the need to support researchers in their activities within the context of the Consortium of European Social Science Data Archives (CESSDA), a pan-European research infrastructure that coordinates and strengthens data services for the social sciences. CESSDA provides harmonised metadata standards, controlled vocabularies, and best practices for data management and curation, all of which are essential for ensuring interoperability and FAIR compliance. Its central role in facilitating data discoverability, reuse, and cross-national comparability makes it a key reference point for our work, particularly in the design of interoperable survey representations and the alignment of our knowledge graph with established European standards. As discussed in [6], the way survey questions are formulated substantially affects the quality and reliability of the collected data. This motivated us to investigate whether enabling researchers to search, compare, and reuse questions and response options from previous studies could improve survey quality. Reusing standardized variable names and response categories facilitates questionnaire design, enhances dataset comparability, promotes interoperability and reusability, and supports cross-study analyses.

Existing approaches typically provide either semantic models (e.g. DDI, Survey Ontology) or repositories of questionnaire metadata (e.g. Euro Question Bank), but not an integrated workflow supporting semantic survey import, ontology-based representation, reusable questionnaire components, and survey reconstruction through a knowledge graph.

The remainder of the paper is organised as follows. Section 2 discusses related work. Section 3 presents the system architecture, including the ontology design and implementation. Section 4 describes the evaluation methodology and results. Finally, Section 5 outlines the conclusions and directions for future research.

2. Related work

2.1. Current Survey Publishing Practices and the EQB Initiative

For researchers, whether questionnaire designers or survey methodologists, integrating and analysing survey data is often time-consuming, as search, discovery, and retrieval require navigating multiple data sources and heterogeneous file formats. Several European infrastructures—such as the European Social Survey (ESS) [7], the Leibniz Institute for the Social Sciences (GESIS) [8], and the UK Data Service (UKDS) [9]—publish structured questionnaires alongside study-level metadata. However, these instruments are typically distributed in formats such as PDF, commonly used during survey authoring but misaligned with the FAIR principles. Non-semantic formats hinder interoperability and reuse, and although PDFs may be accessible, their discoverability and machine readability remain limited.

CESSDA has taken steps toward greater interoperability through the Euro Question Bank (EQB) project [10]. EQB aggregates metadata from several European archives and enables users to search, browse, and download variables, questions, and related materials. It is built upon the Data Documentation Initiative (DDI)¹, an XML-based standard for describing surveys, data files, and study metadata, supporting both DDI-Lifecycle and DDI-Codebook. Unlike many existing infrastructures, EQB also publishes survey questions in DDI format. Nonetheless, it remains under development, currently hosts metadata from a single project, and its XML foundation limits direct integration with semantic technologies.

2.2. Semantic Approaches to Survey Modelling

Improving reusability and enabling cross-national comparison requires making survey data and metadata available in RDF, the dominant standard for interoperable structured information. RDF facilitates explicit linking within and across datasets, enabling the construction of rich knowledge graphs. As a result, semantic models for representing survey instruments and for transforming non-RDF sources into machine-actionable formats have attracted growing interest.

Within the DDI ecosystem, the DDI-RDF Discovery Vocabulary (Disco) [11] provides an RDF representation for a core subset of DDI metadata and supports data discovery across Linked Open Data. Its coverage of questionnaire structures, however, remains limited. Disco has been applied, for example, in the GESIS Panel knowledge graph, which integrates heterogeneous sources into a unified RDF model exposed through SPARQL endpoints [12].

The Survey Ontology [4] offers an OWL-based, component-oriented model for representing questionnaire structures and respondent answers. Originating in the Coney conversational survey system [13], it has supported research on adaptive, hypothesis-driven survey administration [5]. Nevertheless, it is tailored to conversational formats, presupposing sequential question flows and providing limited support for alternative organisational structures such as thematic grouping or branching logic. Its emphasis on text-based interaction constrains the representa-

¹<https://ddialliance.org/>

tion of other modalities, and the model lacks semantic mechanisms for explicitly linking questions and answers beyond the generic `sur:leadsTo` relation.

Overall, existing ontologies do not offer a sufficiently modular framework for reusing components across questionnaires, nor do they address the extraction challenges required to convert unstructured survey documents into structured semantic representations.

2.3. Survey Authoring Tools

Widely used survey-authoring platforms, such as Qualtrics and SurveyMonkey, provide mature environments for designing and deploying questionnaires, including advanced logic and integration features. However, they prioritise operational authoring and data collection, with limited support for ontology-driven design or knowledge-graph workflows.

Among open-source solutions, LimeSurvey is widely adopted and frequently used within the community of *Fostering Open Science in Social Science* (FOSSR) project², which is funding this work. It offers a stable editing interface, structured import/export functionalities, and well-documented APIs, making it a practical integration point for prototyping semantic workflows. Yet, mainstream authoring systems focus on questionnaire creation rather than semantic modelling, while question-bank services prioritise search and reuse over semantic interoperability.

Despite increasing interest in structured, standards-based representations of survey instruments, workflows grounded in knowledge graphs or ontology-driven assembly are not yet well supported in current authoring ecosystems. This gap motivates the architecture introduced in the next section.

3. System architecture

The proposed software architecture adopts two complementary architectural styles, i.e. Representational State Transfer (REST) [14] and Component-Based Architecture, to ensure modularity, scalability, and interoperability [15]. In our system, RESTful APIs mediate communication between the Survey Builder interface and the LimeSurvey Semantic Converter, enabling the seamless transfer of survey data for processing and semantic transformation. Instead, the Component-Based style structures the system as a set of loosely coupled, highly cohesive modules, each encapsulating dedicated functionality. This design supports independent development and maintenance, while enabling the replacement or enhancement of modules without affecting the rest of the system.

The architecture comprises four main components, illustrated in Figure 1 and available on GitHub³. These are: (i) the *Knowledge Graph*, which stores survey structures and metadata as RDF; (ii) the *Survey Builder* that enables the creation of new surveys by reusing and composing existing questions in the knowledge graph according to given criterion; (iii) the *Semantic Controller*, which transforms survey data from the survey management platform into RDF; (iv) the *Survey*

²<https://www.fossr.eu/>

³<https://github.com/fossr-project/survey-builder>

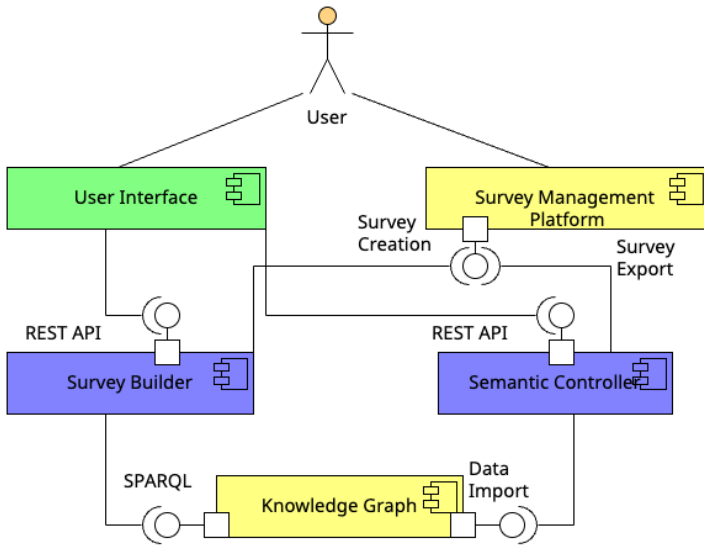


Figure 1. System architecture. External systems are shown in yellow, back-end components in blue, and the front-end in green.

Management Platform, which is a conventional system (e.g. LimeSurvey [16]) to support survey deployment, response collection, and administration; (v) the *User Interface*, a web-based graphical interface that enables users to interact with both the Survey Builder and Semantic Controller. This modular architecture is not tightly coupled to a specific survey management platform, allowing alternative survey tools to be adopted while preserving the semantic conversion and knowledge graph capabilities.

In the next subsections we provide more details for each component comprising the ontology (i.e. Survey-ont) that models the data in the knowledge graph.

3.1. Ontology and knowledge graph

The *Knowledge Graph* is the module of the architecture responsible for managing the semantic data. In our case, it is based on Ontotext GraphDB⁴. It supports the upload of RDF graphs, the execution of SPARQL queries, and the separation of ontology definitions from survey instances through named graphs, enabling versioning, traceability, and integration with external semantic resources. The semantic data are represented in the knowledge graph according to the *Survey-ont*. Survey-ont, is the ontology we modelled to represent knowledge about surveys following a pattern-based approach grounded in the eXtreme Design (XD) methodology [17]. XD is an iterative and collaborative approach that emphasises the identification of competency questions (CQs) [18], the reuse of Ontology Design Patterns [19] (ODPs), and the continuous validation of the ontology against real-world requirements.

⁴<https://graphdb.ontotext.com/>

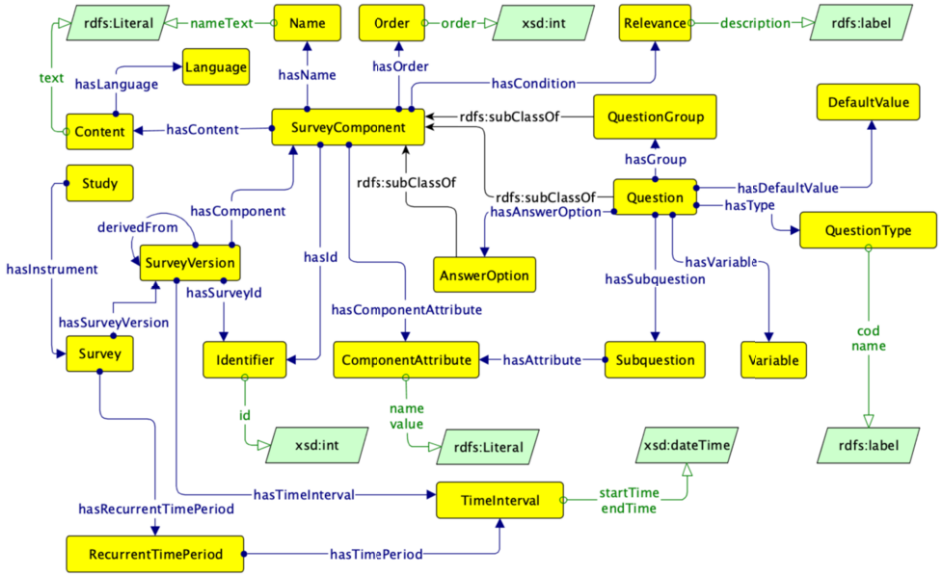


Figure 2. Graffoo diagram depicting the Survey-ont schema.

Domain Analysis and Requirements Elicitation. The ultimate goal of our ontology and knowledge graph is to support questionnaire reuse, comparison, and harmonisation by maximising FAIRness. For example, a scenario can be focused on a researcher that intends to investigate the use of tablets among older adults. Rather than designing a new questionnaire from scratch, the researcher might benefit from questions gathered from existing surveys that have already addressed similar topics in previous studies, such as “How often do you use a tablet?”, “How confident do you feel when using a tablet?”, or “What do you use your tablet for?”. These questions are associated with additional characteristics that might include: (i) the questionnaire and question group in which a question appears; (ii) the variable associated with the question; (iii) the question type (e.g., single choice, multiple choice, free text); and (iv) the complete set of answer options and their ordering.

This enables researchers to compare how similar questions have been operationalised across different surveys. For example, tablet usage may be measured through a frequency scale (e.g., “Never”–“Daily”), a proficiency scale (e.g., “Very difficult”–“Very easy”), or a binary behavioural indicator (e.g., “Yes”–“No”). Although these questions target related concepts, they capture different dimensions of technology use and therefore provide alternative operationalisations of the same research domain. Moreover, questions originating from different surveys can be connected through common variables and shared conceptual categories, enabling researchers to identify semantically related items even when their wording differs. This capability facilitates the comparison of survey instruments and supports the harmonisation of questionnaire components across studies. Instead of manually inspecting multiple questionnaires, researchers can rapidly identify reusable survey components, compare alternative formulations, and select the most appropriate questions and response scales for a new study. By exposing questions, vari-

ables, groups, and answer scales as interconnected semantic entities, the knowledge graph provides the semantic infrastructure required for questionnaire comparison, component reuse, and survey harmonisation.

Additionally, to capture the conceptual structure of surveys in the social sciences, we conducted a systematic analysis of the LimeSurvey internal data model and the survey design practices commonly adopted in research projects. We selected LimeSurvey as, in the implementation of our architecture, it serves as the reference survey management platform. This analysis focused on identifying the essential components of surveys, such as studies, surveys, survey versions, question groups, questions, sub-questions, answer options, variables, and temporal structures, as well as the relationships among them. Representative survey instruments from multiple social-science projects were inspected to extract recurring modelling patterns. Based on our analysis and following the XD principles, we formulated a comprehensive set of user stories that were used to derive competency questions (CQs) aimed at guiding the modelling process. The CQs address structural aspects (e.g., survey versions, component ordering), logical aspects (e.g., relevance conditions), multilingual representation, and data-collection semantics (e.g., variables and default values). Table 1 reports the full list of identified CQs, whilst the user stories can be found in the project's GitHub repository⁵. Figure 2 presents the Graffoo diagram [20] that illustrates the complete ontology we designed.

Reuse of Ontology Design Patterns. To ensure semantic interoperability and modelling robustness, we reused well-established patterns from the Ontology Design Patterns (ODPs) repository. In particular, we adopted the following ODPs: (i) *PartOf*⁶ for representing survey components and their containment hierarchy; (ii) *Classification*⁷ for modelling question types and language assignments; (iii) and *Versioning* for representing survey versions, their identifiers, and their derivation relations. Pattern reuse ensured design consistency and alignment with semantic-web best practices.

Iterative Refinement, validation and release. The modelling process consisted of repeated cycles of refinement, where candidate ontology fragments were validated against (i) the competency questions; (ii) concrete survey examples implemented in LimeSurvey; and (iii) domain requirements emerging from social-science research workflows. Each modelling iteration addressed the gaps identified in previous cycles and incorporated additional structural or semantic requirements. This iterative approach ensured that the ontology faithfully captures the complexities of the survey instruments, supports multilingual content, models temporal structures for longitudinal studies, and enables robust reuse of the survey components. Then, the resulting ontology is encoded in OWL and published openly on

⁵<https://github.com/fossr-project/survey-builder/blob/master/ontology/Ontology%20requirement.pdf>

⁶<http://ontologydesignpatterns.org/cp/owl/partof.owl>
<http://ontologydesignpatterns.org/cp/owl/partof.owl>

⁷<http://www.ontologydesignpatterns.org/cp/owl/classification.owl>

Table 1. Competency Questions for the Survey

ID	Competency Questions
<i>1. Survey Structure and Organization</i>	
CQ1	What surveys are associated with a specific study?
CQ2	What are the different versions of a particular survey?
CQ3	Which survey version was a particular version derived from (version history)?
CQ4	What is the recurrent time period for longitudinal surveys?
CQ5	What are the start and end dates for each collection period?
<i>2. Survey Components</i>	
CQ6	What components does a survey contain?
CQ7	What is the order of components within a survey?
CQ8	What are the identifiers associated with each survey component?
CQ9	What attributes does a specific component have?
CQ10	What is the name and value of a component attribute?
<i>3. Questions, question groups and Answer Options</i>	
CQ11	What questions belong to a specific question group?
CQ12	What is the condition (relevance) for displaying a particular question or question group?
CQ13	What answer options are available for a given question?
CQ14	What answer options are available for a given question group?
CQ15	What is the default value for a specific question?
CQ16	What type does a question have?
CQ17	What variable is associated with a question?
CQ18	Does a question have subquestions, and if so, what are they?
CQ19	What is the order of answer options for a question or question group?
<i>4. Variables and Data Collection</i>	
CQ20	What variable name is associated with each question?
CQ21	What variables are collected in a survey?

GitHub⁸ with a Creative Commons Attribution 4.0 International license. It provides a machine-actionable, FAIR-aligned semantic representation of surveys designed for integration into knowledge graph infrastructures and automated survey construction workflows.

3.2. Semantic Controller

To enable the systematic import of LimeSurvey questionnaires into the ontology and to support the creation of new FAIR-by-design surveys, we designed and implemented a software system equipped with a user interface that covers the entire data lifecycle: from the acquisition of raw survey data, to their transformation and publication within the knowledge graph, and finally to their querying through SPARQL to support the creation of new surveys.

The Semantic Controller counts of different modules. The first module is the *Survey Manager*. For the purposes of the project, we rely on LimeSurvey REST APIs, which, through the RemoteControl API⁹, enable the retrieval of structured representations of survey elements, including groups, questions, multilingual con-

⁸OWL ontology available at: <https://github.com/fossr-project/survey-builder/blob/master/ontology/limesurvey.rdf>

⁹https://www.limesurvey.org/manual/RemoteControl_2_API

tent, relevance conditions, and response options. The client preserves the hierarchical organisation and ordering of each questionnaire, exporting the resulting data in JSON or CSV formats ready for semantic processing.

The second module is the *Converter*, which applies mapping rules based on RML [21] declarative mappings to transform the exported data into RDF triples compliant with the Survey-ont. The converter generates semantic entities such as question types, variables, answer options, and relevance conditions, while reconstructing the hierarchical structure of the questionnaire (study, survey, survey version, and components), thus ensuring interoperability and semantic consistency.

3.3. Survey Builder

The *Survey Builder* enables users to search, modify, and reuse questions, question groups, and answer options available as RDF in the knowledge graph and previously defined in other surveys. This component supports the reuse and exploration of solutions adopted by other researchers when studying specific aspects of reality, thereby facilitating the analysis, management, and design of new surveys in LimeSurvey.

3.4. User interface

The semantic conversion process and its outcomes are made available to end users through a web-based interface developed in Python. The application enables interaction with both the Survey Builder and the Semantic Controller. In particular, the interface allows researchers to upload the knowledge graph and survey data from JSON and CSV files via the Semantic Controller, and to create new questionnaires using the data stored in the knowledge graph through the Survey Builder. Through the Survey Builder, users can select and recombine existing components stored in the graph, as well as export data to the survey management platform.

The interface was developed following an agile methodology, with the support of a Large Language Model, namely Anthropic's Claude¹⁰, and inspired by Extreme Programming principles. It is structured around three main views. Figure 3 shows the user interface to interact with the Survey Builder. The latter is divided into the left section for browsing and managing questionnaires components (i.e. groups, questions, etc.) and the right section is used for their visualisation. Users can view a questionnaire under construction in different representations, including a hierarchical view as well as JSON and CSV formats. At the bottom of the panel, additional actions are available, such as deleting selected elements to restore an initial configuration. It is also possible to export the created questionnaire directly to the Management Survey Platform, in our case Limesurvey.

A typical workflow involves connecting to the knowledge graph via the triplestore, browsing and searching question groups and individual questions, selecting relevant items for a new questionnaire, modifying texts or group structures when needed, previewing and validating the structure, and finally exporting the survey to LimeSurvey. Once exported, a questionnaire can be distributed.

¹⁰<https://claude.ai/>

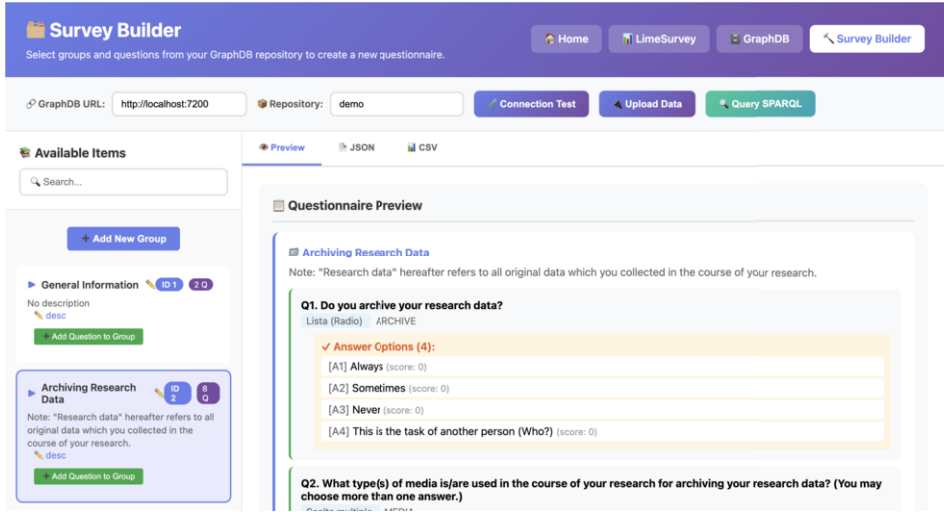


Figure 3. The User interface.

4. System Evaluation

To validate ontology and the system we conducted two evaluation activities: (i) a functional evaluation of the ontology and (ii) a usability evaluation of the system. The first evaluation focused on assessing whether the ontology correctly satisfies the modelling requirements and supports the intended competency questions, while the second aimed at analysing the usability and user experience of the Survey Builder platform during two realistic tasks. The ontology evaluation was particularly important because the system enables interaction with, and storage of, survey data through a knowledge graph. Accordingly, assessing how such knowledge is modelled within the knowledge graph is crucial for determining the system's ability to provide an appropriate level of semantic representation, consistency, interoperability, and support for survey management and reuse. The second evaluation was equally important because it assessed how end users may benefit from using the system to create FAIR-by-design surveys, particularly in terms of usability.

4.1. Ontology Evaluation

We evaluated the ontology following state-of-the-art methodologies [22,23], we evaluated the ontology along the functional dimension. Namely, competency questions (CQs) were used as unit tests. The unit testing phase focused on validating the ontology against the defined competency questions. For each CQ, a SPARQL query was designed¹¹ and executed against a dedicated set of synthetic test dataset specifically created for validation purposes. This dataset, available on

¹¹The designed SPARQL queries are available, along with their corresponding CQs, in the project's GitHub repository: The CQs converted to SPARQL are available in the project's GitHub repository <https://github.com/fossr-project/survey-builder/blob/master/ontology/survey-tests/sparql-query.sparql>

GitHub¹², includes representative individuals and relationships covering the main modelling scenarios addressed by the ontology. The unit testing was performed through OWLUnit [24].

4.2. Usability Evaluation

This evaluation was conceived as a pilot study: a preliminary small-scale investigation conducted to assess the feasibility of the proposed methodology and to obtain initial evidence before a larger-scale study. We assessed whether the proposed application improves the efficiency and usability of survey reconstruction workflows compared to LimeSurvey, which served as the baseline platform. Participants were asked to complete a questionnaire reconstruction task on both platforms and subsequently rate their experience using the System Usability Scale (SUS) [25]. The SUS is a widely adopted metric for assessing system usability: it is technology-independent and has been shown to yield reliable results even with relatively small sample sizes [26], as in our case 12 participants. Moreover, it incorporates a two-factor orthogonal structure that enables the scale to be scored along independent Usability and Learnability dimensions [27].

Study design and participants. We adopted a within-subject experimental design [28] in which each participant performed two reconstruction tasks: one using our application and one using LimeSurvey. To mitigate order and learning effects, the order of the platforms to use during the experiment was balanced across participants, and each participant worked on a different questionnaire-platform pairing.

The usability study was conducted in two phases involving a total of twelve participants, with six participants in each phase. The first phase evaluated the initial version of the system, whereas the second phase was carried out after extending the Survey Builder interface to support the reconstruction of complete response sets for closed-ended questions (e.g., all answer options in multiple-choice questions). This extension enabled the assessment of a more advanced and comprehensive level of questionnaire reconstruction functionality.

Participants were recruited among researchers with little or no prior experience with LimeSurvey. This choice reduced expertise-related bias and ensured a fairer comparison of task completion times across the two platforms.

Tasks. Participants were first introduced to both platforms and their core functionalities. They were then asked to reproduce two ten-item questionnaires derived from an original thirty-three-item source instrument already stored on both platforms. *Task A* required participants to reproduce a questionnaire containing modified group labels and item texts, thus requiring them to edit and adapt the material. Instead, *Task B* consisted in reproducing a questionnaire containing a direct subset of the original items. For each platform involved in the study, participants were provided with: (i) a PDF copy of the target questionnaire; (ii) access to the thirty-three-item source questionnaire stored in the repository; and

¹²<https://github.com/fossr-project/survey-builder/blob/master/ontology/survey-tests/survey-tests.ttl>

(iii) instructions to reconstruct group and question structure and adjust item text when required.

In case LimeSurvey was used as the tool, participants duplicated the source questionnaire and applied the necessary modifications within the platform. In the new application, by contrast, they loaded the corresponding data from the triple-store and reconstructed the questionnaire through the Survey Builder interface. Two demonstration videos of the tasks performed using both systems are available for viewing on Open Science Framework¹³ (OSF).

Measures and results. We collected two indicators: (i) efficiency, measured as task completion time; (ii) perceived usability, measured through the System Usability Scale (SUS), administered independently after the task on both platforms. Figures 4a and 4b report completion times for both reconstruction tasks, separately for the two study phases. Across both phases, the Survey Builder consistently outperformed LimeSurvey in both Task A and Task B. In Phase 1, the Survey Builder achieved lower mean completion times across both tasks, and a similar pattern was observed in Phase 2, where the new system again enabled faster questionnaire reconstruction than LimeSurvey. These results suggest that structured components and integrated search and reuse mechanisms reduce manual editing effort and interface navigation overhead, which in LimeSurvey typically requires navigating multiple panels and adjusting item properties individually. Participants also benefitted from the integrated search and reuse functionality of Survey Builder, particularly in Task B, where the reuse of existing components was more straightforward.

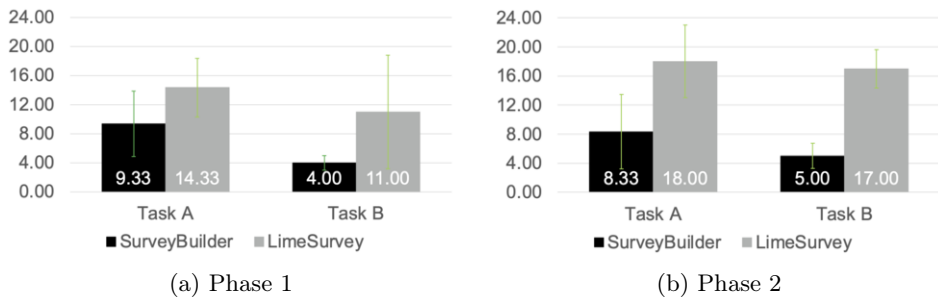


Figure 4. Task completion times across the two study phases. Green vertical bars indicate the standard deviation.

Figures 5a and 5b report the results obtained through the SUS for SurveyBuilder and LimeSurvey during both phases. These results further reinforce our findings. Considering that a SUS score of 68 is generally regarded as the average usability benchmark for interactive systems [26], the Survey Builder achieved a markedly higher average score of 86.25 in Phase 1, placing it well within the “excellent” usability range. By contrast, LimeSurvey obtained an average score

¹³The videos are available at https://osf.io/8qkyz/overview?view_only=166f71013cf14ec38772dff16203c0e1.

of 54.58, corresponding to the lower end of the “OK/acceptable” range. Similar trends were observed for Phase 2, where the Survey Builder achieved an average SUS score of 79.58, while LimeSurvey obtained an average of 55.

The breakdown of Learnability and Usability scores shows a consistent pattern across both study phases, with the Survey Builder outperforming LimeSurvey in all dimensions. Both Learnability and Usability subscales consistently favour the Survey Builder, confirming the trends observed in the overall SUS scores.

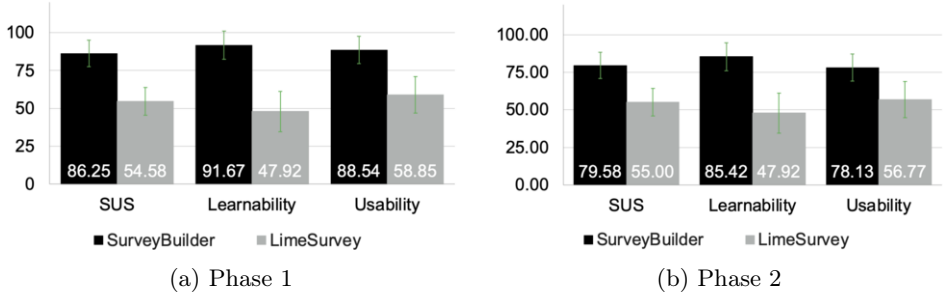


Figure 5. SUS, Learnability, and Usability scores across the two study phases. Green vertical bars indicate the standard deviation.

differences between Survey Builder and LimeSurvey across SUS, Learnability, and Usability scores. In particular, all pairwise comparisons yielded $p < 0.01$, supporting the conclusion that participants consistently perceived the Survey Builder as more usable than LimeSurvey.

Furthermore, several participants informally reported that LimeSurvey required more exploratory actions and menu navigation, whereas the Survey Builder provided a more guided workflow centered on component reuse. Although some participants initially needed to familiarize themselves with the semantic search panel, this did not appear to negatively affect task performance. Their feedback further suggests that increased familiarity with the interface could improve both efficiency and perceived usability.

Discussion. The results of this pilot study suggest that the proposed application streamlines survey reconstruction workflows through semantic technologies and knowledge graph-based reuse mechanisms. Quantitative findings indicate improvements in both efficiency and perceived usability, enabled by structured components and integrated search functionality that reduce the manual editing and navigation overhead typical of traditional survey authoring tools such as LimeSurvey. The consistency observed across participants, in terms of both completion times and SUS ratings, further supports these preliminary findings.

Several considerations are relevant for interpreting these results. First, the study involved participants with limited prior experience with LimeSurvey, which reflects a realistic onboarding scenario for many users but may not fully capture performance differences that could emerge with experienced LimeSurvey users. Second, although the Survey Builder supported more efficient reconstruction workflows, some of its semantic mechanisms may initially be less intuitive for

users unfamiliar with knowledge graph-based systems. Broader adoption would likely benefit from additional onboarding support and usability refinements aimed at improving the transparency of semantic functionalities for non-expert users.

Finally, these results suggest several promising directions for extending this approach. The semantic modelling of questionnaires may support more advanced forms of automation, such as recommending reusable components, detecting inconsistencies with source instruments, or providing adaptive interfaces tailored to different user roles. Integration with additional knowledge graphs or domain-specific ontologies may further enhance the precision of search and reuse mechanisms.

In summary, despite the limited sample size, this pilot study provides encouraging evidence that knowledge graph-based reuse of questionnaire components can reduce manual effort while improving workflow efficiency and perceived usability.

5. Conclusions and future work

In this work, we presented a system for constructing interoperable, reusable, and semantically enriched questionnaires, enabling both automated reuse and user-driven adaptation while preserving semantic consistency and traceability.

The usability evaluation suggests that questionnaire reconstruction with the proposed application is more efficient and better perceived than LimeSurvey. Across both study phases, participants consistently completed tasks faster and reported higher SUS scores, indicating that semantic support can streamline reconstruction workflows. Nevertheless, these findings should be validated on a broader population, including experienced survey designers. From a system perspective, the platform already supports survey structure modelling and answer option representation. However, more advanced features, such as complex conditional logic and richer survey orchestration, represent natural directions for further development.

Additionally, future work will focus on extending the platform's capabilities and interoperability. Key directions include integrating additional European question banks to expand the pool of reusable components, improving export functionalities for survey items (including metadata, multilingual content, and structural constraints), and supporting panelist selection for targeted survey deployment. We also aim to strengthen compatibility with established social science standards through alignment with DDI 3.3, and to explore semantic and AI-based techniques, including LLM-assisted transformations, for adaptive modification and harmonisation of survey instruments.

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Declaration on generative AI

In the preparation of this work, the authors used GPT-5 in order to: Grammar and spelling check. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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Knowledge Graph Construction, Integration & Interoperability

Declaratively Populating Wikibases with WBML

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Abstract. Purpose: This paper addresses the lack of a general declarative approach for generating Wikibase content from heterogeneous data sources. Existing declarative mapping approaches mainly target Resource Description Framework (RDF), while Wikibase-oriented solutions are often limited in scope or procedural in nature. We therefore propose the Wikibase Mapping Language (WBML), a declarative mapping language tailored to the Wikibase statement model.

Methodology: WBML was designed by reusing selected RML concepts for logical sources and iterations, while introducing Wikibase-specific constructs. We implemented a Python prototype that translates WBML into intermediate RDF Mapping Language (RML) and RDF representations, executes them with Morph-KGC, and ingests the results into Wikibase.

Findings: The study shows that declaratively generating Wikibase statements from heterogeneous sources is feasible. We also reported on a demonstration using complex JSON and a simpler, real-world use case for our institution, using CSV files. WBML supports the generation of entities, fingerprints, statements, qualifiers, references, and ranks.

Value: This work provides a basis for declarative construction of Wikibase knowledge graphs from heterogeneous sources. It demonstrates how established ideas from the RDF mapping community can be adapted to other types of knowledge graphs. Future work includes supporting newer RML features not yet supported by reused components and managing updates.

Keywords. Wikibase, KG Construction, Declarative ETL

1. Introduction

Knowledge Graphs (KGs) have become an important technology (and technology stack) for modern data management, and Wikibase², the open-source software underlying Wikidata [1], has emerged as a widely adopted platform for building them, spanning domains from public administration [2] to life sciences [3]. The authors in [2] have also presented several other such bespoke deployments of Wikibase. As Wikibase deployments grow, so does the need for tooling to populate instances from the heterogeneous data sources that organizations typically work with.

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²<https://wikiba.se/>

As we will discuss in Section 2, existing approaches have limitations. Declarative languages such as the Relational Databases to RDF Mapping Language (R2RML) and the RDF Mapping Language (RML) target RDF and do not naturally accommodate Wikibase’s “statement-centric” model, in which each statement can carry qualifiers, references, and ranks. And Wikibase also has its own approach for generating opaque identifiers for things (called Items) and properties. And Wikibase-specific tools either rely on “low-level” operational syntax or require custom imperative code. Only T2WML [4,5] offers the closest declarative solution, but it is limited to tabular data and requires that item and property identifiers be known in advance.

To address this gap, we propose WBML (Wikibase Mapping Language), a declarative mapping language that generates Wikibase statements from heterogeneous sources, including JSON, CSV, and SQL. WBML reuses RML’s abstractions for logical sources and iterations while introducing dedicated mapping constructs for entities, fingerprints, statements, qualifiers, references, and ranks. We implement a Python prototype and demonstrate its capabilities on a fairly complex dataset and a real-world institutional use case. Both datasets are presented in this paper, as the second use case relies on simpler data to be ingested into a Wikibase instance.

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 presents the WBML design. Sections 4 and 5 describe execution and implementation. Section 6 presents the demonstration and use case. Section 7 discusses the approach and its limitations, and Section 8 concludes with directions for future work.

2. Related Work

2.1. Wikibase as RDF

RDF graphs model knowledge as sets of triples that connect resources and values via properties, optionally organized in (named) graphs. In Wikibase, however, the data model³ that conceptually represents the graph is statement-centric: entities are described through statements that not only express a claim (property and value for an entity), but may also include qualifiers, references, and ranks. Thus, where the unit of RDF is a triple or quad, Wikibase was conceived to be more fine-grained and to support statements about statements.

While not important for this paper, it is interesting to note that the RDF representation of Wikibase instances is fairly convoluted.⁴ Wikibase SPARQL endpoints provide both “simple” triples for the claims, and triples representing the reified statements with subjects, predicates, objects, references, and qualifiers.

2.2. RDF Generation

A large body of work has addressed the problem of declaratively generating RDF from existing data sources. [6] The goal is to describe, at the mapping level, how data in a source should be transformed into RDF triples using a chosen vocabulary, rather than implementing the transformation imperatively in custom code. This problem is closely

³<https://www.mediawiki.org/wiki/Wikibase/DataModel>

⁴https://www.mediawiki.org/wiki/Wikibase/Indexing/RDF_Dump_Format/pl

related to Wikibase statement generation, since both involve expressing how source data should be transformed into graph-structured knowledge, but using different graph types. The goal of this section is not to provide an overview of the state of the art in RDF generation; an excellent overview is presented in [6], but to present some important initiatives and their rationale.

R2RML⁵ is the W3C Recommendation for expressing customized mappings from relational databases to RDF datasets. It provides a declarative mechanism for defining how rows in relational tables are transformed into RDF resources and triples, and its mappings are themselves represented as RDF graphs.

RML [7,8] extends R2RML to heterogeneous sources, including CSV, JSON, XML, and databases, enabling the transformation and integration of RDF from multiple sources into a single mapping. While conceived for generating RDF, its abstractions for source access, iteration, and term construction provide useful building blocks for a declarative Wikibase generation pipeline. YARRRML [9] is a human-readable serialization for expressing declarative mappings that can be translated into RML. Its main contribution is not a new target model, but making mappings easier to write and maintain.

SPARQL-Generate [10] extends SPARQL's syntax with source-specific generator functions to produce RDF directly from non-RDF sources. Facade-X [11], more recently, takes a different approach; it first projects a source into a generic RDF-shaped meta-model, which unmodified SPARQL 1.1 CONSTRUCT queries can then transform into the target graph. Despite this difference, both require that the source be "queryable" as RDF before generation can occur. This works well when the target itself is RDF, since the query language and the target data model coincide.

2.3. (Declarative) Approaches to Wikibase Statement Generation and Population

T2WML [4,5], which stands for Table-to-Wikidata Mapping Language, is a mapping language for transforming tabular data into Wikibase statements. T2WML can directly produce Wikibase statements; however, it is not general-purpose and is limited to tabular data, making it less suitable for heterogeneous sources such as JSON, SQL, or NoSQL.

RaiseWikibase [12] addresses the problem of bulk data insertion into Wikibase, focusing on efficient integration and transaction management. Unlike the Wikibase API, RaiseWikibase supports transactions, allowing rollback of multiple write operations. Its main limitation is that it provides abstractions for writing directly to the relational database underlying Wikibase, rather than mapping sources to Wikibase statements. Consequently, RaiseWikibase is better understood as a system-oriented ingestion solution that can be adopted through declarative approaches.

QuickStatements⁶ is used for batch creation and editing of statements in Wikibases. It offers a mechanism for bulk insertion via a compact, tool-specific command syntax, making it well-suited for straightforward import and curation tasks. However, this syntax is relatively low-level and operational, relying on Wikibase identifiers and edit commands rather than on a human-readable declarative description of how source data should be mapped. QuickStatements can encode complex Wikibase statements, but its linear command syntax makes complex statement structures less explicit and harder to

⁵<https://www.w3.org/TR/r2rml/>

⁶<https://www.wikidata.org/wiki/Help:QuickStatements>

manage than in a declarative mapping approach. In other words, QuickStatements, like RaiseWikibase, can be used by declarative approaches as a component for data ingestion.

WikibaseIntegrator⁷ is a Python library that provides an interface to the Wikibase API. Unlike direct database-level approaches, it follows the official application interface, thereby aligning better with the Wikibase approach. Its main drawback is that the integration logic must be implemented programmatically. I.e., it does not provide the declarative abstraction needed to specify mappings of heterogeneous sources onto Wikibase statements.

We furthermore note that only QuickStatements, WikibaseIntegrator, and WBML support qualifiers, references, and ranks. As for T2WML and RaiseWikibase, they do not support ranks.

2.4. Conclusion

Several declarative approaches to KG construction already exist. However, these approaches primarily focus on generating RDF from heterogeneous sources. By contrast, declarative approaches that directly target the Wikibase statement model remain limited; among them, T2WML is a notable example, although it is primarily designed for tabular data and does not support ranks. This observation motivated the study that led to this paper.

3. Wikibase ML

We developed the Wikibase Mapping Language⁸ (WBML), a declarative mapping language designed specifically to map (semi-)structured data sources onto Wikibase instances. The design of WBML was guided by two goals: adopting concepts from existing mapping languages where possible and introducing new constructs where the Wikibase data model requires them.

3.1. The Gist of WBML

The mapping in Figure 1 illustrates the core constructs of Wikibase Mapping Language.

A `wbml:StatementsMap` is the central construct: it binds a logical source (adopted from RML) to a Wikibase entity and declares what statements to generate for that entity. The `wbml:entityMap` identifies each entity via an `entityTemplate` (an internal key resolved to a Wikibase QID). A `wbml:labelMap` produces multilingual labels directly on the entity fingerprint. Each `wbml:statementMap` generates one Wikibase statement.

This mapping generates: 1) one Wikibase item per person as well linking it to an item representing its type, 2) an English label for each item drawn from the “name” field, 3) a “nationality” string statement for each person; and 4) “knows” item statement linking persons to one another, illustrating how WBML handles entity-to-entity relationships across sources (here the same source).

As discussed above, the Wikibase data model differs from RDF. As such, we cannot adopt RML constructs (triples maps, term maps, etc.) for generating Wikibase state-

⁷<https://github.com/LeMyst/WikibaseIntegrator>

⁸<http://w3id.org/dre/wbml#>

ments. Moreover, whereas one generates IRIs and blank nodes for RDF, Wikibase uses opaque entity identifiers (which are generated by a separate component). Rather than “abusing” RML to generate Wikibase statements, we introduce dedicated constructs such as `wbml:statementMap`, `wbml:qualifierMap`, and `wbml:referenceMap` to declaratively capture Wikibase’s data model.

There are parts of RML that we can reuse as is. WBML adopts RML’s notion of logical sources for data iteration. Constructs such as `rml:logicalSource`, `rml:source`, `rml:referenceFormulation`, and `rml:iterator` are used unchanged, allowing WBML mappings to iterate over JSON, CSV, and SQL sources in the same way RML does. This reuse keeps the language familiar to practitioners already knowledgeable in RML and avoids duplicating established solutions.

This choice was not merely one of familiarity. Alternative RDF-generation paradigms based on SPARQL, such as SPARQL-Generate and Facade-X, require that the source be already queryable as RDF. Applied to Wikibase, this would mean first designing an RDF vocabulary to represent Wikibase’s statement-centric metamodel (items, statements, qualifiers, references, and ranks) so that SPARQL could then query and recon-

File

```
[
  { "id": 1, "name": "Bob", "nationality": "British", "knows": 2 },
  { "id": 2, "name": "Jane", "nationality": "American", "knows": 3 },
  { "id": 3, "name": "Zach", "nationality": "Canadian" }
]
```

WBML Mapping

```
<#PersonMapping> a wbml:StatementsMap ;
  rml:logicalSource [
    rml:source "demonstrators/test/input_data/persons.json" ;
    rml:referenceFormulation ql:JSONPath ;
    rml:iterator "$[*]"
  ] ;
  wbml:entityMap [
    wbml:entityTemplate "Person_{id}" ;
    wbml:class [
      a wbml:Class ;
      wbml:entityTemplate "Person" ;
      wbml:label "Person"@en
    ]
  ] ;
  wbml:labelMap [
    wbml:reference "name" ;
    wbml:language "en"
  ] ;
  wbml:statementMap [
    wbml:propertyMap [
      a wbml:Property ;
      wbml:entityTemplate "nationality" ;
      wbml:label "nationality"@en ;
      wbml:propertyType wbml:string
    ] ;
    wbml:valueMap [ wbml:reference "nationality" ]
  ] ;
  wbml:statementMap [
    wbml:propertyMap [
      a wbml:Property ;
      wbml:entityTemplate "knows" ;
      wbml:label "knows"@en ;
      wbml:propertyType wbml:item
    ] ;
    wbml:valueMap [
      wbml:entityTemplate "Person_{knows}"
    ]
  ] .
```

Resulting Wikibase KG

Bob (Q2)			
Item	Discussion	Read	View history Tools
No description defined			
+ to more languages			
Language	Label	Description	Also known as
default for all languages	No label defined	--	
English	Bob	No description defined	
French	No label defined	No description defined	
Dutch	No label defined	No description defined	
American English	No label defined	No description defined	
Statements			
nationality	British	+ 0 references	
knows	Jane	+ 0 references	
instance of	Person	+ 0 references	

nationality (P4)			
Property	Discussion	Read	View history Tools
No description defined			
+ to more languages			
Language	Label	Description	Also known as
default for all languages	No label defined	--	
English	nationality	No description defined	
French	No label defined	No description defined	
Dutch	No label defined	No description defined	
American English	No label defined	No description defined	
Data type			
string			
Statements			

Figure 1. Example of a simple WBML mapping to transform data from JSON into a Wikibase KG.

struct it. This reintroduces exactly the kind of reified, complex representation discussed in Section 2.1, only shifted from Wikibase’s RDF export format to the mapping author’s input vocabulary. Moreover, this new vocabulary should also include a mapping between identifiers and Wikibase identifiers for entities and statements. RML’s logical source and iterator constructs, by contrast, operate directly on the heterogeneous sources. In other words, RML lets us abstract Wikibase’s data model at the mapping level, rather than first representing that model in RDF. A further practical reason for this choice is that RML-IO and RML-LV are well-defined; using those constructs unchanged means adopting an already standardized mechanism.

3.2. WBML Features

We assume the reader is familiar with RML and thus proceed with the description of the WBML constructs. We start with `wbml:EntityMap` that are connected to a Statements Map with the predicate `wbml:entityMap`. Much like `rml:SubjectMap`, an Entity Map is responsible for generating the Wikibase entities used in statements. We chose “EntityMap” as both items (Q) and properties (P) are the subjects of statements.

```
<#PokemonMapping> a wbml:StatementsMap ;
  rml:logicalSource [
    # Omitted for brevity
  ] ;
  wbml:entityMap [
    wbml:entityTemplate "Pokemon_{id}" ;
    wbml:entityType wbml:Item ; # Default wbml:Item, can also be wbml:Property
    wbml:class [
      wbml:entityTemplate "Pokemon" ;
      wbml:label "Pokemon"@en ;
      wbml:description "A fictional creature"@en
    ] ;
  ] ;
# To be continued
.
```

Wikibase generates opaque identifiers for Items of the form Q_n where n is a positive integer. With `wbml:entityTemplate`, we will associate values with such identifiers, which are then stored in an associative array on disk for reuse at later times.

Items can be associated with one or more classes via `wbml:class`. Similarly, the `wbml:entityTemplate` for classes is used to find an existing Item that represents the class. If no such item exists, one is created with the labels, descriptions, and aliases provided in the mapping. Labels, descriptions, and aliases are also updated if they change. Whenever a user knows which are the classes to use, i.e., the Q identifier, then `wbml:classId` can be used to provide an explicit Q-identifier. Finally, Wikibase has no built-in “instance of” relationship, and one is created and stored in the associative map if necessary.

In a Wikibase, an entity can have at most one label and one description per language, whereas multiple aliases may be provided in the same language. The labels, aliases, and descriptions make up an entity's *fingerprint*.⁹

```
wbml:labelMap [ wbml:reference "name.french" ; wbml:language "fr" ] ;
wbml:descriptionMap [ wbml:reference "description" ; wbml:language "en" ] ;
```

Generating statements is fairly straightforward. The generation of a property-value pair for an entity is based on RML-like predicate and object maps, which we call property and value maps, to conform to Wikibase's nomenclature. WBML also allows properties to be declared via an entity template, with the difference that the entity type for property maps is `wbml:Property`, which yields P-identifiers. Value maps can generate values for which data types can be provided and, in the case of an entity, even declare the value's type using the same `wbml:class` construct.

```
wbml:statementMap [
  wbml:propertyMap [
    wbml:entityTemplate "PokedexNumber" ;
    wbml:label "Pokédex number"@en ;
    wbml:propertyType wbml:quantity ;
  ] ;
  wbml:valueMap [ wbml:reference "id" ; wbml:datatype xsd:integer ]
] ;
```

In Wikibase, a qualifier is a property-value pair attached to a statement to refine its meaning, whereas a reference is a set of one or more property-value pairs attached to a statement to record its provenance. While we avoided discussing “snaks” in the previous section, it is worth noting that, in Wikibase terminology, the property-value components that make up a reference are usually called snaks. However, to avoid this terminology, since users are not confronted with it when using a Wikibase, we propose using the term *reference component*. The example below shows how we add a qualifier to a statement (“is hidden ability”, which can be true or false), and one reference that consists of two components: the source-based and the date.

```
wbml:statementMap [
  wbml:propertyMap [ wbml:entityTemplate "HasAbility" ; ] ;
  wbml:valueMap [ wbml:entityTemplate "Ability_{ability_name}" ; ] ;
  wbml:qualifierMap [
    wbml:propertyMap [ wbml:entityTemplate "IsHiddenAbility" ; ] ;
    wbml:valueMap [ wbml:reference "ability_flag" ; wbml:datatype xsd:boolean ]
  ] ;
  wbml:referenceMap [
    wbml:referenceRecord [
      wbml:propertyMap [ wbml:entityTemplate "StatedIn" ; ] ;
      wbml:valueMap [ wbml:constant "pokedex.json" ; ]
    ] ;
  ] ;
]
```

⁹If multiple descriptions are needed for an item, we presume that these would have (potentially) different references and qualifiers, and therefore should be represented with statements. But as this is about Wikibase data modeling, this is out of the scope of this study.

```

wbml:referenceRecord [
  wbml:propertyMap [ wbml:entityTemplate "StatedOn" ; ] ;
  wbml:valueMap [ wbml:constant "2023-01-01" ; wbml:datatype xsd:date ]
]
] ;
]

```

A rank is a statement-level annotation that indicates the relative status or priority of a statement among multiple statements for the same property (preferred, normal, or deprecated). WBML also provides support for a rank map, but we refer the reader to the source code for examples.

Now that we have covered the most important constructs of WBML, we will describe additional constructs and their rationale for design.

3.3. Joining Statements Maps

Similar to RML, we allow Statements Maps to be joined. To this end, WBML adopts the idea of a *Referencing Object Map*, but translates this for Statement Maps. This approach allows us to generate interrelated Wikibase statements from data that comes from different sources.

```

<#ItemMapping> a wbml:StatementsMap ;
# rml:logicalSource [...]
# wbml:entityMap [...]
# ...
wbml:statementMap [
  wbml:propertyMap _:HasItemCategory ;
  wbml:valueMap [
    wbml:parentStatementsMap <#ItemCategoryMapping> ;
    wbml:joinCondition [ wbml:child "type" ; wbml:parent "type" ] ;
  ]
] .

```

This mapping creates statements for items whose value is not read directly from the item source but is obtained by joining to another Statements Map. Concretely, it links each item to its corresponding item category by matching the child field “type” in the item data with the parent field “type” in the category mapping.

3.4. Property Ranges and XSD Datatypes

The reader might have noticed that in WBML, mapping authors use the RDF terms `wbml:item`, `wbml:quantity`, `wbml:string`, ... in the declaration of new properties, but that the value maps use XSD data types. Our rationale is that XSD datatypes are richer and provide more control. For example, integers, doubles, and floats are all mapped onto Wikibase’s quantity type. Using XSD datatypes to generate values allows us not only to validate the data but also to format the lexical representation for ingestion.

3.5. Reusing Existing Entity Identifiers

We have introduced `wbml:entityTemplate`, which may contain one or more references enclosed in curly braces for string generation. If no such references are present, then we

have a constant “key.” Those strings are then used to populate a dictionary that relates those values to P and Q identifiers.

There are cases where one knows the Items and properties beforehand and thus can provide them as constants. To this end, we support using `wbml:propertyId`.

```
wbml:propertyMap [ wbml:propertyId "P42" ; ] ;
```

At the time of writing, we have chosen to use strings rather than IRIs, as the availability of a Wikibase’s SPARQL endpoint depends on whether the extension is provided, and the mapping would rely on convoluted approaches where the IRI needs to be parsed as a string for a particular token to be looked up via Wikibase’s API.

3.6. Reusing IRIs from Existing Vocabularies and Ontologies

WBML also supports the “reuse” of existing vocabularies for generating properties. This was motivated by the fact that there are quite a few prevalent ontological relationships that are not “natively” supported by Wikibase, such as `rdf:type`, `rdfs:subClassOf`, etc. We allow mapping authors to use these IRIs as constants using `wbml:property`.

```
wbml:statementMap [
  # wbml:propertyMap [ wbml:entityTemplate "type" ; ] ;
  wbml:property rdf:type .
  wbml:valueMap [ wbml:entityTemplate "Pokemon" ; ] ;
] ;
```

Currently, the only “built-in” property that is supposed to be supported is `rdf:type`, whose correspondence must be provided as input; otherwise, a new P will be generated for this property. We assume this supports the syntactic sugar offered by `wbml:class`. This supposition also allows mapping authors to create type declarations with references and ranks using a Statement Map.

4. Executing WBML Mappings

A WBML processor requires a connection to a Wikibase instance, a WBML mapping, and, optionally, a dictionary (generated by a previous run) that acts as a look-up file. WBML iterates over records selected by a Logical Source and applies the mapping rules to each record. For every iteration, it generates Wikibase statements and fingerprints. In this sense, the behavior of WBML is similar to that of RML, but there are a couple of important differences.

First, Wikibase instances work with opaque identifiers. We use the values generated by `wbml:entityTemplate` to find an existing entity identifier (P or Q), or create a new one, and relate this identifier to the value. In that sense, the behavior is similar to the R2RML Term Generation Rule for a blank node.¹⁰ In WBML, we return a Q or P-identifier that is unique to the natural RDF lexical form corresponding to the value.

¹⁰See Section 11.2 of <https://www.w3.org/TR/r2rml/>

Secondly, if no look-up file is provided, then new items and properties are generated. This is why we also allow mapping authors to execute the mapping with a flag to look for existing items and properties using exact matching, given a preferred language. Such matches are then stored in the look-up file. Algorithm 1 details how WBML currently manages fingerprints. Indeed, we raise errors when the generated fingerprints are not compliant with Wikibase. When values already exist in Wikibase but differ, we do not update them; instead, we show a warning. This feature is merely a convenience, though declarative approaches to entity reconciliation will be considered for future work.

Algorithm 1 Managing fingerprints of an entity

```

for all generated entities (P or Q) s do
  if |labels for a language| > 1 then
    Raise Error
  end if
  if |descriptions for a language| > 1 then
    Raise Error
  end if
  id ← find id in look-up file
  if id = null then
    if label ≠ null ∧ search_wikibase = true then
      id ← find id with exact match label and preferred language in Wikibase
    end if
    if id = null then
      id ← id of newly created entity
    end if
  end if
  e ← getEntity(id)
  if e already has a (different) label for a language then
    Ignore, show warning
  end if
  if e already has a (different) description for a language then
    Ignore, show warning
  end if
  Add missing labels, descriptions, and aliases
end for

```

As for the statements, we verify their existence for reuse and, where necessary, complement them with ranks and references.

5. Implementation

The implementation of WBML follows from its design choices and partial reuse of RML concepts. The process is illustrated in Figure 2, and the pipeline works as follows:

1. First, WBML mappings are translated into an intermediate RML mapping of SPARQL CONSTRUCT queries implemented in Python using RDFLib. For example, the node template "Pokemon_id" is used to create an RML term map that relies on generating the URN "urn:wikibase:item:Pokemon_id". Here, the important point is that all mappings are conceptually transformed into RML triple

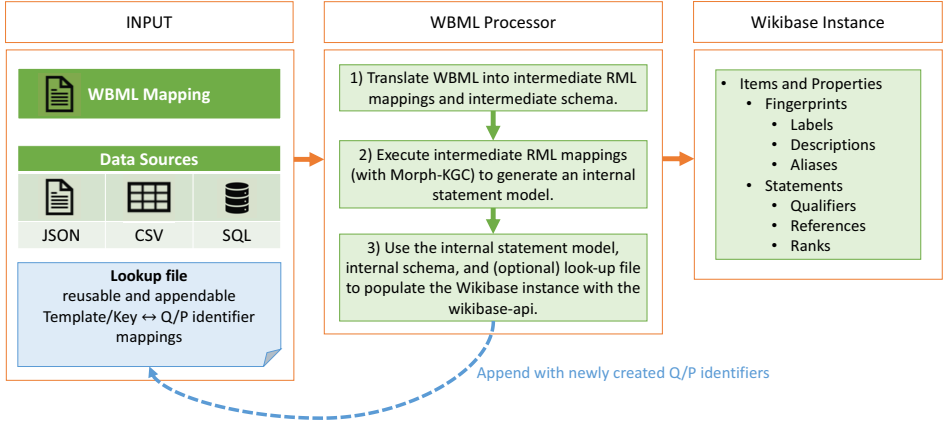


Figure 2. WBML Implementation's Execution Pipeline

maps that generate reified statements, i.e., a triples map for generating instances of statements with a subject, predicate, object, rank, qualifiers, and references.

2. This intermediate RML is not exposed to the user, but allows us to generate RDF with existing RML technologies. In the second step, we use Morph-KGC [13] to execute the resulting mappings and generate RDF in this internal format.
3. Finally, a Python ingestion component, using wikibase-api¹¹, transforms this intermediate output into Wikibase content by creating and updating statements, fingerprints, Items, and Properties where required. This component also maintains and reuses an associative array that stores the Wikibase identifiers generated for keys and templates, thereby ensuring consistent reuse of newly created entities (Items and Properties) across mapping executions.

While WBML is currently executed through an internal translation to intermediate RML and RDF representations, its declarative nature is preserved because these representations are not exposed at the WBML level. Users specify only the correspondence between source data and Wikibase structures. Unlike state-of-the-art approaches, which even declarative approaches rely on prior knowledge of P and Q identifiers, our approach enables users to include ontological information in the mapping (or in a separate graph combined with the mapping).

The prototype, together with demonstration examples, is available on GitHub and released under the MIT License.¹²

5.1. Discussion

Our implementation currently relies on Morph-KGC, mainly because its Python implementation made it convenient to use in a prototype. However, this choice also has its limitations: Morph-KGC supports R2RML and older RML versions, but full compliance with the latest RML specification is not guaranteed. This may become relevant if

¹¹<https://pypi.org/project/wikibase-api/>

¹²<https://github.com/uliege-dre-lab/wbml-impl>

WBML is extended with features such as RML Logical Views (LV) [14], which would allow combining multiple sources into a single view and applying functions over that view. Supporting such extensions is left for future work. Conceptually, adopting RML LV should be trivial, since the logical source would point to an RML Logical View.

6. Demonstration and Use Case

In this section, we report on a demonstration and a use case.

6.1. Pokédex Demonstrator

As a demonstrator, we use data from the Pokédex’s JSON representation. We have chosen this dataset as it allows us to demonstrate practically all of WBML’s capabilities on a complex, multivalued dataset with deeply nested JSON structures.

The dataset covers four interrelated entity types: Pokémon, Moves, Types, and Items. Each Pokémon entry includes multilingual labels (English, Japanese, Chinese, and French), a description, a Pokédex number, base stats such as HP, one or more elemental types, abilities with a qualifier indicating whether the ability is hidden, and evolution chains with qualifiers that capture the evolution condition. Move entries include numeric properties (e.g., accuracy) and a categorical property for move category, alongside a type association. Type entries describe type effectiveness relationships (effective against, ineffective against, no effect against) as item-valued statements linking types to one another. Item entries include multilingual labels, descriptions, an item number, and an item category, all resolved via a join.

The mapping illustrates the range of WBML’s capabilities: item-valued statements, multilingual fingerprints (labels, descriptions), qualifiers on statements, references on statements, and joins via `wbml:parentStatementsMap`. All Wikibase properties and “class” items are declared in the mapping and created on demand, so no manual setup of the target Wikibase instance is required before running the pipeline.

The mapping is available in the repository, and can be run by following the instructions in the README file. The result of running this mapping for a local Wikibase instance can be seen in Figure 3. It is important to note, however, that the entity IDs (Q and P identifiers) may differ across Wikibase instances.

6.2. An Institutional Research Register

At the authors’ institution, the Technology Transfer Office (TTO) aims to build a register of tools, platforms, and equipment, along with their associated research units. The problem is that this information is governed independently by individual research units and laboratories, and this information needs to be integrated using a common model. The goal is to populate a Wikibase instance that serves as an authoritative source, so that when external parties inquire, for example, about the availability of specific equipment for an experiment, the TTO can efficiently direct them to the appropriate unit.

The project proceeds in several phases. The first phase focused on designing a common ontology. Ontology constructions consisted of defining the classes, properties, labels, and descriptions needed for the register. Domain experts “modeled” the ontology in

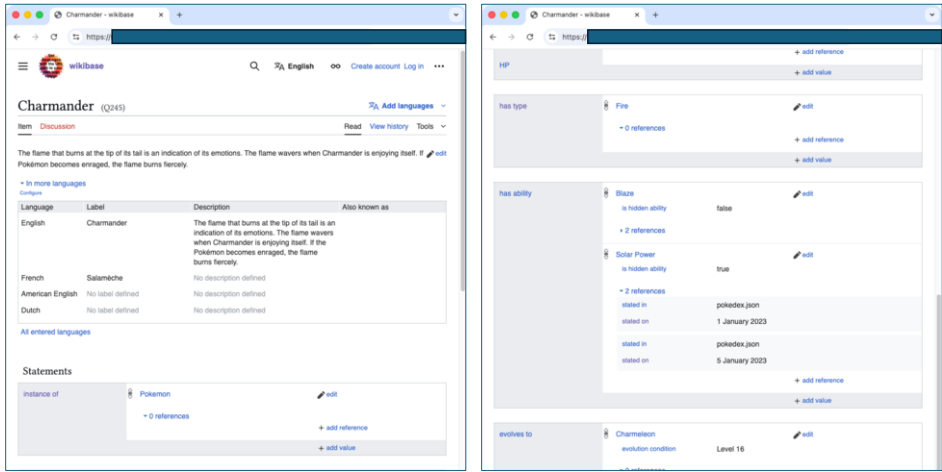


Figure 3. Local Wikibase instance updated after running the Pokédex demonstrator.

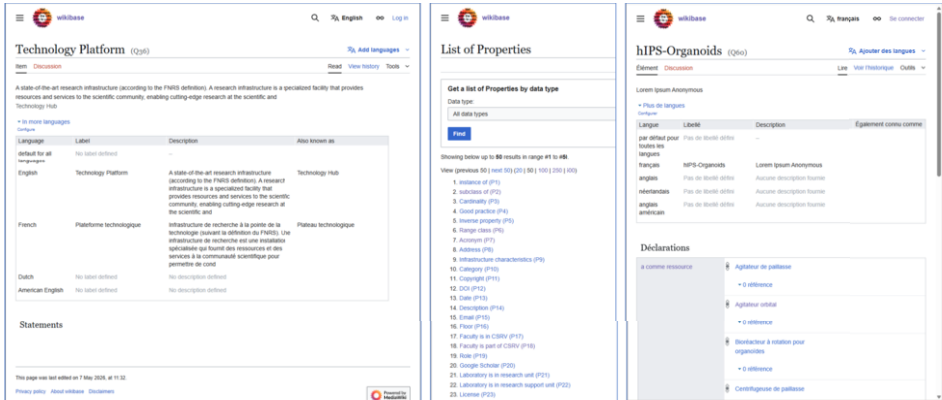


Figure 4. Local Wikibase for the Institutional Research Register (anonymized). The first two screenshots show an example of a “class” and its properties generated from CSV files in phase 1. The third screenshot shows an example of a technological platform and its equipment, generated with data from phase 1. This screenshot is in French as no translated labels and descriptions were provided.

Notion¹³, producing tables of classes with their metadata, inter-class relationships, and class-to-value relationships. For this demonstrator, the CSV exports from Notion served as the data sources from which WBML generated the corresponding Wikibase items and properties.

This demonstrator mapping converts a domain model into a Wikibase KG. It reads CSV files exported from Notion, which contain both French and English content. Six statement maps handle the different entity types: one creates Wikibase items for each class with multilingual labels, aliases, and descriptions; a second builds the “class hi-

¹³<https://www.notion.so/>

erarchy” by creating and using a property based on `rdfs:subClassOf`. Three further maps handle data properties, splitting them by their declared range into string, URL, and time property types, and creating corresponding Wikibase properties with labels and descriptions in both languages. The final map handles object properties, including statements on cardinality, good practices, and their range class (linked as a reference to the corresponding class item).

In the second phase, some people have also started gathering information about the technological platforms of certain research units in Notion. Dumps were transformed into Wikibase statements using a similar approach, and the same lookup file was used to reuse the “classes” and properties.

7. Discussion

Currently, we have adopted RML for logical iterations via logical sources, i.e., pulling data from files and databases. Our approach can be easily extended to RML Logical Views (LV) [14]. RML LV is an extension to RML that lets one define a flattened (“relational”-like) and format-agnostic view over one or more existing data sources. Functions can even be applied to such views in order to preprocess the data. What is interesting is that views yield logical iterations, which are used to generate RDF. As such, our approach could incorporate RML LV and functions applied to views. In theory, support for Logical Views relies solely on adopting a compliant engine. While RML-compliant processors exist, such as SDM-RDFizer [15], our prototype was built in Python using Morph-KGC for purely pragmatic reasons.

7.1. Limitations

We currently assume that values are generated through logical iterations. Wikibase supports special “snaks” to indicate that a statement has no value or no known value. This requires including special keywords in WBML. Related to that, not all Wikibase types are supported. We currently support Wikibase strings, times, quantities, and URLs. But we do not (yet) support Wikibase-specific data types, such as LaTeX formulas and linguistic data types. This requires research beyond the widely used XSD datatypes.

From the implementation’s perspective, we did not rely on Wikibase extensions (such as the SPARQL endpoint) or on modules developed by others. Our approach relies on a Python library that wraps Wikibase’s PHP API. As such, there is no support for transaction management. A solution would have been to directly engage with the database, but that would have required a database connection and not a Wikibase account, which we deem dangerous in practice. Therefore, we believe it is more appropriate to connect to a Wikibase instance via a Wikibase account, though we should investigate data ingestion with transaction management. This also means that qualifiers and references are only pushed alongside newly created statements. On re-runs or after partial failure, an already existing statement’s qualifiers and references are not reconciled. To the best of our knowledge, RaiseWikibase [12] was the only tool that explicitly considered transaction management.

8. Conclusions

This paper presents WBML, a declarative mapping language for populating Wikibase instances from heterogeneous data sources. By reusing established RML concepts for logical sources and iterations, and introducing dedicated constructs for the Wikibase statement model, WBML addressed the lack of a flexible, declarative approach for generating Wikibase statements from heterogeneous sources and for ingesting them into a Wikibase instance. The feasibility of WBML was demonstrated through two use cases: a fairly complex JSON dataset with multivalued references (i.e., keys containing multiple values) and a simpler real-world institutional use case using CSV files. Together, these cases illustrate WBML's ability to handle joins across sources, multilingual fingerprints, and statement-level annotations.

As for future work, we will look at extending WBML to generate statements that fall outside RDF and to manage Wikibase instances. More concretely, we want to propose constructs for such as “no value” and “unknown value” snaks. We deem it important to seek out use cases where these make sense rather than propose them, as we believe there are practical implications when data needs to be managed. E.g., what happens when an unknown value becomes known? We will also investigate the problems of declarative entity reconciliation and the declarative expression of updates and deletions. For the former, we could look into related work on entity resolution, such as SILK [16]. SILK proposes a Link Specification Language (LSL) that allows one to specify how entities from different Linked Data datasets should be compared for similarity. As for the latter, prior work has already combined LDES [17] with RML to generate evolving datasets. However, LDES's underlying model is append-only: it supports the continuous addition of new elements and tagging elements as deleted, but does not natively support modifying or removing *members*¹⁴ from a graph. When the source data explicitly signals a deletion (e.g., a status flag), the RML mapping can detect it and emit a delete-typed member accordingly. Implicit deletion, by contrast, only works if diffs are computed between complete snapshots of the source. Where LDES focuses on evolving datasets and communicating diffs, we work on a whole knowledge graph. It might nonetheless be interesting to explore how constructs for updating and deleting statements could be incorporated into WBML, and how Wikibase's statement metadata (e.g., ranks) could support them.

To finalize, WBML demonstrates that declarative mapping approaches need not be confined to RDF and that established ideas from the RDF mapping community can be meaningfully adapted to other types of KGs, in this case, Wikibase.

Declaration on Generative AI

During the preparation of this work, the authors used ChatGPT and Grammarly to: Grammar and spelling check, paraphrase, and reword. While the initial prototype was developed mostly by hand for the MEng thesis, Claude was used for debugging, restructuring the codebase to separate the project from the demonstrator, and ensuring data anonymization for the review. After using these tools, the authors reviewed and edited the content as needed, and take full responsibility for the publication's content.

¹⁴In LDES, the unit is called a member and a single member can group many triples.

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Surprising Effectiveness of Self-Demonstrations in Enhancing Schema-Ontology Mapping with LLMs

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Abstract. Integrating heterogeneous relational databases into a centralized ontology remains a persistent challenge in enterprise knowledge representation, primarily due to semantic heterogeneity, cryptic schema naming, missing metadata, and the abstraction gap between relational schemas and ontological models. Although large language models (LLMs) offer strong semantic reasoning capabilities, we show that directly applying them through one-shot prompting or naïve multi-stage pipelines leads to poor performance for schema–ontology mapping. This paper presents a self-demonstration-driven approach that combines a neuro-symbolic task decomposition with a novel mechanism for automatically generating pattern-guided, dependency-aware demonstrations to address this integration challenge. Our approach incorporates two key strategies to achieve substantial accuracy gains over existing LLM-based schema integration methods: (i) a neuro-symbolic decomposition of the task into cascaded sub-tasks, where symbolic constraints structure the search space and LLMs perform semantic reasoning within each focused sub-task, and (ii) self-generated demonstrations guided by domain-agnostic patterns to supervise each sub-task. Experiments on three of the most challenging scenarios from the RODI benchmark show that our approach achieves state-of-the-art performance, substantially outperforming (~25 percentage points F1 improvements) both traditional schema-to-ontology mapping techniques and recent LLM-based schema-to-ontology and schema matching approaches. Ablation studies further reveal the significant benefits of pattern-guided self-demonstrations and the complementary benefits of neuro-symbolic task decomposition.

Keywords. Database-Ontology mapping, data integration, enterprise search, R2RML generation

1. Introduction

Enterprises increasingly seek to represent organizational knowledge through a single unified ontology so that analytics, search, and reasoning can operate over a semantically consistent enterprise view [1,2,3]. In practice, however, enterprise data is dis-

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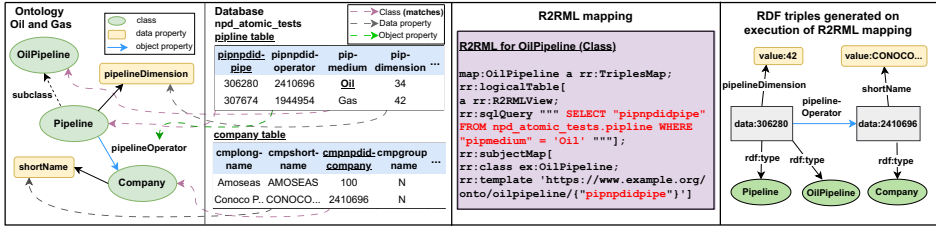


Figure 1. RDB-to-Ontology Mapping Example

tributed across multiple heterogeneous and independently evolving relational databases. As a result, querying across these sources requires mapping each database schema to a centralized ontology (Example illustrated in Figure 1). Doing this manually is a time-consuming, error-prone, and cognitively demanding task at enterprise scale, given the large number of ontology entities, database tables, and columns involved.

Traditional automated mapping systems [4,5] rely heavily on lexical similarity and hand-crafted templates; consequently, they often miss deeper semantic correspondences, struggle with cryptic or noisy schema elements, and fail when key metadata is incomplete. Large language models (LLMs) are a natural fit for this task because they offer broad semantic reasoning capabilities [6,7,8,9] that can bridge representational gaps between schemas and ontologies. Yet our early experiments show that directly prompting an LLM to produce complete DB-to-ontology mappings in one shot yields poor results. When the overall task is organized in a neuro-symbolic manner, where symbolic structure decomposes the task into stages and constrains the search space while LLMs perform semantic reasoning within each stage, the model makes better local decisions and produces better mappings. However, the benefits yielded by this staged LLM driven approach in zero-shot setting leads to sub-optimal results as inaccuracies at each LLM stage accumulate, leading to a decline in overall performance. To improve accuracy, we sought to harness the in-context learning (ICL) [10] capability of LLMs to adapt to new tasks or domains using demonstrations with a few input-output exemplars. We observe that manually curated demonstrations substantially improved accuracy, but they are labor-intensive and not scalable due to the need for domain and task expertise.

Matchmaker [11, 12], an approach for schema matching, demonstrates the capability of LLMs to automatically generate few-shots for self-improvement. However, extending their approach for our pipeline does not yield us performance on par with the manually curated few-shots. This was mainly because matchmaker’s approach tend to generate similar-looking demonstrations, with limited semantic variations in the matches.

This leads to key insight of this work: We allow LLM to generate self-demonstrations while constraining generation using typical reusable, domain-agnostic patterns observed across real data integration cases. We define pattern families that capture common regularities and guide the LLM to instantiate them, for each stage, to generate self-demonstrations. For example, a class in an ontology matches a DB table following either an exact string-based, syntactic, semantic, domain-based, or categorical-column-value-pair-based match. An object property connecting two classes, in contrast, matches with either the foreign key column of the matched table of the domain class or the columns of the junction table that links the tables matched with the domain and range classes. The demonstrations are created in a dependency-aware manner: outputs from earlier

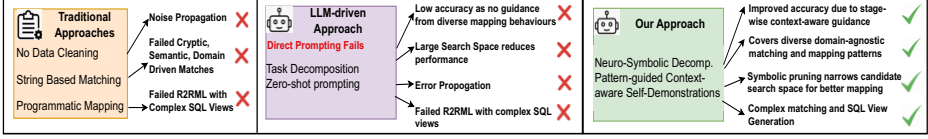


Figure 2. Motivation behind Our Approach of DB-to-Ontology Mapping

stages (such as class/table alignments) are carried forward while generating demonstrations for downstream sub-tasks (e.g., property matching and SQL-view mapping). We show that our staged process of automatic generation and utilization of pattern-grounded, dependency-aware self-demonstrations, which are diverse and less noisy, is effective and is the primary driver of performance gains for LLM driven database to ontology mapping task. Although our approach invokes LLMs at multiple stages, database to ontology mapping is typically a one-time integration process whose resulting mappings are reused extensively across downstream queries and applications. As a result, we prioritize accuracy and semantic correctness over execution efficiency or compute cost in enterprise settings. The key contributions of our approach are summarized as follows:

- We propose a neuro-symbolic decomposition based approach that combines the strengths of traditional symbolic approaches such as Milan [13] with the power of modern LLMs. The database-to-ontology mapping task is systematically decomposed into multiple stages and fine-grained sub-stages, each executed using self-demonstration-guided LLMs.
- We introduce a pattern-guided context-aware self demonstration method that automatically constructs diverse, stage-specific in-context exemplars, with one-time domain agnostic pattern curation.
- Our approach achieves state-of-the-art results on the three most challenging scenarios from the RODI benchmark [4] (geographical, conference and oil and gas), demonstrating substantial improvements over direct prompting and prior automated baselines (~ 25 percentage points in F1 score) and even slightly better results than manually curated few-shots.
- We show via detailed ablations that our approach benefit from neuro-symbolic task decomposition and self demonstrations.

2. Related Work

DB-Ontology Mapping. Early systems such as BootOX [14], Ontop [15], MIR-ROR [16], and D2RQ [17] derive a bootstrapped ontology from a database and then try to map this ontology with the schema, which is not a realistic setting. Subsequent tools like COMA++ [18], IncMap [19], A4MO [5], and MILAN [13] rely mainly on lexical/structural similarity with heuristic pruning and template-based mapping. These approaches degrade under cryptic nomenclature, incomplete metadata (e.g., missing FKs), and when semantic correspondences transcend string similarity. Recent LLM-based methods (Table-to-KG [20] and LLM4VKG [21]) introduce learned semantic matching with LLMs but often stop at alignment, keep mapping template-driven, or do not provide targeted, stage-specific guidance to the LLM. In contrast, our approach combines

symbolic decomposition and pruning with LLM-based semantic reasoning, and further provides targeted, stage-specific guidance through pattern-guided self-demonstrations.

Schema Matching. In the DB-to-ontology mapping task, the initial correspondence identification step closely resembles schema matching. LLM-based schema matching commonly follows “retrieve candidates then re-rank with an LLM,” as in Re-Match [22] and Magneto [23]. Matchmaker [11,12] further explores automated few-shot self-improvement. Yet schema matching is typically simpler than DB-ontology mapping given the abstraction and granularity mismatch between schemas and ontologies. Matchmaker’s few-shot generation approach when adapted to our setting leads to sub-optimal results. In our setting, automated few-shots help only when they are diverse, scenario-specific, and dependency-aware, characteristics we enforce via self-demonstrations.

Takeaway. Prior literature indicates that LLMs can capture semantics beyond strings, but it underplays how to combine semantic reasoning with symbolic decomposition and how to feed LLMs the right in-context signals. We operationalize this through a neuro-symbolic design with pattern-guided self-demonstrations, automatically synthesized, diverse, and scenario-specific exemplars that steer the model across mapping stages.

3. Background and Problem Definition

We formalize the problem and recall essential background on ontology structure and schema-to-ontology mappings. We are given as input: (i) an Ontology $\mathcal{O}$ consisting of a set-of classes C such as Pipeline in Figure 1, subclasses $C_S \rightarrow C$ (e.g. OilPipeline) for a subset of classes C , data properties D_C (e.g. pipelineDimension) with domain as a class (e.g. Pipeline) and range as its data type (such as ‘float’), and object properties O_{C_D, C_R} (e.g. pipelineOperator) linking two classes, a domain class C_D (e.g. Pipeline) and a range class C_R (e.g. Company), and (ii) Relational Database schema $\mathcal{D}$ consisting of entities such as tables T (e.g. pipeline), columns belonging to those tables C_T (e.g. pipmedium), and data values of those columns V_{C_T} (e.g. ‘Oil’).

An Ontology-DB mapping entails associating ontology concepts (class, data properties and object properties) with corresponding database entities (tables, columns). Typical regularities include: classes map to table primary keys or filtered table subsets via categorical columns; subclasses map to column-value filters in or adjacent to superclass tables; data properties map to columns in class tables or one-hop neighbors; object properties maps to foreign keys or junction-table keys connecting domain and range.

A snippet of such a mapping task between a subset of oil and gas ontology made for the Norwegian Petroleum Directorate (NPD) and a relational DB is shown in Figure 1. Mappings are expressed in R2RML (Relational to Resource Description Framework Mapping Language) [24]. Our goal is to generate a set of R2RML mappings consisting of SQL views, with subject and object columns as illustrated in Figure 1 such that execution of the R2RML mapping leads to RDF triples, populating $\mathcal{O}$ with the data in $\mathcal{D}$.

4. Self-Demonstration Driven Approach

4.1. Overview of Neuro-Symbolic Decomposition

Motivated by the observation that despite long context support, current LLMs struggle to produce accurate R2RML mappings in a single, one-shot pass, we decompose the DB-

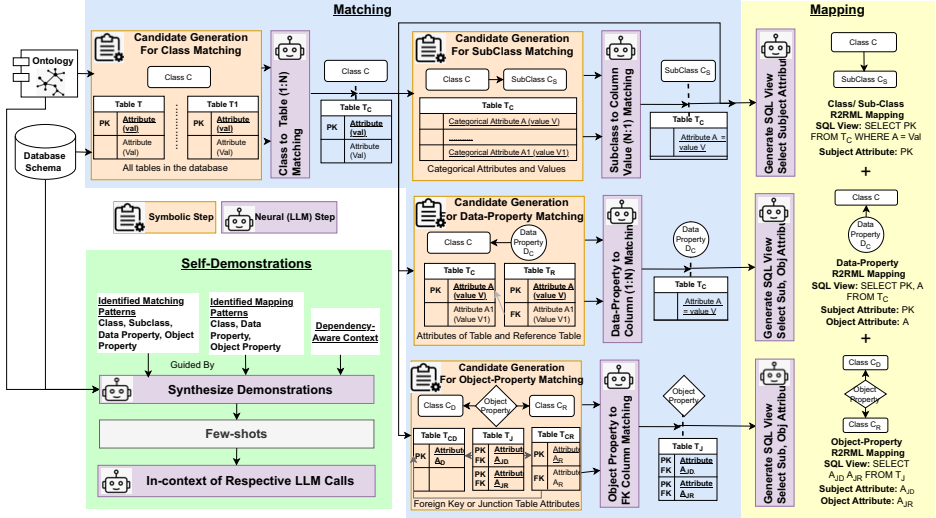


Figure 3. Overview of Self-Demonstration Driven Mapping Approach. Decomposition: (i) Matching, (ii) Mapping and Self-Demonstrations. T_C : Table matched with class C , C_S : Subclass of Class C , D_C : Data-property of Class C , T_K : Table referring to T_C , T_J : Junction Table

to-Ontology mapping task into two stages: (i) Matching: Identifying the most relevant target entities in the DB schema $\mathcal{D}$ that correspond to a source concept in the ontology $\mathcal{O}$, (ii) Mapping: Constructing R2RML mappings for each ontology concept $\mathcal{O}$ by generating SQL views over matched schema entities. We infer missing foreign keys in $\mathcal{D}$ and domain or range declarations in $\mathcal{O}$ when such metadata is absent, before serving them as an input to the above stages. Both stages are further divided into sub-stages for aligning ontology concepts, preserving their dependencies. Similar to Milan [13], we address the vast schema–ontology search space by performing matching in a top-down, stage-wise manner, progressively pruning candidates using symbolic constraints. We decompose the overall task into sub-stages based on ontology entity types and construct focused candidate sets using rule-based reasoning, substantially reducing the number of database entities taken into consideration as candidates compared to full database schema. Unlike Milan, we tightly integrate this symbolic decomposition with neural component by invoking LLMs at each sub-stage, equipped with carefully scoped contextual information. Operating over these focused candidate sets allows the LLM to reason within a constrained and relevant context, leading to more accurate and robust mappings [25,26,27,28].

(i) *Matching*. Matching proceeds in four sub-stages in the order: class, subclass, data property, and object property. Class matching is a two-step process that first selects the best matching table, then additional matching tables if present, are identified. Subclass matching is invoked only when the superclass is matched and searches categorical column–value pairs in the superclass tables and in one hop related tables. Data property matching draws candidates from tables matched to the domain class and from one hop related tables. Object property matching runs only when both domain and range classes are matched, considering candidates from the domain and range tables, one hop related tables, and keys of the junction table, when applicable. For example, in the NPD database–ontology pair from the RODI benchmark (see Section 5.1), symbolic pruning reduces the

candidate search space by approximately 95% for data property matching and 85% for object property matching stages, thereby enabling more focused and accurate matching.

(ii)*Mapping*. Mapping is decomposed into class, data property, and object property mapping. For each ontology concept, the model generates a SQL view and identifies the subject and object columns required by R2RML using the matches obtained in the matching stage for that concept. Class mapping selects subject identifiers from primary keys of the matched class tables. For subclasses, the mapping involves generating SQL filter conditions (e.g., category predicates) that distinguish subclasses when they match to a shared table. Data property mapping combines the matched value column with the identifiers of the domain class, while object property mapping links identifiers of the domain and range classes through the matched relation columns, including junction table paths when required. Per concept outputs are aggregated into the final R2RML mapping file. The matching and mapping prompts, along with the symbolic steps, are provided in the repository². Figure 3 provides an overview of this decomposition.

This sequential stage-wise structure improves mapping quality but the decisive gains arise when each stage is augmented with self-generated, pattern-guided demonstrations. These demonstrations are instantiated on the input DB–ontology pair and supplied in-context at the point of use, improving decision fidelity across the above stages.

4.2. Why Pattern-guided Context-aware Self-demonstrations.

Our initial zero-shot LLM experiments for the matching and mapping sub-tasks yield sub-optimal results. We observe that manually curated exemplars for In-Context Learning (ICL) lead to substantial performance improvements. However, this manual curation is labor-intensive and not scalable for distinct DB–Ontology pairs since it requires domain and task expertise. Hence, we seek methods of auto-generation of few-shots.

One of the Schema Matching approaches, Matchmaker [11,12], utilized automatic few-shot generation for self-improvement in schema-to-schema matching. We observe that adapting Matchmaker for schema-to-ontology mapping, often produce exemplars concentrated around a narrow style of semantic similarity, with limited coverage of the broader matching and mapping behaviors required for DB-to-ontology integration (Details in Section 5.3). Database to ontology mapping problem is fundamentally harder than schema matching because databases and ontologies represent knowledge at different levels of abstraction and granularity. Relational schemas are designed for efficient storage and query execution, and therefore expose low-level, fine-grained structures such as tables and columns. Ontologies, in contrast, encode high-level domain concepts, constraints, and relationships. This mismatch is amplified in enterprise settings where naming conventions vary across systems, semantically related entities use dissimilar labels, and similarly named entities may carry different meanings depending on context. This leads to distinct scenarios or patterns of this mapping task, with much higher variations than schema matching. Thus, diversity in few-shots is critical to cover varied matching and mapping scenarios [29,30]. Further, each sub-stage of schema-to-ontology mapping task relies on information produced by previous ones and thus, synthesizing demonstrations for any sub-stage must incorporate dependency-aware context (i.e, the outputs from earlier stages) to ensure coherence and correctness of the demonstrations. Thus, unlike

²<https://anonymous.4open.science/t/Schema-Ontology-Mapper-4BC1/>

schema-matching pipelines, the schema-to-ontology mapping stages exhibit strong sequential dependencies to avoid error-propagation through stages.

To avoid manual intervention but at the same time retain the quality and diversity of few-shots, we devise the self-demonstrations method to auto-generate few-shots for any given DB-ontology pair with minimal one-time guidance using a reusable library of domain-agnostic patterns and dependency-aware context.

4.3. Synthesis of Self-Demonstrations

We provide the LLM with an exhaustive set of predefined domain-agnostic input patterns that demonstrate distinct strategies of matching and mapping (patterns are provided in the repository). With these patterns in context, we task the LLM to instantiate them on the given DB-ontology pair. Thus, starting from generic patterns, we generate a comprehensive and diverse set of few-shots for any given DB-ontology pair. With this approach, we ensure that the generated few-shots cover a wide array of matching and mapping cases.

A pattern is a template that captures a matching/mapping behaviour. The pattern library encodes recurring correspondence behaviors seen across DB-ontology integration tasks: lexical and semantic alignment patterns (e.g., exact, paraphrased, or abbreviation-based matches), structural patterns (e.g., object properties realized through foreign keys or junction tables), and value-constrained patterns (e.g., subclasses represented as filtered subsets based on categorical column values). We also include mapping-side patterns for SQL view construction, such as projection-only mappings, join-based mappings, and mappings requiring value normalization or conditional filtering. This pattern-oriented organization makes each demonstration explanatory rather than merely illustrative: the few-shot not only states the final match or SQL fragment, but also reflects the underlying pattern that justifies it. Patterns also act as a controllable diversity mechanism. During demonstration synthesis, we explicitly prompt the LLM to cover different pattern families instead of selecting random examples. As a result, the final in-context set spans both straightforward and complex behaviors, improving robustness when the target concept exhibits ambiguous naming, implicit relations, or non-trivial SQL transformations.

Few-shots are generated for each matching and mapping sub-stage sequentially by providing stage specific patterns (prompt is provided in repository). For each sub-stage, LLM leverages the outputs of the relevant prior stage(s) to further improve the quality of the demonstrations. For example, for generation of few-shots for subclass, data and object property matching step, class matching output serves as additional input. Whereas for data and object property mapping, the output of data and object property matching serve as an input (details are provided in Figure 4).

This dependency-aware context improves demonstration quality for two reasons. First, it guides the LLM to focus on candidates that are structurally consistent with earlier

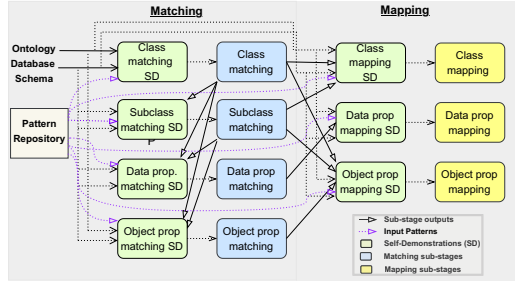


Figure 4.: Dependency-Aware context for Self-Demonstrations

decisions (e.g., data-property examples are grounded in tables already selected for the domain class), which lowers semantically plausible but invalid matches. For example, consider the data property `hydrocarbonType`, whose domain is the class `Discovery` and whose range is string. From the earlier class matching stage, the class `Discovery` has been aligned with the `discovery` table. During the self demonstration generation step for property matching, this prior alignment is included as contextual information. By making the class to table correspondence explicit, the context directs the LLM’s attention toward columns associated with the `discovery` table, leading it to focus on candidates such as `discovery.dshctype` rather than exploring other semantically plausible but structurally less aligned columns such as `dscarea.dshctype` and `fldarea.dshctype`.

Second, because established upstream decisions (e.g., class-table alignments) are passed into downstream demonstration construction, demonstrations across downstream tasks (e.g., data property alignment / mapping) become more consistent and correct. For example, consider the data property `wellboreStratumBottomDepth`, whose domain is the class `WellboreStratum` and whose range is decimal. In an earlier class-matching stage, the `WellboreStratum` class is aligned with the `wellbore_formation_top`, `strat_litho_wellbore` tables using richer contextual signals, giving these class table alignments a higher likelihood of being correct. When these higher confidence upstream decisions are incorporated as contextual input during the self demonstration generation step for property matching, they provide reliable structural anchors for downstream reasoning. As a result, the LLM constructs property-level demonstrations under the assumption that `wellboreStratumBottomDepth` should be interpreted within the context of the `wellbore_formation_top` and `strat_litho_wellbore` tables, leading to consistent demonstrations that align the property with `wellbore_formation_top.lsubottomdepth` and `strat_litho_wellbore.lsubottomdepth`. In contrast, without incorporating these upstream alignments, the LLM may either arbitrarily select a single table or treat multiple semantically plausible columns as equally valid, increasing the likelihood of generating inconsistent or incorrect property-level demonstrations. Thus, dependency-aware few-shot generation improves example quality and consistency while limiting error propagation across sub-stages. Figure 4 summarizes the context used for self-demonstration synthesis across stages.

In manual few-shot curation, an expert selects demonstrations, each corresponding to a distinct matching or mapping pattern in the DB-ontology pair. Identifying representative examples for new pairs is challenging and time-consuming. In contrast, our automatic approach instantiates at most one example per applicable pattern. Both settings use the same number of few-shot examples (one per pattern). We use the same fixed domain-agnostic pattern set across all input DB-ontology pairs; new domains require no expert-authored templates, though experts may optionally add rare missing patterns.

5. Experimentation

5.1. RODI Benchmark

We use RODI (Relational-to-Ontology Data Integration) [4], benchmark to test the R2RML gen-

Name	Classes	Props.	Tables	Cols.	FKs	Queries
Conference	23	77	66	125	–	39
Mondial	49	71	42	160	60	50
Npd	300	350	70	1000	100	439

Table 1.: RODI Scenarios

eration task. This benchmark has 18 test scenarios belonging to three domains. Each scenario consists of a DB, an ontology and set of evaluation queries to test expected results.

We experiment with one representative scenario from each domain, viz., Mondial for Geographical Data, NPD-Atomic for oil and gas, and Conference-no-FK, the most challenging scenario within each domain [13]. Table 1 presents the statistics for the DB entities and ontology concepts of these scenarios. These span large DB schema and ontology of significant complexity, providing a realistic simulation of enterprise-level integration challenges. The Conference-no-FK scenario is especially difficult because DB schema lacks foreign key definitions, thus introducing ambiguity and noise into the task. For example, `was_a_program_committee_of` column of the `Committee` table should ideally be a foreign key of the `Conference_volume` table, but this relationship is missing in RODI. The mapping of data and object properties of the Mondial scenario exhibits a high degree of complexity, involving domain and range definitions, with logical combinations of multiple classes. For example, the domain class of `type` data property is defined as a complex expression involving unions, intersections and negations of classes, such as "`((GeographicalThing or Membership) and (not (City)) and (not (Continent)) and (not (Estuary)) and (not (River)) and (not (Source)))`". It also involves object properties sharing identical domain and range classes. For example, the object properties `neighbor` and `dependentOf` have the same domain and range class as `Country`. The NPD-Atomic scenario represents the most complex case with very large ontology and DB schema. This scenario is characterized by a large number of one-to-many (1:n) mapping and individual ontology concepts must often be mapped to a union of several DB entities, to retrieve complete and correct results. For example, the mapping of `productionYear` data property is defined as a union over columns of 5 different tables.

Metric: We adopt the evaluation framework of the RODI benchmark. For each scenario a series of query pairs test a range of mapping challenges which are the result of distinct naming conventions of DB and ontology, structural conflicts caused by process of DB normalization, varied implementations of class hierarchies in an ontology, or semantic heterogeneity stemming from the conflict between a database’s closed-world and an ontology’s open-world assumption (object-relational gap). Every query pair consists of (i) a SPARQL query (test query) that runs against the RDF ontology data, resulting after execution of the generated R2RML, and (ii) a semantically equivalent SQL query (reference query) evaluated against the corresponding DB. For each query pair, the local per-query F-measure is computed by comparing reference tuples (SQL execution results) with result tuples (SPARQL execution results). The overall score reported for each scenario is the average of F-measures across test queries. More details are provided in [4]. We additionally report an average score (Avg), defined as the mean of F-measures computed over all queries across the selected test scenarios.

5.2. Baselines

Following approaches serve as our baselines: (i) traditional top-performing DB-to-Ontology mapping (Milan [13] and A4MO [5]) (ii) LLM-based DB-to-Ontology mapping (LLM4VKG [21] and table-to-KG [20]) and (iii) LLM-based state-of-the-art

schema matching (Magneto [23]). **AM4O**: Automap [5] converts the DB schema into a putative ontology, where tables become classes, columns become data properties, and foreign keys become object properties. The classes and data properties of the putative and target ontology are matched using string similarity metrics. The alignment extension step refines these initial matches by analyzing the structural relationships between the DB and ontology concepts. Finally, SQL queries for logical tables are programmatically derived using the DB paths identified during the alignment extension step. Automap’s alignment extension requires defined foreign keys, rendering it ineffective when absent. String similarity struggles with cryptic names or complex domains, and initial matching errors can propagate to subsequent alignment extensions. We use the results provided in [4]. **MILAN**: MILAN [13] uses Levenstein distance³ to match DB-ontology correspondences and to identify optimal matches using combinatorial optimization. MILAN also generates the R2RML mappings programmatically using a set of predefined templates for the SQL views. It is currently the state-of-the-art traditional approach for the DB-to-ontology mapping task. We use the results provided in [4]. **Table-To-KG**: Table-to-KG [20] uses LLMs to match table columns with semantic concepts in a knowledge graph. For each DB column, the top-k (k=50) ontology concepts are ranked with the highest cosine similarity of text embeddings. We generate the embeddings with Gemini-embedding-001⁴. The top-k matches are re-ranked with an LLM and a sliding window approach (step size = 10, window size = 20). Chain-of-Thought with self-consistency [31] (n = 3 samples) is used to retrieve multiple rankings, which are subsequently fused by reciprocal rank fusion to select the top-ranked match. Although the original work uses Llama-3-70B⁵ for re-ranking and GPT-4o for completion, for a fair comparison, we use Gemini-2.0-flash for all stages. Also, this method only provides matching, to obtain mapping, we use the programmatic mapping approach like Milan. **LLM4VKG**: LLM4VKG [21] is a recent LLM-based approach to construct Virtual Knowledge Graphs (VKGs), consisting of DB, ontology and mappings. It performs the alignment with three modules. Retriever finds semantically similar ontology candidates using a sentence similarity model. Matcher refines matches and assigns a matching degree (High, Medium, Low) via an LLM. Finally, Namer generates new ontology terms when no suitable match exists. As a baseline, we use the results provided in [21]. **Magneto**: Magneto [23] is the state-of-the-art approach that leverages LLMs for the schema matching task, with the assumption of having a single table in the source and the target schema. To use this approach as one of the baselines, we customize it for matching ontology concepts to DB entities and execute mapping with Milan’s programmatic approach. For each concept in the ontology, we rank all entities in the DB using a score provided by a retriever (MPNet [32]), performing a semantic match. Then, for each concept in the ontology, the top-k ranked DB entities are re-ranked by prompting the LLM. We customize their prompt, originally designed for schema matching to schema-ontology matching. They use GPT-4o-mini⁶ for LLM re-ranking. However, for a fair comparison, we keep the LLMs consistent with those we use for our approach. **Matchmaker**: Matchmaker [11,12] is an LLM-based approach for schema matching. It involves a multi-step pipeline comprising candidate generation (using semantic retrieval and LLM reasoning),

³https://en.wikipedia.org/wiki/Levenshtein_distance

⁴<https://developers.googleblog.com/en/gemini-embedding-available-gemini-api/>

⁵<https://huggingface.co/meta-llama/Meta-Llama-3-70B>

⁶<https://platform.openai.com/docs/models/gpt-4o-mini>

LLM-based candidate refinement, and confidence scoring. A key feature of Matchmaker is LLM self-improvement through automated in-context example selection. It identifies a set of ‘easy’ (having a semantic similarity score > 0.95 with target schema entities) and ‘challenging’ (exhibiting the lowest semantic matches) source schema entities. The complete matching pipeline is then executed for each identified entity, with all intermediate outputs stored. An LLM evaluator assesses the final target schema entity match, assigning a relevance score from 0 to 5. The intermediate traces from the top-n source entities with the highest evaluation scores are selected as in-context examples. To assess the effectiveness of our self-demonstration generation strategy, we compare it with Matchmaker’s few-shot example selection strategy adopted and integrated in our pipeline. Specifically, for each stage in our approach, we obtain DB schema matches under zero-shot conditions, for ontology concepts selected using Matchmaker’s method. These results are then evaluated by an LLM evaluator, and the top-n scoring ontology concepts and their matches are chosen as in-context examples for each stage.

5.3. Results and Discussion

RQ1: How do LLM-driven approaches compare with traditional approaches? Direct single-pass LLM prompting is insufficient for obtaining complete DB-to-ontology mapping, as evidenced by the *-Matching+Mapping* ablation in Table 2, which achieves only 0.10 average F1 score and underscores the combined challenge of semantic alignment and R2RML mapping generation when attempted jointly in a one-shot setting. In comparison, traditional baselines such as Milan and A4MO demonstrate better performance (Table 2) but remain constrained by dependence on lexical similarity and predefined mapping templates.

Prior LLM-driven systems do not outperform the strongest traditional baseline. Table 2 shows that LLM4VKG (0.21), Table-to-KG (0.18), and Magneto (0.23) remain below Milan (0.31). This pattern indicates that simply introducing an LLM is not enough, performance depends on how the LLM is embedded in the pipeline.

Scenario	Conf	Mondial	NPD	Avg
Traditional Baselines				
AM4O [5]	0.41	0.44	0.23	0.26
Milan [13]	0.46	-	0.30	0.31
LLM Driven Baselines				
LLM4VKG [21]	0.51	0.18	0.19	0.21
Table-to-KG [20]#	0.33	0.10	0.17	0.18
Magneto [23]#	0.41	0.28	0.21	0.23
LLM Driven Demonstration Based Baselines				
Matchmaker [11, 12]	0.56	0.38	0.19	0.24
Manually Curated FS	0.72	0.82	0.48	0.53
Our Approach	0.79	0.79	0.51	0.56
Ablations				
- Self-Demonstrations (FS)	0.54	0.46	0.36	0.38
- Matching	0.51	0.50	0.36	0.38
- Mapping#	0.62	0.36	0.43	0.44
- Matching+Mapping†	0.46	0.08	0.07	0.10

Table 2.: Results on RODI Benchmark with Gemini-2.0-Flash. Conf: Conference-no-fk; NPD: Npd-Atomic; FS: Few-Shot; #: With programmatic mapping; †: No Decomposition.

The observed gap with Milan

stems from limitations in earlier LLM-based approaches. Without decomposing the mapping task into sequential symbolic sub-stages that progressively constrain candidate sets using rule-based reasoning, the model faces a large, weakly constrained search space, making it more prone to imprecise or incorrect mappings. In addition, the lack of stage specific supervision in the form of high quality few shot exemplars provides insufficient guidance on the expected behavior of different mapping sub-types. Element wise matching without sufficient contextualization further degrades performance; for example, when the model attempts to align a property of a given class without knowledge of other properties of that class, the table matched to the class, or relevant candidate columns along with representative example values, it may select a plausible but incorrect candidate due to the absence of sufficient context. Finally, reliance on a template based or programmatic mapping layer as in Milan limits expressiveness and cannot support multiple joins, unions, filtering conditions, or normalization rules required for complex mapping scenarios. Unlike prior LLM-driven approaches we exploit the complementary benefits of symbolic search space reduction and LLM-guided (beyond syntactic or semantic) candidate matching and SQL view generation to achieve substantial improvements.

RQ2: How does our symbolic decomposition-based LLM approach compare with traditional approaches? Our neuro-symbolic pipeline consistently outperforms traditional systems across all scenarios. In Table 2, our method achieves the best results on Conference-no-fk (0.79), Mondial (0.79), and NPD-Atomic (0.51), and substantially exceeds Milan, A4MO in average performance. The key reason is the complementary design of symbolic structure and neural reasoning. The symbolic layer decomposes the task into two stages: (i) matching (class, subclass, data property, object property) and (ii) mapping (SQL view generation with subject/object selection). This decomposition reduces task complexity and focuses each LLM call on a narrower, better-contextualized decision. The neural layer (LLM) complements this symbolic structure by going beyond name similarity to provide semantic and domain-specific disambiguation for cryptic schema elements using ontology context, schema structure, and representative column values, while also generating expressive SQL for non-trivial mappings.

Ablation results in Table 2 highlight the complementary roles of the matching and mapping stages. Dropping the matching stage substantially reduces performance, while replacing LLM-based mapping stage with programmatic mapping also results in a drop. These degradation indicate that both stages contribute essential, distinct capabilities to the pipeline, and their integration is critical for achieving high mapping quality.

RQ3: How do prior few-shot-based LLM approaches perform compared with prior approaches? Few-shot supervision is beneficial, but its effectiveness depends on the quality and breadth of the demonstrations. As shown in Table 2, manually well-curated few-shots substantially improve performance over traditional and LLM-driven baselines, underscoring the importance of in-context learning for this task. However, manually curating high-quality demonstrations is challenging and time-consuming at enterprise scale, requiring substantial domain expertise. As the number of ontology concepts and schema elements grows, identifying representative, diverse, and non-redundant few-shots becomes harder, making manual curation difficult to scale.

We evaluate a prior automatic few-shot selection strategy proposed in Matchmaker, originally developed for schema-to-schema matching. When adapted to our pipeline, Matchmaker-style automatic few-shot selection yields limited, inconsistent improve-

ments and continues to perform below the strongest traditional baseline, Milan. A key limitation of automatically selected few-shots is their lack of coverage across the full range of mapping challenges. The selected examples tend to focus on straightforward semantic similarity cases, while failing to capture harder yet more consequential mapping patterns. In addition, Matchmaker’s selection strategy does not incorporate dependency-aware context across stages, further degrading the coherence and correctness of the demonstrations. As a result, the selected few-shots fail to convey the behaviors needed for difficult cases, leading to reduced robustness and lower overall mapping quality.

RQ4: How does our dependency-aware, pattern-based self-demonstration approach compare with prior approaches? Our self-demonstration strategy addresses the above limitations by synthesizing stage-specific exemplars from reusable, domain-agnostic pattern families while respecting inter-stage dependencies. This produces demonstrations that are both diverse and structurally consistent with the pipeline’s execution order. As shown in Table 2, removing self-demonstrations lowers average performance to 0.38, underscoring the importance of high-quality supervision for the final outcome. An important and somewhat surprising observation is that, while the model performs poorly in zero-shot settings, enabling it to condition on self-generated few-shots guided by patterns and prior stage outputs leads to substantial performance gains. With self-demonstrations enabled, our full approach achieves an average score of 0.56, establishing state-of-the-art performance on Conference-no-fk, Mondial, and NPD-atomic.

Notably, our automatic self-demonstration strategy matches or slightly exceeds the performance of manually curated few-shots while fully eliminating manual curation overhead. The improvement is not due to simply increasing the number of examples, but due to generating the right demonstrations that are diverse, grounded in defined pattern families, and aligned with inter-stage dependencies in the pipeline.

RQ5: Is our approach generalizable across LLMs? We perform a comparative analysis using GPT-4o to assess the influence of LLM choice. The LLM-driven baselines are compared with our approach on the Conference-no-FK and Mondial scenarios. The NPD scenario, being significantly larger, is omitted from this specific evaluation due to budget limitations. Table 3 presents these results, indicating that performance with GPT-4o is lower than Gemini-2.0-Flash (Table 2). With GPT-4o, our approach consistently surpasses Table-to-KG and Magneto across scenarios, demonstrating its generalizability across LLMs.

Overall, the answers to RQ1–RQ5 converge on a clear conclusion. Strong performance on DB-to-ontology mapping does not emerge from LLM capability alone, it depends on how the task is organized and supervised. In particular, symbolic decomposition improves tractability and dependency-aware pattern-guided self-demonstrations improve decision quality at each stage, and expressive mapping generation is essential for resolving structurally complex correspondences.

Error Analysis: We analyze errors for the most complex NPD-Atomic scenario. Its 439 test query pairs evaluate mapping outcomes for distinct ontology concepts: 134 classes, 213 data properties, and 92 object properties. Our approach yields incorrect mappings

Scenario	Conf	Mondial	Avg
Table-to-KG [20]#	0.21	0.13	0.17
Magneto [23]#	0.49	0.34	0.40
Our approach	0.56	0.65	0.61

Table 3.: Results on RODI Benchmark with GPT4o. Conf: Conference-no-FK; #: With programmatic mapping.

for 46 classes, 157 data properties, and 63 object properties. We randomly sample and analyze 20 test queries for each ontology concept.

For classes, 12 errors stem from incorrect matches, 6 from incorrect SQL views, and 2 from incorrect subject/object column selection. Incorrect matches mainly arise when a class corresponds to multiple tables or when predicted and actual tables are ambiguous. For example, the Jacket4LegsFacility class matches both `facility_fixed` and `facility_moveable` tables (where `fclkind = 'JACKET 4 LEGS'`), but our approach identifies only `facility_fixed`, relying on the definition of 'Jacket' as a fixed offshore structure anchored to the seabed. Similarly, `ParcelBAA` class is incorrectly mapped to `baaarea` table instead of `bsns_arr_area` table, because both tables share similar column names and contain `baakind` column with value 'PARCELL', making them hard to distinguish. SQL-view errors involve (i) incorrect or missing categorical filter values and (ii) incorrect column selection. For example, although `Onshorefacility` class is correctly matched to column `fclkind`, the generated SQL filters by 'LANDFALL' instead of 'ONSHORE FACILITY'. Likewise, for `LithostratigraphicUnit` class, correctly matched to `wellbore_formation_top` table, the SQL view selects `lsuname` and `lsulevel` columns instead of all primary keys of the table.

For data properties, 10 errors are due to incorrect matches and 10 to incorrect SQL view generation. Most incorrect-match errors stem from incorrect domain class inferences during the prior metadata inference step. For example, the data property `sourcePressure` is inferred to belong to `SeismicSurvey` class. This inference is semantically plausible, since source pressure relates to seismic acquisition, but the ground-truth mapping assigns it to `SurveyArea` and `SurveyMultilineArea`. These more specific domain classes are difficult to infer from the property name alone. SQL-generation errors involve incorrect column selection or missing information. For example, although `dateStatusTo` data property is correctly matched to the `field_activity_status_hst` column, the generated SQL view does not filter out the placeholder date '9999-12-31T00:00:00', leading to errors. A particularly challenging case is `coreIntervalTop` property of `WellboreCore` class, correctly matched to `wlbCoreIntervalTop` column in `wellbore_core` table. The ground-truth SQL view is as follows:

```
(SELECT wellbore_core_id, wlbnpdidwellbore,
wlbcorenumber, wlbcoreintervaltop FROM
wellbore_core WHERE wlbcoreintervaluom = '[m]')
UNION ALL (SELECT wellbore_core_id,
wlbnpdidwellbore, wlbcorenumber,
wlbcoreintervaltop * 0.3048 FROM wellbore_core
WHERE wlbcoreintervaluom = '[ft]')
```

The ontology does not specify that all values must be normalized to meters using `wlbCoreIntervalUom` column, making the correct SQL view impossible for the LLM to generate.

For object properties, 16 errors are due to incorrect matches and 4 to incorrect SQL views. Among incorrect-match errors, 6 stem from wrongly inferred domain/range classes and 3 from incorrect domain/range matches. SQL-view errors usually involve incorrect column selection or missing information. For example, the LLM fails to correctly

map the object property `productionForField` because the reason for excluding value ‘44576’ is not specified. The ground-truth SQL view is:

```
SELECT prfnpdidinformationcarrier, prfyear
FROM field_production_yearly
WHERE prfnpdidinformationcarrier <> ‘44576’.
```

6. Conclusion

Enterprise-scale schema-to-ontology mapping remains a challenging problem due to severe semantic heterogeneity, noisy and incomplete metadata, cryptically named schema elements, and the fundamental mismatch in abstraction between relational data and ontological models. This work shows that while LLMs offer strong semantic reasoning capabilities, they are ineffective when applied in a naïve or weakly structured manner. Our study establishes that a neuro-symbolic approach is essential for achieving reliable and accurate mappings. By decomposing the task into structured stages and using symbolic constraints to systematically narrow the search space, the approach improves tractability and enables LLMs to operate within focused contexts leading to improved accuracy.

At the same time, decomposition alone is insufficient to fully address the challenges of schema-to-ontology mapping. An important finding of this work is the surprising effectiveness of self-demonstrations in supervising each stage of the pipeline. Self-demonstrations refer to automatically generated in-context examples produced by the language model itself, showing how specific matching or mapping decisions should be made for a given sub-task. These examples are guided by reusable, domain-agnostic mapping patterns, ensuring coverage of common and complex correspondence behaviors observed in real integration settings. By tailoring self-demonstrations to individual sub-tasks, the model receives explicit and diverse guidance that clarifies expected reasoning and outputs. Furthermore, generating these examples in a dependency-aware manner, where downstream stages are conditioned on the outputs of earlier ones, preserves consistency across stages and limits error propagation. Together, these properties enable more reliable local decisions, which accumulate to produce more accurate mappings. Leveraging the neuro-symbolic decomposition together with pattern-guided context-aware self-demonstrations, our method establishes state-of-the-art performance on the three most challenging scenarios from the RODI benchmark, demonstrating ~25 percentage points improvements in the F1 score over prior automated baselines.

As future work, we plan to reduce cost by partially replacing LLM-based reasoning with programmatic and template-driven mapping for well-structured and repetitive correspondence patterns. In addition, instead of producing a single mapping decision, we will output ranked candidate matches accompanied by confidence scores, enabling expert-in-the-loop workflows where domain experts can efficiently validate, adjust, or correct mappings by inspecting a small set of plausible alternatives rather than performing exhaustive manual verification.

Declaration on Generative AI

During the preparation of this work, the author(s) used Microsoft Copilot in order to paraphrase and reword portions of human-authored text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

Tools and services: Microsoft Copilot.

Tools' contributions: Paraphrasing and rewording (linguistic refinement) of human-authored text.

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Ontology-Aware Prompting for Knowledge Graph Construction from Text

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Abstract. Constructing Knowledge Graphs (KGs) from unstructured text has traditionally relied on multi-stage pipelines combining Named Entity Recognition, Relation Extraction, and Entity Linking components. These architectures are costly to adapt to new domains and suffer from error propagation across stages. While Large Language Models (LLMs) enable end-to-end triple generation from text, their direct application introduces two key challenges: (1) hallucination, where generated facts are not supported by the input text, and (2) ontology misalignment, where extracted entities and relations violate the constraints of a target schema. To address these limitations, we propose an ontology-aware multi-prompt framework that integrates the target ontology into the prompting process and applies hierarchical verification after triple generation. Specifically, we combine Tree-of-Thoughts-style reasoning for structured exploration, schema-constrained OpenIE, and relaxed triple generation to produce a diverse set of candidate triples while maintaining ontological awareness. A rule-based three-filter filtering module then refines these candidates using cross-prompt agreement, evidence grounding, and surface-form alignment, without relying on external toolchains. Experiments on Text2KGBench, including DBpedia-WebNLG and Wikidata-TekGen, show improvements of up to 2.3 times over LLM-based baselines, together with hallucination rates below 0.20 in most domains and ontology conformance above 0.90 on most DBpedia-WebNLG domains. The framework is released as open-source and available at <https://github.com/dice-group/ontology-aware-kg-construction>.

Keywords. Knowledge Graph Construction, Large Language Models, Ontology-Aware Prompting

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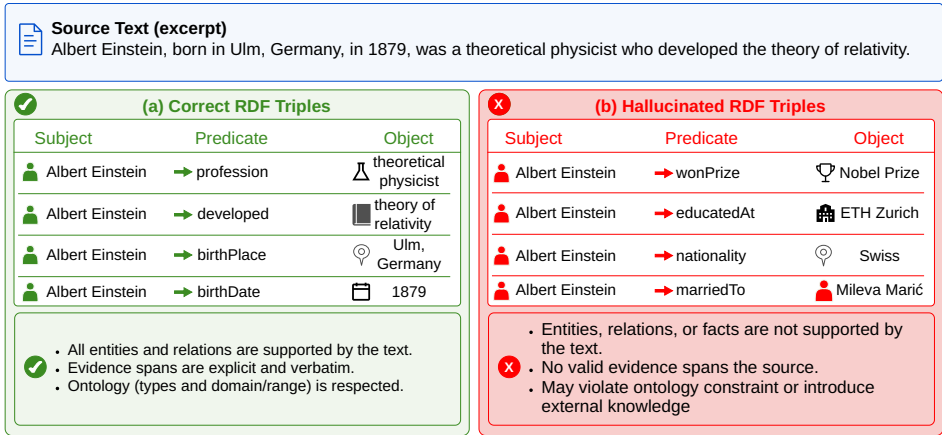


Figure 1. Examples of correct (a) vs. hallucinated (b) RDF triples extracted from text.

1. Introduction

Knowledge graphs (KGs) represent entities, concepts, and their relationships as Resource Description Framework (RDF) triples. They support a wide range of downstream tasks, including semantic search [1], question answering [2], and recommendation systems [3]. Traditionally, KG construction is decomposed into a sequence of specialized tasks: Named Entity Recognition (NER), Entity Linking (EL), and Relation Extraction (RE) [4,5]. These tasks transform unstructured text into structured RDF triples in one pipeline that populate a target ontology. However, these pipelines are costly to engineer and tightly coupled to fixed schemas, which has motivated a shift toward generative approaches [6].

Recent advances in Large Language Models (LLMs) introduce a new paradigm for automated KG construction [7]. Instead of chaining NER, EL, and RE, an LLM can generate RDF triples directly from text in a single step. Figure 1(a) illustrates this on the sentence “*Albert Einstein, born in Ulm, Germany, in 1879, was a theoretical physicist who developed the theory of relativity.*”. From this input, the LLM produces the triple $\langle \text{Albert_Einstein}, \text{profession}, \text{theoretical_physicist} \rangle$, where *Albert_Einstein* is the subject, *theoretical_physicist* is the object, and *profession* is the relation that links them. Despite this promise, LLM-based extraction suffers from two persistent limitations. The first is *hallucination*, where the model generates a triple that is plausible but not supported by the source text [8]. The second is *ontology misalignment*, where the generated entities or relations violate the type, predicate, or domain/range constraints of the target ontology. Figure 1(b) illustrates the first failure with the triple $\langle \text{Albert_Einstein}, \text{nationality}, \text{Swiss} \rangle$. This statement is historically true, but it is not present in the source sentence. Plausible but unsupported claims of this kind are particularly damaging in high-precision domains such as healthcare, law, and scientific knowledge management. Both limitations have received attention in the literature. However, to the best of our knowledge, no prior work tackles them jointly. For instance, COMET [9] and AutoKG [10] extract triples directly from text, yet provide no mechanism to enforce factual grounding or ontology compliance. More recent work begins to address the ontology side. Norouzi

et al. [11] and ODKE+ [12] guide the LLM through ontology-aware prompts, while Aljabari et al. [13] apply the same idea to ontology alignment. However, all three rely on a single prompting strategy, limiting the available signals to detect possible hallucinations. Among them, only ODKE+ introduces explicit safeguards against hallucination, and even there, the coverage at the entity and relation level remains partial. Consequently, to the best of our knowledge, no existing approach combines multi-prompt extraction with explicit hallucination control under an ontology.

In this work, we present an ontology-aware, multi-prompt framework for end-to-end KG construction. Our framework includes two stages: i) a multi-prompt extractor that produces candidate triples through three complementary strategies, each occupying a different point on the precision–schema–recall trade-off. ii) a three-filter validator that decides which candidates to keep. Each filter applies a different acceptance criterion. *Filter 1* accepts triples on which multiple prompts agree and whose arguments are textually anchored in the source. *Filter 2* accepts triples that arrive with an explicit evidence span and pass a weighted evidence score. *Filter 3* accepts the remaining candidates through a surface-grounding fallback. We evaluate the framework on the Text2KGBench benchmark [14], covering the DBpedia-WebNLG and Wikidata-TekGen datasets across 29 ontologies. The results show consistent improvements up to 2.3 times in precision, recall, and F1-score over fine-tuned baselines such as ALPACA-LORA-13B and VICUNA-13B. In addition, the framework reduces the three Text2KGBench hallucination metrics — Subject Hallucination (SH), Object Hallucination (OH), and Relation Hallucination (RH) — on both datasets, with values below 0.20. An ablation study further isolates the contribution of each component to the overall performance. We summarize the main contributions of this work as follows:

1. A multi-prompt extraction stage that generates candidate triples through three complementary strategies, *Tree-of-Thoughts Search*, *Schema-Constrained Open IE*, and *Relaxed Extraction*, each occupying a different position on the precision–schema–recall trade-off.
2. A three-filter, evidence-driven validator that accepts a triple only when one of three criteria is satisfied: cross-prompt agreement with lexical anchoring (*Filter 1*), an explicit evidence span passing a weighted score (*Filter 2*), or a surface-grounding fallback (*Filter 3*).
3. A comprehensive evaluation on the Text2KGBench benchmark across 29 ontologies and two domains (DBpedia-WebNLG, Wikidata-TekGen), showing consistent F1 improvements over fine-tuned baselines (ALPACA-LORA-13B, VICUNA-13B) and reduced Subject, Object, and Relation Hallucination rates.

2. Related Work

2.1. Traditional KG Construction

Traditional KG construction relies on pipeline-based methods that combine NER [4], RE [5], Event Extraction (EE) [15], and EL [16] from unstructured text. These pipelines are implemented using either rule-based or machine learning approaches. Rule-based systems employ pattern matching and dependency path heuristics, offering interpretability and high precision in domain-specific settings [17,18]. Statistical and neural methods,

including supervised learning and distant supervision, improve scalability and generalization to unseen data [19,20]. Open Information Extraction (OpenIE) methods [21,22] enable open-domain extraction without predefined ontologies but produce noisy and redundant outputs due to limited semantic constraints. Despite high-quality results when carefully tuned, these pipelines require substantial human effort, domain expertise, and labeled data, and they suffer from error propagation across components [23,24].

2.2. *Generative KG Construction Using LLMs*

Recent advances in LLMs have enabled direct triple generation from unstructured text without intermediate NLP components [24]. This end-to-end approach offers scalability but introduces two key challenges: hallucination, the generation of plausible but unsupported facts [8], and ontology misalignment when triples do not adhere to target schemas [24]. Several studies focus on open-domain extraction without explicit ontology constraints. For instance, ChatIE [25] formulates information extraction as a multi-turn question-answering problem to extract entities and relations in a zero-shot setting. AutoRE [26] introduces a Relation-Head-Facts paradigm for document-level relation extraction without requiring predefined relation types. Similarly, Papaluca et al. [27], and KGGen [28] evaluate zero-shot triplet extraction. Recent work embeds chain-of-thought reasoning into LLM prompts for KG construction [29], though without ontology context to guide reasoning steps or dedicated mechanisms to filter hallucinated outputs. Early generative efforts include COMET [9] for commonsense inference. Text2KGBench [14] established baselines (Alpaca-LoRA [30,31], Vicuna [32]) on ontology-aligned datasets. AutoKG [10] combines LLMs with external tools via agent coordination. iText2KG [33] enables zero-shot incremental construction. KARMA [34] employs multiple agents for extraction and validation in biomedical domains.

2.3. *Ontology-Aware KG Construction*

Our work belongs to the category of ontology-aware KG construction, where schema information is explicitly incorporated to guide LLM-based extraction and improve ontology compliance. Kommineni et al. [35] proposed a semi-automatic approach that first extracts an ontology from text using user-defined competency questions and then uses the extracted ontology to retrieve triples. Feng, Wu, and Meng [36] extended this direction by using an LLM to generate competency questions automatically for ontology extraction, followed by guided triple extraction on Wiki-NRE [37], SciERC [38], and WebNLG [39]. In the medical domain, Bhattarai et al. [40] proposed an ontology-guided prompting approach that extracts concepts from the UMLS ontology to personalize prompts for triple extraction. Their evaluation on the n2c2 [41] and ADE [42] datasets demonstrated improvements over standard retrieval-augmented generation techniques. Recent work systematically investigates the role of ontology in LLM-based KG construction. Norouzi et al. [11] studied LLM-based ontology population, highlighting the importance of prompt design and ontology structure for improving extraction accuracy and reducing hallucinations. Their modular ontology method dynamically builds context by summarizing knowledge base texts and retrieving relevant information through embedding-based search. Khorshidi et al. [12] proposed ODKE+, a production-grade system that dynamically generates ontology snippets tailored to each entity type to align

extractions with schema constraints. The system employs post-extraction validation using additional LLM calls to verify and correct extracted triples against the ontology, demonstrating effectiveness in reducing schema violations.

2.4. Positioning of Our Work

Our work differs from prior ontology-aware approaches in two key respects. *First*, while existing methods such as ODKE+ [12] and Norouzi et al. [11] inject ontology terms as unstructured context into prompts, our framework structures the generation process using Tree-of-Thoughts reasoning to explore candidate entity types and relation constraints before committing to triple outputs. Moreover, we employ additional prompting strategies, including OpenIE and one-shot prompting, to extract more RDF triples in different domains. *Second*, unlike approaches that rely on external toolchains (e.g., AutoKG [10]) or multiple LLM calls for validation (e.g., ODKE+ [12]), our method employs a lightweight rule-based three-filter filtering module that operates at the entity, relation, and triple levels without requiring additional API calls.

3. Approach

We propose an ontology-aware, multi-prompt framework for end-to-end RDF triple extraction from unstructured text, as shown in Figure 2. Given input text x and a target ontology Ω , our framework works in two stages: (1) a *multi-prompt extractor* that generates a diverse pool of candidate triples P using three distinct LLM prompting strategies, and (2) a *hierarchical, rule-based validator* that filters these candidates through a three-tier process to ensure factual grounding and ontology compliance. Ontology awareness in our framework means that the target ontology is used to guide prompting, candidate generation, and lightweight validation. The ontology provides canonical predicates, type descriptions, and domain/range information that influence which triples are proposed and accepted. The validator, however, does not perform full reasoning over OWL axioms or exhaustive checks of class membership and logical consistency.

3.1. Stage 1: Generating Candidate Triples

Instead of relying on a single extraction method, we use three prompting strategies that vary in reasoning depth, schema strictness, and recall orientation. This diversity provides the downstream evaluator with three complementary candidate pools ($P_1, P_2, P_3 \subseteq P$), respectively: high-precision triples from structured reasoning, ontology-aligned triples with explicit evidence, and high-recall triples from a more relaxed extraction process.

Strategy 1: Tree-of-Thoughts Semantic Extraction The first strategy focuses on precision by framing RDF extraction as a search problem, following the Tree-of-Thoughts paradigm [43]. The search begins from an empty state S_0 and proceeds over partial triple sets $S \subseteq T$, where each state S represents a set of candidate triples for the current input x under ontology Ω . At each step, a generator $\text{Gen}(p_{\theta_g}, S)$ proposes up to k new triples, where p_{θ_g} is the generator prompt. Each resulting state $S' = S \cup \{(s, r, o)\}$ is then scored in $[0, 1]$ by an evaluator $\text{Eval}(p_{\theta_e}, S')$, where p_{θ_e} is the prompt used for evaluation. This score reflects two properties: how well the triple is grounded in the source text x , and

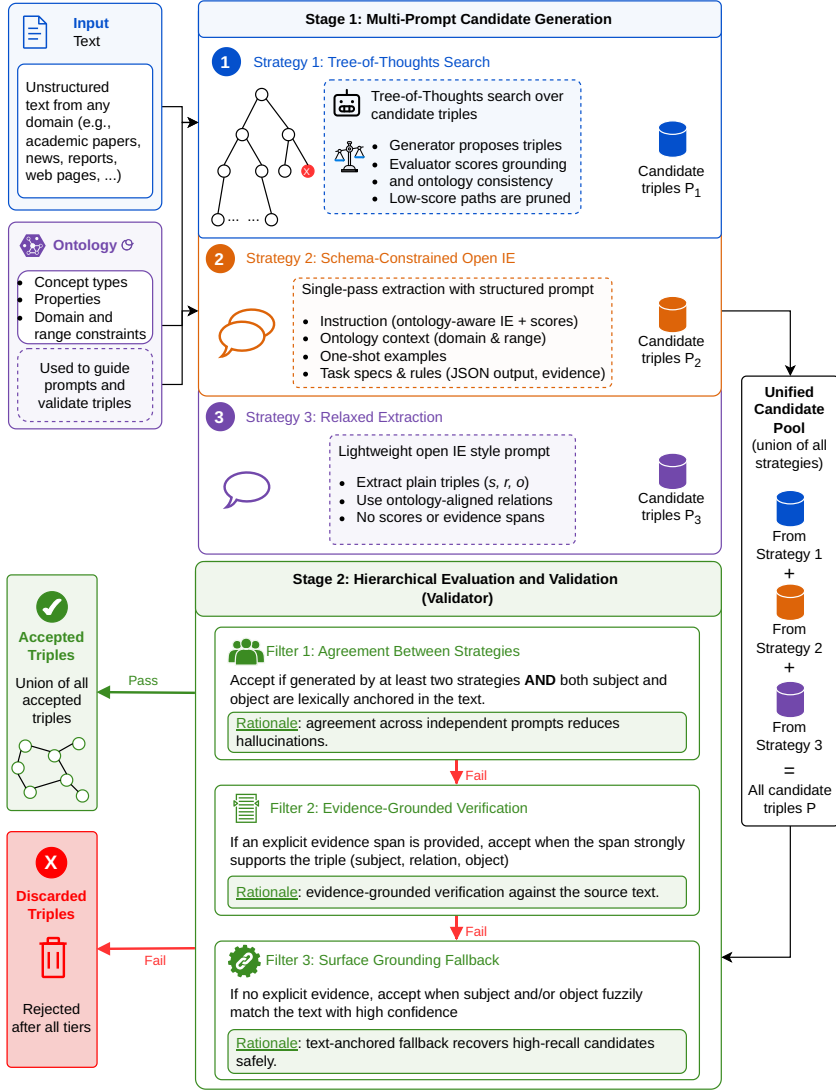


Figure 2. Ontology-aware multi-prompt framework for constructing knowledge graphs from text. Stage 1 combines three LLM prompting strategies to generate candidate triples, and Stage 2 applies a three-tier validator to filter hallucinations and enforce ontology compliance.

how well it aligns with the ontology Ω . If the score falls below a threshold τ , the branch is pruned, removing weak candidates early on. Expansion continues until a maximum depth $D_{\max}$. The highest-scoring deep paths form the output S^* , and each accepted triple is stored alongside its supporting textual span $\text{support}(s, r, o) \subseteq x$.

Strategy 2: Schema-Constrained Open IE The second strategy aims for schema-aligned extraction in a single pass by explicitly binding the model to the ontology. The prompt includes four parts. (i) *instructions* that frame the task as Open IE with confidence scores in $[0, 1]$. (ii) an *ontology block* that lists the domain and range for each predicate

as short textual descriptions rather than raw RDF/OWL syntax. (iii) a *one-shot example* showing a valid triple alongside its evidence span, and (iv) a *task specification* that presents the input text x and requests strict JSON output containing exact quotes from the input text. This produces ontology-aligned triples paired with direct textual support.

Strategy 3: Relaxed Extraction The third strategy maximizes recall by removing structural constraints. The model receives only the input text (x) and triples set P , and generates plain triples (s, r, o) . It omits confidence scores, reasoning traces, and evidence spans. Because rigid formats can cause the model to miss valid triples, relaxing these constraints recovers additional candidates. Although this increases noise, the trade-off is acceptable because Stage 2 filters it.

3.2. Stage 2: Three-Tier Validation

Stage 1 produces a mixed pool of candidates P . To separate reliable triples from noise, the validator asks the question for each candidate triple t : What kind of evidence supports this triple? Since different triples carry different types of evidence (cross-prompt agreement, explicit spans, or none), we use a three-filter cascade approach. A triple is accepted as soon as it satisfies one of the filters, following from stronger evidence to weaker ones. Filter 1 requires both multi-prompt agreement and lexical anchoring for the subject and object, and therefore tends to admit fewer but highly supported triples. Filter 2 focuses on explicit evidence spans and admits triples whose supporting span achieves a high evidence score E_B . Filter 3 serves as a permissive fallback: it recovers additional candidates based on fuzzy grounding and overall text–triple coherence when no explicit span is available. This cascade prioritizes high-confidence triples first while still recovering recall through softer conditions at the final stage. Algorithm 1 summarizes this process. Here, $\text{sim}(\cdot, \cdot) \in [0, 1]$ denotes a string-matching score between an entity surface form and the source text x , implemented as a normalized edit similarity.[44]. The functions $E_B, E_C \in [0, 1]$ are evidence scores and used consistently throughout the paper; Section 4.2 specifies the exact thresholds used in our experiments.

Filter 1: Cross-Prompt Consensus with Lexical Anchoring. Filter 1 of Algorithm 1 (lines 4–5) exploits agreement across prompting strategies. The intuition is: if multiple independent prompts generate the same triple, that triple is unlikely to be a shared hallucination. We achieve this with the two complementary conditions in line 4, both of which must hold for a candidate to be accepted. The first condition, *multi-prompt agreement*, requires the candidate t to appear in at least two of the three candidate pools, that is, $|\{j \mid t \in P_j\}| \geq 2$. This extends the self-consistency principle [45] from a single model’s sampled reasoning paths to a multi-prompt ensemble. However, agreement alone is not enough to filter all hallucinated triples, since independent prompts may still converge on a plausible but text-absent triple. To rule out this case, the second condition, *lexical anchoring*, requires the subject and object of t to each match the source text x above the grounding thresholds τ_s^{anc} and τ_o^{anc} ; concretely, $\text{sim}(s, x) \geq \tau_s^{\text{anc}}$ and $\text{sim}(o, x) \geq \tau_o^{\text{anc}}$. Whenever both conditions hold, line 5 adds t to the validated set T_1 .

Filter 2: Evidence-Grounded Verification. A triple that fails Filter 1 may still be reliable if its originating prompt provided a verbatim span support $\subseteq x$. The cascade therefore falls through to Filter 2 of Algorithm 1 (lines 6–7). We evaluate the triple against its own evidence by computing a normalized score E_B , defined as the weighted sum of four

Algorithm 1 Hierarchical Ontology-Aware Validation

Require: Candidate pools P_1, P_2, P_3 ; source text x ; anchoring thresholds $\tau_s^{\text{anc}}, \tau_o^{\text{anc}}$; fuzzy-match thresholds $\tau_s^{\text{fuz}}, \tau_o^{\text{fuz}}$; acceptance thresholds τ_B, τ_C .

Ensure: Validated triple set $\hat{G}$

```

1:  $\hat{G} \leftarrow \emptyset$ 
2:  $T_1 \leftarrow T_2 \leftarrow T_3 \leftarrow \emptyset$ 
3: for all  $t = (s, r, o) \in P_1 \cup P_2 \cup P_3$  do
4:   if  $|\{j \mid t \in P_j\}| \geq 2$  and  $\text{sim}(s, x) \geq \tau_s^{\text{anc}}$  and  $\text{sim}(o, x) \geq \tau_o^{\text{anc}}$  then
5:      $T_1 \leftarrow T_1 \cup \{t\}$  ▷ Filter 1: agreement + lexical anchoring
6:   else if  $t$  carries supporting span  $\text{support}(s, r, o) \subseteq x$  and  $E_B(t, x) \geq \tau_B$  then
7:      $T_2 \leftarrow T_2 \cup \{t\}$  ▷ Filter 2: evidence score  $E_B$ 
8:   else if  $E_C(t, x) \geq \tau_C$  then
9:      $T_3 \leftarrow T_3 \cup \{t\}$  ▷ Filter 3: surface fallback score  $E_C$ 
10:  end if
11: end for
12:  $\hat{G} \leftarrow T_1 \cup T_2 \cup T_3$ 
13: return  $\hat{G}$ 

```

heuristic features: complete-match, subject-match, object-match, and span-conciseness. Respectively, they are weighted by $\alpha_B, \beta_B, \gamma_B$, and δ_B . The complete-match returns 1 if the subject, predicate, and object all appear within the evidence span; this carries the highest weight, since joint surface presence is the strongest single indicator of textual support. The subject-match and object-match check whether each argument appears in the evidence span. Finally, the span-conciseness is computed as the Jaccard similarity [46] between the evidence span and the source text, penalizing overly broad spans that would otherwise inflate the score. When $E_B \geq \tau_B$, line 7 admits t into the Filter 2 set T_2 . This design follows the claim–evidence pairing principles established in automated fact-verification research [47].

Filter 3: Surface Grounding Fallback. Candidates that fail both earlier branches reach the Filter 3 of Algorithm 1 (lines 8–9). They have neither multi-prompt agreement nor an explicit evidence span, a situation that is common for Strategy 3 outputs. To verify whether they are nonetheless supported by x , we apply a surface-level fallback score E_C , computed as a weighted sum of three heuristic features: Subject Grounding, Object Grounding, and Triple Coherence. Respectively, they are weighted by α_C, β_C , and γ_C . The dominant term, *Subject Grounding*, returns 1 if the subject matches a span in the text above the strict threshold τ_s^{fuz} , measured by the normalized Levenshtein ratio [44]; the strict threshold and higher weight reflect that subjects are typically stable named entities, and that a hallucinated subject would corrupt an entire entity node in the resulting graph. The *Object Grounding* feature applies the same test to the object, but against the more relaxed threshold τ_o^{fuz} , since objects often contain free-text descriptions. The final feature, *Triple Coherence* is computed as the Jaccard similarity between the concatenated triple elements and the source text, supplying a weak overall coherence signal. When $E_C \geq \tau_C$, line 9 admits t into the Filter 3 set T_3 , closing the cascade.

3.3. Final Knowledge Graph

Let T_i denote the set of triples accepted by Filter i in Stage 2. The final knowledge graph is the union $\hat{G} = T_1 \cup T_2 \cup T_3$. The two stages work complementarily: Stage 1 maximizes recall through diverse prompts, while Stage 2 safeguards precision by demanding grounded evidence at one of three levels. This design separates the task of *finding* candidate triples from the task of *validating* them.

4. Experiments

We design our experiments to answer the following research questions.

- **RQ₁**: *How effectively can our framework construct KGs from unstructured text?*
- **RQ₂**: *To what extent can our framework mitigate hallucinations during KG construction?*

4.1. Evaluation Setup

Datasets. We evaluate on *Text2KGBench* [14], an ontology-guided benchmark for text-to-KG construction. It contains two datasets: *DBpedia-WebNLG*, comprising 4,860 sentences across 19 ontologies with multiple gold triples per sentence, and *Wikidata-TekGen*, containing 13,474 sentences across 10 Wikidata-derived ontologies with a single annotated triple per sentence. The datasets differ substantially in quality. *Wikidata-TekGen* is constructed through distant supervision and therefore contains incomplete and noisy sentence–triple alignments. For example, a sentence may express multiple relations while only one is annotated, or may omit explicit mentions of entities contained in the associated triple. To reduce this limitation, we enrich the training data before fine-tuning.

Wikidata-TekGen Enrichment. Following prior data-enrichment approaches [31,48,49,50], we use *Qwen2.5-14B-Instruct* [51] to generate additional ontology-consistent triples. The original gold triple is used as an anchor during generation to preserve factual grounding. Generated triples are retained only if their relations belong to the target ontology.

Baselines. We compare against four open-source baselines that report results on *Text2KGBench*. *Alpaca-LoRA-13B* [14] is an instruction-tuned LLaMA-13B fine-tuned with LoRA [30] on GPT-4-generated data. *Vicuna-13B* [14] is a LLaMA-13B fine-tuned on multi-turn ShareGPT conversations. *T5-large* [52] is a 738M-parameter encoder–decoder model that frames triple extraction as sequence-to-sequence generation. *ChatGLM-6B* [29] is an open bilingual LLM; we adopt its CoT-with-ontology variant, which reports the best results. We exclude Norouzi et al. [11] and ODKE+ [12] from this comparison: although both are conceptually closer to our framework, neither releases full prompts or code, which makes faithful reimplementing infeasible.

Evaluation Metrics. For **RQ₁**, we report Precision, Recall, and F1-score using exact matching between normalized predicted and gold triples. Normalization includes lowercasing, removal of non-alphanumeric characters, and whitespace stripping for all labels. Moreover, we also perform underscore replacement and WordNet lemmatization for subject and object labels. For **RQ₂**, we use four hallucination-related metrics from

Text2KGBench: Subject Hallucination (SH), Object Hallucination (OH), Ontology Conformance (OC), and Relation Hallucination (RH). SH and OH measure the proportion of extracted subjects and objects unsupported by the source text after stemming and tokenization, defined as $SH = |HS|/|\hat{G}|$ and $OH = |HO|/|\hat{G}|$, where HS and HO denote hallucinated subject and object sets in the generated graph $\hat{G}$. The OC metric measures the proportion of extracted triples whose relations belong to the target ontology, $OC = |R_{ont}|/|\hat{G}|$, where $R_{ont} \subseteq \hat{G}$. Meanwhile, the RH metric is defined as its complement, $RH = 1 - OC$. We highlight that neither OC nor RH metrics validate complete class constraints, domain/range compatibility, or logical consistency.

4.2. Implementation Details

Base Model and Training. We use *LLaMA-3-8B-Instruct* as the base model across all framework components and set decoding temperature to 0 for reproducibility. The model is fine-tuned using 4-bit QLoRA on the official Text2KGBench training split, combining DBpedia-WebNLG training data and enriched Wikidata-TekGen training data. The 95/5 split is used only within this training portion for internal training and validation during model selection. All results reported in Tables 1 and 2 are computed on the separate held-out Text2KGBench test split, which is not used for fine-tuning or validation. All experiments run on a single NVIDIA RTX 3090 GPU (24 GB). All details about the prompts, models, and hyperparameters can be found on the GitHub repository for reproducing our experiments and supporting further research.

Strategy and Validator Configuration. The three prompting strategies use different model configurations. *Tree-of-Thoughts Search* employs the fine-tuned model for candidate generation, while retaining the base model for state evaluation and evidence extraction to avoid self-validation effects. We employ $k = 2$, $D_{max} = 2$, and $\tau = 0.7$ for ToT Search hyperparameters. *Schema-Constrained Open IE* uses the base model only, following prior ontology-guided extraction approaches. *Relaxed Extraction* employs the fine-tuned model because structured generation benefits from task adaptation. Validator thresholds were tuned through grid search on a held-out 5% subset of DBpedia-WebNLG and kept fixed across all experiments. The hyperparameters are set as follows. For *Filter 1*, we set $\tau_s^{anc} = \tau_o^{anc} = 0.9$. For *Filter 2*, we set $\tau_B = 0.9$, $\alpha_B = 0.40$, $\beta_B = \gamma_B = 0.25$, and $\delta_B = 0.10$. For *Filter 3*, we set $\tau_s^{fuz} = 0.8$, $\tau_o^{fuz} = 0.55$, $\alpha_C = 0.50$, $\beta_C = 0.40$, and $\gamma_C = 0.10$.

5. Results and Discussion

We evaluate our framework against four baselines across 29 domains on two benchmarks, addressing RQ₁ (extraction quality) and RQ₂ (hallucination mitigation). Our results consistently demonstrate that ontology-aware, multi-strategy extraction with evidence-driven validation outperforms all baselines in extraction quality and substantially reduces hallucinations across most domains.

5.1. Extraction Quality (RQ₁)

Table 1 reports per-domain Precision, Recall, and F1-score. Our framework achieves the highest F1-score in all 19 DBpedia-WebNLG domains and in 7 of 10 Wikidata-TekGen

Table 1. Comparative performance analysis (Precision, Recall, and F1) across **DBpedia** and **Wikidata** domains (**RQ₁**). Higher is better. Best values are highlighted in **bold**.

Domain	ALPACA-LoRA			VICUNA			T5-LARGE			CHATGLM			OUR APPROACH		
	P	R	F1	P	R	F1	P	R	F1	P	R	F1	P	R	F1
DBpedia-WebNLG															
University	0.29	0.16	0.20	0.31	0.19	0.23	0.38	0.38	0.38	0.32	0.14	0.19	0.79	0.63	0.68
MusicalWork	0.18	0.15	0.15	0.20	0.18	0.18	0.39	0.40	0.40	0.22	0.15	0.17	0.59	0.56	0.56
Airport	0.28	0.21	0.20	0.32	0.23	0.23	0.40	0.40	0.40	0.23	0.16	0.18	0.77	0.66	0.69
Building	0.35	0.27	0.29	0.48	0.33	0.38	0.40	0.40	0.40	0.40	0.23	0.28	0.72	0.71	0.71
Athlete	0.38	0.30	0.32	0.39	0.26	0.29	0.38	0.39	0.39	0.28	0.22	0.24	0.79	0.76	0.76
Politician	0.39	0.27	0.30	0.39	0.28	0.32	0.39	0.39	0.39	0.29	0.12	0.16	0.66	0.63	0.64
Company	0.35	0.21	0.25	0.48	0.30	0.34	0.35	0.36	0.36	0.40	0.25	0.30	0.86	0.67	0.73
CelestialBody	0.52	0.41	0.47	0.48	0.46	0.46	0.38	0.38	0.38	0.59	0.49	0.52	0.91	0.94	0.92
Astronaut	0.34	0.24	0.27	0.48	0.36	0.40	0.39	0.39	0.39	0.45	0.21	0.27	0.89	0.81	0.84
ComicsCharacter	0.52	0.41	0.45	0.45	0.41	0.42	0.32	0.32	0.32	0.17	0.14	0.15	0.70	0.68	0.68
Transport*	0.13	0.07	0.08	0.14	0.09	0.10	0.40	0.40	0.40	0.33	0.23	0.25	0.69	0.61	0.63
Monument	0.11	0.07	0.08	0.05	0.05	0.05	0.37	0.37	0.37	0.13	0.12	0.12	0.58	0.57	0.55
Food	0.38	0.28	0.31	0.40	0.32	0.34	0.34	0.34	0.34	0.51	0.37	0.41	0.76	0.81	0.77
WrittenWork	0.39	0.35	0.36	0.34	0.34	0.34	0.38	0.38	0.38	0.52	0.31	0.38	0.90	0.90	0.90
SportsTeam	0.41	0.29	0.31	0.40	0.32	0.34	0.34	0.34	0.34	0.16	0.14	0.14	0.74	0.71	0.71
City	0.10	0.09	0.09	0.12	0.12	0.12	0.50	0.50	0.50	0.12	0.09	0.10	0.61	0.62	0.60
Artist	0.35	0.33	0.34	0.52	0.43	0.46	0.39	0.40	0.39	0.30	0.21	0.24	0.54	0.56	0.54
Scientist	0.42	0.33	0.35	0.53	0.43	0.46	0.39	0.40	0.39	0.33	0.23	0.25	0.81	0.80	0.80
Film	0.18	0.15	0.13	0.23	0.14	0.14	0.39	0.39	0.39	0.14	0.09	0.11	0.45	0.44	0.44
Count as Best	0	0	0	0	0	0	0	0	0	0	0	0	19	19	19
Wikidata-TekGen															
Movie	0.28	0.14	0.17	0.33	0.23	0.25	0.27	0.29	0.28	0.34	0.57	0.43	0.46	0.41	0.41
Music	0.33	0.19	0.22	0.42	0.28	0.32	0.29	0.30	0.29	0.29	0.47	0.36	0.54	0.51	0.50
Sport	0.51	0.44	0.45	0.57	0.52	0.54	0.50	0.56	0.53	0.21	0.37	0.27	0.57	0.64	0.58
Book	0.25	0.21	0.21	0.35	0.28	0.30	0.35	0.29	0.37	0.32	0.55	0.41	0.42	0.46	0.42
Military	0.21	0.21	0.21	0.24	0.23	0.23	0.39	0.41	0.40	0.50	0.79	0.62	0.49	0.60	0.52
Computer	0.29	0.27	0.27	0.28	0.25	0.26	0.33	0.34	0.33	0.33	0.57	0.42	0.48	0.51	0.48
Space	0.58	0.55	0.52	0.67	0.63	0.63	0.52	0.54	0.53	0.53	0.74	0.62	0.76	0.78	0.76
Politics	0.22	0.21	0.21	0.32	0.32	0.32	0.42	0.42	0.42	0.27	0.39	0.32	0.59	0.65	0.60
Nature	0.15	0.15	0.15	0.27	0.24	0.25	0.42	0.53	0.47	0.22	0.29	0.25	0.54	0.59	0.55
Culture	0.15	0.16	0.15	0.31	0.31	0.31	0.45	0.45	0.45	0.32	0.32	0.32	0.42	0.48	0.44
Count as Best	0	0	0	1	0	0	1	0	1	1	4	2	8	6	7

domains. The largest margins arise where the ontology declares a rich predicate set and fine-grained class types: *CelestialBody* (F1 = 0.92 vs. 0.52 for ChatGLM, the strongest baseline), *WrittenWork* (0.90 vs. 0.38 for T5-Large), *Astronaut* (0.84 vs. 0.40 for Vicuna), *Food* (0.77 vs. 0.41 for ChatGLM), and *Airport* (0.69 vs. 0.40 for T5-Large). In each of these domains, our F1 exceeds twice that of the best baseline—a pattern directly attributable to ontology grounding. When the schema provides discriminative domain and range information, our framework uses it to guide candidate generation and vali-

dation, differently from the baselines, which do not include an explicit evidence-driven validation stage. On Wikidata-TekGen, the framework still wins in most domains, but with smaller margins. The strongest gains here are in *Space* ($F1 = 0.76$), *Politics* (0.60), *Nature* (0.55), and *Sport* (0.58). We attribute this to the distant supervision artifacts described in Section 4.1, which limit the achievable F1 on this benchmark. All results are obtained using *LLaMA-3-8B-Instruct* (8B parameters), while *Alpaca-LoRA* and *Vicuna* both use 13B parameters. Ontology-aware prompting, therefore, appears to provide an inductive bias that partially offsets the reduced model scale.

5.2. Hallucination Mitigation (RQ_2)

Table 2 reports the hallucination-related metrics OC, SH, RH, and OH. We compare only against *Alpaca-LoRA-13B* and *Vicuna-13B* baselines, since published *Text2KGBench* evaluations for *T5-large* and *ChatGLM-6B* do not report these metrics. On DBpedia-WebNLG, OC is consistently high, reaching 0.97–1.00 in domains such as *University*, *WrittenWork*, *SportsTeam*, and *CelestialBody*, with RH close to zero. Two design choices contribute to this outcome: *Schema-Constrained Open IE* restricts the model’s relation vocabulary to ontology predicates, while *Filter 2* of the validator accepts a triple when the evidence span explicitly contains the predicted relation. OH decreases substantially even in surface-ambiguous domains, particularly for *City* (OH decreases from 0.68 under *Alpaca-LoRA* to 0.26) and *Artist* (from 0.26 to 0.09). This effect is largely driven by the joint-presence term ($\alpha_B = 0.40$), where an object is added to $\hat{G}$ only if it co-occurs with its subject and relation within the cited evidence span, preventing generation plausibility alone from validating a triple. Wikidata-TekGen exhibits less consistent behavior across domains. OC remains high and OH remains low in well-defined domains (*Politics*, *Culture*, *Music*). However, RH increases in domains with sparse or ambiguous ontology coverage, particularly *Space*. Examination of failure cases shows that source sentences contain relations not represented in the underlying Wikidata-derived ontology; consequently, the evaluator labels these otherwise valid extractions as RH. This effect is largely attributable to benchmark limitations rather than systematic failures of the framework, and explains the remaining RH observed on this dataset.

The main exception on DBpedia-WebNLG is *MusicalWork*, where SH increases to 0.54 (*Alpaca-LoRA*: 0.32, *Vicuna*: 0.32). This apparent degradation is due to the *MusicalWork* structural characteristics, which contain several fine-grained subject types (e.g., `dbo:MusicalWork`, `dbo:Album`, `dbo:Single`). *Text2KGBench* computes SH using exact type matching against these fine-grained classes. Our framework may predict a semantically correct sibling class for the domain, which remains ontologically plausible but is counted as a hallucination. In contrast, fine-tuned baselines such as *Alpaca-LoRA* and *Vicuna* generate fewer distinct entity-type predictions and therefore incur fewer type mismatches. Importantly, the high OC (0.97) and low RH (0.03) for *MusicalWork* indicate that relation extraction and overall OC remain strong, while only fine-grained subject-type resolution is affected.

5.3. Ablation Study (RQ_1 and RQ_2)

Table 3 reports an ablation of the three prompting strategies and the hierarchical validator. Configurations are denoted by the strategies included: T = *Tree-of-Thoughts Search*,

Table 2. Performance evaluation of all models in mitigating hallucinations, averaged per domain (RQ₂). Best values per metric are highlighted in **bold**. OC $\uparrow$: higher is better. SH, RH, OH $\downarrow$: lower is better.

Domain	ALPACA-LoRA				VICUNA				Our Approach			
	OC	SH	RH	OH	OC	SH	RH	OH	OC	SH	RH	OH
DBpedia-WebNLG												
University	0.89	0.13	0.11	0.26	0.92	0.11	0.08	0.21	0.97	0.00	0.03	0.02
MusicalWork	0.85	0.32	0.15	0.33	0.89	0.32	0.11	0.27	0.97	0.54	0.03	0.23
Airport	0.94	0.12	0.06	0.45	0.92	0.03	0.08	0.27	0.97	0.04	0.03	0.16
Building	0.93	0.17	0.07	0.32	0.98	0.02	0.02	0.22	0.98	0.02	0.02	0.07
Athlete	0.94	0.11	0.06	0.27	0.92	0.01	0.08	0.13	0.97	0.11	0.03	0.13
Politician	0.92	0.15	0.08	0.38	0.89	0.11	0.11	0.29	0.97	0.16	0.03	0.19
Company	0.95	0.17	0.05	0.44	1.00	0.09	0.00	0.36	0.97	0.01	0.03	0.15
CelestialBody	0.96	0.12	0.04	0.49	0.97	0.05	0.03	0.46	1.00	0.01	0.00	0.34
Astronaut	0.87	0.08	0.13	0.58	0.87	0.06	0.13	0.28	0.98	0.00	0.02	0.28
ComicsChar.	0.93	0.62	0.07	0.42	0.97	0.64	0.03	0.28	0.98	0.41	0.02	0.08
Transport*	0.97	0.18	0.03	0.36	0.94	0.14	0.06	0.41	0.98	0.17	0.02	0.16
Monument	0.97	0.13	0.03	0.31	0.94	0.18	0.06	0.31	1.00	0.00	0.00	0.31
Food	0.92	0.12	0.08	0.26	0.94	0.05	0.06	0.20	0.99	0.06	0.01	0.11
WrittenWork	0.90	0.20	0.10	0.50	0.92	0.12	0.08	0.33	0.99	0.05	0.01	0.15
SportsTeam	0.90	0.11	0.10	0.24	0.91	0.04	0.09	0.11	0.99	0.03	0.01	0.08
City	0.96	0.07	0.04	0.68	0.98	0.04	0.02	0.67	0.95	0.28	0.05	0.26
Artist	0.83	0.07	0.17	0.26	0.89	0.03	0.11	0.13	0.99	0.37	0.01	0.09
Scientist	0.92	0.14	0.08	0.43	0.95	0.05	0.05	0.30	0.97	0.02	0.03	0.23
Film	0.82	0.08	0.18	0.33	0.94	0.30	0.06	0.19	0.98	0.01	0.02	0.14
Count as Best	0	1	0	1	3	8	3	3	16	10	16	17
Wikidata-TekGen												
Movie	0.92	0.25	0.08	0.24	0.89	0.26	0.11	0.26	0.90	0.23	0.10	0.06
Music	0.91	0.18	0.09	0.24	0.94	0.16	0.06	0.22	0.94	0.13	0.06	0.04
Sport	0.96	0.18	0.04	0.11	0.85	0.22	0.15	0.13	0.90	0.14	0.10	0.06
Book	0.94	0.17	0.06	0.19	0.92	0.16	0.08	0.23	0.88	0.12	0.12	0.05
Military	0.93	0.21	0.07	0.26	0.80	0.19	0.20	0.26	0.87	0.16	0.13	0.08
Computer	0.83	0.17	0.17	0.13	0.85	0.15	0.15	0.11	0.81	0.13	0.19	0.02
Space	0.90	0.14	0.10	0.10	0.93	0.15	0.07	0.08	0.79	0.09	0.21	0.04
Politics	0.90	0.19	0.10	0.15	0.92	0.17	0.08	0.15	0.95	0.15	0.05	0.04
Nature	0.93	0.22	0.07	0.20	0.68	0.10	0.04	0.14	0.91	0.08	0.09	0.06
Culture	0.54	0.16	0.46	0.14	0.59	0.15	0.39	0.12	0.83	0.11	0.17	0.06
Count as Best	4	0	4	0	3	1	4	0	4	10	3	10

S = Schema-Constrained Open IE, and R = Relaxed Extraction. The suffix *w/o Evaluator* indicates that the candidate pool is added directly to $\hat{G}$ without the three-filter validator, while *Full* denotes the complete pipeline.

Effect of the Validator. The validator’s contribution is isolated by comparing *Full* with *TSR w/o Evaluator*, since both use all three prompting strategies. On DBpedia-WebNLG, the validator improves F1 from 0.64 to 0.69 and OC from 0.92 to 0.98, and reduces RH

Table 3. Average F1, OC, SH, RH, and OH scores across ablation variants of our approach. Full denotes the complete pipeline, including the evaluator. The remaining variants use subsets of prompting strategies: *T* (Tree-of-Thoughts), *S* (Schema-constrained Open IE), and *R* (Relaxed Extraction). Best results are highlighted in **bold**.

Variation	DBpedia-WebNLG					Wikidata-TekGen				
	F1 ↑	OC ↑	SH ↓	RH ↓	OH ↓	F1 ↑	OC ↑	SH ↓	RH ↓	OH ↓
Full	0.69	0.98	0.12	0.02	0.16	0.54	0.91	0.17	0.09	0.05
T w/o Evaluator	0.65	0.99	0.13	0.01	0.20	0.54	0.94	0.17	0.06	0.12
S w/o Evaluator	0.42	0.86	0.05	0.14	0.11	0.29	0.77	0.02	0.23	0.13
R w/o Evaluator	0.66	0.99	0.11	0.01	0.20	0.50	0.94	0.17	0.06	0.10
TS w/o Evaluator	0.62	0.91	0.09	0.09	0.17	0.51	0.83	0.01	0.17	0.14
TR w/o Evaluator	0.68	0.99	0.12	0.01	0.20	0.50	0.93	0.17	0.07	0.13
SR w/o Evaluator	0.62	0.91	0.08	0.09	0.17	0.48	0.81	0.08	0.19	0.14
TSR w/o Evaluator	0.64	0.92	0.09	0.08	0.18	0.52	0.83	0.10	0.14	0.15

from 0.08 to 0.02. Similar gains on Wikidata-TekGen, where F1 increases from 0.52 to 0.54, OC from 0.83 to 0.91, and OH decreases substantially from 0.15 to 0.05. These results show that the validator converts a diverse but noisy candidate pool into a graph with stronger ontology alignment and fewer hallucinations. In contrast, SH increases slightly after introducing the validator (DBpedia-WebNLG: 0.09 $\rightarrow$ 0.12; Wikidata-TekGen: 0.10 $\rightarrow$ 0.17). We interpret this as a recall–precision trade-off induced by *Filter 3*. The validator preserves additional triples that would otherwise be discarded, some of which rely on fuzzy subject matching. Although this introduces more subject mismatches, the overall effect remains beneficial because F1 and OC improve simultaneously.

Strategies in Isolation. Among individual strategies, *T* performs best, achieving F1 scores of 0.65 on DBpedia-WebNLG and 0.54 on Wikidata-TekGen, with corresponding OC values of 0.99 and 0.94. This suggests that the iterative ontology-guided search effectively filters low-quality candidates during generation. *R* also performs competitively (0.66 and 0.50), indicating that relaxed output constraints improve recall by recovering otherwise discarded triples. In contrast, *S* performs worst in isolation (0.42 and 0.29) and exhibits the highest RH values, suggesting that schema constraints alone are insufficient without downstream validation.

Strategy Combinations. The interaction between strategies further highlights their complementary roles. On DBpedia-WebNLG, *TR w/o Evaluator* achieves F1 = 0.68, approaching the full system (0.69), indicating that *T* contributes precision while *R* improves coverage. Adding *S* without the validator does not improve performance, as additional candidates primarily introduce noise. On Wikidata-TekGen, *TSR w/o Evaluator* performs best among non-validator configurations (F1 = 0.52), but still falls below the complete pipeline in ontology conformance (0.83 vs. 0.91). These results suggest that prompt diversity alone can improve recall, whereas the validator is required to convert this diversity into reliable graph construction.

Summary. Overall, *T* is the primary contributor to extraction quality, *R* provides additional recall, and *S* contributes diversity that becomes beneficial only when combined

with structured validation. Together, these findings answer RQ₁ and RQ₂: prompt diversity increases candidate coverage, while evidence-driven validation transforms this diversity into a reliable knowledge graph.

5.4. Limitations

We acknowledge three limitations of this work. First, performance depends on the quality and completeness of the supplied ontology, and robustness to incomplete or noisy ontologies remains untested. Performance may also vary with the model, prompt, decoding settings, ontology structure, and dataset. A broader robustness study across model families, prompt variants, and ontology types is complementary to quantify the robustness. Second, both *Text2KGBench* datasets contain independent short sentences, limiting conclusions regarding longer contexts such as paragraphs or documents. Third, the generated knowledge graph is not linked to external resources such as DBpedia or Wikidata. Integrating entity linking could enable bidirectional enrichment with existing KGs.

6. Conclusions and Future Work

We presented an ontology-aware, multi-prompt framework for RDF triple extraction that decouples candidate generation from validation. Three prompts balance precision, schema adherence, and recall, while a three-filter, evidence-driven validator admits only triples supported by cross-prompt agreement, evidence spans, or surface grounding. On the *Text2KGBench* benchmark across 29 ontologies, the framework outperforms four baselines in F1-score on all DBpedia-WebNLG domains and most Wikidata-TekGen domains, despite using an 8B-parameter backbone. It substantially reduces object and relation hallucinations while improving ontology conformance, and ablation results identify the validator as the key component that transforms prompt diversity into a high-precision graph. Future work will evaluate the framework’s robustness to ontology quality, assess its performance under longer contexts, such as paragraphs and full documents, and integrate entity linking with public knowledge graphs, including DBpedia and Wikidata.

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Declaration on Generative AI

During the preparation of this work, the authors used *ChatGPT*, *Gemini*, and *Grammarly* for grammar and spelling correction, paraphrasing, and linguistic polishing of human-written text. The authors carefully reviewed and edited the content and take full responsibility for the integrity and content of the publication.

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Generative AI, LLMs & Retrieval-Augmented Knowledge Graphs

OntoExtend: A Framework for Requirement-Driven and Scalable Ontology Extension with LLMs

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Abstract. Ontology extension refers to the process of enriching an existing ontology in response to emerging requirements, making it more complete. This task is a resource-intensive and error-prone process. Large Language Models (LLMs) have shown promising performance on generating ontologies from scratch, but current approaches rarely tie ontology extension explicitly to requirements or reusable core models, and offer limited, systematic evaluation of LLM outputs. This paper introduces *OntoExtend*, a requirements-driven framework for ontology extension with LLMs. It uses retrieval-augmented generation (RAG) over relevant input ontologies and requirements in the form of competency questions to propose grounded extensions. We evaluate OntoExtend on 39 CQs from two use cases: a public EU-project ontology, Onto-DESIDE, and an industrial ontology from Bosch. The generated fragments show few structural issues, satisfy all functional evaluation tests, and are rated by ontology engineers as requiring minor to moderate revision before integration. These results suggest that OntoExtend is useful as a drafting assistant for requirement-driven ontology extension in real world scenarios, while remaining sensitive to CQ specificity and modelling profile.

Keywords. Ontology extension, Ontology generation, Ontology engineering, Large Language Models

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1. Introduction

In recent years, Large Language Models (LLMs) have been increasingly integrated into various stages of the ontology engineering pipeline. Existing work includes ontology generation with LLMs [13,14,16,25], where an ontology is constructed from a set of requirements, and ontology evaluation [15,17,28], where the goal is to determine whether a given ontology correctly models those requirements. However, to the best of our knowledge, no prior study has examined how LLMs can extend an existing (input) ontology by reusing ontology elements while modelling additional requirements provided in the form of competency questions (CQs), i.e. natural language questions outlining and constraining the scope of an ontology. This task, which we call *ontology extension*, poses several challenges. For instance, input ontologies often contain hundreds of classes and properties, exceeding the input context limitations of current LLMs. Even when an ontology is small enough to fit within the large context window of state-of-the-art LLMs, they may be misguided by irrelevant details and produce off-target or inconsistent outputs [25].

Ontology extension is thus the systematic enrichment of an existing ontology to satisfy new functional requirements [22]. Using LLMs to produce these extensions could enable incremental maintenance of ontology modules as new requirements arise. To this end, we introduce the *OntoExtend* framework (Fig. 1), a retrieval-augmented approach for ontology extension that takes new CQs and the existing ontology as input. For each CQ, *OntoExtend* retrieves from the input ontology the named classes and properties, along with their axioms, that are relevant to the CQ. The CQ, together with this retrieved fragment, are then provided to an LLM as relevant context without exceeding its context window or overburdening it. In the rest of this paper, we use the term *fragment* to denote the self-contained set of RDF/Turtle axioms retrieved or generated for a specific extension step associated with a single CQ. This enables the LLM to generate highly contextualised ontology extensions that can be consistently integrated into the input ontology to model the new requirement.

The research questions (RQs) driving this work are: (i) Which embedding configuration is the most suitable for building a retrieval model that extracts a compact fragment of the ontology most relevant for a given CQ? (ii) Which LLMs are most effective at extending the retrieved fragment so that the CQ is correctly modelled? (iii) What evaluation criteria are required to assess the quality of the generated ontology fragments? (iv) What are the strengths and weaknesses of the extended ontologies produced by *OntoExtend*? To answer these questions, we propose the following contributions:

1. *OntoExtend*: our proposed framework that extends ontologies based on new requirements²;
2. Experiments comparing retrieval and prompting strategies to identify effective configurations for LLM-based ontology extension on two domain-specific, real-world case studies, demonstrating the approach’s applicability across different domains and requirements;
3. An evaluation methodology that assesses both structural validity and functional adequacy of generated extensions against the requirements in real-life settings through a user-based study.

²The code and the experiment data (ontologies and CQs), along with examples that could not fit in the paper, are available at <https://github.com/dersuchendee/OntoExtend>

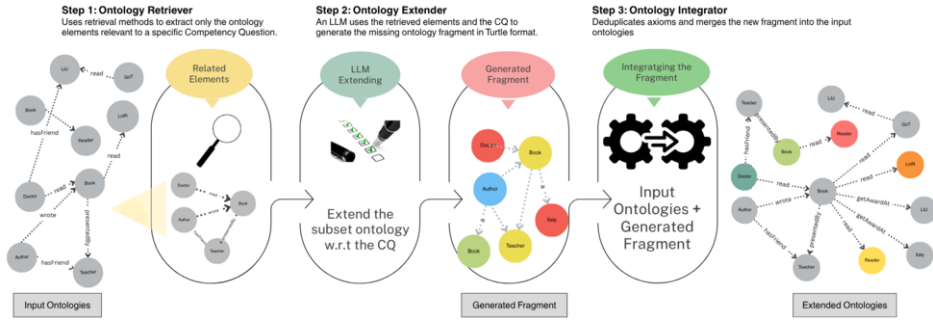


Figure 1. Overview of Ontoextend: 1) The *Ontology Retriever* extracts the relevant ontology elements of the input ontologies given a new competency question. 2) The *Ontology Extender* uses the retrieved ontology elements together with the competency question to prompt an LLM to generate missing ontology fragments. 3) The *Ontology Integrator* integrates the fragments into the extended ontologies.

The rest of the paper is organised as follows. Section 2 presents the related work; Section 3 describes OntoExtend; Section 4 describes the experimental setup; Section 5 shows the evaluation, and Section 6 reports the experimental results. Finally, Section 7 discusses the evaluation metrics, the impact and extensibility of the benchmark, and Section 8 concludes the paper and outlines future research directions.

2. Related Work

LLM-based ontology extension, at times called *ontology enrichment*, remains in an early phase with respect to other LLM-assisted knowledge engineering tasks, with approaches ranging from interactive tools to automated enrichment systems. Early approaches demonstrate the potential and limitations of semi-automated methods: Soares et al. [26] extended the Agricultural Product Types Ontology (APTO) using ChatGPT-4 for interactive generation of OWL taxonomic axioms. Similarly, Matieu and Groza [18] developed a Protégé plugin using a fine-tuned GPT-3 model to translate controlled natural language into OWL Functional Syntax. Work in Phrase2Onto [22], a prototype for ontology extension, showed promising results in this direction but was limited to toy ontologies. In addition to the mentioned limitations, these studies do not apply to larger ontologies.

Automated extension systems push toward greater autonomy while exposing new challenges: Taxoria [8] focuses on taxonomy enrichment by having LLMs propose child terms for existing nodes, which are then validated for semantic relevance before integration with provenance tracking, though control over hallucinated nodes and implicit requirement capture remain concerns. Wu et al. [29]’s online clustering framework extends this automation further, using LLM agents to propose and cluster new classes around existing concepts in an evolving medical ontology. Still in the biomedical domain, Dong et al. [3] focus on enriching OWL ontologies as formal KBs, finding that the baseline LLM-based methods are yet to achieve satisfying performance on the benchmark.

Comprehensive extension frameworks address broader ontology engineering needs: Kholmska et al. [12]’s work extends the input ontology through a multi-LLM workflow supporting concept search, extraction, alignment, and the generation of CQs and SPARQL

queries, though this exposes LLM limitations in highly specialised domains and requires manual repair of shallow or incorrect suggestions. Joachimiak et al. [10] developed the Artificial Intelligence Ontology using an Ontology Development Kit workflow with AI-driven curation support. In García Fernandez et al. [7], during the LLM-based extension process, two important limits were found: LLMs hallucinated when asked about existing standards and reusable ontologies, and the LLM-generated extension was shallower than the manual gold standard. Human review stayed essential at every step.

These works reveal that current LLM-based ontology extension is predominantly semi-automatic enrichment of existing inputs with a bias toward taxonomic growth, facing recurring constraints around reproducibility, reliance on interactive tools, limited support for complex axioms, and continued necessity of human judgment for requirement elicitation and validation. However, none of these methods focuses on the retrieval of existing ontology elements from baseline ontologies, while our approach integrates retrieval mechanisms directly into the extension process. Furthermore, most of the works use minimal to no formal evaluation.

Several related works sit nearby but are less about extending a mature input ontology and more about generating or reconstructing ontologies from scratch [1,4,9,17,21]. A parallel line of work looks at ontology transformation and general LLM-based assistance for ontology engineering [16,24,25,27]. These works reinforce the usefulness of LLM support throughout the ontology lifecycle, but they do not yet address retrieval-aware, requirement-driven extension of a mature input ontology.

3. The OntoExtend framework

The proposed system implements a retrieval-based pipeline for ontology extension, organised into three principal subsystems: the Ontology Retriever, the Ontology Extender, and the Ontology Integrator, illustrated in Fig. 1. Throughout this paper, input ontologies and reference ontologies refer to the same artefacts: the ontology (or ontologies) the user wants to extend, which are also indexed by the Ontology Retriever for element retrieval. The RAG component indexes the same ontology files that constitute the extension target, so that generated fragments are grounded in and consistent with the existing modelling choices. Optionally, previously generated fragments can be re-indexed so that later CQs build on earlier ones.

3.1. Ontology Retriever

Since the whole set of input ontologies may not fit into the context size of current LLMs, the Ontology Retriever is responsible for constructing and querying a semantic index over the input ontologies (i.e., the ontologies to be extended).

At preprocessing time, the retriever component parses each ontology file specified as input by the user and iterates over all declared OWL entities, including classes, object properties, data properties, annotations, and their SHACL shapes if available. For each entity, it constructs an *OntologyElement* record containing: the entity IRI; human-readable labels and comments if present; domain and range declarations (where applicable); super- and sub-class (or sub-property) relations; and a verbatim Turtle snippet capturing the canonical declaration of the entity.

Table 1. Comparison of the related works discussed in Section 2. Columns indicate whether a work explicitly reports ontology pitfall analysis (OOPS!), syntax or well-formedness checking (Syntax), semantic or logical consistency validation (Consist.), verification against requirements (Req. verif.), explicit presence of superfluous elements (Superfl. el), formal user evaluation (User eval.), expert-based assessment (Expert), evidence across multiple domains (Domain gen.), evaluation on an actively used ontology (Real-world Onto.), and evidence of scalability (Scal.). Symbols: Y = yes, P = partial, N = not reported.

Ref.	Approach	Structural			Functional		User eval.	Expert	Domain gen.	Real-world Onto.	Scal.
		OOPS!	Syntax	Consist.	Req. verif.	Superfl. el.					
[26]	Interactive taxonomy extension	N	N	N	N	N	Y	Y	Y	Y	N
[18]	Protégé plugin / CNL to OWL	N	N	P	N	N	N	N	N	N	N
[22]	Prototype ontology extension	N	N	N	N	N	Y	Y	Y	Y	N
[8]	Taxonomy enrichment (Taxoria)	N	N	N	N	N	N	N	N	Y	P
[29]	Online clustering framework	N	N	N	N	N	N	N	N	Y	Y
[3]	Biomedical enrichment benchmark	N	N	N	N	N	N	N	N	Y	P
[12]	Multi-LLM extension workflow	N	P	P	N	N	N	N	N	Y	P
[10]	AI Ontology curation support	N	P	Y	N	N	N	N	N	Y	P
[7]	Human-reviewed ontology extension	N	Y	N	Y	N	N	N	Y	Y	N
[9]	RAG ontology construction	N	N	N	N	N	N	N	N	N	Y
[1]	Research ontology construction	N	N	N	N	N	N	N	N	Y	Y
[17]	Ontology generation	N	N	N	N	N	N	N	N	Y	N
[21]	Ontology-Toolkit	N	N	N	N	N	N	N	N	N	N
[4]	Ontology generation from seeds/corpora	N	N	Y	N	N	N	N	N	N	N
[27]	Ontology transformation support	N	N	N	Y	N	N	Y	N	Y	N
[25]	Ontology engineering assistant	N	Y	N	Y	N	N	N	Y	N	N
[16]	Ontology engineering assistant	Y	Y	N	Y	Y	Y	N	Y	Y	N
Ours	OntoExtend	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y

These *OntologyElements* form the input knowledge base used for subsequent retrieval. To support semantic search, each *OntologyElement* is embedded into a dense vector representation using a configurable sentence embedding model. Each element is serialised as a pipe-delimited string combining its URI local name, human-readable label, comment, element type, and domain/range before embedding, e.g., ‘hasMaterialComponent | has material component | ... | Type: object property | Domain: Material | Range: MaterialComponent’. The resulting vectors are normalized and stored in a FAISS index [5] configured for inner-product similarity. This choice enables efficient nearest-neighbour search over large ontology collections while preserving cosine-like similarity semantics. An example of the element to be embedded with the pipe format is shown in Figure 2.

At query time, a user-supplied CQ is embedded into the same vector space and used to interrogate the FAISS index. The retriever returns the top-k (the default is 20) most similar *OntologyElement* instances, effectively selecting those ontology entities whose textual and structural descriptions are most relevant to the CQ. This retrieval step ensures that the subsequent generation is tightly anchored in the terminology and modelling pat-



Figure 2. Example of the element to be embedded.

terns of the reference ontologies. Moreover, it is the key for promoting reuse of existing ontology elements.

3.2. *Ontology Extender*

The Ontology Extender assembles the final prompt for the LLM with the task to propose consistent extensions. Retrieved elements are grouped by their source ontology and rendered as Turtle snippets, which are injected into the prompt as read-only context. In addition, it constructs a unified prefix block that merges the namespace declarations of all input ontologies, ensuring that every element IRI retrieved from any source ontology resolves correctly in the generated output and can be directly reused by the LLM without namespace conflicts. OntoExtend also enables ontology engineers to configure different prompts so that users can select among the most suitable templates at run-time.

Before a fragment is accepted, it is passed through a two-stage validator: a Turtle parser verifies syntactic correctness, and a constraint checker verifies whether generated properties satisfy the modelling conventions required in the selected use case. In the two profiles used in this evaluation, this includes explicit ‘`rdfs:domain`’ and ‘`rdfs:range`’ declarations, but this is a configurable governance constraint and not a universal criterion for ontology correctness. Fragments failing validation are either retried or flagged.

Given the assembled prompt, this Ontology Extender calls the LLM service and extracts the returned Turtle code block, which is treated as a candidate TBox fragment.

Prompt variants and deployment profiles: To support heterogeneous deployment requirements, the system maintains different prompt templates that configure the extender stage. These prompts are cleanly decoupled from the input retrieval, generation, and integration pipeline.

In our work, we used two distinct templates according to the domain of the data we experimented with (see Section 4). For the industrial ontology, we develop a SHACL-based prompt template, whereas for the EU ontology we defined a prompt template for the generic use of restrictions and reuse of classes policy. The former instructs the LLM to output SHACL `NodeShape` and `PropertyShape` definitions following specific naming conventions and including `sh:name` labels; this is essential for downstream compliance workflows that consume SHACL artefacts. Its modelling guidance largely stops after requiring appropriate domain and range declarations. In contrast, the latter prompt template requires that every newly introduced class and property be annotated with both `rdfs:label` and `rdfs:comment`. Both impose strong reuse constraints, instructing the LLM to reuse existing ontology elements without redeclaring it. By externalising these prompt configurations, the system can switch between companies and use cases without any changes to the underlying system components.

3.3. *Ontology Integrator*

The Ontology Integrator module is responsible for incorporating the generated fragment into the input ontologies. Its operation is limited to deduplication (removing repeated axioms, named classes, properties, and prefixes) followed by concatenation of the cleaned fragment with the input ontologies.

The Ontology Integrator additionally forwards each generated fragment to the Ontology Retriever for indexing, thereby reducing the likelihood that the Ontology Extender would recreate similar elements (classes or properties) when subsequent CQs were processed. Re-indexing allows each generated fragment to serve as a reference for subsequent CQs, promoting cross-fragment consistency. However, this was disabled in our evaluation so that each CQ fragment could be assessed independently.

4. Experimental setup

In this section, we describe the experimental setup, including the dataset we created, parameter tuning for the retrieval and the extender module, and the evaluation criteria.

Dataset Creation: The evaluation covers two settings: four modules of the Onto-DESIDe ontology network³, treated jointly as one setting (20 CQs), and an internal Bosch ontology (19 CQs). Onto-DESIDe addresses circular economy data interoperability, whereas the Bosch setting follows a more manufacturing industrial profile. The dataset was manually created by systematically removing selected classes and their related properties from each ontology, and then formulating CQs about the removed elements. The procedure is outlined below:

1. Select a random set of classes C from the ontology.
2. For each class c , remove c and add to the set P all properties whose domain or range restrictions references c .
3. Iteratively expand the removal process by adding c all subclasses of each removed class to C , and include in P any properties that reference these subclasses. This step is repeated iteratively until no additional subclasses are found.
4. Finally, construct a set of CQs associated with C and P . Each CQ is formulated by asking what each corresponding class with its properties, were intended to represent or accomplish within the original ontology.

This procedure yields a set of CQs that systematically probe the missing ontology components, enabling an effective evaluation of the OntoExtend framework.

Dataset statistics: The EU-project parts have 20 CQs in total, while the Industry ontology comprises 19 CQs. Detailed counts of tokens, the sum of classes and properties, and axiom counts for each subset are provided in the GitHub repository. The input ontologies were converted to Turtle with compact prefix declarations to reduce size. Token counts are measured using the GPT-4o tokeniser (<https://platform.openai.com/tokenizer>), see Table 2.

Parameter and configuration tuning

³<https://cordis.europa.eu/project/id/101058682>

Table 2. Token counts of input ontologies after conversion to Turtle.

Ontology fragment	Size (tokens)	Axioms	Classes+Properties
EU-project — part 1	75 000	2920	405
EU-project — part 2	22 000	958	242
EU-project — part 3	25 000	1172	270
EU-project — part 4	6 000	228	54
Industry use case	22 000	979	134

We ran the experiment on a separate subset to find out the best embedding model and parameter configuration to test on the main experiment, along with the best prompt to minimise the common mistakes by LLMs in ontology generation.

Best retrieval configuration

To embed each ontology element into a vector with good settings, we first ran a small experiment to choose the embedding model and text configuration before the main study to find a good set of parameters and embedders. Basing on the results of previous research [14,25], we compared three OpenAI embedding models: `text-embedding-3-small`, `text-embedding-3-large`, and `text-embedding-ada-002` under four different styles of formulating ontology elements as a raw text before sending them into embedding mode. In particular, we varied the delimiter used to separate consecutive elements, comparing the use of a pipe character (`|`) with the use of a newline character (`\n`). For each configuration, we manually judged relevance on five CQs not present in the test data and computed precision at 3 and 20 (focusing on top-3 retrieved for answer quality in case of creating a small ontology subset, but still monitoring top-20 to have an estimate of recall). In particular, we computed precision at cutoffs 3 and 20 for the retrieved elements being tagged as similar to the input CQ by the Ontology Retriever. Evaluations are done manually by two ontology engineers cross cross-checking each other's work, letting P_3 and P_{20} denote precision at 3 and 20 (often referred to as $P@3$ and $P@20$). Our primary selection metric is a weighted average $M_w = 0.7 P_3 + 0.3 P_{20}$ which emphasises the quality of the very top-ranked results (those most likely to be consumed by the LLM in Ontology Extender) while still rewarding configurations that retrieve more relevant items in the top 20. As an additional check, we also monitored the multiplicative average of them $M_{\text{prod}} = P_3 \cdot P_{20}$, which penalises configurations that perform well at only one cutoff. Both metrics agreed on the same best configuration per model.

Across all 12 configurations, the best overall result came from `text-embedding-ada-002` with pipe-separated element axioms and comments included. On this basis, we selected that as the embedding model for the main experiment. Inputs of the top configurations are shown in Table 3.

Table 3. Embedding model comparison results by examining whether to include comments in the embedding and how to separate axioms in the raw text.

Embedding model	Separator, Comments	M_{prod}	M_w
<code>text-embedding-3-small</code>	Newline, with Comments	0.22	0.62
<code>text-embedding-3-large</code>	Newline, without Comments	0.20	0.54
<code>text-embedding-ada-002</code>	Pipe, with Comments	0.23	0.63

Best prompt and LLM extender

For the LLMs, we selected o1-preview, reported as the best-performing model in previous work [16], and GPT-5, the latest LLM from OpenAI. We did not select other models or families because previous studies reported lower performance relative to o1-preview for what concerns ontology generation tasks [16].

Prompt development followed the same tuning procedure used for retrieval. After fixing the retrieval configuration and integrating it into the extension framework, we evaluated several prompt variants on a small development set of CQs (disjoint from the main evaluation set) to identify formulations that minimised typical LLM errors in ontology construction. The initial prompt was modelled closely on the prompt used in previous work [16] and was augmented with an explicit list of observed pitfalls not to be included in the output. This baseline was refined iteratively: after each pilot run, two ontology engineers inspected generated axioms for recurrent problems and adjusted the prompt wording, constraints accordingly.

The final prompt comprises (i) a general template that specifies the task and required output format, (ii) a section into which the Ontology Retriever's elements are inserted, (iii) a placeholder for the given CQ, and (iv) a directive that indicates the expected modelling formalism. The framework populates all components automatically, while the style directive may be supplied by the user to constrain output style. For instance, in the EU-Project, this section requested OWL restrictions for creating restrictions in the generated ontologies; in the industry use case, we disallowed OWL restrictions and required only SHACL shapes. Consequently, the two use cases should not be interpreted as testing the same level of OWL expressivity. The EU-project setting evaluates OWL-style fragment generation, whereas the industry setting evaluates extension under a SHACL-oriented engineering profile used in that deployment context. The final prompt template is available on the Github repository.

4.1. Evaluation criteria

Building on previous studies in LLM-assisted knowledge engineering [16], we adopt a multidimensional evaluation setup that combines structural and functional metrics along with human evaluation.

Structural Metrics. We employ the Ontology Pitfall Scanner (OOPS!) [23] for reporting on pitfalls, Pellet reasoner and syntax checking through the RDFLib Python library⁴.

Functional Metrics. To assess the generated ontologies, we employ CQ verification [2] and counting of superfluous elements based on Lippolis et al. [16] by modifying the definition of the latter for the specific task of ontology extension. CQ verification evaluates whether a given CQ is actually represented in the ontology by attempting to formulate a SPARQL query that retrieves an answer for that CQ. If no such SPARQL query can be written to obtain an answer, the CQ is considered *not modelled* in the ontology. Likewise, [16] defines a superfluous element in a way that treats some generated classes and properties as superfluous because they cannot be referenced directly in the SPARQL query used for CQ verification. In our work, we adopt a more specific definition: a *superfluous element* is a named class or property that: 1) is not mentioned in the verification SPARQL used to in CQ verification, and 2) is not connected to any component appearing in that query by a `subClassOf` or `subPropertyOf` relation. Under this definition, ele-

⁴<https://rdflib.readthedocs.io/en/stable/>

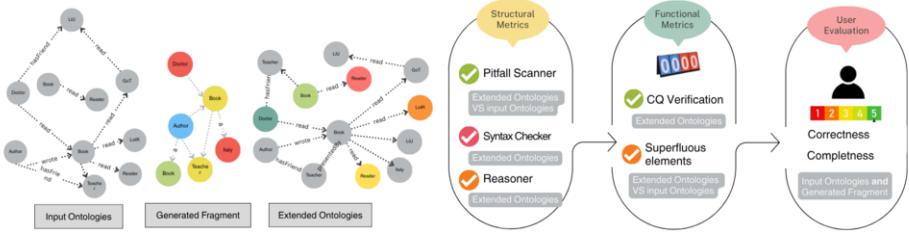


Figure 3. Evaluation workflow: input ontologies, the generated fragments, and the extended ontologies are fed to different evaluation metrics to create the final report.

ments that are absent from the verification query but are hierarchical relatives (subclasses or subproperties) of query components are *not* considered superfluous, while genuinely disconnected named classes and object properties are flagged as superfluous.

Survey. Finally, we created a survey to ask six ontology engineers from both academia and industry to evaluate the extended ontologies. We adopted evaluation criteria adapted from Monka et al. [19], where two dimensions for evaluating *SPARQL* generation are proposed: *Correctness* and *Completeness*. In our work, *Correctness* measures the syntactic and semantic quality of the generated fragment in isolation, and *Completeness* measures the sufficiency of the generated fragment after it has been combined with the input ontology; it captures the amount of effort required by an ontology engineer to make the fragment usable. Users rated the generated ontology fragments on a five-point Likert scale along two dimensions, namely *Correctness* and *Completeness*.

For *Correctness*, a rating of 1 indicates a generation failure, for example because the output exceeded token limits. A rating of 2 denotes a syntactically erroneous fragment, such as one containing issues in the generated code. A rating of 3 is assigned when the fragment is syntactically correct but semantically incorrect, for instance due to an inappropriate taxonomy. A rating of 4 corresponds to a fragment that is both syntactically and semantically correct, but does not fully satisfy the intent of the competency question (CQ). Finally, a rating of 5 indicates a fragment that is syntactically and semantically correct and fully aligned with the intended meaning of the CQ.

For *Completeness*, a rating of 1 means that the generated ontology fragment is not useful and would require complete manual reworking. A rating of 2 indicates that significant changes are needed, corresponding roughly to 25–40% of the fragment requiring modification. A rating of 3 reflects the need for moderate changes, with approximately 11–25% of the fragment needing revision. A rating of 4 denotes that only minor changes are required, typically affecting around 1–10% of the fragment. A rating of 5 indicates that the generated fragment, in combination with the input ontology, is complete and correct.

5. Evaluation

In this section, we present the methodology used to evaluate the generated ontology extensions based on the criteria introduced in Section 4 in detail.

5.1. Structural Evaluation

As shown on the right side of Fig. 3, each generated extension module is validated for correct Turtle syntax via the RDFLib Python library. To assess common modelling pitfalls, we run OOPS! on the input ontology both before and after integrating the generated extension, compare the reported pitfalls, and document any new issues introduced by the extension. In addition, we execute the Pellet reasoner to perform a consistency check on the input ontology after the generated extension has been integrated.

5.2. Functional Evaluation

Functional adequacy (CQ verification). As described in Section 4.1, we perform CQ verification on each generated extension after it is added to the input ontology to ensure that each CQ is correctly modelled (see Figure 3). Two ontology engineers independently verify the CQ modelling decisions and then cross-check each other's annotations; any disagreement is resolved through discussion. This adjudication process follows the same general workflow as in the work by [14].

Superfluous elements. After CQ verification, the same pair of engineers inspects the integrated file to identify and count superfluous elements introduced by the generated extension by comparing the number of superfluous elements before and after adding the extension. As detailed in Section 4.1, each decision is cross-checked, and we report the number of superfluous elements that were generated but deemed unnecessary by the OntoExtend framework.

5.3. Evaluation by Ontology Engineers

In the experiment, we recruited six ontology engineers from both industry and academia to evaluate the generated extension modules. After a short briefing that explained the task and the survey form, each evaluator received a set of CQs, the input ontologies, and the generated ontology fragments for each CQ. For every generated extension, the evaluators answered the two survey questions regarding correctness and completeness (see section 4.1). The evaluators were also encouraged to provide free-text comments. Finally, the evaluators also participated in a debriefing session to share general feedback. Participants were free to use any tools they preferred to visualise the ontology artefacts (e.g., Protégé, Topbrail EDG, VSCode).

After data collection, we computed the mean correctness and completeness metrics from the surveys. We also computed the *Fleiss* weighted observed agreement P_o [20] to quantify the inter-annotator agreement among our evaluators.

6. Results

In this section, we present the results of our experimental setup according to the evaluation criteria outlined in Section 5 of the paper.

Table 4. Summary of structural (syntax and OOPS!) and functional (CQ-verification and percentage of superfluous elements) evaluation for the EU-Project and the Industry ontologies, all LLMs combined and separated.

Use case	OOPS! P2,P4&P8 (o1-prev,GPT-5)	CQ-verification (o1-prev,GPT-5)	Syntax errors (o1-prev,GPT-5)	Superfluous elements (o1,GPT-5)
EU-Project	13 (7,6)	100% (100%, 100%)	0% (0%,0%)	2% (3.8%,0%)
Industry	141(70,71)	100% (100%,100%)	2.5% (5%,0%)	0% (0%,0%)

6.1. Results of Structural Evaluation

Firstly, the generated ontology fragments almost did not exhibit any syntax issues in Turtle. Second, the analysis of the integrated ontologies with OOPS! shows that the generated extension fragments do not introduce any new *critical* or *important* modelling pitfalls after being integrated into their input; only a small number of minor issues were observed in specific cases. For the EU-project ontology, we found two types of minor issues. First, P02 (creating synonyms as classes) appears in a single EU-project only: in that case, each generated extension contributed one P02 instance. Second, P04 (unconnected ontology elements) was detected in just two of the four EU-project use cases; in those two cases, each extension added at most one additional P04 instance.

For the industry ontology, OOPS! reported P08 (missing annotations). This behaviour is likely due to the annotation-free style of the input ontology: OntoExtend appears to have replicated that style as it is asked in the prompt, yielding missing annotation warnings when annotations were expected by OOPS!. On average, P08 occurs approximately 3.7 times per CQ-based extension for both LLMs.

Comparing OntoExtend to existing work on ontology generation [6,13,16], which showed a considerable number of *important* and *critical* OOPS! pitfalls, OntoExtend exhibits only a few minor pitfalls. Overall, the results indicate that the generated extensions are structurally acceptable and do not systematically introduce new modelling defects; the limited minor issues observed are confined to specific use cases and are straightforward to fix.

6.2. Results of Functional Evaluation

The results of CQ verification show that all CQs in both use cases are correctly modelled, with no minor issues of the type identified by Saeedizade and Blomqvist [25]. Both GPT-5 and o1-preview produce extension fragments with only negligible amounts of superfluous elements. A comparison of the number of superfluous elements with the work in [16] further highlights this improvement. Using our refined definition of superfluous components, the ontology fragments generated in this work contain fewer than 2% unnecessary elements, whereas [16] reported around 30% superfluous elements in their generated ontologies (35% with [16]’s original definition of superfluous elements).

6.3. Results of survey from Engineers’ Evaluation

Six ontology engineers, comprising three engineers for the EU-Project and three who worked on the industry ontology, evaluated the generated ontology fragments by filling out the survey, results shown in Table 5.

Engineers who evaluated fragments of the EU ontologies identified some systematic deficiencies. First, they observed unconnected elements within fragments that ought

Table 5. Survey results on evaluating ontology extension fragments for the EU-Project and industry ontologies. The ratings are between 1 and 5 (see section 4.1)

Model	Metric	Industry		EU-Project	
		Mean	Fleiss P_o	Mean	Fleiss P_o
o1-preview	Correctness	4.91	0.97	3.69	0.80
o1-preview	Completeness	4.56	0.89	3.11	0.85
GPT-5	Correctness	4.96	0.98	3.66	0.87
GPT-5	Completeness	4.54	0.87	2.94	0.87

to have been linked to the input ontologies, e.g. via a `subClassOf` relation. Second, class naming was often suboptimal (for example, excessively specific names), indicating recurrent poor modelling patterns. Model-specific issues were also reported. Outputs produced by GPT-5 frequently omitted explicit domain and range declarations for newly introduced properties, imposed incorrect semantic restrictions (e.g., inappropriate use of `allValuesFrom` or `complementOf`), and employed overly simplified names. Conversely, fragments generated by o1-preview commonly redefined existing classes, contained syntactically incorrect axioms, which were not shown by the tools used for the Structural evaluation, assigned incorrect domains to some properties, and created object properties that appeared too specific compared to individual CQs. Overall, the evaluators with high agreement concluded that the generated fragments require moderate revision before they can be safely integrated into the input ontologies.

Regarding the industry ontology, the main errors concerned that in few cases, only SHACL property shapes were added without the corresponding object property plus proper `rdfs:domain` and `rdfs:range` definitions, and in a few cases the `sh:datatype` was missing. Furthermore, it was noted that sometimes LLMs omitted the required `rdfs:subClassOf` axioms and lacked comment annotations for classes and properties. This is valid for both models. Specific only to o1-preview, it tends to introduce many additional Property shapes and object properties that are not needed to answer the CQs. There is also concern in two comments that extensions were generated without grounding from the CQ or the input ontologies. In this case, the LLM anticipated some extensions purely based on patterns available in the input ontology. Overall, however, the results regarding the industry ontology show a high degree of user satisfaction (minor or no changes needed to the fragments) and strong observed agreement among the evaluators.

7. Discussion

In this section, we analyse the results, outline the main limitations of our study and experimental setup, and discuss directions for future work.

Overall results: Overall, our results show that OntoExtend can reliably produce high-quality ontology extensions across both domains we have considered. The generated fragments are almost always syntactically valid, introduce only a small number of minor OOPS! pitfalls, especially when compared with existing related work. They correctly model all evaluated CQs, and contain fewer than 2% superfluous components, while the six ontology engineers generally judged them to be syntactically and semantically adequate and requiring at most minor to moderate edits. Taken together, these findings indicate that OntoExtend already offers a practically useful level of automation for real-

world ontology extension and maintenance. In practice, *OntoExtend* functions as a drafting assistant: it generates a complete extension module that users typically only need to review and lightly edit. This keeps human–tool interaction minimal, shifting effort from manual axiom authoring to quick validation and targeted refinement. This significantly lowers the barrier to entry for domain experts, allowing them to expand ontologies with less reliance on often lacking ontology engineers.

Formulation and quality of the CQs affecting the results: Industry ontology engineers assessed the industry ontology and the EU-project ontologies (the latter developed by academic partners), and inspected the generated extension for the EU-project. Respondents were encouraged to add free-text notes on each generated module.

Our evaluation covered two distinct usage scenarios: (i) extending an ontology from highly specific, pre-determined CQs (the tool acts as an assistant for writing axioms), and (ii) constructing or extending ontologies from more general, open CQs (the tool must make substantive modelling choices). Although the same overall procedure was used to create the datasets, the EU-project CQs differ qualitatively from the industry CQs. The latter were deliberately very focused and targeted: they were authored jointly by domain experts and ontology engineers so that high-level modelling choices were already fixed and each CQ primarily required the tool to generate a missing class or a small set of properties from the natural-language question. By contrast, the CQs for the EU-Project ontology were defined in a more open manner. I.e., the authors, mostly domain experts, left many of the modelling decisions to the tool. Consequently, the tool in this case has to make the open modelling decisions when generating the ontology fragments.

This difference in CQ style affects the evaluation. For the EU-project CQs, the LLMs often produced reasonably structured extensions, but ontology engineers judged some outputs as less complete because additional manual refinement was needed to align the results with modelling choices implied by the full specification. Thus, lower correctness and completeness scores in this setting reflect rather the limitations of the CQ definition (i.e., missing clarity on the expectations and contextual detail), not necessarily a weakness of the *OntoExtend* framework. In the industry setting, the tool mainly needed to render valid axioms according to an input ontology and hence achieved much higher scores for completeness and correctness. The lower scores on correctness and completeness observed in the EU-project scenario from the evaluators' survey results can therefore be explained as a result of the different task complexity of the ontology generations.

LLM behaviour as an implicit quality measure: As discussed previously, the same framework is applied to two qualitatively different CQ sets and associated ontologies: one in which CQs are tightly scoped, and another in which CQs lack clear expectations and context, leaving many modelling decisions open. Our analysis shows that *OntoExtend* tends to perform better in terms of correctness and completeness, alignment with expert expectations, and reduced need for post-editing when the CQs and background ontologies are well structured, precise, and internally coherent. These findings in turn suggest that LLM performance can serve as a proxy indicator for the quality of requirements (CQs) and/or ontological artefacts: if a given set of CQs and input ontologies systematically yields higher quality extensions with fewer modelling errors, then this combination is in a pragmatic sense “better” for downstream ontology engineering with LLMs (and probably also humans). Conversely, when the same solution struggles, this most likely is a result of ambiguity, under-specification, or hidden modelling assumptions in the input. In line with recent discussions on using LLMs as auxiliary evaluators in knowledge en-

gineering, our results therefore support the interpretation of LLM behaviour as a novel, tool-centric dimension of ontology and requirement quality.

Practical impact: Sending very large ontology fragments (e.g., ~75k-token) directly to an LLM resulted in substantially longer end-to-end response times and higher API costs in our experiments. By contrast, *OntoExtend* retrieves a compact subset of the ontology and provides only that subset to the LLM; this greatly reduces the typical LLM turnaround and lowers the amount of data transmitted and the associated cost.

Limitations and future work: Despite the promising aspects of this work, it also has limitations. First, our evaluation is restricted to two domain-specific use cases; a broader set of domains and modelling scenarios would be necessary to draw more general conclusions. Second, we experimented with only a small number of LLM configurations, and a more systematic comparison across different models and providers could reveal important performance differences. A further potential threat to the validity of our results is data leakage, i.e., the possibility that the underlying LLMs have been exposed during pre-training to ontologies, CQs, or related documentation that are similar to (or overlap with) our experimental material. In our setup, for the EU-Project ontology, the ontologies have been publicly released, but the material concerning CQs and their coverage is not publicly available. For the industry ontology, it was not disseminated in public code or data repositories, which reduces the risk that related artefacts have been seen during model training. Future work will explore more rigorous safeguards against leakage, such as on-the-fly benchmark generation, experiments with open-weight models trained on controlled corpora, and dedicated checks for overlap between evaluation data and known pre-training sources. Finally, our findings indicate that the quality and formulation of requirements (in our case, CQs) have a substantial impact on the outcome. Future work should explicitly analyse and improve CQ quality itself, for example by (semi-)automatically generating CQs, or mapping them to existing CQ templates (see the work by Keet et al. [11]).

8. Conclusion

In this work, we introduced *OntoExtend*, a scalable requirement-driven framework that extends one or more input ontologies according to the requirements specified in a set of CQs. *OntoExtend* processes each CQ individually and extracts, for each CQ, the relevant existing ontology elements as context for an LLM to generate the missing ontological structures necessary to fully answer the CQ in a coherent manner. To answer our research questions, we found that using `text-embedding-ada-002` as an embedder produced the most effective retrieval subsets for our experiments (RQ1). Furthermore, both `GPT-5` and `o1-preview` yielded comparable performance when supplied with the retrieved context (RQ2): the generated fragments contained almost no Turtle syntax errors, did not introduce any new *critical* or *important* OOPS! pitfalls, only a negligible amount of minor pitfalls, and exhibited only a small fraction of superfluous elements. We propose a multi-faceted evaluation (RQ3), combining syntactic checks, OOPS! analysis, CQ verification, measurement of unnecessary classes/properties (superfluous), and an analysis of the generated outputs by six ontology engineers. This evaluation setup shows that *OntoExtend* is able to extend ontologies that are both correct and complete according to ontology engineers. Overall, for what concerns the benefits and weaknesses of the extended

ontologies produced by OntoExtend (RQ4), the extended fragments are structurally and syntactically correct, accurately model all evaluated CQs, and add fewer than 2% superfluous elements, with engineers often needing only minor edits in the industry use case. The positive evaluation from ontology engineers shows high potential for the usage of OntoExtend in real scenarios. However, we still observe weaknesses such as occasional missing domain/range declarations, suboptimal or overly specific names, and some unconnected elements. Moreover, performance degrades when CQs are open-ended or underspecified, showing that OntoExtend is effective when requirements are precise but sensitive to the quality of the input CQs.

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Use of Generative AI. ChatGPT was used to enhance the readability of some of the text and improve the language of this paper, after the content was first added manually. After using this service, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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KGXL-Query: Extending SPARQL with Semantic Search and LLM Functionality for Question Answering over Heterogeneous Data

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Abstract. The quest to answer complex questions, those requiring both structured facts and contextual evidence from unstructured data, has shown a gap in knowledge systems: the incompatibility between symbolic and semantic inference. Knowledge Graphs (KGs) support formal, logically grounded querying via structured languages such as SPARQL. Conversely, LLMs (Large Language Models) and semantic text search offer broad knowledge coverage through vector-based similarity metrics. However, beyond using KGs for refining text or LLM results, and complementing missing information in KGs via LLMs or text search, there is less emphasis on accessing LLMs & semantic text search functionality through SPARQL. Moreover, existing Question Answering (QA) methods on heterogeneous data sources use RAG, data unification, and decompose-and-aggregate methods. The unifying approaches suffer from irreversible information loss; decomposition-driven methods are prone to cascading errors; and RAG methods lack explainability and are limited to shallow textual integration. Therefore, we introduce KGXL-Query, a new SPARQL functionality extension that embeds LLMs and semantic search directly into SPARQL, enabling hybrid inference where symbolic and non-symbolic methods coexist within a single, executable, and explainable framework. Additionally, KGXL-Query paves the way by transforming the question to leverage the model rather than altering the data to fit it. Our experiments on four heterogeneous QA datasets show that KGXL consistently outperforms existing methods across all four benchmarks.

Keywords. Knowledge Graph, Question Answering, SPARQL, Hybrid QA

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1. Introduction

Answering complex questions that require facts and contextual evidence from heterogeneous knowledge sources remains challenging in modern knowledge systems [9]. Such questions often require information spanning Knowledge Graphs (KGs), unstructured textual data, and Large Language Models (LLMs), each with distinct data organization and access [13]. A systematic mechanism is therefore required to jointly query these sources, complement their information gaps, and support inference across them. KGs store knowledge as triples (subject, relation, object) and support formal, logical querying through structured languages such as SPARQL² and graph queries [8]. In contrast, textual sources are inherently unstructured, and access to their knowledge relies on semantic search techniques based on vector similarity [21]. LLMs, in turn, encode statistical patterns from large-scale training data within billions of parameters and predict the next token given a prompt [20]. KGs provide structured and verifiable facts that can ground and constrain LLM inference, while LLMs offer flexible, context-aware inference that can identify implicit or missing knowledge [18]. Textual sources extend coverage beyond the boundaries of predefined schemas and ontologies [9]. Despite these mutual benefits, their heterogeneity introduces a semantic and structural gap that hinders seamless interoperability [18]. In hybrid QA, there has been no attempt to integrate all the aforementioned three sources within a single, unified data access and inference structure. This work challenges that separation by embedding LLMs and semantic search directly into the SPARQL query. We refer to this as the KGXL (KG teXt LLM)-query.

As shown in Figure 1, existing QA (Question Answering)³ systems over heterogeneous data sources, referred to as hybrid QA, address the heterogeneity through unification, decomposition & aggregation, or retrieval-augmented generation (RAG). Unification-based methods flatten KG triples into text, convert text into a KG, or transform all sources into a single representation followed by answer extraction [19]. However, such transformations lead to loss of the data's intrinsic structure, omission of relevant information, and redundancy. Decompose-and-aggregate approaches split a complex question into sub-questions and execute them sequentially or in a tree-based structure [25,27,29]. These methods are prone to cascading errors and offer limited opportunities for heterogeneous reasoning [11]. RAG-based methods retrieve information via search tools, filter results using embedding similarity, and rely on LLMs for answer synthesis [35,31,6]. RAG pipelines often lack explainability and integration with structured data.

SPARQL, by design, supports structural querying over KGs through a well-defined syntax and semantics. Furthermore, it supports logical filtering, aggregation, string and numeric functions, and result ordering. It also allows federated querying of different KGs via the SERVICE operation [2]. However, SPARQL operates on exact symbolic values, leaving room for extensions in semantic similarity and LLM-based inference. Building on these strengths and overcoming the limitations, we extend SPARQL with three additional capabilities: semantic search, LLM invocation, and RAG. Rather than transforming data to fit a model, our approach allows us to convert the question into an executable query, called KGXL-Query.

²SPARQL is a query construct used for data retrieval from KGs

³Question Answering (QA) is the task of answering questions over certain knowledge sources.

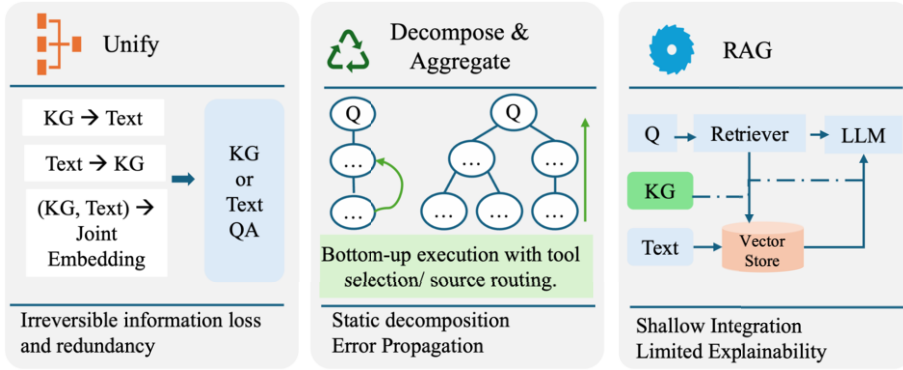


Figure 1. QA over heterogeneous data approaches and their limitations: data unification methods lead to information loss, decompose-and-aggregate causes cascading errors, and RAG pipelines lack deep data integration and limited explainability.



Figure 2. KGXL-Query framework: Interlinking Text, LLM, and KG via SPARQL function expansion. Where KGXL-Query leverages SPARQL (built-in) functions and also supports semantic search, LLM inference, and RAG.

As shown in Figure 2, KGXL-Query introduces a new framework that extends SPARQL to support LLM inference, semantic text search, and RAG. As a result, SPARQL is no longer limited to static, schema-bound querying but can infer over rich contextual information from external sources. For example, the KGXL-Query in Figure 2 begins by identifying all entities in Wikidata whose occupation (P106) is defined as Astronaut (Q11631). From these, it further filters those who have a significant event (P793) record ‘walk on the Moon’ (Q42882411). For each identified astronaut, the query retrieves their English name and corresponding Wikipedia URL. Then the query leverages the `xfn:llm()` function to seamlessly bridge symbolic and neural inference within the same query. The `xfn:llm()` function is invoked with a prompt. Using the provided Wikipedia URL, the `xfn:llm()` function fetches the corresponding Wikipedia page, extracts the relevant textual content, and summarizes it to generate a biography. Despite the

advantage of accessing all source data via a single query, formulating KGXL queries for complex questions can be challenging. This requires a robust method that understands the question and aligns the information request to the proper built-in or newly introduced functions and structures. Especially, the proper arrangement of the operations also matters in handling the intrinsic data structure differences. Where KGs only allow literals or URIs, unlike LLMs & text, which are strings. Therefore, we apply Chain-of-Thought (CoT) with a self-correction prompt that guides the generation of KGXL queries for hybrid questions. Our experiments show that the approach is viable and can be further improved through more robust query generation mechanisms.

In general, our method avoids data transformation and duplication, preserves information fidelity, and enables explainable execution. By embedding LLMs and semantic search directly into SPARQL, KGXL-Query represents a necessary evolution of knowledge systems, moving beyond data unification and sub-question decomposition toward a true fusion of symbolic and subsymbolic intelligence [9]. Our contributions are:

- We design and implement KGXL-Query, an extension of SPARQL that integrates semantic text search and LLM invocation into the KG query execution pipeline, enabling unified symbolic and subsymbolic inference within a single and explainable framework.
- We demonstrate the feasibility of KGXL-Query by extending Apache Jena with semantic text search, LLM, and RAG functionalities. Where KGXL-Queries can be written and executed directly in the Apache Jena Fuseki SPARQL engine interface, allowing users to interactively formulate and evaluate hybrid SPARQL queries without any additional tooling.

The KGXL-Query SPARQL functions extension implementation in Apache Jena is available on the GitHub page at <https://github.com/semantic-systems/jena/tree/llm-calls>. Additionally, the KGXL-Query implementation is also found at <https://github.com/semantic-systems/kgxl-query>.

2. Related Works

2.1. SPARQL Extensions and Custom Functions

SPARQL was designed as a declarative query language for RDF data with a well-defined set of built-in functions for triple matching, aggregation, and logical filtering [2]. Most SPARQL engines provide extensibility mechanisms in the form of user-defined functions and engine-specific extensions [33]. For example, Apache Jena ARQ allows developers to register custom functions that are evaluated during query execution. Extensions to SPARQL can be augmented without modifying its core grammar or execution model [16]. For instance, Virtuoso's `bif:contains` operator or Jena's text-indexing extensions enable lexical matching across large RDF graphs. However, these extensions remain limited to keyword matching and do not capture semantic similarity or contextual meaning. KGXL-Query builds on this line of work by extending SPARQL's function space with semantic and generative operators. Unlike traditional extensions, which are restricted to deterministic and symbolic computation, KGXL-Query introduces functions

for vector-based semantic search and LLM-based inference while preserving SPARQL's declarative execution semantics.

SPARQL 1.1 introduces federation capabilities through the `SERVICE` clause, enabling queries to be distributed across multiple remote SPARQL endpoints [1]. Federated querying allows data integration at query time and has been widely adopted for querying Linked Data sources on the Web. The `SERVICE` mechanism expects remote endpoints to expose a SPARQL interface, and intermediate results cannot be easily combined with fine-grained query expressions or non-SPARQL computations. Moreover, it does not support querying unstructured text or invoking external generative model services [1,2]. KGXL-Query differs from federated SPARQL approaches by embedding external computation directly into the query language as functions. Rather than delegating subqueries to remote SPARQL endpoints, KGXL-Query allows semantic search and LLM-based reasoning to be invoked at the expression level and composed with symbolic graph patterns within a single query.

2.2. QA over Heterogeneous Data

QA methods over heterogeneous data sources can be categorized into three broader classes (see Figure 1). The first approach converts the sources into either a question-specific subgraph or a textual context, then provides an answer using text- or KG-based QA. For example, UniK-QA [17] converts KG triples, table rows, and infobox facts into linearized text and jointly indexes them with unstructured textual sources. Next, retrieves relevant textual chunks using a dense retriever and a fusion-in-decoder reader that returns an answer. Similarly, Quasar [6] unifies the sources by verbalizing them into text. Guided by a structured question-intent representation, iteratively re-ranks the retrieved textual evidence using graph neural networks [34] and returns an answer. On the other hand, graph-based methods like PullNet [23] and Uniqorn [19] construct question-specific context graphs by retrieving and pruning evidence from KGs and text, followed by answer computation using Group Steiner Trees [24] graph algorithms. Another line of work bridges heterogeneous sources by learning joint embedding-based representations. For example, DRAGON [32] performs deep bidirectional pretraining over text and KG triples using a cross-modal encoder with a unified self-supervised objective. Similarly, JointGT [10] aligns KG triples and text in a shared embedding space via contrastive learning and further incorporates structure-aware transformers with linearization objectives for triple-to-text mapping. These methods support retrieval and alignment across sources, but they rely exclusively on embedding-based inference and do not provide executable symbolic inference. UniHGKR [14] follows a similar strategy by linearizing tables, infoboxes, and KGs into text and learning unified embeddings alongside unstructured content. Overall, embedding-driven unification approaches integrate knowledge primarily during model training, which limits flexibility and controllability at query time. In contrast to these unification-driven approaches, KGXL-Query addresses heterogeneity by preserving structured querying, embedding LLM invocation and semantic search directly into SPARQL rather than flattening all data into a single representation.

Unlike the unification methods, the decompose-and-aggregate approach divides the complex questions into sub-questions and aggregates intermediate answers. SearChain [30] constructs a reasoning tree that iteratively routes sub-queries to a retriever and an LLM. Likewise, ProbTree [3] employs probabilistic tree-of-thought reasoning, executing

inference bottom-up from leaf nodes to the root. These methods rely heavily on LLM-based orchestration and do not support symbolic execution. HumanIQ [11] similarly combines human-designed tool chains, including Wikipedia search, SPARQL query generation, KG execution, and LLM-based reading, but performs reasoning through explicit action sequences rather than within a unified query framework. Recently, SH-CoDE [27] parses complex questions into a Hybrid Question (HQ) representation and executes them as a binary tree, routing sub-questions to KGQA or TextQA modules and propagating intermediate results using operators such as JOIN and COMP. HySQA [25] follows a similar decomposition strategy, identifying the appropriate source for each sub-question and aggregating results via the RAG pipeline. Although these approaches support multiple heterogeneous sources, their sequential, pipeline-driven execution makes them vulnerable to cascading errors. In line with this approach, AtomR [29] introduces atomic operations (search, relate, and filter) that allow LLMs to access KGs, text, and the web at a fine-grained level. However, AtomR simulates query operators via prompts and deduces an answer by executing the atomic reasoning tree in post-order traversal. In contrast, KGXL-Query embeds LLM invocation and semantic search as executable query functions within SPARQL itself, rather than emulating symbolic operators through prompting.

The third class, QA over heterogeneous methods, leverages RAG’s capability to retrieve relevant facts from different sources and uses an LLM to deduce an answer [31]. In line with this approach, SPAGHETTI [35] retrieves evidence from KGs, text, tables, and LLM-generated claims in parallel. Then generates an answer via an LLM. Although this design improves coverage through redundancy and verification, SPAGHETTI repeatedly relies on LLMs to unify information at each decision point. By contrast, KGXL-Query integrates symbolic and subsymbolic reasoning within a single declarative query. SparqLLM [15] is the closest work to KGXL-Query, as it enables retrieval-augmented SPARQL queries via user-defined functions that generate named graphs from search engines and LLMs. However, SparqLLM primarily treats LLMs as text extractors that populate auxiliary graphs and explicitly avoids extending SPARQL semantics. KGXL-Query generalizes this idea by embedding LLM invocation and semantic search as part of the query constructs, enabling tighter integration between symbolic and subsymbolic reasoning.

In general, existing hybrid QA approaches either rely on data unification, construct external reasoning structures, or decompose questions into pipeline-driven execution plans. KGXL-Query differs primarily by adopting a query-first design that integrates symbolic inference, text semantic search, and LLM-based inference into a single, executable, and explainable SPARQL query.

3. Proposed SPARQL Extensions in KGXL-Query

KGXL-Query extensions are designed as SPARQL functions and do not require changes to the SPARQL grammar or algebra. The underlying SPARQL engine handles query parsing, algebraic plan generation, and symbolic graph pattern operator execution. All the new functions are registered with the SPARQL engine’s `FunctionRegistry`. During query execution, these functions are invoked in the same manner as built-in SPARQL functions.

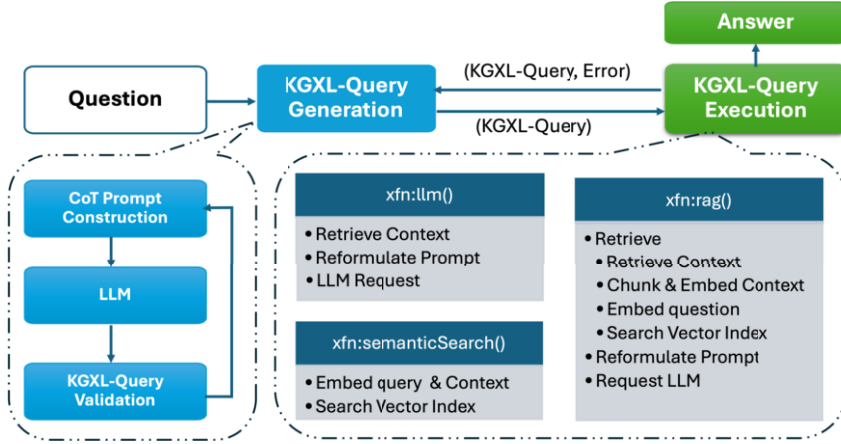


Figure 3. KGXL-Query Generation and Execution

1. `llm(prompt, format)`. The `llm` function enables invocation of LLMs from within SPARQL. It accepts a prompt and an optional format parameter that constrains the expected output (e.g., `literal`, `json`, or a schema-constrained value). The function delegates the prompt to an external LLM service and returns the generated result as a SPARQL-compatible literal. By requiring an explicit prompt and output format, `llm` supports both text generation and LLM-driven inference while keeping the invocation transparent at the query level. For example, `BIND(llm("History of Hamburg", format="literal") AS ?description)`, returns a textual description of Hamburg that can be directly projected in the query result.

2. `semanticSearch(text, query, top_k)`. The `semanticSearch` function performs vector-based semantic matching over unstructured text. It takes a text argument representing the context to be searched (e.g., an abstract or description), a query, and an integer `top_k` specifying the retrieval bound. Internally, the function invokes an external semantic search service that computes dense embeddings and similarity scores. In its predicate form, `semanticSearch` returns a boolean value indicating whether the query is semantically present in the given text above a configurable similarity threshold, making it suitable for use in `FILTER` clauses. A related variant exposes the maximum similarity score to support ranking and threshold-based selection. This design allows semantic similarity to be integrated into SPARQL without modifying the active KG data.

3. `rag(prompt, context)`. The `rag` function implements semantic retrieval with LLM-based answer generation. It takes an explicit prompt, which specifies the task or question, and a context argument containing textual evidence. The function delegates execution to an external RAG service, which synthesizes an answer grounded in the provided evidence. The result is returned as a literal and can be bound to a variable using `BIND` or further processed by downstream query clauses. By separating evidence retrieval from answer generation and making both inputs explicit, `rag` supports transparent and explainable hybrid QA within SPARQL.

KGXL-Query Generation Chain-of-Thought Prompt**System Instructions**

Your task is to generate a **KGXL-Query** that combines symbolic knowledge graph traversal with neural extensions. Follow the three blocks below strictly and produce a valid query.

Block 1: Schema-Aware Intent and Entity–Predicate Mapping

Analyze the user question and identify all relevant entities, predicates, and constraints. Use the class and property definitions of the target knowledge graph(s) to:

- Ground question terms to KG classes, entities, and predicates.
- Resolve ambiguities using schema semantics.
- If the question spans multiple KGs, determine how entities and predicates should be linked across graphs.

Block 2: KG Triple Pattern and Navigation Specification

Construct the symbolic part of the KGXL-Query by:

- Defining the required triple patterns to retrieve all deterministic facts from the KG.
- Specifying structural navigation paths needed to reach relevant entities.
- Restricting this block to information that is directly represented as KG triples.

Block 3: Neural Function Selection and Input Construction

For any part of the question that cannot be answered from KG triples alone, select and invoke the appropriate KGXL-Query extension function. Use the following rules:

- `xfn:llm()`: Use when the requested information requires free-text reasoning or synthesis. Construct the prompt by concatenating:
 - * the KG-retrieved entity name,
 - * the specific information request,
 - * the textual context source (e.g., abstract, description, Wikipedia, or an external URL obtained from the KG),
 using the SPARQL `CONCAT` function.
- `xfn:rag()`: Use when external documents must be retrieved. Create a separate context variable containing the candidate textual source URI. The prompt string must contain only the information request and instruction.
- `xfn:semanticSearch()`: Use when semantic similarity over textual data is required. Provide only the query and the associated context.

Output Requirement

Return only the final **KGXL-Query**, combining the symbolic triple patterns and any necessary neural function calls. Do not include explanations or intermediate reasoning.

Table 1. KGXL-Query Generation Chain-of-Thought Prompt

4. KGXL-Query Generation and Answering

As shown in Figure 3, given a complex question that is answerable over a KG and text, the KGXL-Query generator parses the question into an executable SPARQL via a few-shot CoT prompting. During KGXL-Query execution, the custom KGXL functions trigger an external API call. In case an error occurs, it sends the query with the error back to the KGXL-Query generator. The following subsections provide details on how the query generation and execution tasks are performed.

KGXL-Query Error Correction Prompt

SYSTEM INSTRUCTIONS: Your task is to correct a KGXL-Query, a special type of SPARQL, that has special functions in it to invoke LLM inference, RAG, and semantic search. Properly analyze the input query, the KG schema, and the error. (Look in detail at the examples of how to use the external functions.)

EXAMPLES

INPUT: KGXL-Query: FINAL_QUERY

KG schema: KG_Schema

ERROR: error

CHECKLIST: Could you check:

- All predicates exist in the schema.
- All triple patterns are syntactically valid.
- Neural functions are used only when necessary.
- Neural function inputs follow KGXL-Query rules.
- No redundant or conflicting clauses.

OUTPUT FORMAT: [VALIDATION_RESULT]

OK OR [List of violations] [CORRECTED KGXL_Query]

Table 2. KGXL-Query Error Correction Prompt

4.1. KGXL-Query Generation

KGXL-Query generation is implemented using a prompt-based approach that incorporates a *self-corrective CoT* mechanism. Given the hybrid nature of KGXL-Query, which integrates symbolic reasoning with neural components, query generation begins by identifying entities and predicates in the input question. Schema-aware instructions embedded in the prompt enable the model to perform dynamic entity–predicate mapping using the target KG’s class and property definitions. Based on this mapping, the prompt provides symbolic guidance for determining the structural navigation paths within the KG required to retrieve all relevant entities. When a query involves constraints spanning multiple KGs, an optional prompt block specifies how these graphs are linked.

As shown in Table 1, the KGXL-Query Generation CoT prompt is organized into three interdependent blocks. The first block focuses on schema-aware intent mapping, instructing the system to align question terms with KG entities and predicates. The second block defines the triple patterns that the LLM can use to extract structured facts from the KG. The third block introduces the KGXL-Query extension functions `xfn:llm()`, `xfn:semanticSearch()`, and `xfn:rag()` to handle information that is not directly represented as triples in the KG.

When delegating subtasks to neural components, the CoT prompt explicitly specifies how inputs should be constructed. For `xfn:llm()`, the prompt encapsulates the retrieved entity name, the specific information request, and the relevant textual context source (e.g., Wikipedia or another URL) using the SPARQL `CONCAT` function. In the case of `xfn:rag()`, a separate context variable is created to store the potential textual source URI. At the same time, the prompt itself contains only the information request and instructions. In contrast, the `xfn:semanticSearch()` interface receives only the query along with its associated context.

4.2. KGXL-Query Error Correction

To handle errors caused by incorrect KGXL-Queries, we apply a self-corrective CoT (see Table 2). It checks whether the error is due to SPARQL syntax, schema incompliance, or incorrect use of KGXL-Query extension functions. Then, revise the query with only minimal, targeted updates to address the identified errors. Meanwhile, it preserves correct symbolic triple patterns and adds neural extension functions only when explicitly indicated by the validation. Constraining the self-correction to validation-based changes helps prevent uncontrolled query drift.

5. Experimental Setup

5.1. KGXL-Query Extensions Implementation Prototype

The prototype builds on Jena ARQ’s extensibility mechanisms for user-defined functions. The KGXL functions are implemented as subclasses of `FunctionBase` and registered with Jena’s `FunctionRegistry`.

Semantic Search Implementation: The `xfn:semanticSearch` functionality is realized as a lightweight Java client that invokes an external semantic search service over HTTP. During query execution, the function receives the current variable binding (e.g., a textual description such as an abstract) and a query string, constructs a JSON request containing the query, the text, and a configurable `top_k` parameter, and sends it to the semantic search endpoint. The semantic search endpoint implements a standalone FastAPI service backed by a Dockerized Elasticsearch instance. The service exposes a REST endpoint that accepts a natural-language query, a textual context, and a retrieval bound (`top_k`). For retrieval, it computes dense sentence embeddings using a transformer-based model (“all-mpnet-base-v2”) and combines vector similarity with keyword-based matching in Elasticsearch through a weighted scoring (`semantic=1.0, keyword=0`) function. The service supports both scalable retrieval over a pre-indexed text corpus and semantic matching over runtime text supplied by the SPARQL engine. Search results are returned as a ranked list of text fragments with similarity scores, serialized as JSON.

RAG Service is implemented as a separate FastAPI service that exposes a simple REST interface. The corresponding SPARQL function, `xfn:rags`, accepts two arguments: an explicit natural-language instruction and a contextual evidence string, typically the JSON-encoded output produced by `xfn:semanticSearch`. The Java function forwards these parameters to the RAG service, which applies evidence-aware reasoning to produce an answer. The generated answer, together with the evidence used, is returned as a string literal and bound to a SPARQL variable.

Data Exchange and Execution Model: All communication between the SPARQL engine and external components is via RESTful APIs, with JSON as the interchange format. Semantic search results and RAG outputs are represented as JSON-encoded string literals within SPARQL result bindings. This choice avoids side effects on the active dataset, does not require access to Jena’s execution context or dataset graph, and ensures that results are directly visible in standard SPARQL clients such as Apache Jena Fuseki.

5.2. Datasets

We use QALD-7 [28], CompMix [5], HySQA [25], and BlendQA [29] hybrid QA datasets for the evaluation of our method. BlendQA contains human-verified LLM-generated questions that require facts from Wikipedia text, Web sources, and Wikidata triples. During the experiment, we design a CoT prompt that reads the explicit question decomposition trees (KG-Web and Text-KG) in the dataset to formulate executable KGXL-Queries. Likewise, the HySQA dataset is an LLM generated with a human-verified test set. HySQA questions need facts from the DBLP and SemOpenAlex KGs, as well as textual context from Wikipedia. Unlike BlendQA and HySQA, QALD-7 and CompMix question-answer pairs are manually created. QALD-7 questions rely entirely on the DBpedia KG and entities' textual descriptions, whereas CompMix questions require facts from Wikipedia text, tables, infoboxes, and Wikidata. CompMix provides retrieval results for Wikipedia and Wikidata. Hence, we use the script⁴ to download the pickle file and set up our retriever on it. Besides, the questions include the QID of entities, and we leverage that when generating the KGXL queries. In the case of QALD-7, each question has an associated pseudo-query, which we use in the CoT prompt to instruct the LLM to generate the KGQL-Query⁵.

Dataset	Metric	Best Baseline	KGXL (Best)	Improvement
CompMix	P@1%	SPAGETTI (56.50)	61.90	↑ 5.40
BlendQA	KG Web (F1)	AtomR (45.63)	50.68	↑ 5.05
	Text KG (F1)	AtomR (39.69)	46.68	↑ 6.99
HySQA	EM	HySQA (adaptive decomp) (72.50)	79.91	↑ 7.41
	F1	HySQA adaptive decomp. (76.35)	83.11	↑ 6.76
QALD7	F1 (CoT)	Qwen 3.6 (50.08)	61.52	↑ 11.44
	F1 (CoT+SC)	Qwen 3.6 (61.78)	71.52	↑ 9.74

Table 3. Summary of KGXL Performance Across Datasets compared to State-of-the-Art Baselines.

5.3. Evaluation Results

We evaluate the KGXL model's performance using the metrics specified in their benchmark dataset. Precision (P) measures the ratio of correctly predicted positive observations to the total predicted positives, while Recall (R) calculates the ratio of correctly predicted positive observations to all actual positives. The F1-score is the harmonic mean of P and R. Notably, for QALD-7, it is computed based on entities DBpedia URI matching, whereas for BlendQA and HySQA, it is calculated based on token-level overlap between the generated text and the ground-truth answers. Additionally, Exact Match (EM) is a strict binary metric indicating whether the model's output exactly matches the ground truth, and P@1% measures Precision for the top 1% of ranked predictions or retrieved candidates.

⁴<https://github.com/PhilippChr/QUASAR/blob/main/scripts/download.sh>

⁵For DBLP, SemOpenAlex, and DBpedia, we set up a Jena SPARQL endpoint using their respective RDF dumps. For Wikidata, we use a publicly available SPARQL endpoint.

Model	CoT			CoT + SC		
	P	R	F1	P	R	F1
KGXL (Qwen 3.6)	52.00	50.04	50.08	63.04	66.04	61.78
KGXL (Mistral Large)	<u>57.24</u>	<u>54.40</u>	<u>54.70</u>	<u>71.24</u>	<u>68.40</u>	<u>68.70</u>
KGXL (GLM 4.7)	63.50	61.14	61.52	73.50	71.14	71.52

(a) Our method (KGXL) Results on the QALD7 dataset with different LLMs.

Model	P@1%
UNIK-QA	44.00
CONVINSE	40.70
EXPLAIGNN	0.442
GPT-3 (text-davinci-003)	50.20
QUASAR	56.40
SPAGETTI (SoTA)	<u>56.50</u>
Ours: KGXL (Qwen 3)	43.88
Ours: KGXL (Mistral)	(↑ 3.21) 59.71
Ours: KGXL (GLM 4.7)	(↑ 5.40) 61.90

(c) Results on the CompMix dataset.

Model	KG Web	Text KG
Chain-of-Knowledge	22.66	37.55
ProbTree	24.48	38.81
AtomR	<u>45.63</u>	<u>39.69</u>
KGXL (Qwen 3.6)	29.22	28.59
KGXL (Mistral Large)	37.80	41.25
KGXL (GLM 4.7)	50.68	46.68

(b) F1-Scores on the BlendQA dataset.

Model	EM	F1
Contri(e)ve [22]	32.0	40.7
Fondi and Jiamekong [7]	43.59	45.05
Bridge-Generate [26]	61.25	67.33
SH-CoDE [27]	70.51	71.83
HySQA (SOTA) [25]	<u>72.50</u>	<u>76.35</u>
Ours: CoT (Qwen 3.6)	50.14	53.28
Ours: CoT+SC (Qwen 3.6)	60.11	64.71
Ours: CoT (Mistral Large)	68.66	71.50
Ours: CoT+SC (Mistral Large)	75.78	78.25
Ours: CoT (GLM 4.7)	73.78	76.77
Ours: CoT+SC (GLM 4.7)	79.91	83.11

(d) Results on the HySQA dataset.

Table 4. Performance comparison of KGXL against state-of-the-art baselines across the BlendQA, CompMix, QALD7, and HySQA datasets. (a) Experimental results: Precision (P), Recall (R), and F1-Score on the QALD7 test set using KGXL-Query generation using CoT and CoT-SC (Self Correction). (b) F1 Score Comparison Across Methods on KG Web (132 Questions) and Text KG (163 Questions) from the BlendQA dataset. (c) Performance comparison of the KGXL method and baseline models on the CompMix dataset (test set size: 2,674). Metrics include P@1%. (d) Exact Match and F1-Score performance of our method (KGXL) on the HySQA dataset in Chain-of-Thought (CoT) mode alone and with Self-Correction (SC), using different LLMs.

Table 4 presents the experimental results of KGXL⁶, comparing it against existing baselines across four datasets: QALD7, BlendQA, CompMix, and HySQA using three different LLMs (Qwen 3.6⁸, Mistral Large⁹, and GLM 4.7¹⁰). On the QALD7 dataset (see Table 4 (a)), we evaluate KGXL using P, R, and F1-score under two settings: Chain-of-Thought (CoT) and CoT with Self-Correction (SC). Across all experiment settings, adding SC consistently improves performance. For KGXL, using Qwen 3.6 improves P/R/F1 from 52.00/50.04/50.08 using CoT to 63.04/66.04/61.78 using CoT + SC. Mistral Large also shows a similar trend, where scores rise from 57.24/54.40/54.70 to 71.24/68.40/68.70. GLM 4.7 achieves the highest overall performance, with CoT yield-

⁶In our experimentation, we use the LLMs: LLAMA 3 (70B-Instruct), Qwen 3.6-35B-A3B, Mistral Large 3 (675B Instruct 2512), and GLM 4.7, which are hosted by GWDG⁷.

⁸<https://huggingface.co/Qwen/Qwen3.6-35B-A3B>

⁹<https://huggingface.co/mistralai/Mistral-Large-3-675B-Base-2512>

¹⁰<https://huggingface.co/zai-org/GLM-4.7>

ing 63.50/61.14/61.52 and CoT + SC further increasing results to 73.50/71.14/71.52. These results indicate that the structured enhancement provided by SC is beneficial across different model backbones. For BlendQA, we compute F1 scores using the evaluation script in ATOMR [29] across two input settings: KG Web and Text KG. We compare KGXL against baselines including Chain-of-Knowledge [12], ProbTree [3], and AtomR (see Table 4 (b)). AtomR serves as the strongest baseline, achieving F1 scores of 45.63 on KG Web and 39.69 on Text KG. KGXL shows substantial improvements, particularly with the GLM 4.7 backbone. While KGXL with Qwen 3.6 reaches 29.22 (KG Web) and 28.59 (Text KG), and Mistral Large achieves 37.80 and 41.25, respectively, the GLM 4.7 experiment outperforms all baselines in both configurations with scores of 50.68 on KG Web and 46.68 on Text KG.

The CompMix results (see Table 4 (c)) focus on Precision at 1% (P@1%). We compare KGXL against prior systems such as UNIK-QA [17] and CONVINSE [4], as well as stronger methods like QUASAR [6] and SPAGETTI [35] (SoTA), achieves a P@1% of 56.40 and 56.50, respectively. KGXL demonstrates competitive performance that varies significantly depending on the backbone choice. KGXL with Qwen 3 achieves 43.88, while Mistral yields 59.71. Notably, KGXL with GLM 4.7 achieves a P@1% of 61.90, exceeding the stated SoTA baseline and indicating that the language model choice significantly impacts top-ranked Precision. The HySQA dataset evaluates QA performance using Exact Match (EM) and F1-score, comparing prior methods to KGXL under CoT and CoT + SC prompting (see Table 4 (d)). The strongest baseline, HySQA (SOTA), reports an EM of 72.50 and an F1 of 76.35. KGXL demonstrates clear gains when transitioning from CoT alone to CoT + SC. With Qwen 3.6, performance rises from an EM/F1 of 50.14/53.28 (CoT) to 60.11/64.71 (CoT + SC). Using Mistral Large, values increase from 68.66/71.50 to 75.78/78.25. With GLM 4.7, KGXL achieves an EM/F1 of 73.78/76.77 under CoT and improves to 79.91/83.11 with CoT + SC, representing the best result in the table. Overall, these results indicate that the self-correction component consistently improves performance, increasing EM and F1 across all evaluated LLMs and yielding the strongest performance on HySQA with KGXL (CoT + SC) using GLM 4.7.

5.4. Discussion

The experimental results summarized in Table 3 show the competitive performance of KGXL across a range of heterogeneous QA datasets. A key observation is the consistent performance improvement achieved by KGXL across all evaluated datasets. Notably, our method outperforms strong baselines, including the current SoTA in CompMix, BlendQA, and HySQA, resulting in significant F1-score and Exact Match gains. For instance, on the HySQA dataset, KGXL improves the Exact Match score by 7.41 points over the previous best, while on BlendQA, it achieves a 6.99-point increase in F1-score in the Text KG setting.

KGXL's ability to achieve SoTA results across QALD-7 and CompMix datasets highlights its generalizability. It effectively handles the structured, predictable nature of manually curated questions (as seen in QALD-7) while also demonstrating the flexibility required to solve the more diverse and challenging phrasing found in LLM-generated, human-verified scenarios (as seen in BlendQA). Furthermore, the results validate the effectiveness of our proposed framework. Across all LLMs, the transition from CoT to CoT with Self-Correction consistently produces higher performance. This structured en-

hancement is a critical component allowing the models to refine their reasoning paths and minimize errors. Finally, it is worth emphasizing that all our experiments utilize exclusively open-source LLMs (Qwen, Mistral, and GLM). This contrasts with many SoTA methods in CompMix, BlendQA, and HySQA, which often rely on proprietary models. The fact that KGXL achieves superior performance using only open-source models not only lowers the barrier to entry for hybrid QA research but also demonstrates that SoTA reasoning capabilities can be realized without depending on commercial LLM APIs.

To analyze the impact of our reasoning framework, we evaluate KGXL with and without Self-Correction (SC) across different language model backbones. The results show a clear, consistent pattern: adding SC improves performance over CoT, irrespective of the underlying LLM. For example, on the HySQA dataset, incorporating SC with Qwen 3.6 increases the F1-score by over 11 points. Similarly, Mistral Large and GLM 4.7 gain approximately 6.7 and 6.3 points, respectively. This consistency indicates that SC effectively addresses reasoning errors produced by the base models, enabling them to revise intermediate steps and reach more accurate KGXL-Queries. Moreover, although absolute performance varies with backbone strength, with GLM 4.7 generally achieving the highest scores, the relative benefit from SC remains stable across models. This demonstrates that our method’s gains are robust and transferable to diverse open-source architectures.

6. Conclusion, Limitations and Future Work

The advent of empowering QA systems over heterogeneous data sources shifts the focus to adapting the data to the model. Despite the advantage of using existing SoTA models for each data source, fitting the data to the model results in irreversible data loss and redundancy. On the other hand, decomposing the question and directing each sub-question to the respective data source has its own limitations. First, the decision on when to stop the decomposition is vague, and directing sub-questions to their respective sources is inherently problematic when the sources share too many commonalities. To address this, we have shifted our focus to the query. We introduce KGXL-Query, which enables access to all sources via a single query. We achieve this capability by extending SPARQL functionality with LLM and semantic search. Furthermore, our experiment shows that KGXL consistently outperforms other methods across four heterogeneous QA benchmarks. We do not conduct experiments to determine the latency and scalability of our method. Hence, these remain open empirical questions for future work, where systematic benchmarking under varying query complexities and data scales would be necessary to quantify the real-world performance overhead of embedding LLM calls and semantic search within SPARQL. In line with that, future work can explore cost-based query optimization to balance symbolic and neural invocations.

Declaration on Generative AI

During the preparation of this work, the author(s) used Gemma-3.0 and GLM for generating code samples and correcting errors. Additionally, we used Grammarly to correct grammar and spelling. After using these tools, the authors reviewed and edited the content as needed and took full responsibility for the publication’s content.

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GraphRAG Best Practices: A Modular Evaluation Framework and Regulatory Case Study

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Abstract. Graph Retrieval-Augmented Generation (GraphRAG) enhances Large Language Models by integrating knowledge graphs into retrieval and generation. Despite growing adoption, no systematic study has isolated the contribution of individual pipeline components to end-to-end performance, leaving practitioners without evidence-based configuration guidance. We address this gap with a component-level empirical study spanning three GraphRAG stages, namely graph-based indexing, graph-guided retrieval, and graph-enhanced generation, and identify which design decisions most significantly influence answer quality. Across three datasets (MSMARCO, HotpotQA, and a domain-specific EU banking regulation corpus), we benchmark 17 embedding models, six retriever types with varying traversal depths, nine reranking models, and four graph serialisation formats for prompt construction. Our analysis reveals three key findings: (1) retriever combination strategy and traversal depth are dataset-dependent, with domain-specific corpora favouring shallow traversal and separate reranking; (2) a dual-retriever pairing hypothetical-question embeddings with entity-based retrieval, pruned by a cross-encoder reranker, yields the most robust performance across domains (F1 up to 89.9); and (3) code-like graph serialisation (GraphML) provides the best quality-latency trade-off for generation, likely due to LLM familiarity with structured syntax. Applied to 500 questions over EU banking regulations (CRR + CRD IV), the optimised GraphRAG pipeline achieves win rates of 58–67% against Naive, HyDE, and Hybrid RAG baselines across comprehensiveness, diversity, empowerment, and correctness. All generation components use open-source LLMs (Llama-3.3-70B-Instruct), and datasets are publicly available, supporting reproducibility.

Keywords. Graph Retrieval-Augmented Generation, Knowledge Graph, Retrieval-Augmented Generation, Regulatory Compliance, Evaluation Framework

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1. Introduction

Large Language Models (LLMs) have demonstrated remarkable capabilities in natural language understanding and generation, yet they remain prone to hallucinations and factual inaccuracies when addressing domain-specific or knowledge-intensive queries [1,2]. Retrieval-Augmented Generation (RAG) mitigates this limitation by grounding LLM outputs in externally retrieved evidence [3,4]. However, conventional RAG systems that rely solely on vector-based retrieval struggle with complex queries requiring multi-hop reasoning or an understanding of relational structures among information pieces [5,6]. Graph Retrieval-Augmented Generation (GraphRAG) addresses these shortcomings by integrating knowledge graphs into the retrieval and generation pipeline, enabling structured traversal of entity relationships and more precise context selection [7,8].

Despite the growing adoption of GraphRAG systems, the field lacks systematic, empirically grounded guidance on configuring individual pipeline components for optimal performance. Existing comparisons typically evaluate only end-to-end outputs without isolating the contributions of individual design decisions [9]. While best practices have been investigated for conventional vector-based RAG [10], no analogous component-level study exists for GraphRAG. This gap leaves practitioners without evidence-based guidelines and forces ad hoc configuration choices that may be suboptimal or even detrimental for their target domain.

This paper addresses this gap through a systematic, component-level empirical study of GraphRAG best practices. We decompose the GraphRAG pipeline into three stages, namely *graph-based indexing*, *graph-guided retrieval*, and *graph-enhanced generation*, and conduct experiments that isolate the effect of individual design decisions at each stage. By evaluating across both open-domain benchmarks (MSMARCO [11], HotpotQA [12]) and a domain-specific regulatory corpus (EU Capital Requirements Regulation), we identify which configurations are robust across domains and which are domain-dependent. Specifically, we address the following research questions:

- RQ1** Which design decisions at each GraphRAG pipeline stage most significantly influence end-to-end answer quality, and how can best practices be identified through modular experimentation across multiple datasets?
- RQ2** How do retrieval strategy, traversal depth, reranking, and graph serialisation format interact, and what configurations prove most robust across open-domain and domain-specific settings?
- RQ3** How does an optimised GraphRAG configuration compare against conventional RAG paradigms (Naive RAG, HyDE RAG, and Hybrid RAG) in a high-stakes regulatory domain?

Contributions. This work makes the following three contributions:

1. **Component-level empirical study:** We present the first systematic best-practice investigation for GraphRAG that *isolates individual design decisions across the full pipeline lifecycle* on multiple datasets. Unlike prior work that compares end-to-end systems [9] or investigates best practices for vector-based RAG only [10], our study benchmarks 17 embedding models, two extraction prompting strategies, six retriever types at multiple traversal depths, nine reranking models, and four graph serialisation formats.

2. **Transferable findings with domain-dependent nuances:** We identify which design decisions generalise across domains and which require per-domain tuning, yielding a consolidated best-practice recipe for GraphRAG deployment.
3. **Regulatory domain validation:** We validate the optimised pipeline on a full-scale EU banking regulation corpus (CRR + CRD IV), demonstrating consistent improvements over conventional RAG paradigms in a high-stakes compliance setting.

The remainder of this paper is structured as follows. Section 2 reviews related work on RAG and GraphRAG. Section 3 describes our modular experimental methodology. Section 4 presents the experimental results across all pipeline stages. Section 5 reports the regulatory domain case study. Section 6 discusses findings and limitations, and Section 7 concludes with future directions.

2. Related Work

Lewis et al. [3] introduced RAG to mitigate LLM hallucinations [13] by integrating retrieval with generation. Compared to fine-tuning, RAG enables dynamic knowledge updates without retraining [4] and avoids catastrophic forgetting [14]. Subsequent work enhanced RAG through hybrid sparse-dense retrieval [15,16], query rewriting [17], and query transformation techniques such as HyDE [18] and Query2doc [19]. However, conventional RAG still struggles with multi-hop reasoning, often missing crucial intermediate concepts [8], and retrieved documents frequently contain irrelevant noise [20]. These limitations motivated the development of GraphRAG, which leverages knowledge graphs as structured stores to enable relational traversal and more precise context selection.

Following prior categorisations [7,8], the GraphRAG process comprises three stages. The first stage, *graph-based indexing*, constructs KGs using subject-predicate-object triples [20]. Graph quality directly influences downstream performance: sparse or noisy graphs hinder retrieval, while overly dense ones cause information overload [21]. Construction methods range from traditional open information extraction [22], entity linking [23], and relation extraction [24] to LLM-based approaches including joint extraction [25], iterative zero-shot prompting [26], entity-clustering frameworks such as KGGen [27], and multi-agent ontology population [28]. The second stage, *graph-guided retrieval*, identifies relevant subgraphs for a given query. Similarity-based retrieval uses embeddings and cosine similarity [29,30,31] or keyword matching with graph traversal [32,33], whereas LLM-based retrieval dynamically guides the process. For instance, Microsoft’s GraphRAG [34] generates responses from community summaries, while Fact Finder [35] translates queries into Cypher. Post-retrieval pruning or reranking further refines results [30]. The third stage, *graph-enhanced generation*, serializes retrieved subgraphs into prompts for the LLM. Since prompt format significantly impacts performance [36,37], careful design is essential to avoid input overload and the “lost in the middle” phenomenon [38]. Formats explored in the literature include adjacency tables [39], natural language descriptions [40], and code-like representations such as GraphML [41,42].

Building on these foundational components, several end-to-end GraphRAG systems have emerged. Microsoft’s GraphRAG [34] uses hierarchical community summaries with map-reduce aggregation for global queries. LightRAG [43] adds dual-level

retrieval, and HippoRAG [44] combines LLMs with personalised PageRank for multi-hop QA. Open-source frameworks such as LlamaIndex [45] and LangChain [46] also implement GraphRAG pipelines, while applications span customer support [47] and biomedicine [48], with recent work exploring hybrid vector-graph retrieval [49].

Despite these advancements, existing comparisons remain limited to end-to-end evaluations that do not isolate the contribution of individual component choices. Han et al. [9] compare Microsoft’s GraphRAG against LlamaIndex’s property-graph approach without examining how decisions such as chunking strategy, retriever type, traversal depth, or serialisation format independently affect answer quality. Wang et al. [10] explored best practices for vector-based RAG, but no analogous component-level study exists for GraphRAG. Our work fills this gap by systematically isolating each design decision across the full GraphRAG pipeline on multiple datasets, identifying transferable best practices with domain-dependent nuances.

3. Methodology

This section describes our modular experimental methodology for identifying best practices across the three GraphRAG pipeline stages. Figure 1 provides an overview of all design choices evaluated. The evaluation follows a sequential, greedy strategy: within each stage, methods are individually evaluated while other components remain fixed. Once a method consistently outperforms alternatives, it is adopted as the default in subsequent experiments, enabling tractable evaluation of the large configuration space.

All experiments use Llama-3.3-70B-Instruct [50], deployed locally via Ollama on 2 NVIDIA H100 GPUs (80 GB each) with temperature set to zero. We evaluate on three datasets: (1) **MSMARCO** [11], 200 test entries for general-purpose passage retrieval; (2) **HotpotQA** [12], 200 validation entries for multi-hop reasoning; and (3) **CRR-based QA**, 200 domain-specific questions generated via RAGAS [51] from the EU Capital Requirements Regulation (approximately 35% single-hop, 65% multi-hop).

Graph-based Indexing. The indexing stage constructs a property graph from source documents via LLM-based triplet extraction. We evaluate four design decisions. For *chunking*, we compare sentence-level and hierarchical document chunking (preserving legal structure) across five token sizes (128 to 2048, 20-token overlap). Hierarchical chunking uses a custom EUR-Lex parser that segments by legal hierarchy. Evaluation uses Faithfulness (the proportion of claims in the generated answer that are supported by the retrieved context) and Relevancy (the degree to which the generated answer addresses the input query, penalising incompleteness and redundancy) [51] metrics assessed by Nemotron-70B-Instruct [52] on 200 questions sampled from approximately 800 generated from the CRR corpus. For *embedding*, we evaluate 17 models from six families (BAAI/bge [53], Jina AI [54,55], intfloat/e5 [56], avsolatorio/GIST [57], NovaSearch/stella [58], Alibaba-NLP/gte [59]) on 27 MTEB [60] benchmark datasets across five task categories. Models are ranked by their accuracy-to-efficiency trade-off, since the selected embedder is applied to every node and query and therefore affects both quality and latency. For *extraction prompts*, two strategies are compared: an *Open Prompt* performing generic triplet extraction without schema constraints (LlamaIndex SimpleLLMPathExtractor), and a *Schema-based Prompt* specifying entity and relation

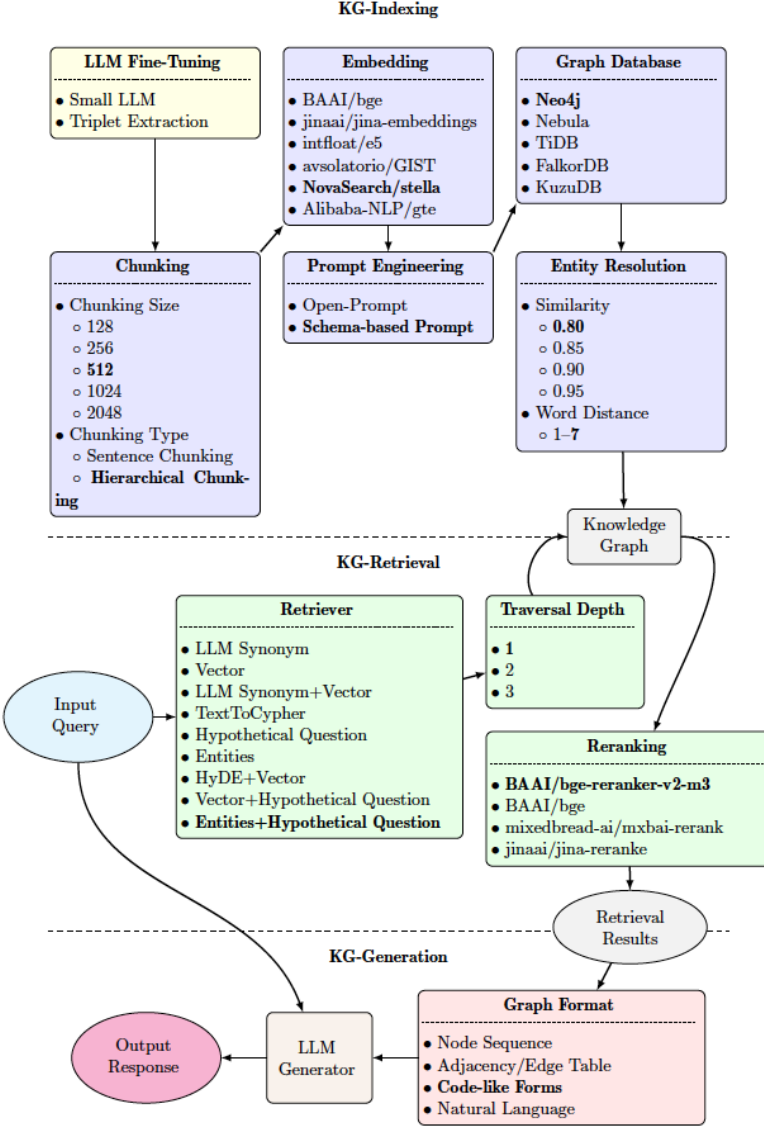


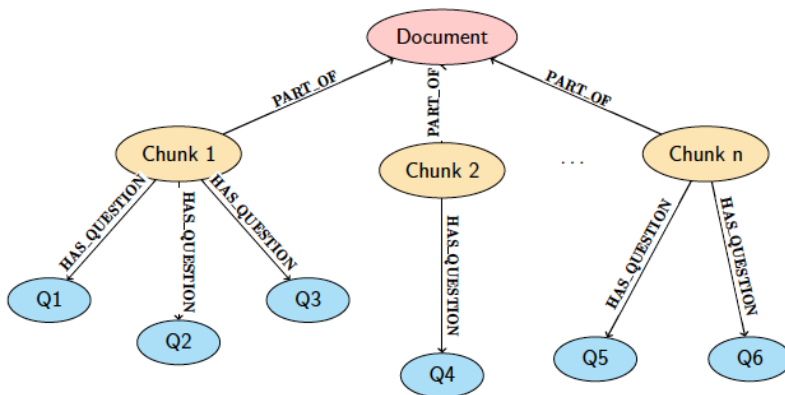
Figure 1. GraphRAG pipeline overview showing all design choices evaluated at each stage. Bold entries indicate best-performing configurations adopted in the final case study.

types while allowing novel types (DynamicLLMPathExtractor), with a schema generated by a reasoning model and manually curated.

For *entity resolution*, we propose a dual-filter algorithm combining embedding-based cosine similarity θ_s with string distance θ_d . The AND-logic requires both semantic and lexical evidence, reducing false positives. For the case study, an additional LLM-based verification step confirms flagged pairs (see Section 5 for details and examples). The algorithm is evaluated on four benchmark datasets [61] (Abt-Buy, Amazon-GoogleProducts, DBLP-ACM, DBLP-Scholar, with 1,000 positive and 5,000 hard nega-

Table 1. Evaluated graph-guided retrieval methods.

Retriever	Category	Strategy
Vector Context	Similarity	Embedding sim. to node content
LLM Synonym	LLM-based	String match via LLM synonyms
TextToCypher	LLM-based	Generated Cypher query
Entities	Hybrid	Entity extraction + embed + keyword
HyDE	Hybrid	Hypothetical document + embed
Hyp. Question	Hybrid	Query $\leftrightarrow$ pre-generated questions

**Figure 2.** Graph structure with hypothetical questions: each document chunk is linked to LLM-generated questions serving as additional retrieval entry points. Adapted from [62].

tive pairs each).

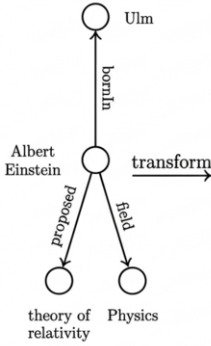
Graph-guided Retrieval. The retrieval stage identifies relevant subgraphs given an input query. Six methods are evaluated (Table 1), spanning similarity-based, LLM-based, and hybrid categories. Among these, the *Hypothetical Question Retriever* (HyQ) enriches the graph during preprocessing by generating questions for each text chunk (Figure 2), so that at query time the input is compared against both entity and pre-generated question nodes.

After initial node retrieval, the system expands via depth-first traversal at depths 1, 2, and 3 hops. Nine reranking models (BAAI, MixedBread, Jina AI) are evaluated on approximately 400 question-context pairs using Hit Rate and Mean Reciprocal Rank (MRR). Retrieval evaluation proceeds in four stages: (1) individual retriever evaluation at depth 1, (2) combination of complementary retrievers, (3) traversal depth analysis for the four most promising retrievers, and (4) end-to-end comparison combining top retrievers with the best reranker under two merge strategies (*early merge*: combine before reranking; *separate rerank*: rerank independently, then merge). Quality is assessed via RAGAS [51] with GPT-4o-Mini², measuring context precision, context recall, and their harmonic mean (F1).

Graph-enhanced Generation. The final stage serializes retrieved subgraphs into text for the LLM prompt. We evaluate four formats (Figure 3): **Node Sequences** (linear listing),

²<https://developers.openai.com/api/docs/models/gpt-4o-mini>, ID: gpt-4o-mini-2024-07-18

Retrieved Graph Data



Adjacency Table

Source	Relation	Target
Albert Einstein	bornIn	Ulm
Albert Einstein	proposed	theory of relativity
Albert Einstein	field	Physics

Natural Language

Albert Einstein was born in Ulm.
He proposed the theory of relativity.
He worked in the field of Physics.

Node Sequence

Albert Einstein → Ulm
Albert Einstein → theory of relativity
Albert Einstein → Physics

Code-like Structure

```

<node id="Albert_Einstein"/>
<node id="Ulm"/>
<node id="theory_of_relativity"/>
<node id="Physics"/>
<edge source="Albert_Einstein"
  target="Ulm">
  <data key="label">bornIn</data>
</edge>
<edge source="Albert_Einstein"
  target="theory_of_relativity">
  <data key="label">proposed</data>
</edge>
<edge source="Albert_Einstein"
  target="Physics">
  <data key="label">field</data>
</edge>

```

Figure 3. Four sub-graph serialisation formats example for prompt construction. Adapted from [7].

Adjacency Tables (source, relation, target tuples), **Code-like Forms/GraphML** (XML-based hierarchical encoding), and **Natural Language** (LLM-converted paragraphs). Evaluation uses RAGAS with GPT-4o-Mini to measure Faithfulness, Answer Relevancy, and latency. The best retriever configuration from the previous stage is kept constant per dataset.

Case Study Setup. To validate best practices in a real-world setting, we construct a full-scale knowledge graph from the complete CRR and CRD IV (approximately 700 articles). The optimised GraphRAG pipeline is compared against three baselines: **Naive RAG** using standard vector retrieval (LlamaIndex default), **HyDE RAG** using query transformation via hypothetical document generation [18], and **Hybrid RAG** combining vector retrieval with the optimised graph-based strategy (HyQ + Entities + bge-reranker-v2-m3) via context concatenation. Evaluation uses 500 question-reference answer pairs generated via RAGAS with GPT-4o-Mini. A custom LLM-as-a-judge approach compares responses across four metrics: *Comprehensiveness*, *Diversity*, and *Empowerment* (adapted from [34,43]), plus *Correctness* (introduced here to assess factual alignment). A “Tie” option reduces forced-choice bias. For each query, both systems generate responses that the judge scores independently per criterion as a win for either system or a tie. Win rates are computed as $\text{wins}(A)/N$ over all $N=500$ questions and serve as the primary comparison metric.

4. Experimental Results

This section presents results across all three GraphRAG pipeline stages, reporting the most significant findings.

Graph-based Indexing Results. Table 2 summarises the effect of chunk size and segmentation method on downstream QA quality. A chunk size of 512 tokens achieves the best balance, as smaller chunks (128 to 256) fragment context and hurt inter-sentence

Table 2. Chunking strategy comparison on the CRR corpus (512-token window). Faithfulness and Relevancy assessed by Nemotron-70B on 200 questions.

Chunking Type	Entities	Triplets	Faith.	Relev.
Sentence (512 tokens)	1,384	1,599	97.0	94.0
Hierarchical (512 tokens)	1,971	2,466	97.5	95.5

Table 3. Prompt engineering comparison on the CRR corpus (hierarchical chunking, 512 tokens).

Prompt Type	Entities	Triplets	Faith.	Relev.
Open-Prompt	1,971	2,466	97.5	95.5
Schema-based	1,983	2,527	98.5	97.5

Table 4. Entity resolution performance (combined across Abt-Buy, Amazon-GoogleProducts, DBLP-ACM, DBLP-Scholar, with 6,000 pairs each).

θ_s	θ_d	Precision	Recall	F1
0.80	7	97.45	54.55	69.95
0.85	7	97.52	53.00	68.68
0.90	7	98.86	50.00	66.41
0.95	7	100.00	44.75	61.83

dependencies, while larger chunks (1024 to 2048) dilute the model’s focus. Hierarchical document chunking, which preserves the legal structure of the CRR corpus (parts, chapters, articles), yields marginal but consistent improvements over sentence-level chunking at the same token budget, extracting substantially more entities (+42%) and triplets (+54%).

Among 17 embedding models evaluated on 27 MTEB benchmark datasets spanning five task categories, NovaSearch/stella.en_400M_v5 achieves the best accuracy-to-efficiency trade-off (overall average: 73.93%), closely trailing the larger stella.en_1.5B_v5 (75.60%) at a fraction of the computational cost. It is adopted for all subsequent experiments. Table 3 compares open-prompt and schema-based extraction strategies. The schema-based approach yields modest but consistent improvements in both QA metrics (+1.0 Faithfulness, +2.0 Relevancy) while extracting slightly more entities and relations, confirming that structured guidance improves triplet quality without sacrificing coverage.

Table 4 reports the performance of the proposed dual-filter algorithm across four benchmark datasets. The best overall F1 (69.95%) is achieved at cosine similarity $\theta_s = 0.80$ and word distance $\theta_d = 7$, maintaining precision near 97% while achieving reasonable recall. The AND-logic strategy ensures conservative merging. Although recall is limited on challenging e-commerce datasets where product titles differ extensively in length and wording, precision remains high, avoiding erroneous merges of semantically distinct entities.

Graph-guided Retrieval Results. Knowledge graphs are constructed from all three datasets using the identified best practices: hierarchical chunking (512 tokens) for CRR, and sentence-level chunking (512 tokens) for MSMARCO and HotpotQA (as these lack inherent legal hierarchy). All graphs use stella.en_400M_v5 embeddings, schema-based extraction prompts, and Neo4j storage. Table 5 reports the resulting graph statistics. Table 6 then compares all six retrievers at traversal depth 1. HyDE achieves the highest

Table 5. Knowledge graph statistics per dataset.

Dataset	Entities	Triplets
MSMARCO	5,438	10,056
HotpotQA	3,716	6,487
CRR (Part 1 & 2)	1,983	5,261

Table 6. Individual retriever F1 scores and latency (seconds) at depth 1.

Retriever	MSMARCO		HotpotQA		CRR	
	F1	Lat.	F1	Lat.	F1	Lat.
Vector Context	80.59	5.2	68.18	2.4	74.32	18.4
LLM Synonym	24.00	2.7	22.74	1.9	43.28	10.0
TextToCypher	3.39	3.9	5.15	2.9	2.00	3.0
Entities	75.91	29.4	54.63	10.3	77.81	85.2
HyDE	83.14	9.2	73.56	5.6	74.07	26.1
Hyp. Question	73.62	2.9	67.88	1.2	72.94	7.7

F1 on MSMARCO (83.14) and HotpotQA (73.56), while Entities leads on CRR (77.81). TextToCypher and LLM Synonym perform poorly as standalone methods. The Hypothetical Question Retriever (HyQ) offers the best latency-to-quality trade-off, with response times under 8 seconds across all datasets.

Regarding traversal depth, deeper traversal (2-3 hops) yields modest gains on open-domain datasets. HyDE improves from 83.14 to 83.75 F1 on MSMARCO at depth 3, and Entities increases from 54.63 to 61.10 on HotpotQA. However, on the domain-specific CRR corpus, depth 1 consistently yields the best or near-best results across all retrievers (for instance, Entities drops from 77.81 to 71.32 at depth 3), indicating that excessive traversal introduces noise in densely interconnected legal texts. Among nine candidate reranking models, BAAI/bge-reranker-v2-m3 achieves the highest Hit Rate (94.42%) and MRR (82.37%) on the CRR evaluation set, improving over the no-reranker baseline (90.05% HR, 74.60% MRR). It is adopted for all subsequent end-to-end configurations.

Table 7 presents the key end-to-end configurations combining top retrievers with bge-reranker-v2-m3. Reranking substantially improves both F1 and latency (for instance, Entities on CRR decrease from 85.2s to 9.0s) by pruning irrelevant nodes before generation. The HyQ + Entities combination proves the most robust across datasets. On MSMARCO, early merging at depth 1 achieves the top score (89.87). On CRR, separate reranking at depth 1 leads (83.20), suggesting that independently pruning each retriever’s results is preferable for densely interconnected legal texts. On HotpotQA, the standalone Entities retriever at depth 3 achieves the highest score (77.30), with HyQ + Entities (SR, depth 3) a close second (76.45). These results confirm that merge strategy and traversal depth are dataset-dependent hyperparameters, but the HyQ + Entities pairing provides the most consistent baseline. However, the performance gains of the best methods come at a notable latency cost. For example, on CRR, HyQ + Entities (SR) at depth 1 achieves an F1 of 83.20 but requires 15.7 s, compared to HyQ + Vector (EM) at depth 1 which achieves 80.01 F1 in 7.7 s - a ~4% F1 improvement at a ~104% latency increase. Practitioners must therefore balance quality and responsiveness based on their requirements.

Table 7. End-to-end retrieval F1 and average latency (seconds) with bge-reranker-v2-m3. Latency represents the average wall-clock execution time per query. Best F1 per dataset in **bold**. Lowest latency per dataset underlined. EM = Early Merge, SR = Separate Rerank, D = Traversal Depth.

Configuration	D	MSMARCO		HotpotQA		CRR	
		F1	Lat.	F1	Lat.	F1	Lat.
HyQ + Entities (EM)	1	89.87	7.1	70.97	3.6	77.94	8.9
HyQ + Entities (EM)	2	88.97	7.1	71.65	3.6	77.50	8.7
HyQ + Entities (SR)	1	84.35	9.8	74.37	4.1	83.20	15.7
HyQ + Entities (SR)	3	84.98	9.7	76.45	4.1	82.73	15.0
HyQ + Vector (EM)	1	88.37	<u>5.3</u>	73.89	<u>2.4</u>	80.01	<u>7.7</u>
Entities	3	87.12	7.2	77.30	3.8	70.02	8.9

Table 8. Generation quality across graph serialisation formats. Best values in **bold**, second-best underlined.

Format	MSMARCO			HotpotQA			CRR		
	Faith.	Rel.	Lat.	Faith.	Rel.	Lat.	Faith.	Rel.	Lat.
Node Seq.	<u>91.50</u>	62.54	7.1	76.13	52.92	3.8	82.70	77.61	15.7
Adjacency	90.69	<u>63.24</u>	7.1	79.37	53.97	4.0	80.20	80.48	16.5
Code-like	91.09	63.76	8.5	82.47	52.14	6.1	<u>83.73</u>	76.68	16.8
Nat. Lang.	93.02	62.06	73.8	<u>80.60</u>	<u>53.78</u>	49.2	89.52	<u>79.30</u>	134.0

Graph-enhanced Generation Results. Table 8 compares four graph serialisation formats using the best retriever per dataset (HyQ + Entities EM depth 1 for MSMARCO, Entities depth 3 for HotpotQA, HyQ + Entities SR depth 1 for CRR). Natural Language (short: Nat. Lang.) achieves the highest Faithfulness on MSMARCO (93.02) and CRR (89.52), while the Code-like format leads on HotpotQA (82.47). However, Natural Language imposes prohibitive latency (49 to 134 seconds per query), making it impractical for real-time applications. Among the remaining formats, Code-like (GraphML) provides the most favourable quality-to-latency trade-off. When averaging Faithfulness and Answer Relevancy, it ranks first on HotpotQA and second on MSMARCO, while maintaining moderate latency (8.5 to 16.8 seconds). A plausible explanation is that LLMs, extensively pre-trained on source-code corpora, parse structured XML syntax with high fidelity. GraphML eliminates repetitive textual scaffolding and provides a single, hierarchically organised view of nodes and edges, facilitating the model’s ability to grasp the subgraph topology.

Summary of Best Practices. Table 9 consolidates the identified best practices across all pipeline stages into an actionable recipe. The indexing stage benefits from hierarchical chunking at 512 tokens, stella_en_400M_v5 embeddings, schema-based extraction prompts, and conservative entity resolution using a dual-filter with cosine similarity ≥ 0.80 and word distance ≤ 7 . For retrieval, the HyQ + Entities pairing with bge-reranker-v2-m3 provides the most robust performance, though the merge strategy and traversal depth remain domain-dependent. For generation, Code-like (GraphML) serialisation offers the best quality-to-latency trade-off across all three datasets.

Table 9. Consolidated best-practice recipe for GraphRAG.

Stage	Decision	Best Practice
Indexing	Chunking	Hierarchical, 512 tokens
	Embedding	stella_en_400M_v5
	Extraction	Schema-based prompt
	Entity Resolution	Cosine ≥ 0.80 AND word dist. ≤ 7
Retrieval	Retriever pair	HyQ + Entities
	Reranker	bge-reranker-v2-m3
	Depth/Merge	Domain-dependent
Generation	Graph format	Code-like (GraphML)

5. Case Study: EU Banking Regulations

To validate the identified best practices in a real-world setting, we apply the optimised GraphRAG pipeline to the complete EU banking regulatory corpus and benchmark it against three alternative RAG paradigms. The corpus comprises the complete texts of the Capital Requirements Regulation (CRR) and the Capital Requirements Directive IV (CRD IV), approximately 700 legal articles that constitute the backbone of the EU’s banking regulatory framework. Documents were fetched from Eurlex and preprocessed using a custom hierarchical parser³ that segments regulatory texts by their legal hierarchy (parts, chapters, and articles).

The knowledge graph was constructed using the best practices identified in Section 4: hierarchical chunking at 512 tokens, stella_en_400M_v5 embeddings, and schema-based triplet extraction stored in Neo4j. Table 10 summarises the graph statistics at each construction stage. Entity resolution reduced the graph by 19.6% (from 9,733 to 7,826 unique entities) using the dual-filter algorithm ($\theta_s = 0.80$, $\theta_d = 7$) followed by LLM-based verification. We set θ_s to ≥ 0.80 because lower thresholds increasingly match semantically related but non-identical entities (e.g., “credit risk” and “market risk”), introducing false merges where terminological precision is critical. The word-distance threshold $\theta_d = 7$ was selected based on preliminary experiments showing that entity labels in our corpus rarely exceed seven tokens in length difference when referring to the same concept (e.g., “European Banking Authority” vs. “EBA”). This aligns with findings by Köpcke et al. [61], who report that string-based thresholds in this range effectively filter length-variant duplicates. The language model (Llama-3.3-70B-Instruct) verified whether each flagged pair genuinely referred to the same underlying legal concept, entity, or term, for instance, distinguishing “Capital Requirements Regulation” from “Capital requirements” as semantically distinct. Subsequently, 9,842 hypothetical question nodes were generated and linked to source chunks, expanding the graph to 17,668 nodes as a prerequisite for the Hypothetical Question Retriever.

The GraphRAG pipeline uses the best-performing configuration identified in Section 4: Hypothetical Question + Entities retriever pair with bge-reranker-v2-m3 (separate rerank), traversal depth 1, and Code-like Form (GraphML) prompt representation. It is compared against three baselines. **Naïve RAG** uses standard vector-based retrieval via LlamaIndex’s built-in vector retriever, followed by LLM generation from retrieved

³ Available at: <https://pypi.org/project/eurlex-parser/>

Table 10. Knowledge graph statistics for the CRR and CRD IV corpus across construction stages.

Stage	Entities	Triplets
Initial extraction	9,733	29,708
After entity deduplication	7,826	28,438
After hypothetical question generation	17,668	48,264

chunks. **HyDE RAG** applies hypothetical document generation [18] followed by vector retrieval, identified as one of the most effective retrieval strategies in conventional RAG [10]. **Hybrid RAG** combines vector-based retrieval with the same optimised graph-based strategy (HyQ + Entities + bge-reranker-v2-m3), concatenating retrieved contexts before generation. Evaluation uses 500 question-reference answer pairs generated via RAGAS’s built-in question generation module with GPT-4o-Mini. A custom LLM-as-a-judge approach (GPT-4o-Mini) compares responses across three metrics adapted from Edge et al. [34] and Guo et al. [43], namely *Comprehensiveness*, *Diversity*, and *Empowerment*, plus a fourth metric, *Correctness*, introduced in this study to assess factual alignment with reference answers. A “Tie” option reduces forced-choice bias, and win rates are reported as the primary metric for comparison.

Table 11 presents GraphRAG’s win rates against each baseline across 500 questions. GraphRAG consistently outperforms all three baselines across all evaluation metrics, achieving overall win rates of 60.6%, 58.0%, and 67.4% against Naive RAG, HyDE RAG, and Hybrid RAG, respectively. Several patterns emerge from these results. Diversity is the strongest differentiator, with GraphRAG achieving 76.2 to 76.4% win rates across all comparisons, indicating that structured graph traversal surfaces a broader range of relevant legal provisions than vector-based retrieval alone. Counter-intuitively, combining vector and graph retrieval (Hybrid RAG) performs worse than graph retrieval alone (67.4% GraphRAG win rate), as the longer combined context risks introducing conflicting or noisy information that causes the model to struggle with prioritising relevant content. Despite generating more comprehensive and diverse answers, GraphRAG maintains higher correctness than all baselines (39.0 to 46.0% versus 26.4 to 34.4%), demonstrating that graph grounding *reduces* hallucination even when producing detailed responses. The Naive RAG baseline often returns a single correct passage but omits broader context, while HyDE RAG occasionally introduces unsupported assumptions from the hypothetical document step.

Table 12 illustrates a representative qualitative comparison. For the query “What role does the Council play in the context of capital requirements and economic cycles?”, GraphRAG synthesises information from multiple regulatory articles, referencing connected entities (EBA, ESRB, ECB) and citing relevant provisions, while Naive RAG provides a narrower answer based on fewer retrieved passages. GraphRAG’s advantage stems from its ability to traverse connected entities in the knowledge graph, whereas vector-based retrieval returns isolated text chunks that may capture only one facet of a multi-dimensional legal concept.

6. Discussion

This study provides the first systematic, component-level investigation of GraphRAG best practices across its full pipeline lifecycle. Our findings can be distilled into three

Table 11. Win rates (%) comparing GraphRAG against three RAG baselines on 500 EU banking regulation questions. Bold indicates the winning system per metric.

Metric	Baseline	GraphRAG	Tie
<i>GraphRAG vs. Naive RAG</i>			
Comprehensiveness	38.2	57.8	4.0
Diversity	13.0	76.2	10.8
Empowerment	36.2	60.4	3.4
Correctness	28.0	46.0	26.0
Overall	36.2	60.6	3.2
<i>GraphRAG vs. HyDE RAG</i>			
Comprehensiveness	40.6	56.0	3.4
Diversity	13.8	76.2	10.0
Empowerment	37.8	59.0	3.2
Correctness	34.4	39.0	26.6
Overall	39.2	58.0	2.8
<i>GraphRAG vs. Hybrid RAG</i>			
Comprehensiveness	28.0	61.8	10.2
Diversity	6.4	76.4	17.2
Empowerment	24.4	67.2	8.4
Correctness	26.4	43.6	30.0
Overall	25.4	67.4	7.2

transferable insights. First, graph construction quality is foundational. Schema-based extraction with hierarchical chunking at 512 tokens consistently outperformed alternatives. While the absolute gains over open-prompt extraction were modest (one to two percentage points), these improvements compound across the pipeline, as a cleaner graph yields better retrieval, which in turn yields better generation. Second, the retriever combination outperforms individual retrievers. The Hypothetical Question + Entities pairing proved robust across domains by combining semantic question matching (fast, broad coverage) with entity-centric retrieval (precise, domain-grounded). Reranking with bge-reranker-v2-m3 both improved quality and reduced latency by pruning irrelevant nodes before generation. Third, optimal configurations are partially domain-dependent. While the retriever pairing and graph format (GraphML) generalise well, the merge strategy and traversal depth require per-domain tuning. Domain-specific corpora with dense entity interconnections, such as legal texts, favour shallow traversal and separate reranking, whereas open-domain corpora benefit from deeper traversal and early merging. The consolidated recipe in Table 9 captures these findings as an actionable starting point spanning all pipeline stages.

These best practices offer guidance for practitioners deploying GraphRAG in domain-specific settings. The modular evaluation approach demonstrated here, fixing components sequentially rather than testing all combinations, provides a tractable methodology for configuration search in new domains. Practitioners can adopt the recipe and adapt it by tuning the domain-dependent hyperparameters on a small validation set. The case study results further demonstrate that pure GraphRAG outperforms hybrid approaches combining vector and graph retrieval. This counterintuitive finding suggests that context concatenation from heterogeneous sources can introduce noise that dilutes

Table 12. Qualitative comparison between GraphRAG and Naive RAG on a representative query (case study).

Query	What role does the Council play in the context of capital requirements and economic cycles?
Naive RAG	The Council receives reports from the Commission, along with any appropriate proposals, regarding the cyclicity of capital requirements and their effects on the economic cycle. Additionally, the Council is a recipient of reviews and reports on specific aspects, such as the application of certain articles related to capital requirements, which inform it and may prompt it to take action.
GraphRAG	The Council plays a role in the context of capital requirements and economic cycles as it receives reports from the Commission, which are drawn up in cooperation with the European Banking Authority (EBA), the European Systemic Risk Board (ESRB), and Member States. These reports examine whether the Capital Requirements Regulation has significant effects on the economic cycle and consider any remedial measures that may be justified. The Council also receives legislative proposals from the Commission, if appropriate, based on an analysis of the cyclicity of capital requirements and the ECB's opinion. Furthermore, the Council is involved in the review process of certain articles, such as Article 33(1)(c), where the Commission submits a report and a legislative proposal if necessary, to ensure financial stability in all Member States.
LLM Judge	GraphRAG wins on all four criteria. Comprehensiveness: "provides a more detailed overview including interactions with multiple entities (EBA, ESRB, ECB)." Diversity: "presents a broader range of perspectives by mentioning various stakeholders and highlighting remedial measures." Empowerment: "empowers the reader with a deeper understanding, emphasizing collaboration and article reviews." Correctness: "accurately reflects the Council's role per the reference, correctly mentioning ECB involvement, comprehensive reporting, and legislative proposals."

the LLM's focus, an important consideration for system architects who might assume that more retrieval sources invariably improve quality.

Several limitations should be acknowledged. The current approach constructs static knowledge graphs from a fixed corpus without support for incremental updates [63], temporal reasoning [64], or version tracking, which limits applicability to evolving corpora. It also uses property graphs without formal ontological constraints, foregoing the reasoning capabilities of RDF/OWL-based representations and dedicated domain ontologies and vocabularies. The best practices identified here, particularly regarding schema-guided extraction and structured serialisation, are likely transferable to ontology-backed GraphRAG pipelines but remain to be empirically validated. The greedy, sequential evaluation strategy does not guarantee a global optimum because interactions among components are not fully captured once earlier decisions are fixed. Both RAGAS-based metrics and the case study rely on LLM judges (GPT-4o-Mini), which may favour verbose or well-structured answers over factual correctness and may not align with human expert judgement in specialised legal domains. The retrieval strategy operates at the node and neighbourhood levels without global structures such as community detection or hierarchical summarisation [34], and entity resolution relies on cosine similarity with string distance, with learning-based matching potentially yielding higher recall. Large subgraphs can also exceed the LLM context window, leading to truncation, which partly explains the diminishing returns of deeper traversal.

Regarding threats to validity, we mitigate the propagation of early-stage choices by using well-established benchmarks (MTEB) for embedding selection and by evaluating across three diverse datasets. Experiments were conducted primarily on English-

language legal corpora, so generalisability to other languages, domains, or graph scales remains to be established.

7. Conclusion and Future Work

Regarding **RQ1**, our modular experiments reveal that retriever choice and graph serialisation format exert the largest influence on end-to-end quality, while chunking and extraction prompt type yield smaller but compounding gains. Regarding **RQ2**, the HyQ + Entities dual-retriever with cross-encoder reranking and GraphML serialisation generalises across domains, whereas the merge strategy (early vs. separate) and traversal depth are dataset-dependent. Regarding **RQ3**, the optimised GraphRAG pipeline achieves win rates of 58–67% against Naive, HyDE, and Hybrid RAG baselines on 500 EU banking regulation questions, confirming practical value in a high-stakes compliance setting.

Several directions for future work emerge from this study. First, extending the pipeline to support dynamic and temporal graphs, including incremental updates, temporal annotations, and version tracking, would better accommodate evolving regulatory corpora. Second, incorporating global graph semantics through community detection and hierarchical summarisation [34] could complement local retrieval for abstractive or comparative queries. Third, investigating ontology-guided construction using formal RDF/OWL schemas may bridge property-graph flexibility with semantic web rigour. Finally, complementing LLM-as-judge metrics with expert human evaluation remains essential for validating legal correctness, and scalability testing on larger corpora, such as the full EU acquis, would assess enterprise readiness. All generation components use open-source LLMs, and the source corpora are publicly available, supporting reproducibility and practical adoption.

Declaration on Generative AI: We declare the use of generative AI in this work: (1) Within the research pipeline, Llama-3.3-70B-Instruct (open-source, locally deployed) served as the primary LLM for all pipeline components. GPT-4o-Mini (OpenAI API) was used exclusively for evaluation, serving as the RAGAS evaluator, question generator, and LLM-as-a-judge Llama-3.1-Nemotron-70B-Instruct was used as the evaluator for chunking experiments only, and DeepSeek-R1 was used once for initial schema generation, which the authors subsequently curated manually. (2) In manuscript preparation, Claude (Anthropic) assisted with structuring and rephrasing. All content was verified against experimental results and revised by the authors, who take full responsibility for accuracy and integrity.

Environmental Footprint Estimate: All local experiments were conducted on a single HPC cluster node equipped with 2 NVIDIA H100 GPUs (80 GB each) and 2.0 TiB RAM, totalling approximately 200 GPU hours. Cloud API usage amounted to approximately 22,000 calls for evaluation and question generation.

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Navigating Schema Drift: Orchestrating Agents and Schema Discovery for Robust, Domain-Specific Knowledge Graph Question Answering

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Abstract. Knowledge Graph Question Answering (KGQA) over Large Language Models (LLMs) increasingly translates natural-language questions into SPARQL, yet most systems are statically optimized for a single benchmark and degrade abruptly under cross-domain transfer or schema evolution. We present HARP (Hybrid Agentic-Retrieval Pipeline), a KG-agnostic Text2SPARQL architecture that operates without task-specific fine-tuning. A meta-controller orchestrates a ReAct agent and an execution-validated, schema-guided fallback pipeline. Both paths are grounded by a runtime FAISS-indexed Schema-Discovery service that absorbs structural change by building temporary indices from live SPARQL probes. The system is evaluated across four heterogeneous KGs: DBpedia, a structurally complex Corporate dataset, and two versions of a proprietary railway metadata KG (TrackVideo). The v1 to v2 transition of TrackVideo represents radical schema refactoring, allowing us to test true schema-drift resilience. The hybrid architecture attains a highly consistent performance across all datasets, neutralizing the degradations typically caused by schema drift. A multi-model Quorum extension over heterogeneous LLM families further increases reliability, demonstrating that hybrid agentic retrieval with execution-validated routing is a highly viable paradigm for portable KGQA in production-grade settings.

Keywords. Knowledge Graph Question Answering, Large Language Models, Text2SPARQL, Portability, Schema Drift, Agentic AI, Hybrid Architectures.

1. Introduction

In a modern, data-driven world, organizations rely on vast amounts of heterogeneous data. While traditional storage models efficiently capture data, they often lack the ability to provide deep semantic context [1,2]. Knowledge Graphs (KGs) address this by storing data and connecting it semantically. However, querying these graphs remains a significant

challenge [3,4,5,6]. Extracting information typically requires mastering structured query languages like SPARQL, whose complexity is compounded by a heavy dependence on the specific structure of the underlying KG.

Recently, the success of LLMs has positioned them as promising tools to bridge this gap. Contemporary Knowledge Graph Question Answering or Text2SPARQL methods leverage LLMs to translate natural language directly into executable queries, achieving encouraging results [7,8,9,10]. Despite these advances, important questions remain regarding the adaptability of these systems. As Perevalov et al. [11] note, the majority of KGQA systems are evaluated on a single, general-domain KG (typically Freebase [12], DBpedia [13], or Wikidata [14]), leaving their actual portability across distinct, heterogeneous domains largely untested.

This lack of cross-domain evaluation is critical. The specific schema representation of a KG significantly influences the performance of a KGQA system [15], meaning that relying on a single graph structure masks architectural rigidities. This was evident in the recent Text2SPARQL Challenge² at the European Semantic Web Conference 2025 [16], where different approaches achieved peak performance on different datasets, highlighting a pervasive limitation in portability. Furthermore, in industrial applications—such as the railway metadata Knowledge Graph [17] maintained by a major German rail-infrastructure operator that underlies our case study and for which a first domain-specific, hybrid QA approach was presented in our previous work [18]—schemas are not static. As new applications capture novel data types, the graph continuously evolves. This phenomenon, known as “schema drift”, forces statically optimized or fine-tuned QA systems to be rebuilt entirely to maintain accuracy.

To address both the research portability gap and the industrial challenge of schema drift, we propose a KG-agnostic Text2SPARQL architecture that operates without task-specific retraining or manual schema alignment. By treating query generation as a dynamic orchestration problem, our system combines the logical depth of a ReAct-based agent with the reliability of a schema-augmented retrieval pipeline.

Guided by these challenges, we investigate the following research questions:

- RQ1** How can an LLM-based Text2SPARQL architecture effectively route queries between agentic reasoning and direct retrieval to sustain answer quality across structurally heterogeneous KGs?
- RQ2** To what extent does a dynamic runtime schema-discovery mechanism neutralize performance degradation during major structural refactoring (schema drift)?
- RQ3** Which specific reasoning failure modes (e.g., entity linking vs. complex aggregation) manifest across distinct LLM families within the same pipeline?
- RQ4** How does a quorum-based execution voting mechanism over heterogeneous LLM families impact the Answer Trustworthiness Score compared to a single-model approach?

To answer these questions, the paper makes four contributions, each mapped to one research question:

- **Cross-Domain Portability (RQ1):** We design and evaluate an LLM-based KGQA architecture that maintains robust performance across structurally and semantically distinct KGs without task-specific fine-tuning.

²<https://text2sparql.aksw.org/2025/>

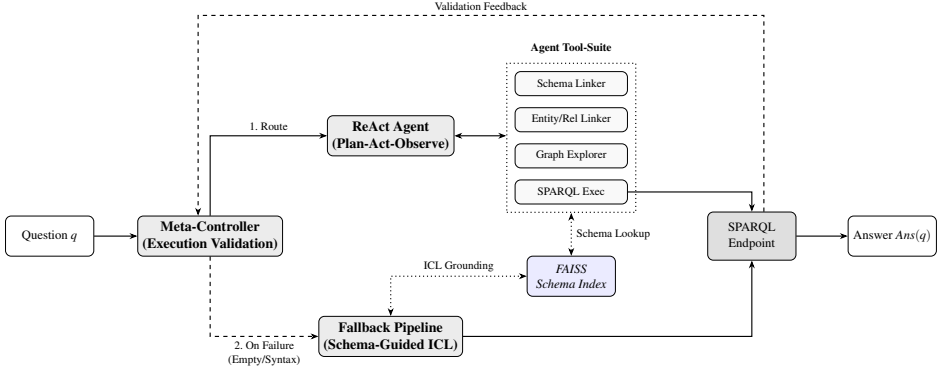


Figure 1. The HARP architecture. The Meta-Controller routes the natural language question to the ReAct agent, which utilizes a specialized tool suite for schema discovery and multi-hop exploration. If the agent’s generated query yields a syntax error or an empty result set, execution falls back to a robust, single-shot retrieval pipeline. Both reasoning paths are grounded in a dynamic FAISS schema index, enabling zero-shot portability across heterogeneous domains.

- **Schema Drift Resilience (RQ2):** We empirically quantify how structural refactoring between KG versions impacts system performance, demonstrating that a hybrid architecture can neutralize the degradations in answer quality and trustworthiness caused by schema drift.
- **LLM Capability Analysis (RQ3):** We benchmark various state-of-the-art LLM families within the same Text2SPARQL pipeline to characterize their reasoning strengths and failure modes across different schema complexities.
- **Multi-Model Orchestration (RQ4):** We introduce and evaluate ensemble strategies, such as Quorum voting, proving that combining multiple LLMs significantly increases the overall portability and reliability of the KGQA system.

To support reproducibility, an anonymized replication package containing prompt templates, evaluation scripts, and analysis artifacts is made available.³

The evaluation prioritizes the accuracy and trustworthiness of the results across four distinct datasets over runtime optimization. Figure 1 provides a high-level overview of HARP, the hybrid Text2SPARQL architecture detailed in Section 3.

The remainder of this paper is structured as follows: Section 2 reviews related work in LLM-based KGQA. Section 3 details the architecture of our hybrid meta-controller and ReAct agent. Section 4 outlines the experimental setting and complexity analysis. Section 5 presents the empirical evaluation against current baselines. Finally, Section 6 discusses the findings, and Section 7 concludes the paper and outlines future work.

2. Related Work

The state of the art in LLM-based KGQA can be organized according to the role the KG plays in the architecture, following the taxonomy of Ma et al. [19]. We review the categories most relevant to portability and schema drift, then examine the direct baselines from the Text2SPARQL Challenge [16].

³<https://github.com/WSE-research/text2sparql-agnostic>

KGs as Background Knowledge. In this category, the KG serves as a static repository that is injected either into model parameters or retrieved at inference time. Parametric approaches such as KG-Adapter [20] use parameter-efficient fine-tuning to bake KG triples directly into LLM weights, achieving strong performance without inference-time retrieval overhead. The critical limitation is rigid coupling to the KG schema: any version update or domain transfer requires costly adapter retraining [15], disqualifying parametric approaches for the schema-drift setting examined here. Retrieval-Augmented Generation (RAG) alternatives [9] keep the KG external and are in principle more adaptive; however, they typically rely on pre-computed indexes or document-to-KG alignments that re-introduce static dependencies and hinder zero-shot portability to structurally novel graphs.

KGs as Reasoning Guidelines: Agentic Exploration. A more expressive paradigm delegates KG interaction to an autonomous reasoning agent. SPINACH [21] was among the first KGQA systems to apply the ReAct prompting framework, equipping an LLM with five graph-exploration tools and discovering the schema through iterative, trial-and-error SPARQL execution rather than from a pre-indexed structure. This design handles incomplete or unfamiliar schemas effectively. However, SPINACH is tightly coupled to Wikidata’s specific API surface, incurs high token costs from extensive exploration histories, and has not been evaluated under schema drift.

Text2SPARQL Challenge Approaches. The first International Text2SPARQL Challenge at ESWC 2025 [16] defines the current state-of-the-art (SOTA) by evaluating portability across two structurally distinct KGs: DBpedia and a Corporate graph. All top submissions extend the agentic ReAct paradigm. **ARUQULA** [8] extends SPINACH with a hybrid vector and graph database toolset for faster entity linking and won the Corporate evaluation track, but requires substantial manual prompt engineering and exhibits high performance variance across graph topologies. **Graf von Data** (IIS-Q) [22] proposes a minimalist three-step agent (*Search, Describe, Query*) over a plain SPARQL endpoint, yielding transparent reasoning at the cost of topology sensitivity. **mKGQAgent** [7], our own prior work, augments the ReAct design with an offline experience pool of execution traces for dynamic few-shot retrieval; it won the overall challenge, but its advantage was driven primarily by multilingual robustness rather than English-only portability.

Research Gap. Synthesizing these observations, three persistent limitations motivate the present work. *Static dependencies:* even dynamic agent-based systems frequently overfit to static retrieval indexes or hard-coded schema assumptions, restricting portability to stable-schema settings. *Opaque model influence:* the literature lacks systematic comparisons of SOTA LLM families within the same pipeline, and multi-model ensemble strategies remain largely unexplored. *Evaluation bias:* all challenge baselines are measured on at most two KGs, one of which (using the DBpedia⁴ knowledge graph) is heavily present in LLM pre-training corpora [11], making it difficult to distinguish genuine portability from parametric recall; such memorization effects have been shown to measurably inflate the perceived SPARQL generation quality of LLMs [23], motivating memorization-resistant evaluation datasets [24]. Moreover, no prior work systematically quantifies performance under intra-KG schema drift. The present paper directly addresses all three gaps.

⁴<https://www.dbpedia.org/>

3. Approach

To address the challenges of portability and schema drift, we propose a KG-agnostic Text2SPARQL architecture that operates entirely without task-specific fine-tuning. Rather than relying on a single, monolithic generative step, the system treats Question Answering as a dynamic orchestration problem, delegating tasks between an exploratory reasoning agent and a stable retrieval pipeline. This orchestration-centric design follows the component-based QA tradition established by the Qanary methodology [25], transferring the principle of composing exchangeable QA components to the runtime routing between LLM-driven reasoning paths.

3.1. Architecture Overview

The final hybrid architecture, which we call **HARP** (Hybrid Agentic-Retrieval Pipeline), is a dual-path system designed to balance the logical depth required for complex, closed-domain reasoning with the structural precision needed for obfuscated or evolving schemas. The architecture processes a natural-language query q through a deterministic meta-controller that routes execution between a primary ReAct-based agent (A_{ReAct}) and a secondary schema-augmented retrieval pipeline as a fallback ($P_{Fallback}$).

The routing logic relies entirely on execution validation rather than predictive confidence. The final generated answer $Ans(q)$ is defined as:

$$Ans(q) = \begin{cases} Exec(A_{ReAct}(q)) & \text{if } Exec(A_{ReAct}(q)) \neq \emptyset \text{ and valid} \\ Exec(P_{Fallback}(q)) & \text{otherwise} \end{cases}$$

where $Exec()$ denotes the execution of the generated SPARQL query against the target endpoint. If the agentic reasoning path results in a syntax error or an empty set (indicating a hallucination or an over-constrained query), the meta-controller automatically delegates the task to the fallback pipeline.

3.2. Dynamic Schema Discovery

A prerequisite for cross-domain portability is the ability to interpret novel graph structures at runtime. Instead of relying on manual ontology alignment or static embeddings, the system features a dedicated Schema Discovery module.

Upon initialization with a new target KG, the module issues a fixed set of exploratory SPARQL queries that capture (i) the class hierarchy (`rdfs:subClassOf` closure over `?c a rdfs:Class`), (ii) property signatures (distinct $\langle \text{predicate}, \text{rdfs:domain}, \text{rdfs:range} \rangle$ tuples), and (iii) up to k_{ex} instance-level examples per class. Each schema element is rendered as a short natural-language description (e.g., “Class Signal; properties: hatSignaltyp, zeigtInfrastrukturelement”), embedded with `sentence-transformers/all-MiniLM-L6-v2`, and stored in an ephemeral FAISS index that is built once per target KG (offline schema discovery; no automatic refresh). During query generation, the system performs a semantic similarity search over this index and injects the top- $k=5$ schema elements most similar to the user’s question into the LLM prompt, grounding the LLM’s parametric knowledge in the factual constraints of the target

graph.⁵ Because the index is rebuilt from live SPARQL probes on every KG initialization, the module automatically absorbs schema drift without offline retraining.

3.3. The ReAct Agent Reasoner

For questions requiring multi-hop reasoning or complex aggregations, the system utilizes an explicit ReAct framework implemented via LangGraph. The agent operates within a structured Plan-Act-Observe loop, guided by a dedicated Planning Component that breaks the natural language question down into a precise list of tasks. To execute this plan, the agent is restricted to a dedicated suite of tools:

- **Named-Entity and Relation Linking Tool:** Retrieves candidate URIs, labels, and confidence scores. The tool first performs a cosine similarity search against an offline vector database containing embedded graph instances and schema elements. To prevent hallucinations, all retrieved candidates are strictly filtered via runtime SPARQL ASK queries against the live endpoint. For entities, the tool verifies their physical existence in a valid triple. For relations, it validates whether the property successfully connects already identified entities or, at a minimum, exists within the active graph topology.
- **Schema Linking Tool:** Discovers relevant classes and properties from the offline schema store. The schema injection includes both the old and the new versions of the ontology, ensuring the agent has a comprehensive structural context to navigate schema drift.
- **Explore Neighborhood Tool:** Retrieves the local graph neighborhood of a given node or edge to provide concrete, instance-level evidence for multi-hop reasoning.
- **SPARQL Execution Tool:** Allows the agent to test its generated queries against the live endpoint and receive deterministic execution feedback before committing to a final answer.

Our implementation follows the prompting strategy proposed by Yao et al. [26] where the agent is forced to emit explicit reasoning traces by outputting a “Thought:” block before any tool invocation. Validation criteria are strictly syntactic and semantic: the agent must verify via the execution tool that intermediate queries are syntactically valid, do not return empty sets, and that variables align with the schema’s domain and range constraints. This explicit trace forces the LLM to ground its logic in auditable tool outputs and prevents structural collapse.

3.4. The Schema-Guided Fallback Pipeline

While the ReAct agent excels at logical decomposition, empirical analysis reveals its vulnerability to entity-linking failures in simple lookup tasks. Entity and Relation Linking remain a primary bottleneck because natural-language surface forms often misalign with obfuscated graph URIs or literals, causing the agent to over-explore or hallucinate identifiers. To neutralize this, the fallback pipeline omits multi-step tool execution. Instead, it relies on an explicit offline schema discovery phase that extracts and embeds the target KG’s TBox (classes, properties, and structural constraints) into a vector database. At

⁵The number of instance-level examples k_{ex} retrieved per class is configurable.

runtime, a direct Entity and Relation Linking step is performed via cosine similarity search between the user’s question and this schema index. The top- k most relevant URIs and predicates, along with dynamically retrieved representative instances from the live endpoint, are injected directly into the LLM’s prompt via In-Context Learning (ICL) for a robust, single-shot query generation.

4. Experimental Setting

To evaluate the proposed approach’s portability and resilience to schema drift, we conducted a systematic empirical analysis across four heterogeneous Knowledge Graphs. The evaluation isolates the impact of cross-domain transfer, structural evolution within a single domain, and the efficacy of multi-model orchestration.

4.1. Knowledge Graphs and Benchmarks

We utilize four KG environments representing a spectrum of scale and structural complexity:

- **DBpedia (EN):** An open-domain KG used to benchmark performance against general-knowledge baselines.
- **Corporate (EN):** A closed-domain KG with high relational density but complex logical requirements, created for the TEXT2SPARQL challenge 2025.
- **TrackVideo (German: *Streckenvideo*) (v1 & v2):** A proprietary railway metadata KG. We employ two distinct snapshots to evaluate adaptivity to schema drift. Version 1 contains 24 classes, while Version 2 underwent significant refactoring, reducing the hierarchy to 13 classes and shifting signal types from an intensional (subclasses) to an extensional (instances) representation.

The test datasets comprise the standard Text2SPARQL Challenge benchmarks, DB25 (100 open-domain questions) and CK25 (50 closed-domain questions). To ensure statistical comparability and a balanced calculation of the overall mean metrics, the proprietary TrackVideo datasets were explicitly sampled to match the volume of DB25 (100 questions each). This strict alignment ensures that aggregate portability metrics are not skewed by dataset size imbalances.

4.2. Structural Complexity and Schema Drift

To contextualize performance and establish a dataset-level indicator of difficulty, we defined a structural complexity metric $C(Q)$ for a SPARQL query Q as:

$$C(Q) = |T(Q)| + |O(Q)|$$

where $T(Q)$ is the set of triple patterns and $O(Q)$ is the multiset of operators (e.g., FILTER, UNION, OPTIONAL). These thresholds were established logically based on the structural depth required for graph traversal. A query is categorized as low complexity ($C(Q) \leq 2$) if it represents a simple one-hop lookup or a basic type constraint, requiring minimal reasoning. The medium tier ($3 \leq C(Q) \leq 4$) captures two-hop relations or single-hop

queries modified by operators, necessitating intermediate entity resolution. The high-complexity threshold ($C(Q) \geq 5$) denotes multi-hop traversals, complex aggregations, or branching logic, which historically represent the primary failure modes for LLM-based reasoning agents. Under this classification, the **Corporate** dataset exhibited the highest complexity (mean $C(Q) = 13.2$), with 88% of its queries classified as high-complexity.

To baseline the difficulty of the TrackVideo v1 to v2 transfer, we operationalized metrics established in ontology matching literature: the Mapping Ratio for semantic continuity and Normalized Graph Edit Distance (GED) for structural schema drift. The Mapping Ratio quantifies the fraction of schema elements from one version that can be aligned with the other via lexical similarity ($\geq 70\%$). For classes and properties separately, it is computed as:

$$\text{Mapping Ratio} = \frac{\text{aligned elements}}{\text{total elements (union of both versions)}}$$

The overall Mapping Ratio is the average of the Class and Property ratios. The normalized GED is the minimum number of node/edge edits required to transform the v1 class hierarchy into the v2 hierarchy, divided by the size of the larger graph. The transition exhibited a low overall Mapping Ratio of **0.255** (11/37 classes, 13/61 properties aligned) and a high normalized GED of **0.750**, confirming a radical simplification of the taxonomy by 59%.

4.3. Implementation and Metrics

The pipeline was implemented in Python using the LangGraph framework. We evaluated a diverse set of LLMs, including gpt-4.1-mini, Qwen2.5-72B-Instruct, and Llama-3.3-70B-Instruct. Local models utilized AWQ or bitsandbytes quantization on dual Nvidia L40S GPUs. Each configuration was executed 5 times to estimate variance. To ensure a rigorous evaluation despite differing data availability, significance ($\alpha = 0.05$) was assessed via two distinct methods. While formal normality testing was precluded by the small sample size per configuration ($N = 5$), approximate normality was presumed given the independent, identically distributed nature of the evaluation runs. A paired two-sided t-test was employed for internal architectural comparisons where full performance distributions are available, whereas a one-sample one-sided t-test was used for external comparisons against state-of-the-art baselines that report only single aggregate scores.

In addition to the standard **F1-Score**, we report the **Answer Trustworthiness Score (ATS)** [27]. For a test set of N questions, ATS is the mean per-question score $s_i \in \{+1, 0, -1\}$, assigning $+1$ to a fully correct answer ($F1_i = 1$), 0 to an abstention (the system returns no result), and -1 to a wrong execution ($F1_i = 0$ with non-empty output): $ATS = \frac{1}{N} \sum_{i=1}^N s_i \in [-1, +1]$. ATS, therefore, penalizes confidently wrong answers more strongly than F1 alone, which is critical for industrial QA deployments where misleading outputs can be worse than no answer. **Mean F1** and **Mean ATS** across all four KGs serve as the primary indicators of system portability.

5. Evaluation

We evaluate HARP against established state-of-the-art (SOTA) baselines from the Text2SPARQL Challenge. Our analysis focuses on performance across heterogeneous

Table 1. Architectural progression: mean F1 across the four KGs for the same `gpt-4.1-mini` backbone. v0.6 omitted (no `gpt-4.1-mini` run). The cross-KG standard deviation collapses from ± 14.6 in the static-prompting baseline to ± 2.0 in HARP.

Version	Mean F1 (%)	Architectural change
v0.1	17.1 $\pm$ 14.6	Static prompting baseline
v0.2	27.4 $\pm$ 10.2	+ few-shot exemplars
v0.3	31.1 $\pm$ 14.8	+ dynamic schema discovery
v0.4	26.9 $\pm$ 5.9	+ agent loop
v0.5	30.6 $\pm$ 16.1	+ tool refinement
v0.7	39.1 $\pm$ 6.1	+ ReAct agent
HARP (v0.8)	47.6$\pm$2.0	+ schema-guided fallback (hybrid)

Table 2. Corporate Dataset: Comparison of F1 and ATS. **Bold** marks the best value per column. F1 of our system is reported as mean $\pm$ std over 5 runs.

System	F1-Score (%)	ATS
ARUQULA [8]	43.6	-0.540
mKGQAgent [7]	32.0	-0.120
IIS-Q [22]	44.4	-0.160
HARP	44.3 $\pm$ 2.2	+0.052

Table 3. DBpedia Dataset: Comparison of F1 and ATS. **Bold** marks the best value per column. F1 of our system is reported as mean $\pm$ std over 5 runs.

System	F1-Score (%)	ATS
ARUQULA [8]	54.6	-0.410
mKGQAgent [7]	52.2	+0.030
IIS-Q [22]	37.7	-0.050
HARP	49.7 $\pm$ 2.9	-0.092

Knowledge Graphs and the resilience of the architecture to structural variation and schema evolution.

5.1. Architectural Progression (Ablation)

Before contextualizing HARP against external baselines, Table 1 reports the F1-score progression of the same `gpt-4.1-mini` backbone across the seven architectural milestones that lead to HARP, averaged over the four KGs. The progression captures the architectural lessons: introducing the schema-discovery service (v0.3) doubles F1 over the static-prompting baseline; adding the ReAct agent (v0.7) provides a further large gain; the final hybrid fallback (HARP) both increases mean F1 and reduces the cross-KG standard deviation by an order of magnitude, directly evidencing the portability gain.

5.2. Comparative Analysis against SOTA

The following tables summarize the F1-Score and Answer Trustworthiness Score of our system compared to leading benchmarks: ARUQULA (ReAct-based), mKGQAgent (multi-agent), and IIS-Q (proprietary optimization).

Table 4. Mean Portability and Stability Comparison Across All Datasets. Metrics marked with an asterisk (*) are calculated based only on the DBpedia and Corporate benchmarks, as the baseline systems were not evaluated on the TrackVideo datasets.

System	DBpedia	Corporate	TV v1	TV v2	Mean F1	Std.	Mean ATS
ARUQULA [8]	54.6	43.6	N/A	N/A	49.1*	7.78*	-0.475*
mKGQAgent [7]	52.2	32.0	N/A	N/A	42.1*	14.28*	-0.045*
IIS-Q [22]	37.7	44.4	N/A	N/A	41.1*	4.74*	-0.105*
HARP	49.7±2.9	44.3±2.2	48.0±2.4	48.6±4.8	47.6	2.33	+0.023

5.3. Mean Portability Analysis

To assess global portability, Table 4 details the mean performance and standard deviation across all tested domains. A low standard deviation indicates high adaptability, as the system maintains consistent quality regardless of the underlying schema.

5.4. Analysis of Results

As demonstrated in Table 2 and Table 3, our system achieves functional parity with highly specialized challenge winners. While baseline architectures often overfit to either open-domain retrieval (e.g., DBpedia) or closed-domain reasoning (e.g., Corporate), HARP remains competitive in both environments. However, the most significant finding is shown in Table 4: our hybrid architecture achieves the lowest **Standard Deviation (2.33)** among all systems. This indicates superior adaptability; while other systems exhibit significant performance swings between open and closed domains, our approach provides a stabilized reasoning core that is less sensitive to KG-specific characteristics.

Furthermore, the system demonstrates remarkable resilience to schema drift. Despite the structural refactoring between TrackVideo v1 and v2, the performance remained almost identical (48.0% vs 48.6%), confirming that the hybrid orchestration can bridge significant version gaps without manual intervention.

5.5. Efficacy of Dynamic Orchestration

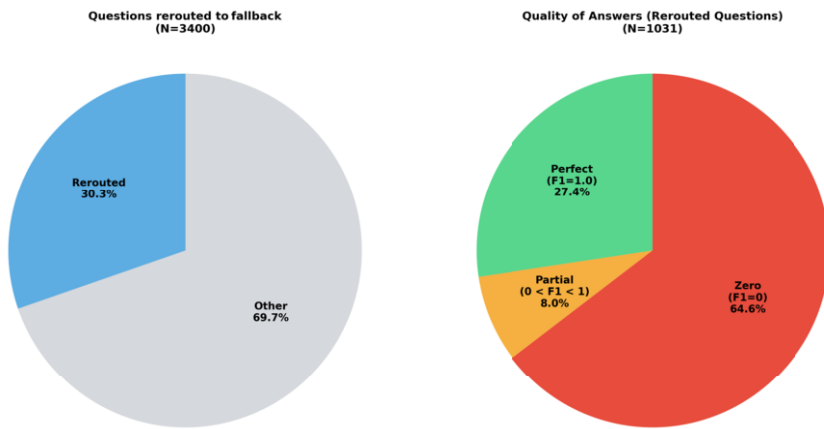
Crucially, the performance recovery on the structurally simpler datasets was achieved without sacrificing the deep reasoning capabilities required for complex ontologies. On the Corporate dataset, the hybrid system increased its F1-score to 44.3% (up from 33.1% in the previous agent-only architecture). This validates the dynamic routing logic: the system successfully deploys ReAct-based reasoning for complex questions and seamlessly shifts to the retrieval fallback when agentic logic fails.

To quantify the impact of this orchestration, we tracked the routing behavior across all evaluation runs. As illustrated in Figure 2, the meta-controller detected agent failures (e.g., empty result sets or syntax errors) and rerouted 30.3% of the queries to the secondary fallback pipeline. Of these rerouted queries, 24.7% subsequently achieved a perfect F1-score. This demonstrates that dynamic orchestration actively recovers queries that would otherwise result in complete failure, directly contributing to the system’s superior mean performance and overall reliability.

Table 5. Model Performance Overview v0.7

Model	Datasets	Mean F1 (%)	Mean ATS
gpt-4.1-mini	4	39.08 $\pm$ 6.06	-0.095 $\pm$ 0.011
Qwen2.5-72B-Instruct	4	27.05 $\pm$ 10.57	-0.445 $\pm$ 0.113
Llama-3.3-70B-Instruct	3	13.30 $\pm$ 1.85	-0.220 $\pm$ 0.284

Qualitative example. A representative rerouted query from TrackVideo v2 is “Welche Infrastrukturaufnahmen zeigen BKVSIG?” (English translation: “Which infrastructure recordings show BKVSIG?”). The ReAct agent failed to recover the intended property pattern and produced an empty result set; the meta-controller then routed the query to the fallback, whose schema-guided ICL retrieved the correct `stk25:hatSignaltyp` (with value `stk25:signaltyp-BKVSIG`) and `stk25:zeigtInfrastrukturelement` properties and produced the gold-standard SPARQL.


Figure 2. Rerouting frequency and answer quality for HARP.

5.6. LLM Capability Analysis and Multi-Model Orchestration

To understand the reasoning capabilities across different LLM families and address RQ3, we evaluated the models on the agentic reasoning core (version v0.7) rather than the final hybrid architecture (HARP). Isolating the ReAct agent in v0.7 prevents the robust fallback pipeline from masking the LLMs’ genuine reasoning and tool-use behaviors. This isolation is crucial to accurately observe their unmitigated failure modes and strengths. As shown in Table 5, while the proprietary gpt-4.1-mini achieved the highest mean F1-score (39.08%), the open-weight Qwen2.5-72B-Instruct exhibited a massive standard deviation (± 10.57).

This high variance confirms that open-weight models possess highly dataset-dependent, orthogonal reasoning capabilities. Instead of a single model dominating universally, models fail and succeed on completely different structural challenges. Because no single LLM perfectly generalizes across all domains without severe performance swings, we evaluated a three-model **Quorum Approach** for the final HARP pipeline. This ensemble runs the full hybrid pipeline under gpt-4.1-mini, gemini-2.5-flash, and Qwen2.5-72B-Instruct, selecting the final SPARQL output by majority vote over

Table 6. Multi-Model Performance Gains (HARP, Quorum over Top-3 models).

Dataset	Single Best (F1)	Quorum Ensemble (F1)	Δ
DBpedia	48.2%	49.6%	+1.4%
Corporate	37.8%	40.2%	+2.4%
TrackVideo v1	34.1%	41.1%	+7.0%
TrackVideo v2	45.5%	47.2%	+1.7%

execution result sets. Filtering and selecting among SPARQL query candidates has previously been shown to substantially improve KGQA quality [28,29]; the Quorum approach extends this principle from validating the candidates of a single model to consensus voting across heterogeneous LLM families.

On ties, the system falls back to the expert model, defined as the configuration with the highest historical F1 on the target dataset (for example, `gemini-2.5-flash` on TrackVideo v2). The objective is to exploit complementary error profiles across LLM families rather than to replace single-model deployment. The “Single Best (F1)” column reports the performance of the highest-scoring individual LLM for each respective dataset when executing the complete HARP pipeline independently. This establishes a strict single-model baseline to measure the empirical advantage (Δ) gained purely by the Quorum ensemble’s majority voting mechanism.

As detailed in Table 6, this strategy consistently improved accuracy. Most notably, it achieved a **7.0% absolute gain** on the TrackVideo v1 dataset. This confirms that majority voting over execution results is an effective mechanism for filtering stochastic reasoning errors and further enhancing the reliability of portable KGQA systems (RQ4).

6. Discussion

6.1. Hybrid Architecture and Schema Drift Resilience

The iterative development history reveals a clear architectural lesson. The static context approach (v0.3) exhibited a significant performance collapse when applied to the evolved TrackVideo v2 schema, despite receiving the updated schema definitions in its prompt. The model failed to bridge the semantic gap introduced by the structural refactoring — specifically, the shift from class-based to instance-based signal-type modeling — confirming that the mere availability of a correct schema is insufficient for robustness. In contrast, HARP (v0.8) maintained statistical parity across the original (48.0%) and evolved (48.6%) graph versions. Robustness was achieved not by memorizing the schema but by decoupling query generation from specific structural assumptions: the agentic layer dynamically inspects the schema at runtime, absorbing the complexity shift that accompanies schema drift.

6.2. Heterogeneous Error Profiles across LLM Families

Multi-model experiments reveal that no single LLM dominates across all evaluation settings. Models with broad pre-training exposure (`gpt-4.1-mini`) consistently outperformed others on open-domain DBpedia, suggesting that parametric memorization rather than zero-shot reasoning drives performance on popular graphs. Smaller generalist models that handled simple lookups adequately failed completely when tasked with maintaining

complex agent state across multi-step plans. The error profiles of different LLM families are therefore largely orthogonal, which makes quorum consensus not merely an optimization but a reliability requirement: majority voting over execution results systematically filters stochastic reasoning errors, yielding the observed **+7.0%** absolute gain on TrackVideo v1.

6.3. Contextualization against State of the Art

The hybrid system achieved a mean F1 of 47.6% across all four KGs, marginally below ARUQULA ($\approx 49\%$ on the two shared benchmarks). This delta requires careful interpretation. Both DBpedia and the Corporate KG are benchmarks that challenge participants explicitly optimize for; DBpedia performance in particular is likely inflated by parametric memorization [11,23]. The critical differentiator of our evaluation is the inclusion of the TrackVideo datasets, which are fully absent from any LLM pre-training corpus and introduce a qualitatively distinct structural challenge. Our system’s stable performance ($\approx 48\%$) on these unseen, narrow-domain graphs establishes portability as an empirically verified property across four structurally and semantically distinct KGs, rather than a two-graph extrapolation.

6.4. Threats to Validity

Dataset limitations. Using 50–100 questions per dataset yields high statistical variance; a single “lucky” run can shift aggregate metrics by several percentage points. The Corporate KG is a structural outlier (mean $C(Q) = 13.2$) that creates a bimodal evaluation landscape. Ground-truth inconsistencies introduce a ceiling effect: a semantically correct SPARQL query can receive a zero F1 score because the gold standard targets a different entity identifier.

Methodological limitations. Failure analysis across heterogeneous KGs is non-trivial because each graph employs distinct naming conventions and modeling patterns (e.g., reification vs. direct properties), preventing a unified causal diagnosis. Component-level accuracy measures extracted from generated SPARQL are proxy measures that may penalize valid alternative query paths.

Technical constraints. The reliance on commercial APIs introduces a reproducibility risk. OpenAI model versions were pinned (gpt-4.1-mini-2025-04-14), but the OpenRouter aggregator does not support precise snapshot pinning; its usage was restricted to the comparative LLM analysis and excluded from primary evaluation results. The multi-model evaluation covered a representative model subset rather than a comprehensive grid search, due to the significant runtime cost of agentic loops.

6.5. Implications

The findings motivate a shift in KGQA system design philosophy toward the **HARP** paradigm—combining the reasoning depth of a ReAct agent with the precision of a schema-augmented retrieval pipeline and dynamically selecting the appropriate path at runtime via execution validation. For practitioners deploying LLM-based QA on enterprise Knowledge Graphs, three practical guidelines follow:

1. **Avoid static prompting** for evolving data: RAG-based and fine-tuned systems fail silently when the schema drifts; agentic schema exploration is required for long-term maintenance.
2. **Implement voting mechanisms**: running three complementary models and taking the majority vote is often more reliable than relying on a single large model, especially for avoiding hallucinations in closed domains.
3. **Hybridize**: deploy a fast retrieval lane for simple lookups and a slow agentic lane for complex analytical queries to balance cost, latency, and accuracy.

7. Conclusions & Future Work

This research demonstrates that achieving high-quality KGQA across heterogeneous domains is possible without the need for computationally expensive task-specific fine-tuning. By shifting the focus from static, dataset-specific optimizations to a modular, agentic architecture, this study addresses the critical challenges of portability and schema drift, providing direct answers to the four research questions raised in Section 1.

Hybrid architectures are essential for robustness (RQ1). Relying solely on agentic reasoning or static retrieval is insufficient for universal KGQA. By combining the logical depth of a ReAct agent with the stability of a schema-augmented retrieval fallback, HARP dynamically balances these paradigms. This hybrid orchestration yields the most reliable performance across diverse graph topologies, achieving the lowest cross-domain F1 standard deviation among all evaluated state-of-the-art systems.

Neutralizing schema drift (RQ2). Static models fail silently when ontologies evolve. However, HARP maintained stable performance (approximately 48% F1) across both the original and radically refactored TrackVideo graphs (v1 and v2). This confirms that decoupling query generation from static structural assumptions and relying on runtime schema exploration effectively neutralizes drift-induced degradation.

No single LLM dominates (RQ3). As detailed in Table 5, component-level analysis reveals that error profiles across LLM families are highly orthogonal. While the proprietary gpt-4.1-mini achieved the highest mean F1-score across domains, open-weight models like Qwen2.5-72B exhibited massive performance variance (± 10.57), indicating they fail and succeed on entirely different structural challenges rather than universally trailing behind. Furthermore, Entity and Relation Linking remain the primary bottlenecks. In “digit-heavy,” low-entropy datasets like TrackVideo, the agent’s tendency to over-reason frequently leads to precision loss, proving that advanced reasoning capabilities can become a liability in straightforward lookup scenarios.

Multi-model orchestration improves trustworthiness (RQ4). Because different LLMs fail in different ways, ensemble strategies are highly effective. Quorum voting over a heterogeneous LLM ensemble produced consistent absolute F1 gains (up to +7.0% on TrackVideo v1). This demonstrates that majority voting based on execution results is a robust mechanism for filtering stochastic reasoning errors and preventing hallucinations.

Taken together, these findings confirm that a Knowledge Graph-agnostic approach—utilizing dynamic schema discovery, agentic reasoning, and runtime exploration—can

achieve performance parity with state-of-the-art baselines while remaining adaptable to entirely new domains, without retraining.

Future research should focus on the dynamic orchestration of multiple models, where a meta-controller intelligently assigns sub-tasks to specialized LLMs based on their observed strengths. Additionally, improving the robustness of linking mechanisms for noisy or obfuscated schemas will be vital for further enhancing the portability of KGQA systems in industrial applications.

Generative AI Disclosure. In accordance with the SEMANTiCS 2026 policy on the fair use of Large Language Models, we disclose the following uses of generative AI in this work. The Text2SPARQL system reported in this paper is itself an LLM-based system; the language models exercised in the experiments (gpt-4.1-mini-2025-04-14, gemini-2.5-flash, Qwen2.5-72B-Instruct-AWQ, and Llama-3.3-70B-Instruct) are documented in Section 4 and produce the system outputs measured in Section 5.

In manuscript preparation. The authors used different versions of Anthropic Claude and OpenAI GPT to assist with text structuring, rephrasing, and stylistic editing. Generative AI was not used to design experiments, fabricate or interpret results, select citations, or make authorship-level decisions. All experimental data, statistical claims, and citations were verified against the underlying logs, source code, and primary literature; the authors take full responsibility for the accuracy and integrity of the final manuscript.

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Neurosymbolic AI, Reasoning & Explainability

Explaining RGNN Predictions on Knowledge Graphs via LRP

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Abstract. Relational Graph Neural Networks (RGNNs) have become powerful tools for learning on relational graph structures such as RDF knowledge graphs (KGs), achieving state-of-the-art performance in tasks such as node classification and link prediction. However, their black-box nature conflicts with a central objective of KGs: enabling transparent and interpretable knowledge-based systems. As a result, predictions produced by RGNNs are often difficult to explain in terms of the graph structure itself. In this work, we present a generalized framework for explaining RGNN predictions using Layer-Wise Relevance Propagation (LRP). The approach traces predictions through the relational message-passing process and re-distributes signed relevance scores to nodes, edges, and multi-hop paths. In doing so, it establishes a semantic predictive loop in which predictions derived from the KG structure can themselves be explained with respect to the underlying computation graph. Beyond extending relevance propagation to relational message passing, the framework unifies structure-oriented and embedding-oriented explanations by linking graph components to embedding-level contributions. Experiments on standard KG learning benchmarks demonstrate that the proposed method yields faithful and interpretable explanations of RGNN predictions across structural levels.

Keywords. Knowledge Graphs, Graph Neural Networks, Explainable AI

1. Introduction

Knowledge Graphs (KGs) have become central to modern data management and reasoning, providing semantically rich representations of entities and their relationships. They are widely used in domains such as search engines, biomedical research, and industrial applications [1–4]. Typically represented as Resource Description Framework (RDF) [5] graphs with labeled directed edges and supported by ontologies as domain conceptualizations, KGs combine machine-readable structure with human-interpretable semantics. This links symbolic knowledge representation with data-driven analysis [6].

Beyond supporting standardized knowledge representation, semantic integration, interoperability, and logical reasoning across heterogeneous data sources [7], KGs increasingly serve as graph-structured inputs for downstream machine learning (ML) tasks [8]. Examples include link prediction and node classification, where models

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generate predictions over relational graph structure. Building on Graph Neural Networks (GNNs) [9–11], Relational GNNs (RGNNs) exploit local KG topology through multi-hop patterns in the graph, achieving state-of-the-art performance on benchmark tasks [12, 13]. However, this integration introduces a conceptual tension: while KGs were designed to support transparent and interpretable reasoning over structured knowledge, RGNNs act as black-box predictors whose internal decision processes remain difficult to explain. Consequently, predictions derived from semantically structured KGs may lose the transparency they were intended to provide. Moreover, this black-box issue is not only of theoretical interest but also increasingly relevant in real-world deployments. For instance, KGs are widely used in industrial decision-support systems [3, 14], where ML models operating on KG data can influence decisions with operational or societal impact. Recent regulatory initiatives and AI governance frameworks emphasize transparency and explainability requirements for ML systems [15, 16]. Ensuring that predictions derived from KG-based learning systems remain explainable is therefore important not only for preserving semantic interpretability but also for supporting accountability and human oversight in practical deployments. Addressing this challenge requires explaining predictions directly in terms of the underlying KG structure. Ideally, predictions produced by KG-based learning models should be traceable to the entities and edges that contributed to them. In this work, we refer to this mechanism as a *semantic predictive loop*: KG structure serves as input to relational graph learning models and the resulting predictions can be explained with respect to the same KG, thereby preserving semantic interpretability throughout prediction. To realize this loop, we employ Layer-Wise Relevance Propagation (LRP), which traces prediction scores backward through the RGNN computation graph and redistributes signed relevance to nodes, edges, and multi-hop paths of the input KG, as illustrated in Figure 1.

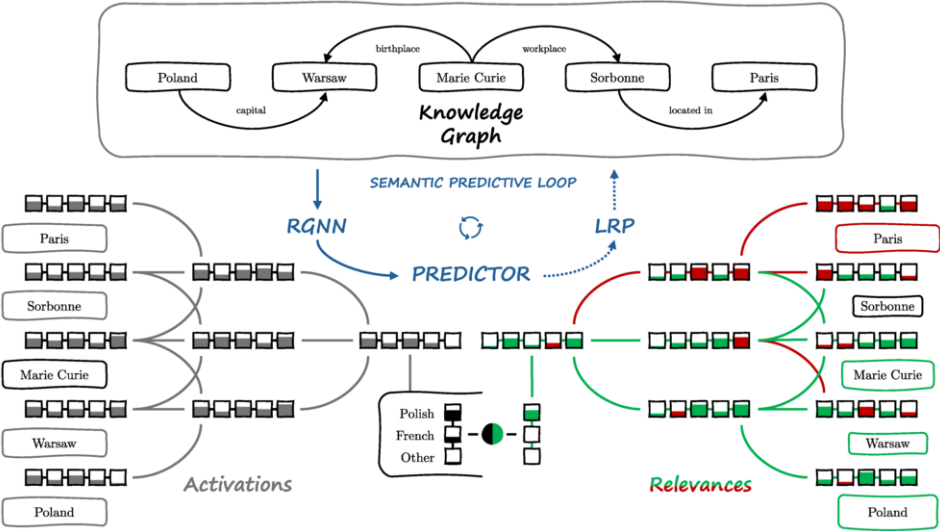


Figure 1. Semantic predictive loop: an RGNN predicts the nationality of Marie Curie from a relational KG, while LRP traces the prediction back through relational message passing to assign signed relevances to nodes, edges, and embedding entries (green: supportive, red: opposing).

2. Preliminaries & Definitions

2.1. Graphs

Graphs provide a general abstraction for representing structured information about vertices (also called nodes) via edges between them [17].

Definition 2.1. Let $\mathcal{V}$ be a non-empty set of vertices and let $\mathcal{M}$ be a non-empty set of labels representing edge metadata. A graph over $(\mathcal{V}, \mathcal{M})$ is a subset

$$\mathcal{G} \subseteq \mathcal{V} \times \mathcal{V} \times \mathcal{M},$$

where $(v, u, m) \in \mathcal{G}$ denotes an edge from $v \in \mathcal{V}$ to $u \in \mathcal{V}$ with label $m \in \mathcal{M}$.

A graph $\mathcal{G}$ is *undirected* if $(v, u, m) \in \mathcal{G}$ implies $(u, v, m) \in \mathcal{G}$, and *directed* otherwise. It is *homogeneous* if $\mathcal{M} = \{*\}$ is a singleton set and *heterogeneous* otherwise. If $\mathcal{R}$ is a finite non-empty set of relation types, then a graph $\mathcal{G} \subseteq \mathcal{V} \times \mathcal{V} \times \mathcal{R}$ is called a *relational graph*. The distinction between generalized and relational graphs is primarily semantic. In relational graphs, edge labels represent relation types and are therefore not permutation-invariant, which is required for the semantic predictive loop proposed in this work. KGs form a special class of relational graphs describing entities and their relationships within a shared conceptualization [6, 8, 18]. In practice, KGs are commonly represented using RDF [19, 20], where facts are modeled as subject–predicate–object triples. RDF Schema (RDFS) [21] and OWL [22] extend RDF with semantic conceptualizations via classes and predicates, i.e., relation types. In graph terms, RDF KGs additionally represent literal attributes as vertices and their corresponding predicates as edge labels. Ignoring literals yields a purely relational RDF KG corresponding to the graph structures processed by relational graph neural models [23].

2.2. Graph Neural Networks

Neural networks (NNs) transform numerical input representations into output representations via compositions of activated affine transformation layers [24, 25].

Definition 2.2 (Dense Layer). Let $d', d \in \mathbb{N}$ denote input and output dimensions. A dense layer $f : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$ with $y = f(x) = \sigma(W \cdot x + b)$ is defined by a weight matrix $W \in \mathbb{R}^{d \times d'}$, a bias vector $b \in \mathbb{R}^d$, and a non-linear activation $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$.

Dense layers are typically optimized with respect to downstream tasks such as classification or regression using gradient-based learning [26]. They form the fundamental building blocks of NNs through compositions of such layers. For example, a multi-layer perceptron (MLP) with $L \in \mathbb{N}$ dense layers f_ℓ is given by

$$F : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d, \quad y = F(x) = (f_L \circ \dots \circ f_1)(x).$$

The layers are evaluated iteratively via $y_\ell = f_\ell(y_{\ell-1}) = \sigma_\ell(W_\ell \cdot y_{\ell-1} + b_\ell)$, where $y_0 := x$, $y_L := y$, and $d'_\ell = d_{\ell-1}$ with $d'_0 := d'$ and $d_L := d$. More complex NN architectures follow the same principle of learning parameters in affine transformations but differ in how these transformations are structured and composed. For instance, Graph Neural Networks (GNNs) aggregate graph context through the edges adjacent to a node, inducing its incident stubs and the node's neighborhood.

Definition 2.3. Let $\mathcal{I} = \{\iota_{\text{out}}, \iota_{\text{in}}, \iota_{\text{id}}\}$ denote the set of edge incidences (outgoing, incoming, and identity for self-loops) and let $\mathcal{G} \subseteq \mathcal{V} \times \mathcal{V} \times \mathcal{M}$ be a graph. If $\mathcal{G}$ is directed, the incidence expansion $\mathcal{G}^+ \subseteq \mathcal{V} \times \mathcal{V} \times \mathcal{M} \times \mathcal{I}$ of $\mathcal{G}$ is defined as

$$\mathcal{G}^+ := \bigcup_{(v,u,m) \in \mathcal{G}} \begin{cases} \{(v,u,m,\iota_{\text{out}}), (u,v,m,\iota_{\text{in}})\}, & v \neq u, \\ \{(v,v,m,\iota_{\text{id}})\}, & v = u. \end{cases}$$

A directed edge between distinct vertices $v \neq u$ induces two entries in $\mathcal{G}^+$, while a self-loop induces a single entry. If $\mathcal{G}$ is homogeneous, then $\mathcal{M} = \{*\}$ holds. Analogously, if $\mathcal{G}$ is undirected, let $\mathcal{I} = \{*\}$, which simplifies $\mathcal{G}^+$ to

$$\mathcal{G}^+ := \{(v,u,m,*) \mid (v,u,m) \in \mathcal{G}\}.$$

For $v \in \mathcal{V}$, tuples (u,m,ι) with $(v,u,m,\iota) \in \mathcal{G}^+$ are the stubs incident to it, i.e.,

$$\mathcal{S}_{\mathcal{G}}(v) := \{(u,m,\iota) \mid (v,u,m,\iota) \in \mathcal{G}^+\} \subseteq \mathcal{V} \times \mathcal{M} \times \mathcal{I}.$$

Given $\mathcal{S}_{\mathcal{G}}(v)$, we define the neighborhood of $v \in \mathcal{V}$ as the vertices adjacent to it:

$$\mathcal{N}_{\mathcal{G}}(v) := \{u \mid (u,m,\iota) \in \mathcal{S}_{\mathcal{G}}(v)\} \subseteq \mathcal{V}.$$

Accordingly, the stubs of a node reduce to its neighborhood in homogeneous and undirected graphs, which are considered by traditional GNN methods [10, 11, 23]. Given a node $v \in \mathcal{V}$ and its neighborhood $\mathcal{N}_{\mathcal{G}}(v)$, these methods consider a base embedding $\psi : \mathcal{N}_{\mathcal{G}}(v) \cup \{v\} \rightarrow \mathbb{R}^{d'}$ to produce an aggregated representation of v . Given a non-linear activation σ , traditional GNN layers can be written as

$$\phi(v) = \sigma(F_{\text{id}}(\psi(v)) + \sum_{u \in \mathcal{N}_{\mathcal{G}}(v)} c(u,v) \cdot F(\psi(u))),$$

where NNs $F_{\text{id}}, F : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$ compute self-messages and intermediate stub embeddings, weighted by coefficients $c : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$. Graph Convolutional Networks (GCNs) [10] consider degree-normalized coefficients and realize intermediate stub embeddings through shared linear transformations of adjacent embeddings:

$$F(\psi(u)) = W \cdot \psi(u) + \beta \quad \text{with} \quad W \in \mathbb{R}^{d \times d'}, \beta \in \mathbb{R}^d.$$

Graph Attention Networks (GATs) [11] consider equivalent stub embeddings but learn attention coefficients for each neighbor from the base embeddings:

$$c(u,v) = \frac{\exp(a(\psi(u), \psi(v)))}{\sum_{u' \in \mathcal{N}_{\mathcal{G}}(v)} \exp(a(\psi(u'), \psi(v)))}.$$

Relational GNN (RGNN) approaches such as RGCN [12], RGAT [13], as well as CompGCN [27] extend traditional GNNs by incorporating relation types via

$$\phi(v) = \sigma\left(\sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{N}_{\mathcal{G},r}(v)} c_r(u,v) \cdot F_r(\psi(u))\right),$$

with relation-specific neighborhoods, coefficients, and NN mappings such as

$$F_r(\psi(u)) = \sigma_r(W_r \cdot \psi(u) + \beta_r), \quad r \in \mathcal{R}.$$

Instead of incorporating explicit self-messages, RGNNs consider the graph closure $\overline{\mathcal{G}} := \mathcal{G} \cup \{(v, v, r_{\text{id}}) \mid v \in \mathcal{V}\}$ with $\overline{\mathcal{R}} := \mathcal{R} \cup \{r_{\text{id}}\}$. Self-messages are thus incorporated through the self-relation r_{id} . While applicable to heterogeneous graphs, RGNN formulations typically assume undirected graphs. In practice, however, many relational graphs are directed and distinguish both edge labels and edge directions. To formally specify how these characteristics are incorporated into message passing, we provide a generalized definition of a GNN layer that covers homogeneous as well as heterogeneous graphs while explicitly accounting for directedness of edges.

Definition 2.4. *Given a graph $\mathcal{G} \subseteq \mathcal{V} \times \mathcal{V} \times \mathcal{M}$ and a node subset $\mathcal{X} \subseteq \mathcal{V}$, we define*

$$\mathcal{N}_{\mathcal{G}}(\mathcal{X}) := \bigcup_{v \in \mathcal{X}} \mathcal{N}_{\mathcal{G}}(v)$$

as the set of neighbors of nodes in $\mathcal{X}$. A generalized GNN layer specifies a mapping

$$\kappa : (\mathbb{R}^{d'})^{\mathcal{N}_{\mathcal{G}}(\mathcal{X})} \rightarrow (\mathbb{R}^d)^{\mathcal{X}}, \quad \kappa(\psi)(v) = \sigma \left(\sum_{(u, m, \iota) \in \mathcal{S}_{\mathcal{G}}(v)} c_{m, \iota}(u, v) \cdot F_{m, \iota}(\psi(u)) \right),$$

where $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is a non-linear activation function. Moreover, $c_{m, \iota} : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}$ with $F_{m, \iota} : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$ are label- and incidence-specific coefficient functions and neural transformations, respectively.

Given a base embedding $\psi \in (\mathbb{R}^{d'})^{\mathcal{N}_{\mathcal{G}}(\mathcal{X})}$, GNN layers can be regarded as constructive graph embedding models since message passing incorporates structural information of $\mathcal{G}$ into embeddings of $\mathcal{X}$. In practice, GNN layers are stacked sequentially, such that a GNN is a composition of GNN layers

$$\mathcal{K} : (\mathbb{R}^{d'})^{\mathcal{N}_{\mathcal{G}}^L(\mathcal{X})} \rightarrow (\mathbb{R}^d)^{\mathcal{X}}, \quad \phi = \mathcal{K}(\psi) = (\kappa_L \circ \dots \circ \kappa_1)(\psi),$$

where $\mathcal{N}_{\mathcal{G}}^n(\mathcal{X})$ denotes the n -hop neighborhood of $\mathcal{X}$ for $n \in \mathbb{N}$, specified as

$$\mathcal{N}_{\mathcal{G}}^n(\mathcal{X}) := \bigcup_{v \in \mathcal{X}} \mathcal{N}_{\mathcal{G}}^n(v) \quad \text{with} \quad \mathcal{N}_{\mathcal{G}}^n(v) = \bigcup_{u \in \mathcal{N}_{\mathcal{G}}^{n-1}(v)} \mathcal{N}_{\mathcal{G}}(u), \quad \mathcal{N}_{\mathcal{G}}^0(v) := \{v\}.$$

The L -hop neighborhood is required since L -layer RGNNs perform L aggregation steps. Analogous to MLPs, these compositions can be decomposed into layers

$$\kappa_{\ell} : (\mathbb{R}^{d'_{\ell}})^{\mathcal{N}_{\mathcal{G}}^{n_{\ell}}(\mathcal{X})} \rightarrow (\mathbb{R}^{d_{\ell}})^{\mathcal{N}_{\mathcal{G}}^{n_{\ell}-1}(\mathcal{X})}$$

with $n_{\ell} = L - \ell + 1$. For instance, for the first GNN layer κ_1 , we obtain $n_1 = L$, and for the final GNN layer κ_L , we obtain $n_L = 1$.

Analogous to MLPs, stacked GNN architectures can be decomposed via $\phi_{\ell} = \kappa_{\ell}(\phi_{\ell-1})$, i.e., $d'_{\ell} = d_{\ell-1}$ with $d'_0 := d'$ and $d_L = d$, $\psi_{\ell} = \phi_{\ell-1}$ with $\phi_0 := \psi$ and $\phi_L = \phi$. In this work, we assume sequential GNNs and node-specific outputs $\phi(v)$ used in downstream ML tasks such as node classification or link prediction. Moreover, we consider $\mathcal{M} = \mathcal{R}$ and apply LRP to RGNN aggregations to propagate predictive relevance scores to graph structures through the computation graph of an RGNN model.

2.3. Layer-Wise Relevance Propagation

Derived from deep Taylor decompositions, LRP explains NN forward passes by conserving output relevance and redistributing it to input features [28, 29]. Signed input relevances indicate which features increase or decrease the output.

Definition 2.5. Let $f : \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$ be a dense layer with component-wise monotonic activation and output $y = f(x)$. Furthermore, let $\hat{y} \in \mathbb{R}^d$ be a relevance vector, where $\hat{y}_j$ quantifies the relevance of the output neuron y_j . LRP computes relevance vectors $\hat{x} \in \mathbb{R}^{d'}$, $\hat{b} \in \mathbb{R}^d$ for the dense layer's input and bias, respectively:

$$\hat{x}_i = \sum_{j=1}^d \frac{W_{j,i} \cdot x_i}{b_j + \sum_{k=1}^{d'} W_{j,k} \cdot x_k} \hat{y}_j, \quad \hat{b}_j = \frac{b_j}{b_j + \sum_{k=1}^{d'} W_{j,k} \cdot x_k} \hat{y}_j.$$

This redistribution preserves total relevance: $\sum_{i=1}^{d'} \hat{x}_i + \sum_{j=1}^d \hat{b}_j = \sum_{j=1}^d \hat{y}_j$.

Accordingly, relevance is redistributed in proportion to each input neuron's contribution to each output neuron. The activation $\sigma : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is called component-wise monotonic if the following holds for any $\tau, \tau' \in \mathbb{R}^d$:

$$\tau_j \geq \tau'_j \text{ and } \tau'_k = \tau_k \text{ for all } k \neq j \implies \sigma(\tau)_j \geq \sigma(\tau')_j.$$

This property allows relevance propagation at the pre-activation level $W \cdot x + b$, such that σ can be treated as the identity during propagation, as illustrated in Figure 2. Standard activation functions such as ReLU, sigmoid, and softmax satisfy this property. In this work, we use the original LRP formulation, although stabilized variants exist to avoid near-zero denominators [29]. While we did not observe numerical issues, such extensions may be required in some applications, typically at the cost of exact relevance conservation.

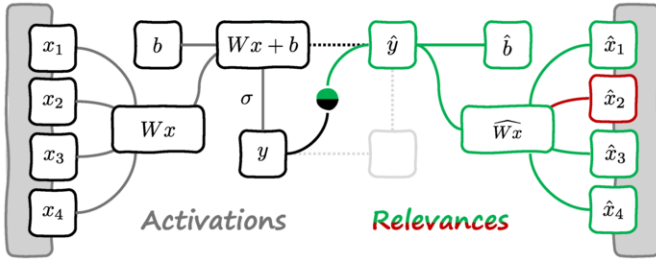


Figure 2. Forward pass and relevance propagation for an output $y \in \mathbb{R}$ in a dense layer. The forward pass computes $y = \sigma(Wx + b)$, while LRP redistributes the output relevance $\hat{y} \in \mathbb{R}$ back to the pre-activation contributions, yielding signed relevance scores in $\hat{x} \in \mathbb{R}^4$ and $\hat{b} \in \mathbb{R}$ (green: supportive, red: opposing).

To apply LRP, an output index $j^* \in \{1, \dots, d\}$ is selected, e.g., corresponding to a predicted class, and the initial relevance is set to $\hat{y}_{j^*} = y_{j^*}$ with $\hat{y}_j = 0$ for $j \neq j^*$. Relevance is then propagated backward according to Definition 2.5, yielding explanations of how input or hidden units contribute to the output. The decomposition into $\hat{x}$ and $\hat{b}$ further separates input-driven and bias-driven contributions. LRP naturally extends to multi-layer NNs through recursive layer-wise propagation, yielding relevance vectors $\hat{y}_{\ell-1} = \hat{x}_\ell$ and $\hat{b}_\ell$ for layers f_ℓ with $\ell = 1, \dots, L$, thereby propagating relevance through the computation graph used within the downstream prediction.

3. Related Work

Explanations of individual GNN predictions target either the embedding space or the underlying graph structure [30]. Model-agnostic methods such as LIME [31] and SHAP [32] can be applied to GNNs by perturbing node features or learned embeddings and observing prediction changes. However, these approaches scale poorly, ignore message passing, and miss structural or relational reasoning. Embedding-oriented methods instead assign semantics to latent dimensions, for example via correlations with structural statistics [33], dimension-level interpretability [34,35], or disentangled representations [36]. While improving transparency in latent space, they do not explain how predictions arise from graph structure. Structure-oriented explainers address this by analyzing the computation graph for a prediction and are grouped into three families.

A: Edge-elimination methods start from the full computation graph and identify explanations by masking edges whose removal minimally changes the model output. GNNExplainer [37] learns such masks via per-instance optimization, which must be repeated for every prediction and downstream task. PGExplainer [38] replaces this with a global surrogate network trained to predict edge relevance from node embeddings, but a separate surrogate must be trained for each task (e.g., each node classification task or relation type in link prediction). Given that both methods operate on homogeneous adjacency matrices, adapting them to RGNNs requires extensions to handle typed message passing.

B: Path-synthesis methods generate explanations by assembling multi-hop relational paths rather than pruning the full graph. RAW-Explainer [39] performs guided random walks whose transition probabilities depend on the model-internal message coefficients $c_{r,\iota}(u, v)$ per stub $s = (u, r, \iota)$ according to Definition 2.4. RelEx [40] learns relation-type importance scores through continuous relation-level masks and then constructs relational paths consistent with these scores. Because relation scores in RelEx are computed for individual link prediction queries, the resulting explanations are not applicable to node classification settings.

C: Propagation-based methods trace contributions through the message-passing architecture and thereby yield structure-oriented explanations. FlowX [41] and GradXKG [42] rely on gradients of the downstream losses, which make the methods sensitive to activation saturation and unable to produce signed attributions. GNN-LRP [43] applies LRP to homogeneous GNNs and produces signed relevance scores reflecting positive or negative evidence. However, its formulation is restricted to untyped message passing and does not support relational aggregation functions. In this work, we extend GNN-LRP to RGNNs to derive signed relevances of KG components and embedding dimensions.

As summarized in Table 1, existing graph-specific explanation approaches fall short of six key requirements for RGNNs: the method must incorporate relation types; explanations should be globally valid without per-instance optimization; relevance scores should be signed to separate positive from negative evidence; the method should apply across downstream tasks; explanations should cover the full computation graph; and embedding dimensions should be covered. While PGExplainer does not explicitly model negative relevance, it distinguishes between relevant and irrelevant edges, and thus its relation-aware extension serves as a baseline for edge- and node-level explanations. For path-level explanations, we consider RAW-Explainer as baseline. Although random walk generation may miss parts of the computation graph, it supports both downstream task scenarios, and therefore provides a benchmark for path-based explainability.

Method	Relational	Global	Signed	NC/LP	Full	Embedding
GNNExplainer (A)	×	×	positive-only	✓/×	✓	×
PGExplainer (A)	×	✓	positive-only	✓/✓	✓	×
RelEx (B)	✓	×	positive-only	×/✓	×	×
RAW-Explainer (B)	✓	✓	positive-only	✓/✓	×	×
FlowX (C)	×	✓	×	✓/×	✓	✓
GradXKG (C)	✓	✓	×	×/✓	✓	✓
GNN-LRP (C)	×	✓	✓	✓/✓	✓	✓

Table 1. Comparison of GNN explanation methods. Columns indicate whether a method supports relational message passing (Relational), global explanations without per-instance optimization (Global), signed relevance computation (Signed), applicability to node classification and link prediction tasks (NC/LP), coverage of the full computation graph (Full), and attribution to individual embedding dimensions (Embedding).

4. Methodology

We extend GNN-LRP to RGNNs operating on relational graphs $\mathcal{G} \subseteq \mathcal{V} \times \mathcal{V} \times \mathcal{R}$. Analogous to prior RGNN works, we treat them as equivalent to their closures, i.e., $\mathcal{G} = \bar{\mathcal{G}}$, implying $v \in \mathcal{N}_{\mathcal{G}}(v)$. The aggregation steps within RGNN layers as per Definition 2.4 can be expressed as compositions of dense layers, thus allowing for the propagation of relevance to edge and node neighborhoods via intermediate embeddings.

As shown in Figure 3, we consider an RGNN output $\phi(v) = \mathcal{K}(\psi)(v)$. Given an initial relevance vector $\hat{\phi}(v)$, we propose two paradigms for distributing relevance: macro LRP aggregates relevances per node and RGNN layer while micro LRP assigns relevance to paths as stub sequences. Intuitively, the goal is to trace how predictive relevance propagates through relational message passing back to the graph structure, thereby enabling the semantic predictive loop. To obtain the initial relevance, an LRP-compatible predictor $\pi : \mathbb{R}^d \rightarrow \mathbb{R}^{d_\pi}$ is assumed which uses $\phi(v)$ as input, i.e., $\pi_\phi(v) := \pi(\phi(v)) \in \mathbb{R}^{d_\pi}$. In line with LRP objectives, we aim to explain predictions by addressing the question:

Why was the j^ -th output entry $\pi_\phi(v)_{j^*}$ of the prediction $\pi_\phi(v)$ high (or low)?*

A typical example is a classifier, where $\pi_\phi(v)_{j^*}$ is the probability of class j^* . Initial relevance $\hat{\pi}_\phi(v)$ is defined as $\hat{\pi}_\phi(v)_{j^*} = \pi_\phi(v)_{j^*}$ and $\hat{\pi}_\phi(v)_j = 0$ for $j \neq j^*$. Since the predictor operates on $\phi(v)$, this relevance is transferred to $\hat{\phi}(v)$.

Analogously, link prediction intends to assign a plausibility score to individual stubs $s \in \mathcal{V} \times \mathcal{R} \times \mathcal{I}$, indicating the likelihood that it is incident to a node. Such scores are computed via corresponding scoring functions $S_\phi : \mathcal{V} \times (\mathcal{V} \times \mathcal{R} \times \mathcal{I}) \rightarrow \mathbb{R}$ with

$$S_\phi(v, s) := S(\phi(v), \phi(u), r, \iota), \quad S : \mathbb{R}^d \times \mathbb{R}^d \times \mathcal{R} \times \mathcal{I} \rightarrow \mathbb{R},$$

where higher scores indicate greater plausibility of the hypothesis $s \in \mathcal{S}_{\mathcal{G}}(v)$.

Accordingly, we consider the downstream prediction $\pi_\phi(v; s) := S_\phi(v, s) \in \mathbb{R}$. Since link prediction scores are single scalars, $\hat{\pi}_\phi(v; s) = \pi_\phi(v; s)$ serves as initial relevance, yielding $\hat{\phi}(v)$. Given an edge (v, u, r) with fixed u and r , we use $\pi_\phi(v; (u, r, \iota_{\text{out}}))$, whereas fixing v and r yields $\pi_\phi(u; (v, r, \iota_{\text{in}}))$. Thus, node classification performs predictions over predefined class labels, whereas link prediction assigns a plausibility score to a stub indicating how likely it is to be incident to a node. In this context, each relation type implies a distinct downstream task.

In the following, we first apply LRP to single RGNN layers and subsequently extend the procedure to multi-layer architectures. Throughout, we assume that no relevance is absorbed by biases, i.e., $\sum_{i=1}^d \hat{\phi}(v)_i = \pi_{\phi}(v)_{j^*}$.

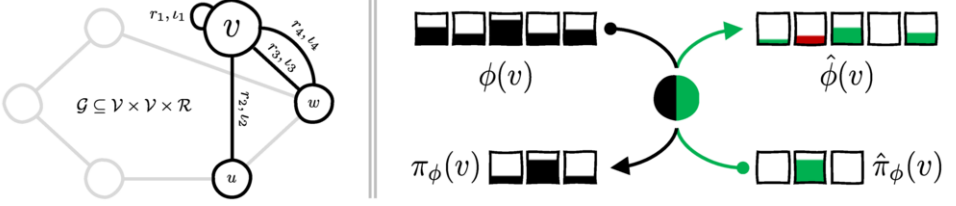


Figure 3. LRP initialization for RGNN pipelines: Initial relevances $\hat{\pi}_{\phi}(v)$ of downstream predictions $\pi_{\phi}(v)$ are redistributed to the RGNN output via $\hat{\phi}(v)$ (green: supportive, red: opposing).

4.1. Applying LRP to Single-Layer RGNNs

Let $\kappa : (\mathbb{R}^{d'})^{\mathcal{N}_{\mathcal{G}}(\mathcal{X})} \rightarrow (\mathbb{R}^d)^{\mathcal{X}}$ be an RGNN layer with an input $\psi \in (\mathbb{R}^{d'})^{\mathcal{N}_{\mathcal{G}}(\mathcal{X})}$ and an output $\phi \in (\mathbb{R}^d)^{\mathcal{X}}$ as per Definition 2.4. Furthermore, we assume an initial relevance vector $\hat{\phi}(v) \in \mathbb{R}^d$. Figure 4 illustrates how $\hat{\phi}(v)$ can be redistributed to stubs $s = (u, r, \ell) \in \mathcal{S}_{\mathcal{G}}(v)$ and the neighborhood $\mathcal{N}_{\mathcal{G}}(v)$.

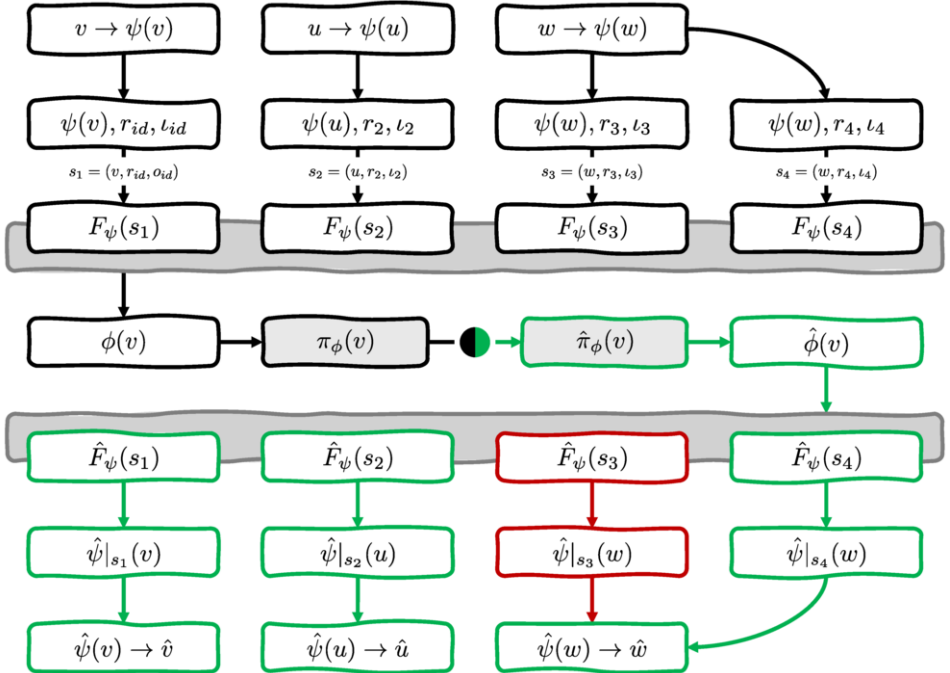


Figure 4. Decomposition of a single-layer RGNN output relevance vector $\hat{\phi}(v)$. Relevance is assigned to stubs $s \in \mathcal{S}_{\mathcal{G}}(v)$ and aggregated into node-level vectors and scalar relevances (green: supportive, red: opposing).

For readability, we consider the abbreviated notations $F_{\psi}(s) := F_{r,\ell}(\psi(u))$ and $c(s, v) := c_{r,\ell}(u, v)$ per stub $s = (u, r, \ell) \in \mathcal{S}_{\mathcal{G}}(v)$ in the following. Moreover,

component-wise monotonic activations are assumed. Hence, the weighted intermediate stub embeddings $\{F_\psi(s) \mid s \in \mathcal{S}_G(v)\}$ yield stub-level relevances²:

$$\hat{F}_\psi(s) = (c(s, v) \cdot F_\psi(s)) \oslash \left(\sum_{s' \in \mathcal{S}_G(v)} c(s', v) \cdot F_\psi(s') \right) \odot \hat{\phi}(v).$$

Since intermediate representations are the output of unbiased and unactivated dense layers $F_\psi(s) = W_{r,\iota} \cdot \psi(u)$ with $s = (u, r, \iota)$ in standard RGNNs, the relevance $\hat{\psi}|_s(u)$ corresponds to the relevance assigned to $\psi(u)$ via s . These stub relevances can be aggregated further to obtain node embedding relevances

$$\hat{\psi}(u) = \sum_{s=(u', r, \iota) \in \mathcal{S}_G(v)} \left(\mathbb{1}_{u=u'} \cdot \hat{\psi}|_s(u') \right) \in \mathbb{R}^{d'} \quad \text{for all } u \in \mathcal{N}_G(v). \quad (1)$$

Scalar relevances $\hat{u} \in \mathbb{R}$ are obtained by component-wise summation of $\hat{\psi}(u)$.

4.2. Applying LRP to Multi-Layer RGNNs

Let $\mathcal{K} = \kappa_L \circ \dots \circ \kappa_1$ be a multi-layer RGNN with $L > 1$ layers. Section 4.1 describes how relevance vectors $\hat{\psi}_L|_s(u) \in \mathbb{R}^{d'_L}$ are computed per stub $s = (u, r, \iota) \in \mathcal{S}_G(v)$. Analogous to single-layer RGNN architectures, these can be aggregated to obtain node-level relevances $\hat{\psi}_L(u) \in \mathbb{R}^{d'_L}$ and scalars $\hat{u}_L \in \mathbb{R}$ for all $u \in \mathcal{N}_G(v)$. Thus, one must decide whether (i) to perform relevance aggregation steps at each layer (macro LRP) or (ii) to assign relevance to paths within the computation graph as sequences of stubs without intermediate node-level aggregation (micro LRP).

4.2.1. Macro LRP for Multi-Layer RGNNs

Given relevance vectors $\hat{\psi}_L(u) \in \mathbb{R}^{d'_L}$ for $u \in \mathcal{N}_G(v)$, macro LRP produces relevance vectors $\hat{\psi}_{L-1}(u)$ for each two-hop neighbor $u \in \mathcal{N}_G^2(v)$ via κ_{L-1} . An iterative application yields sets of relevance vectors $\{\hat{\psi}_\ell(u) \mid u \in \mathcal{N}_G^{n_\ell}(v)\}$. Unlike the output layer κ_L , layers with $\ell < L$ may require propagating multiple relevance vectors. We therefore define the relevance assigned to $\psi_\ell(u)$ from $\hat{\phi}_\ell(w)$ as

$$\hat{\psi}_\ell(u; w) \quad \text{for } (u, w) \in \mathcal{N}_G^{n_\ell}(v) \times \mathcal{N}_G^{n_\ell-1}(v) \quad \text{with } w \notin \mathcal{N}_G(u) \Rightarrow \hat{\psi}_\ell(u; w) = 0.$$

Accordingly, the total relevance $\hat{\psi}_\ell(u)$ is obtained by node-wise summation

$$\hat{\psi}_\ell(u) = \sum_{w \in \mathcal{N}_G^{n_\ell-1}(v)} \hat{\psi}_\ell(u; w).$$

Since $\mathcal{N}_G^0(v) = \{v\}$ holds, this notation also applies to the RGNN output layer, where $\hat{\phi}_L(v)$ is the only relevance vector, implying $\hat{\psi}_L(u) = \hat{\psi}_L(u; v)$.

²Vector operations are elementwise, denoted by the Hadamard product/division $\odot/\oslash$.

4.2.2. Micro LRP for Multi-Layer RGNNs

Instead of aggregating stub relevances at each layer, micro LRP groups stubs into paths $(s_1, \dots, s_n) \in P_n(v) \subseteq (\mathcal{V} \times \mathcal{R} \times \mathcal{I})^n$ with $s_i = (u_i, r_i, l_i) \in \mathcal{S}_G(u_{i-1})$ and $u_0 := v$. Each stub $s_1 \in \mathcal{S}_G(v)$ defines a single-step path $s_1 \in P_1(v)$, which serves as the basis for all two-step paths $(s_1, s_2) \in P_2(v)$ with $s_2 \in \mathcal{S}_G(u_1)$. Iteratively, all paths of length n that originate from v are obtained by means of

$$P_n(v) = \{(s_1, \dots, s_n) \mid s_1 \in \mathcal{S}_G(v), \dots, s_n \in \mathcal{S}_G(u_{n-1})\}.$$

As a result, the total number of paths considered is

$$|P_n(v)| = \sum_{s_1 \in \mathcal{S}_G(v)} \sum_{s_2 \in \mathcal{S}_G(u_1)} \cdots \sum_{s_n \in \mathcal{S}_G(u_{n-1})} 1.$$

In the computation graph, a path $p = (s_1, \dots, s_{n_\ell}) \in P_{n_\ell}(v)$ corresponds to the reverse-ordered chain of intermediate embeddings $\psi_\ell(u_{n_\ell}), \dots, \psi_L(u_1)$. Using LRP, a relevance vector $\hat{p} \in \mathbb{R}^{d_\ell}$ can be propagated to all one-step extensions

$$p' \in \{(s_1, \dots, s_{n_\ell}, s_{n_\ell+1}) \mid s_{n_\ell+1} \in \mathcal{S}_G(u_{n_\ell})\} \subseteq P_{n_\ell+1}(v).$$

Because multi-layer RGNNs are sequential, we have $\phi_{\ell-1}(u_{n_\ell}) = \psi_\ell(u_{n_\ell})$. By setting $\hat{\phi}_{\ell-1}(u_{n_\ell}; p) = \hat{\psi}_\ell(u_{n_\ell}; p) =: \hat{p}$ and $\hat{\phi}_{\ell-1}(u; p) = 0$ for $u \neq u_{n_\ell}$, layer $\kappa_{\ell-1}$ distributes relevance to one-hop extensions $p' = (s_1, \dots, s_{n_\ell}, s_{n_\ell+1})$:

$$\hat{p}' = \hat{\phi}_{\ell-2}(u_{n_\ell+1}; p') = \hat{\psi}_{\ell-1}|_{s_{n_\ell+1}}(u_{n_\ell+1}; p') \in \mathbb{R}^{d_{\ell-1}}. \quad (2)$$

Since the base relevance $\hat{s}_1 = \hat{\psi}_L|_{s_1}(u)$ is directly obtained for any $s_1 \in \mathcal{S}_G(v)$ (see Section 4.1), repeated application of Equation 2 yields relevances for all multi-hop paths $(s_1, \dots, s_n) \in P_n(v)$ with $n > 1$. This ultimately provides the relevances of all full-length paths $P_L(v)$ used in the RGNN computation graph.

5. Experimental Evaluation

We evaluate LRP for RGNNs on KG-based node classification (NC) and link prediction (LP) tasks by analyzing relevance at the level of edges, nodes, and multi-hop paths³. Faithfulness is assessed via top- k deletion experiments measuring how removing the most positive or negative components affects the output, following standard practice in the context of GNN explainability [37, 44].

Experimental Setup. We train sequential RGCN and RGAT models of depth $L = 4$ with embedding dimension $d = 256$. Following standard implementations, identity relations r_{id} are included so that $\mathcal{G} = \bar{\mathcal{G}}$ holds. Model parameters, including initial node embeddings $\psi(v)$, are optimized with Adam (learning rate 10^{-3}). For NC, we use the AIFB and MUTAG KG benchmarks [12, 45] (AIFB: 4 classes, MUTAG: 2 classes). RGNN embedding entries $\phi(v)$ are passed to a two-layer MLP predictor π , whose logits define

³RGNN-LRP code is available: <https://github.com/Krusinaldo9/LRP4RGNN>.

the initial relevance vectors $\hat{\phi}(v)$ via LRP. For LP tasks, we use the FB15K-237 [46] and WN18RR [47] KG benchmarks. Models are trained using negative sampling with 20 corrupted triples per positive triple. Among several scoring functions (DISTMULT [48], TRANSE [49], COMPLEX [50], ROTATE [51]), DISTMULT performed best, providing

$$\pi_{\phi}(v; s) := S_{\phi}(v, s) = w(s)^{\top} \phi(v), \quad w(s) = \phi(u) \odot \lambda(r, \iota),$$

as link scores for $s = (u, r, \iota)$, where $\lambda(r, \iota) \in \mathbb{R}^d$ denotes an auxiliary directed relation embedding. Since the score is a weighted sum over the components of $\phi(v)$, LRP applies directly and yields the initial relevance $\hat{\phi}(v) = \phi(v) \odot w(s)$.

Evaluation Protocol: Each benchmark provides training, validation, and test splits. For NC, these consist of node–label pairs. For LP, they consist of positive triples, with the training triples defining $\mathcal{G}$. Although our focus is on RGNN explainability rather than predictive performance, all models achieve competitive predictive quality on their respective benchmarks. Relevance is assigned to a single output activation (class logit or link score), and signed top- k deletion tests are designed to perturb this specific activation. To ensure comparability across benchmarks, we apply a tanh activation to all logits and scores and ℓ_2 -normalize the base embeddings after each training step. While we focus on single-activation explanations, the framework naturally extends to joint comparisons, e.g., explaining why $\pi_{\phi}(v)_j$ exceeds $\pi_{\phi}(v)_i$ through the difference $\pi_{\phi}(v)_j - \pi_{\phi}(v)_i$.

Evaluation Results: Faithfulness is evaluated by deleting or inverting the highest- and lowest-scoring graph components or embedding dimensions. After updating the computation graph (or embedding input), we recompute the prediction $\pi'_{\phi}(v)$ and measure the resulting change in output activation using the ratio

$$\rho(a, b) = \mathbb{1}_{a < b}(1 - a/b) + \mathbb{1}_{a \geq b}(b/a - 1), \quad \rho(\pi'_{\phi}(v), \pi_{\phi}(v)) \in [-1, 1].$$

Negative values indicate a decrease in the output activation, whereas positive values indicate an increase. Ratios are aggregated across all NC and LP benchmarks. Figure 5 reports two boxplots per method and RGNN architecture: blue plots correspond to removing or inverting the most negatively relevant components (i.e., an increase in activation is expected), while red plots correspond to removing or inverting the most positively relevant components (i.e., a decrease in activation is expected).

5.1. Edge/Stub-Level RGNN Explanations

We begin by evaluating explanations with respect to edges and their associated stubs in the computation graph using macro LRP. For a target node $v \in \mathcal{V}$, we consider the triples participating in paths $P_L(v)$. For every input-layer edge $e = (v, u, r)$ in this computation graph, we compute a relevance vector $\hat{e} \in \mathbb{R}^{d'}$. If $v \neq u$, the directed edge e yields two node–stub tuples, $(v, (u, r, \iota_{\text{out}}))$ and $(u, (v, r, \iota_{\text{in}}))$. Hence, unlike existing approaches, each directed edge receives two relevance vectors, combined as $\hat{e} = \hat{\psi}|_{(u, r, \iota_{\text{out}})}(v) + \hat{\psi}|_{(v, r, \iota_{\text{in}})}(u) \in \mathbb{R}^{d'}$, as well as $\hat{e} = \hat{\psi}|_{(v, r, \iota_{\text{id}})}(v)$ for self-edges $e = (v, v, r)$. Scalar relevance scores are obtained by summing vector components. For LP tasks, we restrict ourselves to tail predictions. Given a test triple (v, u, r) , $u^* = \arg \max_{u' \in \mathcal{V}} \pi_{\phi}(u'; (v, r, \iota_{\text{in}}))$ is predicted and compared to u . However, our goal is not predictive performance but identifying which edges influenced the model's

preference for u^* . As discussed in Section 3, PGExplainer serves as an unsigned baseline. Faithfulness is assessed by deleting the top-10 highest- and lowest-scoring edges or stubs. If a deletion would isolate a node, its weakest edge or stub is retained. Both PGExplainer (PGE) and LRP identify positively relevant edges, but PGE cannot capture negative evidence. LRP, in contrast, reveals both supportive and inhibitory contributions, which become even more pronounced at the stub level (LRP-E / LRP-S).

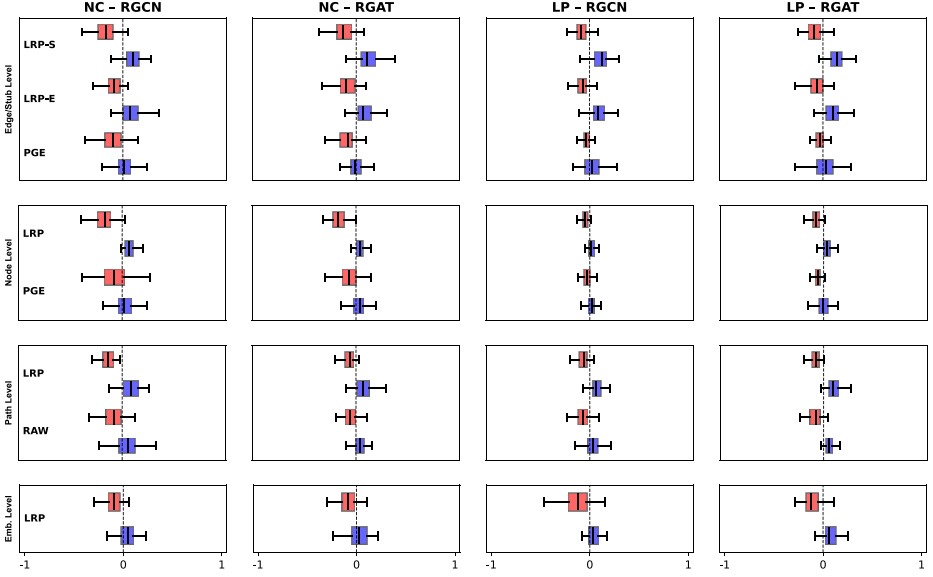


Figure 5. Deletion-based faithfulness w.r.t. activation ratio ρ (x-axis, $\rho \in [-1, 1)$) across edge/stub, node, path, and embedding levels. Red boxplots indicate removal of the most positively relevant components, while blue boxplots indicate removal of the most negatively relevant components.

5.2. Node-Level RGNN Explanations

Node-level relevances are obtained by aggregating edge- and stub-level scores. For macro LRP, we apply the aggregation rule from Equation 1 to compute relevance vectors $\hat{\psi}(u) \in \mathbb{R}^{d'}$ for all $u \in \mathcal{N}_{\mathcal{G}}^L(v)$, whose component sums yield scalar scores $\hat{u}$. PGExplainer scores are aggregated analogously by distributing each directed edge’s relevance to its incident nodes. Since removing a node implicitly removes all adjacent edges, we delete only the single highest- or lowest-scoring node, respectively. Faithfulness is evaluated using the output ratio ρ , comparing $\pi_{\phi}(v)$ with the post-deletion prediction $\pi'_{\phi}(v)$ and aggregating results across all NC and LP benchmarks. LRP again produces more faithful explanations, with the removal of a highly relevant node yielding a stronger activation drop than for PGExplainer. Ratios are generally smaller for LP tasks, which is expected since deleting a single node perturbs fewer paths than deleting individual edges or stubs. Although PGExplainer recovers some negative evidence after aggregation, LRP consistently identifies both supportive and inhibitory nodes.

5.3. Path-Level RGNN Explanations

Path-level explanations assess how multi-hop message passing influences a prediction. As a baseline, we employ RAW-Explainer (RAW), which samples random-walk paths without repeated edges and ranks them to construct explanatory subgraphs. For comparability, micro LRP is restricted to the same set of sampled walks rather than all paths in $P_L(v)$, whose full enumeration would be infeasible in practice. Since our RGNNs have depth $L = 4$, each sampled path contains four distinct edges. Consequently, deleting two distinct paths removes at least five edges from the computation graph. Thus, for both methods, we remove only the top-2 highest- or lowest-scoring paths. We then recompute the prediction and evaluate the resulting activation change using the ratio ρ . Both methods identify supportive and inhibiting paths. RAW-Explainer performs competitively despite lacking explicit signed relevance propagation, while micro LRP retains a slight advantage. The differences remain modest, reflecting that path-level perturbations affect multiple hops simultaneously and produce more stable output shifts.

5.4. Embedding-Level RGNN Explanations

Embedding-level explanations reveal which dimensions of $\psi(v) \in \mathbb{R}^{d'}$ are most influential after LRP propagation. Since individual embedding dimensions do not have predefined semantics, we aggregate relevance by dimension to quantify their contribution to the downstream objective. For LP tasks, each prediction produces a single score. We therefore group predictions by relation type and sum their macro-LRP relevance vectors $\hat{\psi}(u)$ to identify the most positive and most negative embedding dimensions associated with each relation. For NC tasks, each prediction yields multiple logits. We propagate relevance only from the true class and aggregate the resulting relevance vectors per class. To assess faithfulness, we invert the five most or least relevant embedding dimensions and measure the resulting activation change using the ratio ρ . Figure 5 shows that inverting highly relevant dimensions consistently reduces the activation. In contrast, inverting dimensions with low relevance produces only minor changes, indicating that RGNN predictions rely on a relatively small subset of strongly positive embedding dimensions, whereas negatively relevant dimensions play a less dominant role.

5.5. Relevance Distribution Across Computation Graphs

We additionally examine how far the most influential structural components lie from the target node $v \in \mathcal{V}$ in order to understand whether RGNN predictions are primarily driven by local evidence or by information propagated across multiple hops. Node- and edge-level explanations are obtained via macro LRP, whereas path-level explanations rely on unrestricted micro LRP. For each prediction, we determine the hop distance of the most relevant node (self-distance equals 0), the hop distance of the most relevant edge (self-edge distance equals 0), and the number of non-self-edge hops contained in the most relevant path. Figure 6 shows that RGNNs concentrate most relevance on the self-embedding $\psi(v)$, particularly in node-level explanations, while the influence of more distant nodes, edges, and paths decreases with hop distance. This reflects how RGNN models utilize their representations and highlights the strong influence of self-projections.

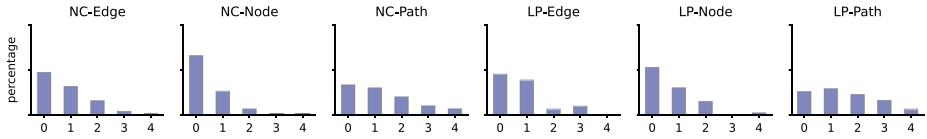


Figure 6. Hop distances (0–4) of the most relevant nodes, edges, and paths across NC/LP tasks. Percentages denote how often the most relevant component occurred at the respective distance from the prediction target. RGCN/RGAT results are combined due to similar distributions.

6. Conclusion

The experimental results demonstrate that LRP provides a coherent and fine-grained framework for explaining RGNN predictions on KGs. The method captures predictive behavior across edges, nodes, embedding dimensions, and multi-hop paths, and it identifies both supportive and inhibiting contributions that unsigned baselines cannot recover. Across node classification and link prediction tasks, LRP consistently reveals that RGNN predictions rely on a comparatively small subset of structural components and embedding dimensions, and it highlights how architectural mechanisms such as attention influence the distribution of relevance across the graph.

Overall, LRP enables a semantic predictive loop by linking model predictions back to the entities, relations, and neighborhood structures of the underlying KG. This provides a principled methodology for analyzing relational models by connecting graph structure, intermediate representations, and final predictions within a unified framework. The propagation of signed relevance makes it possible to trace how information flows through relational message passing and exposes structural influences that remain inaccessible to sampling-based or attribution-agnostic baselines. Moreover, attributing relevance at the level of embedding dimensions extends explainability beyond graph topology and provides a basis for investigating how semantic information is encoded within KG embeddings as latent representations of graph components.

Future work includes extending the evaluation to additional benchmarks, studying interactions with alternative link prediction scoring functions, and exploring applications in temporal or dynamic KG settings. Another promising direction is to move beyond single-output explanations and explore comparative decisions, such as why a model predicts one class over another or prefers one tail entity over its competitors. This requires propagating relevance for combinations of output activations and tracing their joint influence through the computation graph. Finally, integrating embedding-level relevance with methods that ascribe semantic meaning to latent dimensions may further deepen our understanding of how RGNNs encode and transform relational knowledge.

Declaration on Generative AI: During the preparation of this work, the authors used ChatGPT and Gemini for structural feedback, linguistic editing, grammar correction, and minor stylistic improvements. No scientific content, analysis, or conclusions were generated. The authors reviewed and edited the text as necessary and take full responsibility for the final content of the manuscript.

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When Knowledge Graph Relationships Still Matter

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Abstract. Modern language models increasingly recover signals of relatedness from text alone. They can support related-item ranking, entity clustering, and recommendation-style workflows without a separately modeled graph. For the knowledge graph community, this creates a timely and practical question: when do explicit graph relationships still add value beyond what strong text embeddings already capture?

This paper approaches that question carefully. It does not treat knowledge graphs as globally replaceable. Many KG roles — including semantic integration, provenance, governance, validation, querying, and domain modeling — remain distinct from embedding-based retrieval. Instead, we focus on a narrower class of use cases where relationship value is representational: recommendation, discovery, similarity search, related-item ranking, and relation recovery. In these settings, the question becomes measurable: do explicit relationships add recoverable signal beyond node text?

We introduce a relation-dependent diagnostic framework for measuring this boundary. We call the underlying phenomenon language-model encroachment: the growing ability of text-based representations to approximate relational signal traditionally supplied by explicit graph structure. Using an OpenAlex-derived scientific KG case study, we repeat the diagnostic across relation types, dataset slices, publication years, and text encoders. In each setting, text-derived and graph-derived representations are evaluated on the same held-out relationship recovery task. The results show that the graph–text boundary is not fixed. Some relation types are largely recoverable from text for this task. Others retain distinctive graph-derived signal.

The contribution is a practical diagnostic for KG engineering: if a relationship type adds signal beyond text, it may justify graph-engineering effort; if it does not add measurable signal, a text-first system may be sufficient for that task. This does not mean a KG is unnecessary overall, because many KG roles remain justified independently of embedding performance.

Keywords. knowledge graphs; language-model encroachment; relation recovery; text embeddings; graph-derived representations; diagnostic evaluation

1. Introduction

Modern AI has been shaped by two traditions that accelerated at the same historical moment. In 2012, deep learning became visibly powerful after large neural architectures

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transformed image classification [9]. In the same year, large-scale knowledge graphs entered mainstream search and semantic infrastructure, most visibly through the Google Knowledge Graph and the shift from “strings” to “things” [16]. These traditions answered different needs: deep learning learned representations from large data, while knowledge graphs organized explicit entities, relationships, and semantic structure.

For several years, these two directions grew in parallel. Graph neural networks then built a bridge between them: neural representation learning could operate over explicit graph structure, while relationships remained part of the data [1,4]. Transformers and large language models then shifted the question again. Transformer architectures made text representations much stronger [18], and sentence encoders made dense text similarity practical for retrieval, matching, clustering, and ranking without requiring explicit graph infrastructure [15]. As text models improve, they increasingly recover relationship-like signal from text alone, including signals that once seemed to require explicitly modeled relationships.

This raises a practical question for the knowledge graph community: *whether explicit graph relationships still earn their engineering cost for a given task*. The answer is not simply yes or no. Knowledge graphs are not built only for embedding-based retrieval; they also support provenance, governance, semantic integration, validation, querying, and explicit domain modeling. These roles may justify a KG independently of whether graph relationships improve a particular representation-learning task.

This paper focuses on a narrower, measurable capability: whether explicit relations add recoverable representational signal beyond node text for tasks such as recommendation, related-item retrieval, discovery, prioritization, and relation recovery. For organizations deciding whether to build or maintain a KG, this provides a practical diagnostic. If a relation type is nearly recoverable from text embeddings for the target task, a lighter text-first architecture may be adequate for recommendation, retrieval, or discovery, even if the KG remains justified for provenance, governance, or exact querying. If a relation shows a large graph–text gap, the explicit relationship is carrying signal that text misses and may justify continued graph construction and maintenance.

We call this phenomenon *language-model encroachment*: the increasing ability of text representations to recover relational signal traditionally supplied by explicit graph structure. Rather than asking whether graph methods outperform text methods in general, we ask a more specific question: which relations are already recoverable from text, which still carry distinctive graph-derived signal, and how does that boundary vary across relation type, publication year, and text encoder?

Scientific KGs are useful as a case study because they combine rich text with explicit relationships. Papers have titles and abstracts that modern text encoders can process, but they also participate in citations, shared references, topics, venues, authorship, and co-citation structures. These relationships capture different kinds of scholarly relatedness. Two papers may look related because their abstracts use similar language, or because they cite the same foundational work even when their abstracts use different terminology. The first signal may be recoverable from text; the second may still require graph structure. Operationally, the diagnostic asks whether a relation carries information not recoverable from independent textual descriptions of the connected nodes.

We instantiate the diagnostic on OpenAlex-derived scholarly data. The study compares text-derived and graph-derived representation spaces on the same held-out relationship recovery task, repeated across relation types, dataset slices, publication years, and

text encoders. This design lets us ask not only whether graph structure helps, but where it helps and whether that boundary changes under stronger text representations.

The text pathway uses two open sentence-transformer encoders, MiniLM and MP-Net. These models are not intended to represent the frontier of language modeling. They are locally runnable, reproducible research instruments for testing how encoder strength affects the measured graph–text boundary. If stronger text encoders recover more relation-relevant signal, this should appear as stronger text-only recovery and a smaller residual advantage for graph-derived representations.

The paper makes two contributions. First, it introduces a role-specific diagnostic for KG engineering: a way to measure whether explicit relations add recoverable representational signal beyond text for retrieval, recommendation, discovery, and relation-recovery tasks. Second, it provides an OpenAlex-based case study showing that this boundary varies by relation type, data slice, publication year, and encoder strength. The goal is to help KG builders identify which relationship types carry distinctive structural signal for AI retrieval and discovery tasks, while recognizing that many KG functions remain justified independently of embedding performance.

2. Related Work

The paper builds on three lines of work: deep representation learning, knowledge graphs and semantic infrastructure, and bibliometric relations in scientific knowledge graphs.

Deep learning became the dominant paradigm for representation learning after large neural architectures demonstrated that hierarchical features could be learned from large-scale data directly [9,10]. Transformer architectures later made contextual text encoding far more effective [18], and Sentence-BERT showed that dense sentence embeddings can support retrieval, matching, clustering, and ranking from text alone [15]. These developments motivate the central question of this paper: when nodes have rich textual attributes, how much relationship signal can text embeddings already recover?

In parallel, Semantic Web and knowledge-graph research developed explicit relational infrastructure for linking, validation, integration, provenance, and shared semantic modeling [13]. General surveys emphasize that knowledge representation, acquisition, and applications remain central to AI systems that require explicit entities and relations [6]. This broader KG value is not reducible to embedding performance. The present paper focuses on one KG function that can be measured directly: whether explicit relations add recoverable signal beyond text for relation-recovery, retrieval, discovery, and recommendation-style tasks.

Recent graph-augmented AI and retrieval-augmented workflows often place a KG or graph layer beside language models to improve grounding, navigation, traceability, or retrieval context. That design choice is plausible, but it can also obscure whether the graph contributes unique representational evidence or simply duplicates signals already present in text. The present paper is complementary to such graph-augmented systems: it does not propose a new LLM workflow, but offers a relation-level test for deciding when explicit graph links add measurable signal for retrieval- and recommendation-style tasks.

Graph neural networks bridge learned representations and graph structure by making neighborhoods part of representation learning [1]. GNNs are used across node classification, link prediction, and graph classification; recent surveys emphasize that performance

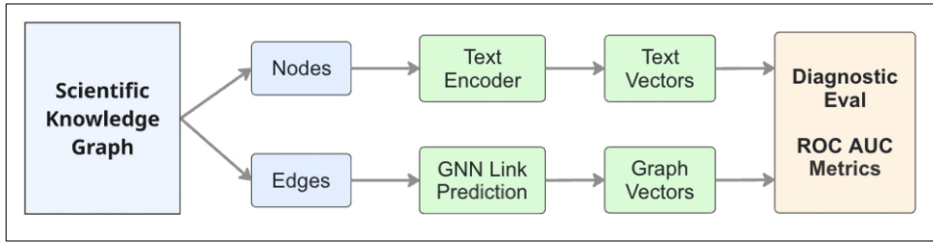


Figure 1. Diagnostic pipeline used in this study. Starting from a scientific knowledge graph, the framework derives text-derived and graph-derived representation spaces. Both spaces are evaluated on the same held-out pair discrimination task using ROC AUC, measuring whether graph structure adds recoverable relation-specific signal beyond text.

depends strongly on graph regime, supervision design, and task definition [8,12]. GraphSAGE is especially relevant here because it provides an inductive formulation for learning node representations from features and neighborhoods [4]. The literature on text-attributed graph representation learning treats the combination of textual node attributes and graph structure as a distinct methodological setting [19]. Our contribution is not a new graph architecture; it is a diagnostic comparison for deciding when graph structure still earns its place in such settings, using GraphSAGE as one reproducible graph-derived representation pathway.

For scientific knowledge graphs, the relations studied here have a long bibliometric history. Bibliographic coupling links papers that share references and captures shared scholarly background [7]. Co-citation analysis links papers frequently cited together as a measure of perceived relatedness in the research community [17]. OpenAlex provides an open scholarly knowledge graph spanning works, authors, venues, institutions, and concepts [14], while broader scientific knowledge graph efforts such as CS-KG 2.0 illustrate the role of structured scholarly data in AI-assisted search and discovery [2].

Work on faithfulness and relational structure distinguishes surface coherence from the evidence or structure supporting an answer [3,5,11]. More broadly, knowledge-graph research studies how graphs are constructed, integrated, and governed [6,13], while graph representation learning and text-attributed graph learning study how topology and text can be used together [8,12,19]. These lines of work generally take the available relation structure as an input to the system. Our contribution instead treats the residual value of an individual relation beyond node text as the object of diagnosis, providing a relation-level test of whether explicit structure adds recoverable signal for a specified task.

3. Methods

3.1. Diagnostic Overview

Figure 1 summarizes the diagnostic pipeline. We assume a text-attributed knowledge graph: entities have textual descriptions, and the graph contains explicit typed relations among those entities. In this paper, the entities are scholarly works, the text is derived from titles and abstracts, and the measured relation types include scholarly relationships such as citation, shared references, shared topic, and related bibliometric links. Observed instances of these relation types are represented as edges between scholarly works.

The diagnostic compares two representation spaces for the same relation-recovery task. The text-derived space captures what can be recovered from node text alone. The graph-derived space captures what can be recovered after learning from the target relation’s graph structure. Both spaces are evaluated on the same held-out positive and negative pairs, so the comparison asks a narrow and measurable question: does explicit graph structure add recoverable signal beyond text for this target relation?

3.2. Data and Relation Graph Construction

We instantiate the framework on OpenAlex-derived scholarly data [14]. Each node represents a scholarly work. The textual description of a node is formed from its title and abstract. Relation-specific edges are derived from scholarly metadata and citation structure, including direct citation, shared references, topic assignments, venues, institutions, authorship, and co-citation structure.

The unit of analysis is a relation type. For each target relation, we construct a relation-specific graph and a corresponding pair-discrimination task. Positive pairs are node pairs supported by the target relation. Negative pairs are sampled non-edge pairs from the same node set. Held-out positive pairs are removed before training graph-derived representations, so evaluation pairs are not directly available to the graph pathway during representation learning. This design keeps the text and graph pathways tied to the same evidence base while allowing us to test whether their induced similarity structures recover the target relation.

3.3. Text-Derived Representation

The text pathway encodes each paper independently from its title and abstract using sentence-transformer encoders. This pathway has no access to citation links, shared-reference structure, topic edges, or any other graph relationship. As a result, two papers are close in the text-derived space when their linguistic content is similar, regardless of whether they are connected in the knowledge graph.

We use two open sentence-transformer encoders: MiniLM (`all-MiniLM-L6-v2`) and MPNet (`all-mpnet-base-v2`). MiniLM serves as a lightweight reproducible baseline, while MPNet provides a stronger open encoder for sensitivity analysis. The goal is not to identify a final best text encoder, but to test whether the measured graph-text boundary changes as text representations become stronger.

3.4. Graph-Derived Representation

The graph pathway learns node representations from the relation-specific graph using a GraphSAGE-style encoder [4]. For each target relation, the model is trained with a link-prediction objective on the training portion of that relation graph. The input node features are the text-derived vectors for the selected encoder, while the relation-specific edges define the graph structure. Thus, the edge set is fixed for a given relation, but the initial node features differ between MiniLM and MPNet. GraphSAGE is used as a reproducible inductive encoder rather than as a claim of architectural optimality. Using the same text embeddings as graph input keeps the semantic starting point constant, while relation-specific graphs isolate the effect of each relation type. Cosine similarity is applied in both representation spaces to keep the downstream scoring rule fixed.

The graph pathway is intentionally relation-aware. It should not be read as a general-purpose KG embedding baseline trained once over all relations. Instead, for each relation it tests whether the observed training portion of that relation graph can reshape node geometry so that held-out instances of the same relation become more recoverable. This makes the comparison diagnostic rather than a claim about the best possible link-prediction model.

Within the primary representation-space diagnostic, link prediction is used only as a training signal. We do not evaluate the final predicted links produced by the link-prediction model. Instead, after training, we extract the model’s pre-final node embeddings and treat them as graph-derived representations. These embeddings are then evaluated as a representation space using the same pairwise similarity procedure applied to the text-derived vectors.

This distinction is important. The diagnostic does not compare a text encoder against a task-specific edge classifier. Instead, both pathways produce node vectors, and both are evaluated through pairwise similarity in their respective embedding spaces. If two papers are close in the graph-derived space, that closeness reflects how the initial text-derived features were reshaped by relation-specific connectivity and neighborhood context, rather than textual similarity alone.

For the complementary encoder-sensitivity analysis in Figure 4, we additionally construct a relationship-aware node representation by concatenating equally weighted, L2-normalized text and GraphSAGE vectors and normalizing the resulting vector. Figure 4 evaluates cosine similarity in this combined space. The GraphSAGE component is trained with the same link-prediction objective, but the reported values are not edge-decoder predictions and are not directly comparable with the graph-embedding cosine gaps in Tables 1–4.

3.5. Pairwise Similarity and Diagnostic Metrics

For each held-out pair, we compute a similarity score in the text-derived space and a similarity score in the graph-derived space. The same labeled pair set is used for both pathways: positives are supported relation pairs, and negatives are sampled non-pairs. Holding the node set, pair labels, and scoring procedure fixed isolates the effect of graph-derived structure on the recovered similarity space.

We use ROC AUC as the diagnostic score. Intuitively, ROC AUC asks how often true relation pairs are ranked above sampled non-pairs. A value near 1.0 indicates strong separation between related and unrelated pairs, while a value near 0.5 is close to random ranking. To compare the two representation spaces, we report the graph–text gap, $\Delta = \text{AUC}_{\text{graph}} - \text{AUC}_{\text{text}}$. A positive gap indicates that graph-derived representations recover relation-specific signal beyond text. A negative gap indicates that text-derived representations recover the relation more effectively under this diagnostic.

We interpret the gap relation by relation. When text-derived similarity matches or outperforms graph-derived similarity, we describe the relation as **text-favorable for relation recovery**. When graph-derived similarity substantially outperforms text-derived similarity, we describe the relation as **graph-favorable** or **graph-dependent** under this diagnostic. A third case occurs when the relation is defined directly by explicit meta-data membership, such as shared topic, venue, or institution. In these cases, high graph-derived performance is expected because the graph representation is learning from the

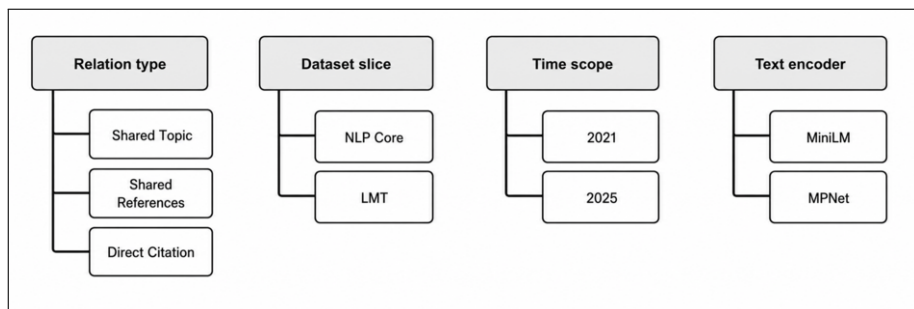


Figure 2. Multi-dimensional evaluation design used in this study. The same representation-space diagnostic is repeated across four dimensions: relation type, dataset slice, publication-year slice, and text encoder. This design makes the graph–text boundary observable as a function of scholarly relation type, domain slice, publication year, and text-model strength.

same membership structure that defines the relation. We refer to these as **structurally protected** relations. They are useful calibration cases, but should not be interpreted in the same way as relations such as shared references, where the graph advantage reflects broader scholarly neighborhood structure.

3.6. Multi-Dimensional Evaluation Design

Figure 2 summarizes the dimensions across which the same diagnostic is repeated. The study varies relation type, dataset slice, time scope, and text encoder. Relation type is the primary axis because edges are the first-class object of the study: different relations encode different kinds of scholarly relatedness and may therefore have different graph–text behavior. Dataset slices test whether the pattern is stable across related scholarly subdomains. Time scopes compare 2021 and 2025 slices descriptively. Text encoders test whether a stronger text representation changes the measured boundary.

Together, Figures 1 and 2 separate the logic of the method from the design of the experiments. The first figure shows how one text-versus-graph comparison is constructed. The second shows where that comparison is repeated. This distinction matters because the paper’s claim is not based on a single aggregate KG score; it is based on the variation of the graph–text gap across relations and evaluation contexts.

4. Experiments

4.1. Experimental Setup

We instantiate the diagnostic on filtered, paper-centered OpenAlex slices [14]. Nodes contain title and abstract text, while relation-specific edges are derived from citation and scholarly metadata. OpenAlex provides a deliberately text-rich setting with structurally different relation types, but it does not represent the construction costs or expert-curated semantics of biomedical, regulatory, or operational knowledge graphs. The study is therefore a diagnostic case rather than an exhaustive OpenAlex benchmark.

For each relation, text-derived and graph-derived representations are evaluated on the same held-out pair-discrimination task. ROC AUC measures separation against sam-

pled non-edges rather than complete reconstruction over all possible pairs. The framework uses public libraries and standard components and can be independently reimplemented from the architecture and protocol in Section 3.

Operationally, for every relation and slice, held-out positive edges are removed before graph training and evaluation negatives are sampled from non-edge pairs within the same retained node set. The graph encoder, pairwise cosine scoring, and ROC AUC computation are kept fixed across relation types so differences are interpreted as relation-level diagnostic signals rather than outcomes of relation-specific tuning. Results are reported as compact diagnostic values rather than multi-seed benchmark estimates; this should be considered when interpreting small differences, especially for sparse citation-derived relations.

Two topic slices are retained. The *Language Models / Transformers Proxy* (LMT) slice focuses on a domain where strong text representations are especially relevant. The *NLP Core* slice serves as a broader robustness setting within related scholarly work. Two publication-year slices, 2021 and 2025, are compared descriptively rather than causally. The year slices are not treated as matched cohorts; they may differ in composition, citation maturity, and edge density. The comparison asks whether the graph–text balance changes under the same diagnostic protocol, not why that change occurs.

The screened relation types are Shared References / Bibliographic Coupling, Direct Citation, Shared Topic, Shared Venue, Shared Institution, Shared Author, and three co-citation variants. These relations encode structurally different notions of scholarly relatedness. Bibliographic coupling captures shared intellectual background across topical boundaries. Direct citation is sparse, directional, and dependent on citation accumulation time. Shared topic, venue, and institution are membership-based groupings. Co-citation variants depend on how the research community clusters papers over time and are sensitive to edge thresholding and citation maturity.

The evaluation is reported in two stages. First, we use a broad relation-type screening to establish the graph–text landscape across the candidate relations. Second, we focus on three relations that represent the most interpretable positions in that landscape: Direct Citation as the text-favorable relation, Shared References as the non-membership graph-dependent relation, and Shared Topic as the structurally protected calibration relation.

4.2. Broad Relation-Type Screening

Before selecting representative relation types, we screen the candidate relation types to establish the landscape of graph–text differences. The screening tables report the initial non-year-partitioned representative screening configuration used to map relation regimes in each topic slice. They are therefore intended to support relation selection; a complementary encoder/year sensitivity analysis using combined text and GraphSAGE representations is reported in Figure 4. Tables 1 and 2 report the two topic slices separately so the patterns are easier to read. Figure 3 provides an illustrative view of the same screening logic in the Language Models / Transformers slice, showing how different relation types occupy different positions in the graph–text landscape. The figure is illustrative rather than exhaustive: it visualizes the main relation behaviors in one slice, while the complete screening, including co-citation variants and the NLP Core slice, is reported in Tables 1 and 2. The large variation in Δ shows that the value of graph structure is relation-dependent rather than global.

Table 1. Broad relation-type screening in the Language Models / Transformers slice. Positive Δ indicates that graph-embedding cosine outperforms text cosine; negative Δ indicates the reverse. Figure 4 reports the complementary combined-representation sensitivity analysis.

Relation	Text	Graph	Δ
Shared References	0.711	0.930	+0.218
Direct Citation	0.893	0.606	-0.287
Shared Topic	0.616	1.000	+0.384
Shared Venue	0.636	0.995	+0.358
Shared Institution	0.584	0.979	+0.395
Shared Author	0.742	0.899	+0.156
Co-Citation Next-Year (≥ 1)	0.837	0.885	+0.049
Co-Citation Next-Year (≥ 2)	0.925	0.813	-0.112
Co-Citation Neighborhood	0.826	0.855	+0.029

Table 2. Broad relation-type screening in the NLP Core slice. Positive Δ indicates that graph-embedding cosine outperforms text cosine; negative Δ indicates the reverse.

Relation	Text	Graph	Δ
Shared References	0.756	0.943	+0.187
Direct Citation	0.894	0.566	-0.329
Shared Topic	0.621	1.000	+0.379
Shared Venue	0.683	0.995	+0.312
Shared Institution	0.602	0.983	+0.381
Shared Author	0.754	0.907	+0.153
Co-Citation Next-Year (≥ 1)	0.832	0.827	-0.005
Co-Citation Next-Year (≥ 2)	0.853	0.643	-0.210
Co-Citation Neighborhood	0.856	0.842	-0.014

Direct Citation is the clearest text-favorable relation in this screening: in both slices, text-derived similarity outperforms graph-derived similarity by a substantial margin. Shared References / Bibliographic Coupling is the clearest non-membership graph-favorable scholarly relation, with a large positive gap in both slices. Shared Topic, Shared Venue, and Shared Institution show near-ceiling graph AUC, but their graph advantage is structurally protected because these relations are membership-based. Co-citation variants are mixed, consistent with their dependence on accumulation time, edge density, and threshold choice.

4.3. Selection of Representative Relation Types

The broad screening shows that relation types behave differently under the same diagnostic. For the focused analysis, we retain three representative relation types that span the most interpretable positions in this landscape.

Shared References / Bibliographic Coupling is retained as the anchor *graph-dependent* case: it captures shared intellectual background that is often only partially visible in titles and abstracts. Direct Citation is retained as the *text-favorable* contrast: under this representation-space diagnostic, text similarity already recovers much of the relation. Shared Topic is retained as a *structurally protected* calibration case: because

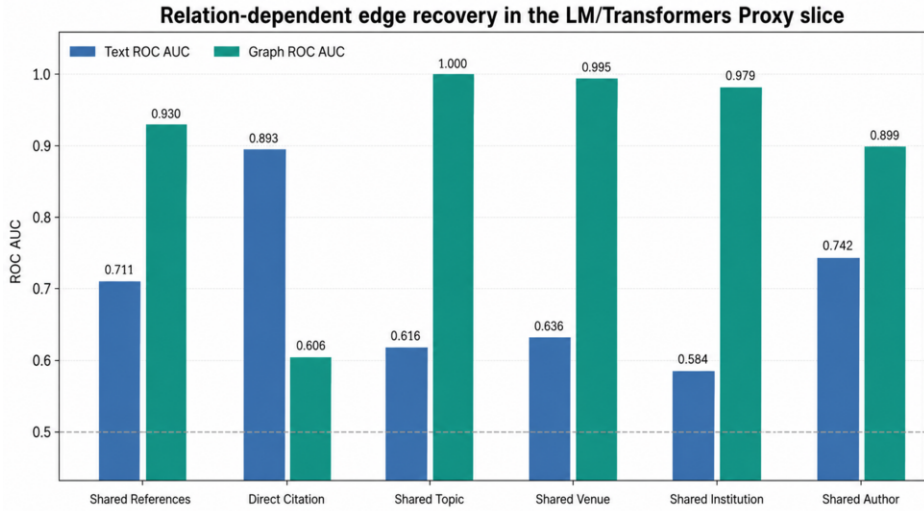


Figure 3. Relation-type screening in the Language Models / Transformers slice. Direct Citation is the text-favorable contrast, Shared References the non-membership graph-dependent case, and Shared Topic the structurally protected calibration case.

topic membership is encoded directly as a graph relation, its graph advantage is expected by construction. As illustrated in Figure 3, these three relation types occupy distinct positions in the screening landscape.

The remaining screened relations are still informative, but they are less useful as representative cases for a compact explanatory paper. Shared Venue, Shared Institution, and Shared Author are additional membership- or proximity-based relations, but they are closer to metadata grouping effects and are therefore less central to the encroachment question. The co-citation variants are also valuable, but they are more sensitive to thresholding and citation maturity, which makes them harder to interpret cleanly in the main text. We therefore keep them in the broad screening while centering the paper on the three relation types that most clearly expose the contrast the diagnostic is intended to measure.

4.4. Relation, Encoder, and Year View

Figure 4 presents a complementary encoder-sensitivity analysis across relation type, dataset slice, text encoder, and publication-year slice. Unlike Tables 1–4, which compare text cosine with cosine in the GraphSAGE embedding space, Figure 4 compares text cosine with cosine in a combined relationship-aware representation containing both text and GraphSAGE components. The residual-gain panels therefore show the incremental value of adding graph-conditioned information to text, while the lower panels show where text alone is already strong. This distinction is especially visible for Direct Citation: it is text-favorable in the graph-only representation-space diagnostic but has the smallest positive residual gain in the combined analysis. Shared References retains graph-derived value under both views, while Shared Topic remains the strongest structurally protected case.

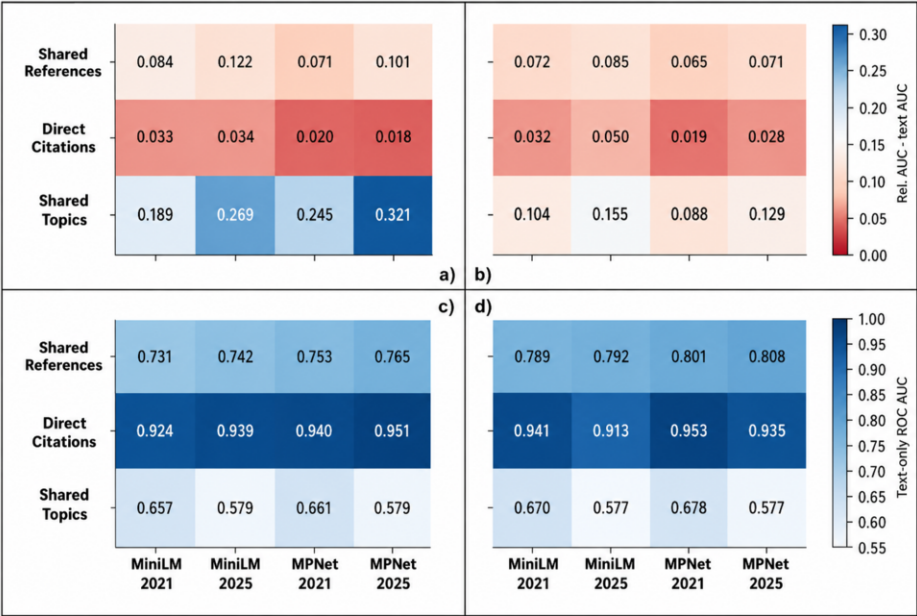


Figure 4. Complementary encoder-sensitivity analysis for the Language Models / Transformers and NLP Core slices. Panels (a,b) show relationship-aware cosine ROC AUC minus text-only cosine ROC AUC, where the relationship-aware representation concatenates normalized text and GraphSAGE vectors; panels (c,d) show text-only ROC AUC. These residuals are not directly comparable with the graph-embedding cosine gaps in Tables 1–4. Direct Citation has the smallest residual gain, Shared References a moderate gain, and Shared Topic the strongest structurally protected gain.

Table 3. Text and graph ROC AUC by relation and year in the Language Models / Transformers slice. Direct Citation 2025 is excluded because the retained relation graph is sparse.

Relation	Year	Text AUC	Graph AUC	Δ
Shared Refs / Bibliographic	2021	0.707	0.933	+0.225
Shared Refs / Bibliographic	2025	0.737	0.914	+0.176
Direct Citation	2021	0.883	0.626	−0.257
Direct Citation	2025	Excluded (sparse)		
Shared Topic	2021	0.653	1.000	+0.347
Shared Topic	2025	0.591	1.000	+0.409

4.5. Focused Relation Outcomes

The focused analysis retains three relations that span the interpretable ends and middle of the spectrum: Shared References / Bibliographic Coupling as the anchor non-membership graph-dependent relation, Direct Citation as the text-favorable contrast, and Shared Topic as a structurally protected calibration case. Table 3 reports these relations by publication-year slice in the LMT setting.

Direct Citation: closest to text-sufficient territory for relation recovery. Direct Citation is sparse, directional, and low-degree, precisely the regime in which graph embedding

Table 4. Shared References / Bibliographic Coupling (≥ 2) by year. The smaller 2025 graph–text gap is descriptive rather than causal.

Slice	2021 Gap	2025 Gap	Change
Language Models / Transformers	0.225	0.176	−0.049
NLP Core	0.198	0.148	−0.050

geometry has limited neighborhood signal to exploit. Meanwhile, papers often cite work that is semantically related in problem, method, or prior context, and that relatedness is frequently visible in titles and abstracts. Under this representation-space diagnostic, text therefore recovers Direct Citation more effectively than the graph-derived embeddings used here. This should not be read as a general claim that text is always better for citation prediction; it is a claim about this diagnostic and this representation comparison.

Shared References: graph-dependent with a narrowing gap. Bibliographic coupling captures shared intellectual background: two papers may draw on the same foundational work even when their abstracts discuss different applications or use different terminology. This signal emerges from the citation neighborhood and is not fully visible in any individual abstract. In the LMT slice, Shared References has a large graph advantage in both years, but the gap narrows from 0.225 in 2021 to 0.176 in 2025.

Shared Topic: structurally protected calibration. Shared Topic reaches graph AUC of 1.000 in both years. This should not be read as evidence that graph learning discovers hidden structure that text cannot recover. Topic assignments are discrete metadata categories encoded directly as graph edges; a graph representation is expected to recover them. The result is useful as a calibration case showing that the diagnostic identifies relations whose edge definition is already encoded as metadata membership.

4.6. Shared References Across Publication-Year Slices

Shared References is the anchor relation because it is structurally interpretable, stable across slices, and not simply a metadata membership edge. Table 4 isolates this relation across both topic slices.

In both slices, the graph–text gap for Shared References is smaller in 2025 than in 2021. This repeated directional pattern is suggestive, but it should be interpreted descriptively because the year slices differ in composition, citation maturity, and edge density. Within those limits, the result is consistent with the moving-boundary view: text representations may recover more bibliographic-coupling signal in some newer slices than in older ones. One possible explanation is that recent LM/NLP papers are more topically homogeneous; another is that abstracts increasingly state shared intellectual context explicitly. The present experiment does not adjudicate between those explanations. Its role is to show why the KG community should monitor this boundary empirically as text encoders and scholarly writing practices change.

5. Discussion

The results support neither the obsolescence nor the universal necessity of knowledge graphs. Instead, explicit relationships have conditional, measurable value that depends on

the relation type and intended use. As text-based systems improve at search, clustering, and recommendation, the relevant question is *which* relationships still carry distinctive structural signal.

5.1. From Model Comparison to KG Design Guidance

The diagnostic turns model comparison into a relation-level KG design signal. For organizations considering whether to build or maintain explicit graph structure, the practical question is whether a relationship improves the target task enough to justify its extraction, validation, storage, querying, and governance costs. This is most relevant to recommendation, retrieval, discovery, prioritization, and relation-recovery systems.

If a relation is text-favorable, a text-first baseline may be sufficient for the target ranking or discovery task, even when the relation remains necessary for provenance, governance, compliance, exact querying, or semantic integration. A large positive graph-text gap instead indicates signal that may justify explicit graph construction and maintenance. The results therefore motivate a “measure before you build” approach, whose practical utility should be evaluated in future organizational and domain-specific deployments.

5.2. Representational versus Decision-Critical Relationships

The results also distinguish relationships that improve exploration from those that determine correctness, risk, or compliance. Approximate relations may support ranking, personalization, or exploratory search well enough through text embeddings. Decision-critical relations, including drug interactions, regulatory dependencies, ownership, supply-chain dependencies, and safety constraints, instead require explicit modeling, traceability, and validation. A text-favorable result should therefore not be read as licensing the silent replacement of a required, evidenced relationship with a model’s prediction of it; this failure mode is evidence substitution.

For graph-augmented AI, augmentation should therefore be justified relation by relation rather than assumed for every LLM workflow. Language models provide semantic generalization, while knowledge graphs provide explicit, traceable, governed, and queryable structure. The question is not “graphs versus LLMs,” but which relationships require explicit structure and how that requirement should be measured.

5.3. Scope and Limitations

The study has several limitations. It uses filtered OpenAlex topic slices rather than the full graph, so results are within-domain diagnostics rather than universal claims, although the framework can be rerun on other text-attributed KGs. The graph-derived pathway uses one GraphSAGE-style model, and alternative graph encoders, symbolic features, or ontology-aware methods may produce different results. The experiments report compact diagnostic values rather than multi-seed confidence intervals, so small differences should not be overinterpreted. Because the diagnostic does not include structured nulls, part of a positive graph-text gap may reflect generic graph structure rather than relation-specific connectivity. The strongest evidence is the repeated relation-level ordering and the large positive or negative graph-text gaps.

The year-wise comparison is descriptive, not causal. The 2021 and 2025 slices differ in composition, citation maturity, and edge density, so the narrowing of the Shared

References gap should be interpreted as consistent with a moving-boundary view rather than proof of a historical trend. Direct Citation in the 2025 LMT slice is sparse and is excluded from the focused year-wise table. Finally, MiniLM and MPNet are open, local encoders selected for reproducibility and cost; frontier commercial or domain-specific scientific encoders may move the boundary further toward text-approximable territory. This is why encoder strength is treated as part of the diagnostic rather than as a nuisance variable.

6. Conclusion

This paper asked where language-model encroachment affects knowledge graph relation types and where explicit structure continues to add measurable value. The answer is relation- and task-dependent. Knowledge graphs remain important for many functions outside the scope of this diagnostic, including provenance, governance, semantic integration, validation, constraints, exact querying, and decision-critical traceability. This paper evaluates one narrower role of graph structure: whether explicit relations improve representation-space recovery for relation recovery, recommendation, discovery, or related-item ranking. For that class of use cases, the representational value of explicit relations can and should be measured.

Applied to an OpenAlex-derived scientific KG case study, the diagnostic shows that there is no single graph-text boundary. Direct Citation is closest to text-sufficient territory for relation recovery under the representation-space diagnostic. Shared References remains the clearest residual graph-dependent scholarly relation, but its graph-text gap narrows across years and under stronger text encoding. Metadata-derived groupings are structurally protected by construction. These findings support a more precise and constructive view of knowledge graph value: graph structure is neither generically required nor generically replaceable. Its value should be evaluated relation by relation and use case by use case.

The broader implication is methodological. As text encoders improve, part of the signal previously supplied by graph structure can be approximated from text. But the remaining residual is strongly relation-dependent and, for relations such as Shared References, still meaningful. The diagnostic framework introduced here provides a practical tool for measuring where graph structure is still earning its place, where text-first systems may be sufficient, and where explicit relationships remain essential for trustworthy semantic infrastructure.

Declaration on Generative AI

During the preparation of this work, the author used ChatGPT and Codex (OpenAI), and Claude (Anthropic), under active author direction, to support literature and source discovery, language editing, rephrasing, structural refinement, code and result inspection, and the design and refinement of diagrams. The research questions, arguments, methodology, experimental decisions, interpretations, and conclusions were developed by the author. The author guided the use of these tools through specific instructions, independently verified relevant sources, results, and factual claims, and critically reviewed and

revised all incorporated outputs. The author takes full responsibility for the publication's content.

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SYNTHOLOGY: Ontology-Based Data Generation for Neurosymbolic Reasoning

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Abstract. Symbolic reasoners offer a simple contract: provide an ontology, receive entailments. Neurosymbolic reasoners promise the same task with added scalability and tolerance to noise, but break this contract by requiring an ontology-specific training dataset, which in most cases does not exist and must be gathered from scratch. This bottleneck blocks practitioners from applying neurosymbolic reasoners to new ontologies. We present SYNTHOLOGY, an ontology-agnostic synthetic data generator that closes this gap by producing the required training data from any OWL 2 RL ontology, with no need for manual data gathering. SYNTHOLOGY engineers data through ontology-guided backward-chaining proof search, which guarantees deep multi-hop derivations by construction, and introduces proof-based negative sampling. Together, these properties force models to traverse the full reasoning chain to correctly classify triples. Our evaluation shows that a model trained on SYNTHOLOGY data outperforms models trained on baseline data; part of this advantage reflects test-set familiarity, though the gain in rejecting near-miss negatives persists under an independently generated test set (Section 6.2). Proof-based negative sampling yields consistent gains over random and constrained sampling on all learning-quality metrics, and SYNTHOLOGY scales to ontologies on which traditional (ontology specific) generators run out of memory. To our knowledge, SYNTHOLOGY is the first ontology-agnostic synthetic data generator that provides multi-hop coverage, near-miss negatives, and derivation metadata by construction, a step toward making neurosymbolic reasoners as readily deployable on a new ontology as their symbolic counterparts.

Keywords. Neurosymbolic AI, Reasoning, Data Generation, Negative Sampling

1. Introduction

Knowledge graphs (KGs) and ontologies form the backbone of structured knowledge representation on the Semantic Web. The task of deriving new, implicit knowledge from these structures is known as *knowledge graph reasoning* (KGR) [1,2], and is the broader task this work is ultimately concerned with. Traditionally, KGR is performed by symbolic reasoners [3,4], which execute the ontology's rules directly and offer a simple contract: provide an ontology, receive sound and complete entailments. While elegant, this approach hits what we call the *reasoning wall* (Section 6.3): symbolic reasoners are sen-

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sitive to noise and computationally expensive, making them infeasible on massive or messy real-world KGs [5].

Neurosymbolic reasoning has emerged as the proposed alternative within the field of KGR [6]. Instead of *executing* the ontology’s rules at inference time, a neural model is trained to *imitate* them. The resulting behavior is called *approximate reasoning*: the model predicts the entailments of the underlying knowledge base in a way that is scalable and tolerant to noise [7]. However, this shift breaks the symbolic-reasoner contract. A symbolic reasoner operates directly on an ontology and immediately produces entailments, whereas a neurosymbolic reasoner first has to be *trained* on a dataset that reflects the target ontology. For most ontologies, such a dataset does not exist and must be gathered from scratch, and this bottleneck blocks practitioners from applying neurosymbolic reasoners to new ontologies at all.

Existing data sources cannot fill this gap, as real-world datasets such as DBpedia or Freebase [8,9] are noisy, incomplete, and tied to specific domains, so they neither cover arbitrary ontologies nor support training a model to learn genuine logical reasoning instead of statistical shortcuts. Synthetic data generators could in principle solve this, but standard approaches typically use procedural generation of base facts [10,11], followed by materialization: a logical reasoner infers the to-be-predicted targets from the previously generated base facts, as in [12]. This methodology often produces structurally shallow graphs, because the generation is not ontology-aware. The resulting graphs contain a lot of trivial facts, while the deep, multi-hop derivations that the model is actually meant to learn occur only by accident. In addition, the resulting datasets lack challenging negative samples, which are critical for training neurosymbolic models: trivial negatives are easy to reject from surface features alone [13], so the training signal tends to collapse to pattern matching rather than reasoning.

To address this gap, we present SYNTHOLOGY, an ontology-driven synthetic data generator. Our novel generator accepts *any* ontology expressed in the supported subset of OWL 2 RL. In a backward-chaining manner, it purposely engineers training data that inherently requires multiple hops to be correctly predicted. Crucially, it also generates near-miss negative samples based on the constructed proof trees. These negatives *look* plausible, but are not logically entailed, forcing models to learn the underlying ontology structure. In summary, our main contributions are:

- **SYNTHOLOGY**: an ontology-agnostic data generator that, given *any* OWL 2 RL ontology, produces a training set sufficient to train a neurosymbolic reasoner over that ontology. This removes the per-ontology data-gathering step that currently blocks deployment of neurosymbolic reasoners on new ontologies.
- **Backward-chaining proof construction**: the core mechanism, which guarantees deep multi-hop coverage by construction rather than incidentally.
- **Proof-based negative sampling**: a method for constructing negative samples based on proof trees that produce near-misses, forcing models to traverse the full reasoning chain rather than rely on surface features.
- **Empirical evaluation**: a comparative study across two ontologies (family tree, OWL2Bench) demonstrating clear advantages in hop distribution, predicate coverage, negative sample quality, and scalability over existing Unguided Deductive Materialization methods.

We also release an open-source (optimized) reimplementation of the Recursive Reasoning Network [14] (the original paper did not release its code), which is used as the evaluation model throughout this paper.

Section 2 introduces the necessary background, Section 3 reviews related work, Section 4 details SYNTHOLOGY’s design, Section 5 describes the experimental setup, Section 6 reports results, and Section 7 concludes the paper and outlines future work.

2. Background

We first define knowledge graphs and ontologies, then describe what we mean by approximate reasoning in a neurosymbolic setting, and finally explain the role of negative sampling in training neural reasoners.

2.1. Knowledge Graphs and Ontologies

We consider a knowledge graph as a collection $\mathcal{G}$ of concrete facts (assertions) about specific entities, referred to as the ABox. To provide semantics and enable reasoning, the KG is paired with an ontology $\mathcal{O}$ (the TBox). Together, the factual KG and the ontology form a knowledge base $\text{KB} = \langle \mathcal{G}, \mathcal{O} \rangle$.

For example, the triples $\langle \text{Alice}, \text{hasParent}, \text{Bob} \rangle, \langle \text{Bob}, \text{hasParent}, \text{Charlie} \rangle$ are explicit, *base facts* in the KG, while a rule like $\langle X, \text{hasParent}, Y \rangle \wedge \langle Y, \text{hasParent}, Z \rangle \Rightarrow \langle X, \text{hasGrandparent}, Z \rangle$ is part of the ontology $\mathcal{O}$. Both are present in the KB. Combining them, one can derive implicit facts, like $\langle \text{Alice}, \text{hasGrandparent}, \text{Charlie} \rangle$. These *inferred facts* are not explicitly stated in the ABox, but logically entailed by the ontology given the existing *base facts*. The *base/inferred* terminology will be used throughout this paper.

In this work, we focus on the OWL 2 RL profile [15] as the target rule language, although SYNTHOLOGY is built for extensibility to other profiles.

2.2. Neurosymbolic Reasoning

Predicting which implicit facts are entailed by a given KB without explicitly applying the ontology’s rules is a form of knowledge graph reasoning (KGR). Previous work refers to this as *approximate reasoning*, trading strict logical completeness for scalable prediction of entailed facts [7]. Standard knowledge graph embeddings (KGEs), such as TransE [16], excel at capturing single-hop patterns through vector-space geometry, but struggle with deep, multi-hop logical chains [1, 14]. For approximate reasoning tasks, this motivates the shift toward neurosymbolic architectures, which combine the scalability of neural networks with the logical precision of symbolic reasoning [2]. A promising example is the *Recursive Reasoning Network* (RRN) [14], which will serve as the evaluation model throughout this paper. The RRN is trained to predict the likelihood $\mathbb{P}(\langle s, p, o \rangle)$ of a triple being logically entailed by the provided base facts. It uses an iterative message-passing mechanism, allowing the model to capture complex, multi-hop logical dependencies. Crucially, the training data must contain sufficient evidence of these deep patterns, and this is precisely what SYNTHOLOGY is designed to provide.

2.3. Negative Sampling

Apart from deep chains, another critical factor in training neurosymbolic reasoners (like the RRN) is negative sampling. Since KGs contain only positive facts, models must learn to distinguish true inferences from false ones by training against corrupted (and thus false) facts. Two standard corruption strategies serve as baselines throughout this paper: *random* corruption replaces the subject or object with an arbitrary individual, while *constrained* corruption additionally requires the replacement to satisfy the corrupted predicate's declared domain or range. Section 3.5 situates both within the broader negative-sampling literature.

3. Related Work

This section reviews existing approaches to generating training data for KGR. We organize them by methodology and, for each family, ask the same question: can it deliver a training set for *any* ontology without per-ontology data engineering?

3.1. Benchmark-Based ABox Generators

Several benchmarks for OWL reasoning provide a procedural ABox generator that creates synthetic data following a fixed schema. The Lehigh University Benchmark (LUBM) [10] populates a university domain, including students, professors, courses, and departments, at configurable scales. Its successor, the University Ontology Benchmark (UOBM) [11], extends LUBM to include additional OWL constructs. OWL2Bench [17], finally, provides a generator paired with ontologies that cover all four OWL 2 profiles (including RL), once again over a university domain. The problem is that each of these generators is hand-built around its target ontology: the type distributions, individual roles, and relation densities are tuned to a specific schema. None of them can be reused on a new ontology without substantial manual re-engineering, which is exactly the bottleneck a deployable neurosymbolic pipeline needs to eliminate.

3.2. Unguided Deductive Materialization

To produce training targets, such a benchmark generator (or a fully random ABox) can be paired with a symbolic reasoner that materializes the deductive closure [12]. We refer to this two-step pipeline as *Unguided Deductive Materialization* (UDM): a procedurally or probabilistically generated set of base facts is first constructed (either tied to a fixed ontology or drawn at random), and a symbolic reasoner then materializes all logically entailed facts to form the training targets. Negatives are typically added via uniform random subject or object corruption [16].

The issue with UDM is that generation *is not guided by the ontology*. The result is a training distribution shaped by what is easy to generate rather than what the model needs to learn. The generator first produces individuals and assertions from statistical templates, and the reasoner is invoked only afterward. As a result, the generated data rarely satisfies the conditions required to fire long rule chains: predicates that depend on the simultaneous presence of multiple different facts produce few, if any, inferences. Typically, shallow schema propagations (like `rdf:type` and `rdfs:subclassOf`) tend to

dominate the closure. Facts that require *multi-hop reasoning* are therefore systematically underrepresented in the resulting training data, regardless of how many base facts are generated, and neural models trained on this output are prone to *shortcut learning* [18].

A second concern is scalability. Forward-chaining reasoners like Apache Jena [4] typically use semi-naïve evaluation, which iteratively saturates the ABox until no new facts can be derived. Although each iteration is bounded, the materialized closure must be retained in memory throughout. In practice, this causes materializers to fail at production-scale ontologies. We refer to this ceiling as the *reasoning wall* and we characterize it empirically in Section 6.3.

3.3. Ontology-Specific ASP Generators

Rather than a benchmark generator + reasoner pipeline, some prior work uses *answer set programming* (ASP); the RRN paper [14] provides an ASP program tailored to a single ontology (family tree). This couples generation to logical structure (unlike UDM), but the ASP program must be hand-written for each new ontology, reintroducing the per-ontology engineering cost we aim to eliminate.

A separate weakness is its handling of negatives, which relies on *exhaustive* rather than *strategic* computation: ASP’s negation constructs let the generator compute *all* possible inferences per sample, guaranteeing negatives but producing heavily skewed datasets dominated by trivial ones.

3.4. LLM-Based Generators

Large language models (LLMs) have recently been explored for KG construction, generating triples from natural-language domain descriptions. LLM outputs lack formal correctness guarantees, making them ill-suited for the logic-consistent data a neurosymbolic reasoner needs [19], and they do not produce the derivation structure (proof trees, hop depths) required to construct hard negatives or verify multi-hop coverage: they generate triples, not the reasoning that supports them.

3.5. Negative Sampling Strategies

Negatives are commonly obtained by *corruption*: replacing the subject or object of a positive triple with another individual, uniformly at random in the simplest case [16]. Such corruptions are usually type-incompatible, and therefore easy to reject. Subsequent work has made them harder in three ways, which recent reviews group into static, dynamic, and adversarial families [13]. The first family restricts which individuals may replace the original: examples are Bernoulli sampling, which chooses the corruption side according to the cardinality of the relation [20], type-constrained sampling, which replaces only within the predicate’s domain and range [21] (our *constrained* baseline), and structure-aware sampling which draws from the k -hop neighborhood of the triple [22]. The second family caches the candidates the model scores highest during training and samples from that cache [23]. The third family is adversarial: a second model is trained to propose the negatives [24], or the model reweights its own random negatives by the scores it assigns them [25].

3.6. What Is Missing

Across these existing families of approaches, three shortcomings recur. (i) Generation is not ontology-aware, so multi-hop inferences appear only by chance and shallow schema propagations dominate. (ii) Negative samples are constructed in isolation from the derivation that produced the positive: static, dynamic, and adversarial corruption strategies alike [13] operate on a single candidate triple and define hardness in the embedding space of the model being trained rather than in the logical structure of the data, yielding trivially rejectable corruptions rather than near-misses that demand genuine reasoning. (iii) Forward-chaining materialization scales poorly on production-sized ontologies. Together, these shortcomings explain why no existing pipeline lets a practitioner go from “I have an ontology” to “I have a trained neurosymbolic reasoner” without substantial per-ontology data engineering.

4. Methodology: Targeted Data Generation

Existing pipelines force a practitioner to choose between a hand-built generator for a specific ontology and an ontology-blind “generate + materialize” approach whose output is shallow and full of trivial negatives, and neither restores the symbolic-reasoner contract. To close this gap, we introduce SYNTHOLOGY. Unlike standard pipelines, our generator designs an ABox through an ontology-guided backward-chaining proof search. In this way, we produce base facts driven by logical *necessity* rather than statistical randomness. The full codebase, including the generator, ontologies, and experimental setups, is open-source² under the MIT license.

SYNTHOLOGY accepts ontologies expressed in a supported subset of OWL 2 RL³. The subset covers the constructs that drive multi-hop inference chains in practice (subclass, subproperty, domain/range, inverse, property chains, symmetry, transitivity); other OWL 2 RL constructs can be added by extending the parser, without changes to the chainer. SYNTHOLOGY executes a three-phase pipeline. A *parsing* phase first compiles the ontology into a set of executable Horn rules $\mathcal{R}$, a schema $\mathcal{S}$ (classes, relations), and constraints $\mathcal{C}$. The *generation* phase (Algorithm 1) then executes a sample-level loop, in which backward-chaining proof search constructs proof trees from grounded rule conclusions. Finally, in the *validation* phase, each candidate graph is checked against $\mathcal{C}$ and injected with negative samples (Section 4.3).

The pipeline is modular by design: new OWL constructs can be supported by adding parser handlers and rule templates, without changing the chainer itself. Users can trade structural properties against generation cost via the configurations introduced in the rest of this section (π_{reuse} , the recursion cap, the depth-bucket weights, etc.), and the framework supports *sweeps* for systematic optimization of these parameters.

²<https://github.com/StreamIntelligenceLab/synthology>

³RDFS schema axioms (`rdfs:subClassOf`, `rdfs:subPropertyOf`, `rdfs:domain`, `rdfs:range`), OWL 2 property axioms `owl:inverseOf`, `owl:propertyChainAxiom`, `owl:SymmetricProperty`, `owl:TransitiveProperty`, and `owl:ReflexiveProperty`, and constraints `owl:IrreflexiveProperty`, `owl:AsymmetricProperty`, `owl:FunctionalProperty`, and `owl:disjointWith`.

4.1. Sample Construction

Data construction (Algorithm 1) assembles one knowledge graph per iteration. A random subset of rules $\mathcal{R}' \subseteq \mathcal{R}$ is selected. For each selected rule $r \in \mathcal{R}'$, k_r independent groundings of its head are created and passed to the backward chainer, which returns a set of proof trees P_r . We therefore refer to the k_r independent groundings of the rule head as *proof roots*. A configurable number of proofs is chosen per rule and merged into a global *proof map* M . This map links every atom appearing in the selected proofs with the list of derivations that produced it. M is then materialized into a candidate KG, completed with class memberships implied by the domain and range declarations of its triples (*schema completion*). This candidate KG is then validated and injected with negative samples.

We have to be careful, since the proof search space grows combinatorially: each premise may admit multiple sub-proofs, and combining them requires a Cartesian product. This repeats recursively down the tree. SYNTHOLOGY therefore uses *bounded* proof engineering: the global proof depth, the number of times a recursive rule may be re-applied, and the number of proof trees retained per atom are all capped.

A key mechanism for producing connected, non-trivial graphs is the *individual reuse pool*. When the chainer binds an unbound variable, it reuses an existing pool member with probability π_{reuse} . If this reused individual violates domain/range constraints, a fresh individual is created and added to the pool. This controlled individual sharing causes derivations to overlap, resulting in interconnected graphs rather than isolated proof trees. The reuse probability is exposed as a configurable hyperparameter, allowing users to trade graph density against derivation diversity for their target ontology.

Because each iteration of the outer loop generates one knowledge graph independently, the memory footprint of generation is bounded by a single sample rather than by the cumulative deductive closure. This is the structural property that lets SYNTHOLOGY scale where forward-chaining materializers cannot.

4.2. Backward-Chaining Proof Search

For each of the candidate rules $p_1 \wedge \dots \wedge p_n \rightarrow h$, the chainer first checks the per-rule recursion cap, and it prunes the branch if the rule's head h is the inverse of the parent predicate (to eliminate infinite inverse loops). Then, the chainer unifies the goal g with h , binds remaining variables (via the reuse pool), and recursively proves the grounded premises $p_1, \dots, p_n$. At most P_{max} combinations of premise proofs are retained per goal. If no rule succeeds, the chainer falls back to a base proof, ensuring every goal predicate has at least one derivation. To counteract the dominance of shallow schema proofs, SYNTHOLOGY supports *depth-stratified sampling*, which distributes retained proofs across easy ($d \leq 1$), medium ($2 \leq d \leq 4$), and hard ($d \geq 5$) buckets using normalized weights. An optional *signature sampling* step additionally groups proofs by their structural signature (the multiset of rule names in the tree) and keeps one representative per signature. This reduces near-isomorphic proofs. The resulting dataset has a strict base-versus-inferred fact distinction, together with derivation metadata for each triple.

4.3. Proof-Based Negative Sampling

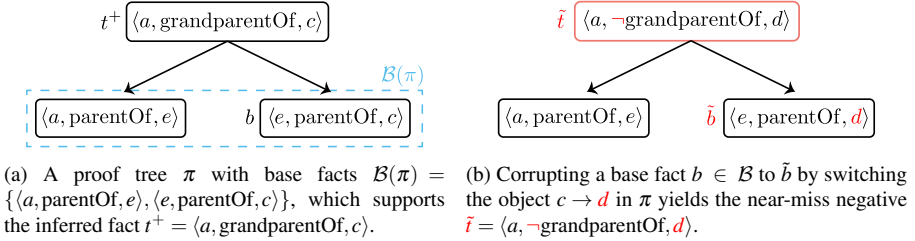
Random and constrained corruption (Section 2.3) both operate on a single triple in isolation and are therefore blind to the derivation that produced it. Whereas the static, dy-

Algorithm 1 Sample Construction**Require:** Rules $\mathcal{R}$, chainer $\mathcal{B}$, target count N_S , configuration Θ **Ensure:** Sample set $\mathcal{D}_S$

```

1:  $\mathcal{D}_S \leftarrow \emptyset$ 
2: while  $|\mathcal{D}_S| < N_S$  and attempt budget not exhausted do
3:   Reset proof map  $M \leftarrow \emptyset$  and individual pool
4:   Select rule subset  $\mathcal{R}' \subseteq \mathcal{R}$ 
5:   for each rule  $r \in \mathcal{R}'$  do
6:     Sample root count  $k_r \sim \mathcal{U}[k_{\min}, k_{\max}]$ 
7:      $P_r \leftarrow \bigcup_{i=1}^{k_r} \text{PROVE}(g_i, 0, \emptyset)$ , with  $g_i$  a fresh grounding of  $r$ 's head
8:     Merge up to  $n_r$  proofs from  $P_r$  into  $M$ 
9:   end for
10:  if  $M = \emptyset$  then continue
11:  end if
12:  Convert  $M$  to a candidate KG; apply schema completion
13:  Perform validation checks
14:  Add negatives using the configured strategy (Section 4.3)
15:   $\mathcal{D}_S \leftarrow \mathcal{D}_S \cup \{\text{KG}\}$ 
16: end while

```

**Figure 1.** Proof-based corruption in action.

namic, and adversarial strategies of Section 3.5 search for hard negatives in the model's embedding space, SYNTHOLOGY introduces *proof-based* negative sampling, which targets the *logical support* of inferred facts. It does this by corrupting a base fact and propagating the error upward along its proof tree.

Let a positive target triple t^+ have at least one proof tree π , and let $\mathcal{B}(\pi)$ be the base facts used to derive t^+ . The proof-based strategy samples a base atom $b \in \mathcal{B}(\pi)$ and perturbs its subject or object, yielding $\tilde{b}$. We note that $\tilde{b}$ is *not* added to the input KG; it exists only as a *conceptual* corruption for generating the negative target. The subject or object substitution is then propagated through the derivation chain, producing a logically flawed goal atom $\tilde{t}$. This triple is only accepted as a negative if it does not coincide with an existing positive triple in the graph.

Consider the rule $\langle X, \text{parentOf}, Y \rangle \wedge \langle Y, \text{parentOf}, Z \rangle \Rightarrow \langle X, \text{grandparentOf}, Z \rangle$ and base facts $\langle a, \text{parentOf}, e \rangle, \langle e, \text{parentOf}, c \rangle$, for example. Figure 1 shows the proof tree π that derives the inferred fact $t^+ = \langle a, \text{grandparentOf}, c \rangle$. If we corrupt the object of the second premise, we obtain $\tilde{b} = \langle e, \text{parentOf}, d \rangle$. Since the variable binding $Y = e$ remains intact, the rule still fires and propagates the error upward to $\tilde{t} = \langle a, \neg\text{grandparentOf}, d \rangle$.

The resulting false conclusion $\tilde{t}$ would be backed by a structurally perfect proof, except that $\langle e, \text{parentOf}, d \rangle$ does not exist in the input KG.

We now illustrate how a model should make a decision about $\tilde{t}$. Because $\tilde{b}$ does not appear in the actual KG, no valid proof of $\tilde{t}$ can exist within it. Yet, the proof structure itself remains sound, except for one corrupted leaf. Indeed, the derivation chain π would produce $\tilde{t}$ if $\tilde{b}$ were true. The model is thus trained to predict $\mathbb{P}(\tilde{t}) \approx 0$ in this case.

Our proof-based negative samples are *difficult* to learn: the model sees a structurally perfect proof, but must recognize that a single leaf premise is missing, therefore rejecting the conclusion. Learning to recognize and reject such near-miss negatives requires the model to traverse the full reasoning chain (and verify that each step is supported by the right facts), forcing *genuine* multi-hop reasoning rather than pattern matching.

To meet the configured negative ratio in finite time, the generator can fall back to constrained corruption when proof-based corruption fails. In practice, however, fallback rarely occurs for dense ontologies (such as our family tree ontology⁴), and the dataset consists almost entirely of hard proof-based negatives.

Finally, SYNTHOLOGY also provides a *mixed* corruption strategy, blending proof-based negatives with constrained corruptions in a user-specified ratio. The intuition is that simpler structural negatives could complement the learning of more complex proof-based ones. Whether this hypothesis holds in practice is examined in Experiment 1.

5. Experimental Setup

We evaluate SYNTHOLOGY along two axes. First, we measure properties of the generated data itself: hop distribution, predicate coverage, generation time, and memory footprint. Second, we measure the downstream effect on a neurosymbolic reasoner trained on that data, using an RRN [14] as the evaluation model. Three experiments test three claims: (i) Experiment 1 isolates the effect of proof-based negative sampling against random and constrained baselines on a fixed ontology; (ii) Experiment 2 compares full SYNTHOLOGY generation against a UDM baseline; (iii) Experiment 3 evaluates generalization to a richer benchmark ontology (OWL2Bench) and probes the scalability ceiling of forward-chaining materialization.

RRN training used a single NVIDIA Tesla V100 GPU (32 GB RAM); dataset generation and scalability experiments ran on a CPU node of an LSF HPC cluster (Intel Xeon Gold 6142, 2.60 GHz, 16 cores).

5.1. The Unguided Deductive Materialization (UDM) Baseline

We treat UDM as a family of approaches rather than a single fixed dataset: an ABox is first constructed under the constraints of the given input ontology, either (1) randomly or (2) procedurally (e.g. via a benchmark). After the ABox is generated in one of the two ways, a symbolic reasoner (in our case Apache Jena [4]) materializes its deductive closure to obtain positive targets. Negatives are then added via 1:1 random subject/object corruption applied uniformly to *all* positive triples.

For Experiment 2, we use the random ABox variant (1): we create an individual pool per sample, assign each individual a random set of ontology classes, and create object-

⁴An OWL 2 RL translation of the `family-tree` ASP program from [14].

property facts by uniformly sampling predicates (under the schema constraints) until the configured base-fact budget is reached. Experiment 3 uses the procedural variant (2), instantiating the ABox via the OWL2Bench Java generator. The Jena reasoning profile differs per experiment: Experiment 2 uses a lightweight profile focused on property-chain inferences (avoiding shallow derivations like disjointness, to isolate deep multi-hop reasoning), while Experiment 3 uses a more expressive profile supporting the richer OWL 2 RL semantics OWL2Bench requires.

To ensure a *matched budget*, we balance both the base-fact count and the positive inferred-target count between the two pipelines. Both generators are configured to exceed the desired budget, after which a post-hoc balancing script trims samples in generation order until the cap is reached. In practice, the resulting budgets agree to within a small margin (see Table 3). Because Jena does not natively provide proof-depth distributions, we compute hop depth via a custom derivation-tracing pass over inferred facts, which introduces additional runtime overhead in Experiment 2 beyond closure computation itself (see Table 4).

For Experiment 3, this derivation logging is disabled during the full-scale baseline generation ($u = 14$ universities) due to its memory cost (Section 6.3). Instead, it is enabled only in a smaller run ($u = 3$ universities), whose hop distribution serves as a representative proxy: OWL2Bench university ABoxes are independent fragments over the same TBox, so the same rules fire at the same proof depths regardless of u . All exact SYNTHOLOGY configurations and RRN hyperparameters are provided in the open-source repository.

5.2. The Evaluation Model: Recursive Reasoning Network

Because the original RRN publication did not provide an official source-code release, we reimplemented the architecture from scratch. To fully utilize GPU parallelization, we grouped message-passing updates into batches. The reimplementation preserves the original architecture and learning objective and reduces training time drastically compared to the unbatched version, although exact convergence parity with the original's reported family-tree figures (99.9% accuracy / 0.976 F1 on inferable relations [14]) was not established within our compute budget of 24 h GPU time.

The RRN outputs a probability score $P := \mathbb{P}(\langle s, p, o \rangle) \in [0, 1]$ indicating the likelihood that the target fact $\langle s, p, o \rangle$ is entailed by the base facts. We evaluate it as a binary classifier and report PR-AUC and AUC-ROC as threshold-independent metrics, alongside F1-score and the False Positive Rate (FPR) at threshold $P > 0.5$. We treat PR-AUC and FPR as the primary metrics, because the central claim of this paper concerns robustness against logically false near-miss negatives, which is a property that PR-AUC and FPR penalize more aggressively than AUC-ROC [26]. Per-class accuracy (Acc^+ , Acc^-) is reported for completeness in Table 1, because it helps diagnose the conservatism or permissiveness of each model.

5.3. Experiment 1: Negative Sampling Strategy Evaluation

Using the family tree ontology, we generate SYNTHOLOGY variants that differ only in the negative-sampling strategy: standard random corruption, constrained random corruption, pure proof-based corruption, and a mixed strategy (blending 50% constrained and

Table 1. RRN predictive relational performance for Experiments 1 and 2. (†) = UDM-generated test set.

Exp.	Dataset	PR-AUC (†)	AUC-ROC (†)	FPR (↓)	F1 (†)	Acc ⁺ (†)	Acc ⁻ (†)
1	Random	50.7%	0.770	20.9%	0.527	58.7%	78.8%
	Constrained	53.6%	0.830	37.7%	0.606	93.5%	62.1%
	Proof-based	54.6%	0.843	36.2%	0.628	96.0%	63.4%
	Mixed	46.6%	0.730	63.8%	0.490	94.9%	36.6%
2	UDM	38.8%	0.594	54.1%	0.457	69.7%	45.9%
	SYNTH.	60.3%	0.825	34.6%	0.636	86.3%	65.1%
2†	UDM	41.8%	0.643	45.3%	0.483	63.3%	54.7%
	SYNTH.	42.9%	0.628	25.7%	0.419	41.2%	74.3%

50% proof-based corruption). Each variant consists of $N_{\text{train}} = 2000$ training, $N_{\text{val}} = 500$ validation, and $N_{\text{test}} = 500$ test knowledge graphs. A separate RRN is trained on each variant and evaluated on the same held-out test set, rich in near-miss targets.

5.4. Experiment 2: Multi-Hop Quality

To test multi-hop reasoning capability, we use the same family tree ontology, since it enables deep inference chains. An RRN trained on the UDM baseline (referred to as RRN_{UDM}) is compared against an RRN trained on SYNTHOLOGY data (referred to as RRN_{Syn}) under the matched-budget protocol described in Section 5.1. The sizes of the input graphs were balanced to have $\sim 200k$ input facts per dataset (see Table 3). Evaluation is performed on two test sets. On the one hand, we evaluate on a held-out test set, generated once by SYNTHOLOGY with elevated recursion/depth limits (reused for both models). On the other hand, to eliminate any potential bias from the SYNTHOLOGY-generated test set, we also evaluate on a test set generated by UDM.

Completely random ABox generation is deliberately the *worst case* for the UDM pipeline, and thus the hardest setting for producing deep inference chains; Experiment 3 re-runs the comparison against a strong, benchmark-provided ABox generator to test the gap from the opposite direction.

5.5. Experiment 3: Generalization to Complex Ontologies

To demonstrate generalization to more complex ontologies, and to validate the ontology-agnostic claim of SYNTHOLOGY, we evaluate both generators (ours and the UDM approach) on the OWL2Bench benchmark ontology. For the UDM baseline, the OWL2Bench Java generator instantiates a university ABox for u universities, Apache Jena materializes its deductive closure, and the resulting graph is partitioned via BFS (initialized at random individuals and capped at k individuals per subgraph) into sample-sized knowledge graphs. Inferred triples that span different subgraphs are discarded.

6. Results and Discussion

6.1. Experiment 1: Negative Sampling Strategy Evaluation

Table 1 reports RRN performance for the four negative-sampling strategies. Proof-based sampling achieves the best score on four of six metrics that capture genuine ranking

Table 2. Per-hop RRN relational-triple performance on two test sets. Hop ≥ 3 is omitted for the UDM test set because Jena produced no negative targets at that depth. **Bold:** best per column within each test-set panel.

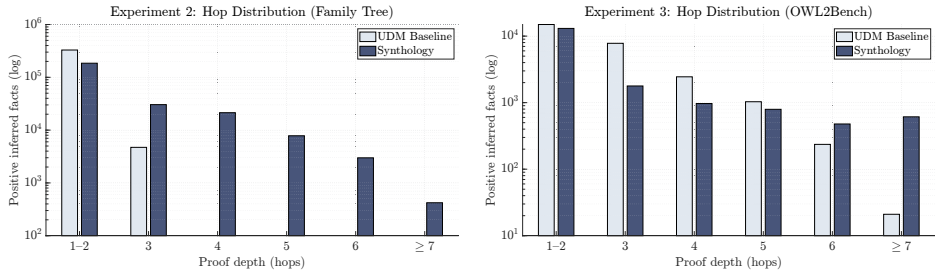
Test set	Method	Hops	Triple Acc. ($\uparrow$)	F1 ($\uparrow$)	FPR ($\downarrow$)
SYNTHOLOGY (2)	UDM	1	42.1%	47.1%	76.9%
		2	69.4%	49.6%	22.6%
		≥ 3	58.5%	43.2%	40.7%
	SYNTH.	1	81.5%	73.9%	21.4%
		2	74.9%	57.4%	17.0%
		≥ 3	79.5%	69.3%	20.5%
UDM-generated (2 $\dagger$)	UDM	1	57.1%	55.4%	45.3%
		2	60.0%	59.3%	44.3%
	SYNTH.	1	65.4%	56.2%	22.3%
		2	57.8%	42.6%	24.5%

and classification quality (PR-AUC, AUC-ROC, F1, Acc^+), with substantial gains over random corruption and smaller but reliable gains over constrained corruption. Random corruption wins on FPR and negative triple accuracy (Acc^-), but reading them in isolation is misleading. The random-trained model achieves Acc^- of 78.8% but only 58.7% Acc^+ . This reflects a classifier that learned *conservatism* rather than reasoning: it rejects most uncertain triples, which trivially produces a low FPR and a high Acc^- , but at the cost of missing a large fraction of true entailments. In contrast, both constrained and proof-based models exhibit the inverse profile: high positive accuracy ($\sim 93\text{--}96\%$) with moderate negative accuracy ($\sim 62\text{--}63\%$). This indicates that they successfully identified most true entailments, but occasionally accepted logically false near-misses, since these are hard to learn by design.

The mixed strategy performs worst overall, falling below even the random baseline on PR-AUC, AUC-ROC, and F1, and yielding a negative accuracy of only 36.6%. This stems from training-signal inconsistency: within a single batch, the model encounters negatives whose detection requires different reasoning (shallow type-compatibility checks *and* deep proof-chain tracing). This prevents a stable decision boundary from being established. Rather than benefiting from the complementary signals, the network oscillates between two incompatible objectives and converges to an aggressive positive bias. This bias is reflected in Table 1: the mixed strategy has a large false positive rate of 63.8%, the worst of all four strategies. A *blending schedule* may implement the complementary-signal method better than the naive 50/50 mix and is worth exploring in future work.

6.2. Experiment 2: Multi-Hop Quality

Despite matched generation budgets, the UDM baseline and SYNTHOLOGY datasets show substantial differences in both reasoning complexity and structural composition. Table 3 summarizes the topological differences. Both methods produce comparable input-graph sizes (225k vs. 206k input facts, an 8.4% gap) and an almost identical total target volume ($\sim 640\text{k}$). The UDM baseline produced a larger number of inferred targets (333k vs. 249k for SYNTHOLOGY). However, as illustrated in Figure 2a, this volume is deceptive: UDM produced an overwhelming majority of its inferences at depth $d \leq 2$, with fewer than 1% of its facts requiring paths of $d \geq 3$ (only 4,716 inferences). In contrast, SYNTHOLOGY engineered a robust distribution of deep multi-hop proofs, creating

(a) Experiment 2: *family tree*.

SYNTHOLOGY outperforms the UDM baseline, producing a deep hop distribution, in contrast with the baseline, which produces 99.3% 1–2 hop facts and only 4,716 facts with ≥ 3 hops.

(b) Experiment 3: *OWL2Bench*.

SYNTHOLOGY creates a deeper hop distribution than the UDM baseline, which gets a ‘head start’ because of the sophisticated OWL2Bench ABox (Section 6.3).

Figure 2. Distribution of positive inference hop depths in the SYNTHOLOGY and UDM datasets for Experiment 2 (family tree) and Experiment 3 (OWL2Bench). The Experiment 3 UDM baseline distribution is measured from the 3-university run, used as a proxy because full-scale derivation logging is infeasible (Section 5.1).

60,310 inferences at $d \geq 3$ and reaching logical chains up to $d = 7$. UDM’s higher inferred target count is thus simply the result of a combinatorial explosion of shallow, trivial inferences, whereas SYNTHOLOGY’s budget is allocated toward deeper derivations.

Furthermore, the UDM baseline’s unguided generation entirely failed to materialize inferences for 18 relations that are either complex (e.g. `secondUncleOnceRemovedOf`) or gender-specific, resulting in exactly zero inferred targets for these predicates. This highlights a critical flaw in relying on unguided materialization: structurally demanding or highly constrained logical derivations occur only incidentally, making the UDM dataset systematically deficient. SYNTHOLOGY achieved complete predicate coverage, confirming that it proactively constructs the prerequisite evidence (base facts) for all defined logical goals (inferred facts).

The influence of these structural differences on model performance is reflected in Table 1. On the SYNTHOLOGY-generated test set, RRN_{Syn} (trained on SYNTHOLOGY data) substantially outperforms RRN_{UDM} , the UDM-trained RRN. PR-AUC improves by 21.5 pp, AUC-ROC by 0.23, and FPR drops by 19.5 pp. To control for test-set bias, we separately evaluated both models on a UDM test set, too. This second test set reveals that RRN_{Syn} ’s advantage is *partially* inflated by test-set familiarity.

On UDM data, PR-AUC converges nearly to the same value (42.9% vs. 41.8%), and AUC-ROC becomes almost identical for both models (0.643 for RRN_{UDM} vs. 0.628 for RRN_{Syn}). Yet crucially, the FPR advantage proves robust: on UDM data, RRN_{Syn} maintains substantially lower FPR (22.3% vs. 45.3% at hop 1 and 24.5% vs. 44.3% at hop 2), as can be seen in Table 2. The same table shows that on the UDM test set, RRN_{Syn} outperforms on triple accuracy at hop 1 (65.4% vs. 57.1%), where both models receive similar input context. At hop 2, RRN_{UDM} exhibits higher F1 (59.3% vs. 42.6%), but this is entirely due to over-prediction: as noted before, its FPR is 44.3%.

These findings indicate that the SYNTHOLOGY test set overstates the PR-AUC and AUC-ROC advantages due to structural familiarity, while the FPR advantage (reflecting the model’s learned precision in rejecting near-miss negatives) generalizes to test data generated by a different system. Finally, we note that SYNTHOLOGY generates the dataset roughly three times faster than UDM at this matched scale (11 min vs. 30 min). Table 4 shows the per-stage breakdown.

Table 3. Comparison of topological dataset statistics across generation methodologies. While the overall generation budgets (base+inferred targets) are matched to $\sim 640k$ triples, and the context graph sizes (input facts) are equalized within a 10% margin via sample trimming, the internal distribution of base versus inferred targets inherently diverges due to the structural differences produced by forward versus backward chaining.

Dataset	Method	Avg #/KG		Input Facts	Base Targets			Inferred Targets			Ratio $\hat{\rho}_{\pm}$
		Base	Inf.		Total	Pos	Neg	Total	Pos	Neg	
Family tree exp. 2	UDM	187.9	532.8	225k	306k	159k	147k	333k	160k	173k	0.998
	SYNTHOLOGY	172.3	532.8	206k	391k	207k	184k	249k	113k	136k	0.998
OWL2Bench exp. 3	UDM	67.4	210	14.5k	18.8k	6.1k	12.7k	26.4k	16.7k	9.7k	1.020
	SYNTHOLOGY	252	824	13.9k	27.7k	13.9k	13.8k	17.6k	9.0k	8.6k	1.020

Table 4. Comparison of computational time breakdown across generation methodologies. *Base Gen.* denotes the initial procedural or random ABox instantiation. *Inference* captures the Java-based forward-chaining materialization (Jena) or backward-chaining search (SYNTHOLOGY). *Overhead* includes BFS subgraph extraction, derivation-tracing for hop-depth computation, and negative sampling mechanics. At $u = 14$ ($\sim 664k$ base facts) only end-to-end wall-clock time was recorded: UDM’s figure is a time-to-failure (out of memory), not directly comparable to SYNTHOLOGY’s time-to-completion.

Dataset	Method	Base Gen.	Inference	Overhead	Total Time
Family tree exp. 2	UDM	<1 min	26 min	2 min	30 min
	SYNTHOLOGY	/	10 min	1 min	11 min
OWL2Bench ($u = 3$) exp. 3	UDM	<1 min	<1 min	4 min	6 min
	SYNTHOLOGY	/	2 min	<1 min	2 min
OWL2Bench ($u = 14$) exp. 3	UDM	–	–	–	OOM at 7.5 h (fail)
	SYNTHOLOGY	/	–	–	70 min (1M targets)

6.3. Experiment 3: Generalization to Complex Ontologies

Hop Distribution. Unlike the *random* ABox generation used in Experiment 2, the UDM baseline in Experiment 3 uses the OWL2Bench Java generator to *procedurally* instantiate realistic university structures, giving the forward-chainer a naturally deeper, more connected starting point. Despite this favorable setup, SYNTHOLOGY still produces a deeper hop distribution: Figure 2b shows a heavier tail of deep inferences than the benchmark-seeded baseline, demonstrating its ability to construct complex inferred facts even on richer schemas.

Dataset Topology. Table 3 shows the structural patterns from Experiment 2 persist at scale. Both generators were constrained to a nearly identical input graph size (14.5k vs. 13.9k input facts) and an equivalent total target count ($\sim 45k$), but the internal split differs: UDM produces 26.4k inferred targets against 18.8k base targets, while SYNTHOLOGY produces 17.6k vs. 27.7k. The structurally rich OWL2Bench ABox triggers a wave of shallow rule executions that dominates UDM’s inference budget, whereas SYNTHOLOGY’s backward-chaining allocates its budget to the specific base evidence its targeted proofs require, yielding a leaner set of inferred targets with a heavier deep-hop tail.

Scalability: the Reasoning Wall. Next, we investigate how each generator scales with ontology size. UDM iteratively saturates the ABox via forward-chaining and must hold every derived fact in memory. Because OWL 2 RL reasoning can be mapped to Datalog [15], its materialization inherits Datalog’s data complexity: in the worst case, the closure size scales polynomially as $O(n^v)$, where n is the number of individuals and v is the maximum number of distinct variables in any rule body [27]. SYNTHOLOGY, by con-

trast, constructs each KG independently, so the per-sample memory footprint is constant in n .

As Table 4 shows, at small scale ($u = 3$ on OWL2Bench, roughly 10k base facts) both approaches complete in comparable time. The gap becomes severe at large scale ($u = 14$, ~664k base facts): Jena’s engine exhausts approximately 112 GB of JVM heap (across 8 cluster nodes) after roughly 7.5 hours, failing to terminate under resource constraints. SYNTHOLOGY generates one million target triples over the same ontology in under 70 minutes on a *single* node of the same cluster, with memory usage bounded by the per-sample proof-depth cap rather than cumulative closure size.

Experiment 3 thus shows that SYNTHOLOGY preserves its structural advantages (deeper hop distribution, complete coverage) on a richer ontology while scaling past the point where forward-chaining materialization fails – with no ontology-specific engineering: the same pipeline that handled the family tree handled OWL2Bench, restoring the scalability the symbolic-reasoner contract requires. Backward-chaining is not universally faster than forward-chaining, but for producing a fixed-size training set for a neurosymbolic reasoner, sampling proofs is a better fit than materializing the full closure.

Generalization Beyond the RRN. Multi-hop coverage by construction, proof-based near-miss negatives, complete predicate coverage, and per-sample bounded memory are evidenced directly through dataset-level metrics in Sections 6.2 and 6.3, independent of which model is trained on the data. We selected the RRN because it is explicitly tailored to multi-hop inference – the architecture for which the marginal value of multi-hop training data is *hardest* to demonstrate. Architectures with weaker inductive biases for multi-hop chains would, if anything, benefit *more* from such data. Confirming this across a broader range of architectures is left for future work.

7. Conclusion & Future Work Directions

Neurosymbolic reasoners break the symbolic-reasoner contract by requiring a training dataset that, for most ontologies, does not exist, and existing synthetic pipelines produce such data only incidentally. We presented SYNTHOLOGY, an ontology-agnostic synthetic data generator that engineers training data by construction: targeted backward-chaining proof search guarantees that every positive target carries a deep proof, and proof-based negative sampling derives near-miss negatives by perturbing a single leaf of one of those proofs. Across two academic benchmark ontologies (family tree and OWL2Bench), SYNTHOLOGY delivers consistent gains over UDM baselines under matched budgets (most robustly in rejecting near-miss negatives; ranking-metric gains narrow on an independently generated test set, Section 6.2), complete predicate coverage where UDM leaves entire relation families uninferred, and tractable generation at scales where forward-chaining materialization fails.

This work can be extended in several directions: proof-tree-based benchmarking of neurosymbolic reasoners with controlled hop depth and calibrated negatives; evaluation on architectures beyond the RRN and a wider range of ontologies; incorporating remaining OWL 2 RL constructs and other OWL 2 profiles; validation on production knowledge graphs; and graduated-curriculum strategies for combining negative-sampling signals.

Declaration on Generative AI

GitHub Copilot, Google Gemini, and Claude were used to automate cumbersome coding tasks and help with paraphrasing and rewording of the paper. The authors have reviewed and edited the content as needed and take full responsibility for the publication's content.

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Applied & Domain-Specific Knowledge Graph Systems

From File Systems to Knowledge Graphs: A Multi-Phase Workflow for the Description of Born-Digital Literary Archives

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Abstract. Background: Born-digital personal archives present novel descriptive challenges that conventional archival methods are not equipped to address. Their scale, format heterogeneity, and multi-layered stratification — physical, logical, and conceptual — require a representational framework capable of capturing document structure, provenance, integrity, and relational contexts. The increasing volume of born-digital archives received by cultural heritage institutions makes the development of automated and scalable description methodologies an urgent practical priority. Methods: This paper presents a replicable five-phase pipeline, grounded in the Born-Digital Ontology (BoDi) — a formal extension of Records in Contexts Ontology (RiC-O) integrating PREMIS, PROV-O, and LRMoo — that transforms born-digital personal archive directories into RDF knowledge bases. The phases are: (1) file system ingestion, comprising SHA-256 integrity verification, parallel multi-tool metadata extraction, and triplestore loading; (2) SPARQL-based validation and rule-based enrichment, including cross-tool metadata alignment, document classification, and duplicate detection; (3) hybrid enrichment combining expert bibliographic curation with automated LLM-based description generation; (4) chain-of-custody reconstruction; and (5) selective entity-level redaction for privacy-aware dissemination. Results: Evaluated on the Valerio Evangelisti Archive — 78,211 files and 11,135 folders totaling 2.1 TB across three storage media — the pipeline generated 61,154,322 RDF triples across 13 named graphs with minimal manual intervention. The resulting knowledge base supports querying across structural, typological, temporal, and relational dimensions, enabling analyses — from duplicate detection across storage media to correlation of archival activity with editorial output — that conventional finding aids cannot support. Conclusions: Born-digital archives present a double challenge: their fragility raises preservation concerns, while their scale makes manual description almost impossible. The pipeline addresses both by inverting the framing: the properties that make born-digital materials resistant to conventional methods become, within a formal framework, sources

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of descriptive richness without equivalent in the analog domain. Ontological modeling combined with automated processing and human-in-the-loop validation offers a viable and empirically demonstrated path toward semantic description of born-digital cultural heritage at scale.

Keywords. born-digital archives, knowledge graphs, RDF, semantic enrichment, provenance, retrieval-augmented generation, archival description, BoDi ontology, Records in Contexts

1. Introduction

Contemporary authorship generates complex, distributed, and stratified documentary ecosystems. Resulting digital materials — manuscripts at successive revision stages, email correspondence, research notes, image libraries, executable scripts, and compressed archives — accumulate across multiple digital storage media over decades, often without deliberate organizational principles and in the absence of any preservation plan [1]. This is particularly true of literary archives, where the documentary ecosystem reflects decades of layered creative activity whose documentary significance is inseparable from its temporal and relational complexity [2]. Born-digital personal archives therefore reach collecting institutions in a state of minimal pre-arrangement, placing the full descriptive burden on archivists and digital preservation specialists.

Traditional archival approach frames the digital as an obstacle, interpreting the technical complexity of the medium for a descriptive problem — a perspective that holds only when born-digital archives are viewed through the lens of analog methodologies. In reality, born-digital archives are computational objects whose digital nature enable automated description with a level of granularity that paper records cannot support. Unlike analog documents, digital records carry intrinsic descriptive layers — format signatures, system metadata, hierarchical file system topologies — that can be accessed and extracted directly [3]. The hierarchical structure of the file system, moreover, encodes organizational and topological relationships that can be captured with full fidelity — something that analog finding aids can only approximate through reconstruction. Born-digital materials carry extensive machine-readable traces of their production and organization. The archival challenge is therefore not to generate descriptive metadata *ex nihilo*, but to extract, formalize, and semantically enrich the information already available in the digital record.

Translating this reframing into practice requires both a robust representational model and a scalable computational framework. Current standards adopted by Libraries, Archives and Museums communities [4,5,6,7,8] still leave a significant gap between accurate technical information and archival description; and a single author's digital heritage may span tens of thousands of files, making manual entity-level description unfeasible at scale.

This paper addresses the following research question: *how can the intrinsic properties of digital materials be leveraged to automate and facilitate archival description processes?* To the best of our knowledge, existing approaches address this question only partially, lacking the integration of a robust representational model and a scalable, reproducible pipeline within a single framework. We address this question with three contributions: (1) a five-phase, openly available pipeline that transforms born-digital archive di-

rectories into RDF knowledge graphs with minimal manual intervention, grounded in the Born-Digital Ontology (BoDi)²; (2) a hybrid enrichment methodology combining rule-based inference, expert curation, and local LLM-based generation; and (3) a large-scale empirical validation on a real-world archive. The pipeline covers structure and metadata extraction and validation, semantic enrichment, automated description generation, chain-of-custody reconstruction, and privacy-preserving redaction.

The remainder of this paper is organized as follows. Section 2 reviews representational requirements, existing models, and the Born-Digital Ontology as the framework that underpins the pipeline. Section 3 describes the phases of the pipeline. Section 4 reports the experimental evaluation on the Valerio Evangelisti Archive. Section 5 discusses findings and limitations. Section 6 concludes and outlines the directions for future work.

2. Background and Related Work

Representing born-digital personal archives requires addressing a set of challenges that arise from the specific nature of digital objects and contemporary documentary production [9]. Born-digital materials exhibit an inherently stratified nature: following Thibodeau [10], every digital object manifests three distinct, yet interconnected, levels of abstraction — physical (bit sequences on storage hardware), logical (file system structure), and conceptual (informational content) — each of which must be explicitly represented. Beyond stratification, digital integrity cannot be assumed and must be cyclically verified: the cryptographic hash of a file provides an objective, machine-verifiable record of its bit-level identity at a given point in time [11]. At the same time, born-digital files accumulate heterogeneous technical metadata automatically during their life-cycle, generated by systems, applications, and users across different technological environments [12,13] — a layer of information whose variability and granularity cannot be anticipated in advance, requiring a flexible, tool-agnostic representational approach. Finally, the provenance of digital materials is multidimensional, encompassing technical, procedural, and curatorial dimensions [14]: the full chain of actors, tools, and operations that produced, transformed, and managed the archive must be documented, as must the relational contexts — bibliographic, social, intellectual — that situate documents within the broader ecosystem of authorial production.

These requirements find a natural match in Linked Data, which the international archival community has progressively adopted as its representational paradigm [15,4] for its flexibility, relationality, and extensibility. The Records in Contexts conceptual model (RiC-CM) and its OWL ontology (RiC-O) [4] represent the most advanced vocabulary for archival linked data, consolidating ISAD(G) and ISAAR(CPF) within a graph-based framework capable of representing complex networks of archival entities, agents, and contexts. RiC-O is progressively affirming itself as the new international reference standard for archival description. Alongside it, domain-specific ontologies such as ArCo [8] and ArchOnto [16] address particular institutional contexts, while the digital preservation community relies on PREMIS [5] for technical metadata and fixity at the object level.

Nevertheless, none of these models satisfies the above requirements simultaneously. As a general-purpose archival ontology, RiC-O does not natively implement the techni-

²<http://w3id.org/bodi>

cal layers required for born-digital description — a gap that EGAD itself acknowledges, noting that RiC-O “is designed with the expectation that it may be extended using cross-domain ontologies specifically addressing technical data needed for the management of instantiations” [4]. PREMIS addresses fixity at the preservation object level, but lacks the conceptual archival description layers needed to situate digital objects within archival, biographical, and intellectual contexts. ArCo is scoped to the Italian cultural heritage context and does not address born-digital peculiarities properly; CIDOC CRM, while internationally adopted, is designed for museum-oriented object description and similarly leaves some of the born-digital technical characterization unaddressed.

The same gap persists at the operational level. Tools such as Archivematica³, BitCurator⁴, and format identification utilities such as DROID⁵ solve the problems of archive ingest, fixity verification, format normalization, and OAIS-compliant packaging. Their output is serialized partly in PREMIS, but its scope is deliberately bounded to preservation-relevant properties, leaving the broader technical characterization of digital objects outside its representational range. More fundamentally, these systems are oriented toward conservation and access rather than description. They identify a digital object and whether it has been altered, but they do not define the contents, how it relates to other objects, or where it sits within the archival, biographical, and intellectual context of its creator.

Closer to the domain of born-digital literary archives, the Pavia Archivi Digitali (PAD) system [1] implements automated metadata extraction from multiple tools and provides browsing functionality, but stores its data in a relational database without modeling according to archival ontologies or standard vocabularies, limiting its capacity for querying and interoperability. The present work proceeds from the preservation layer into the descriptive domain: from identifying the digital object, to the semantic characterization of its contents, to the relation with other objects, and finally where it sits within the archival, biographical, and intellectual context of its creator — a representational depth that preservation systems are not designed to reach.

The Born-Digital Ontology (BoDi) was developed within the broader research program underlying this work to fill this gap; it is taken here as a premise, established prior to and independently of the present contribution. Taking RiC-O as its reference framework, BoDi selectively integrates PREMIS for fixity and storage medium representation, PROV-O [17] for the provenance of processing operations, and LRMoo [6] for the bibliographic layer connecting archival records to literary works. Where these vocabularies do not cover the representational requirements of born-digital description, BoDi introduces new classes and properties grounded in three design principles. First, extracted metadata values are represented as explicit entities in the knowledge graph, while metadata fields themselves (e.g., the File Size) are modeled as entities rather than just scalar attributes.

This design enables provenance tracking, semantic clustering, and cross-tool alignment at the level of metadata concepts while preserving the original terminology used by each extraction tool. Equivalent metadata fields extracted by different tools can therefore be semantically aligned to their original extraction context — a granularity that neither RiC-O nor PREMIS supports natively. The additional graph complexity is a delib-

³<https://www.archivematica.org/en/>

⁴<https://bitcurator.net/>

⁵<https://www.nationalarchives.gov.uk/information-management/manage-information/preserving-digital-records/droid/>

erate trade-off that enables provenance-aware comparison of heterogeneous extraction outputs.

Second, the provenance model is extended to cover computational agents — algorithms, software components, and composite software environments — alongside human agents, with explicit properties for documenting supervisory relationships in hybrid human-machine workflows; unlike RiC-O, whose provenance properties are scoped to archival entities, BoDi’s corresponding properties apply to any domain entity, enabling provenance tracking of computational outputs such as extracted metadata values and automatically generated descriptions.

As part of the broader research programme in which BoDi was developed, the ontology has been formally validated through RDF syntax checking, OWL 2 DL consistency reasoning (HermiT), and structural quality assessment (OOPS! [18]), confirming logical coherence at both the terminological and assertional levels [9].

The design principle is modularity: RiC-O remains fully usable as a standalone framework, and the imported ontologies are employed only where they directly address representational needs that RiC-O alone does not cover.

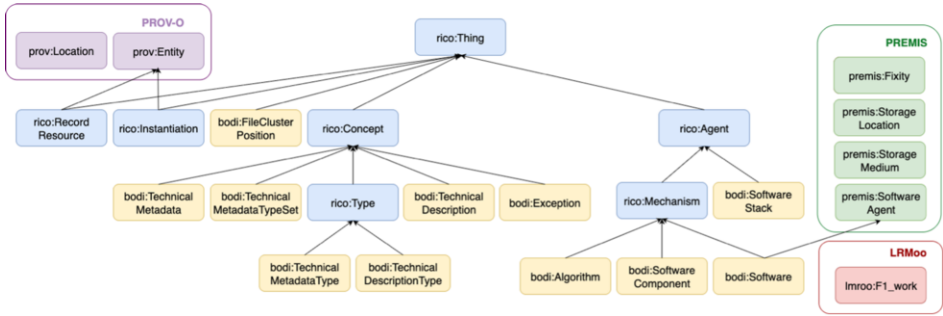


Figure 1. The Born-Digital Ontology (BoDi): modular integration of RiC-O, PREMIS, PROV-O, and LRMoo, with BoDi-native classes and properties highlighted.

3. The Five-Phase Pipeline

The pipeline takes as input one or more directories of a born-digital archive and produces as output a multi-graph RDF knowledge base conformant with BoDi. It is implemented in Python 3.12, orchestrated by a central script, and uses Blazegraph as the target triplestore, accessed via REST APIs. All data is serialised in N-Quads format, ensuring portability across any W3C-compliant triplestore.⁶

The pipeline is designed around three methodological commitments. First, (1) *integrity preservation*: every operation maintains the physical and logical integrity of the source materials through cryptographic verification and preventive backups. Second, (2) *reversibility and transparency*: all transformations are documented and traceable, connecting the files to their semantic representations through verifiable, reproducible processes. Third, (3) *standard compliance*: all generated representations conform to docu-

⁶The code is publicly available at https://github.com/LuciaGiagnolini12/FileSystem_to_BoDi.

mented models — BoDi, RiC-O, PREMIS, PROV-O, LRMoo — guaranteeing portability, interoperability, and reusability. Table 1 summarizes the five phases.

Table 1. Overview of the five-phase pipeline.

Phase	Key Process	Output
1. Archive Ingestion and RDF Baseline Construction	Read-only protection; SHA-256 integrity baseline; RDF conversion of file system hierarchy; parallel metadata extraction via three tools; SPARQL consistency and integrity verification	N structural + $3N$ meta-data named graphs; named graph registry; JSON hash registers
2. Validation & Rule-Based Enrichment	SPARQL validation suite (entity counts, hierarchical consistency, hash integrity, metadata coverage); rule-based enrichment: duplicate detection, cross-tool alignment, metadata clustering, type assignment, title and date formalization	Validated KG; named graph of duplicate links, cross-tool equivalences, type assignments, formalized titles and dates
3. Semantic Enrichment via Specialized Knowledge and Generative Models	Expert bibliographic mapping to literary work entities; deterministic SPARQL retrieval feeding a locally executed LLM for natural-language description generation	Work-entity named graph; named graph of natural-language descriptions with provenance
4. Chain of Custody Reconstruction	Physical media census; storage medium and location modeling; derivation relations between instantiations with temporal and agent anchoring	Named graph of storage media, locations, and derivation sequences
5. Selective Redaction for Publication	Configurable whitelist/blacklist policy by entity type; field-level screening of sensitive metadata; redaction status flag on all entities	Redacted KG with all entities flagged; full pre-redaction backup

3.1. Phase 1: Archive Ingestion and RDF Baseline Construction

Phase 1 transforms one or more file system directories into an RDF knowledge base. It is the only phase requiring direct access to the file system; all subsequent phases operate exclusively within the knowledge base. The phase proceeds through ten sequential steps, each completing across all target directories before the next begins; a dependency system blocks advancement unless all prerequisite artifacts are present, ensuring full traceability.

Read-only protection. All archive directories are set to read-only mode before any operation begins. The system automatically generates a directory-specific recovery script for each target, guaranteeing complete reversibility.

Integrity baseline. A SHA-256 cryptographic fingerprint (hash) is computed for every file using native OS commands (`sha256sum` on Linux). Results are serialized in structured JSON registers recording hash values, file sizes, timestamps, and processing statistics. This baseline serves three purposes: (1) attesting the state of the source mate-

rials at the moment of ingestion; (2) detecting any subsequent modification of those materials throughout the workflow; and (3) enabling identification of duplicate files across directories (see Phase 2).

RDF structure generation. The file system hierarchy is converted into a BoDi-conformant RDF representation. Files and folders are modeled as `rico:Record` and `rico:RecordSet` instances, respectively, each paired with a `rico:Instantiation` representing their filesystem realization and each situated in its hierarchical context according to its position in the file system. Hash values are recorded as `premis:Fixity` entities, file paths as `prov:Location` instances. The resulting triples are stored in a dedicated named graph per directory and loaded into the triplestore via the REST API.

Quality control. Two independent SPARQL-based checks are executed before proceeding to metadata extraction. A *consistency check* verifies that entity counts in the RDF graph match those obtained from Bash file system commands; an *integrity check* recomputes the SHA-256 values and verify the matching with those in the JSON registers of the Integrity baseline.

Multi-tool metadata extraction. Native technical metadata is extracted through three specialized tools. The Python `os` library⁷ extracts file system metadata (timestamps, permissions, inode identifiers). Apache Tika⁸ provides format-agnostic content metadata across over 1000 file formats, mapping extracted fields onto standard vocabularies including Dublin Core and XMP. ExifTool⁹ targets multimedia files, extracting deep technical parameters such as EXIF data, codec specifications, audio frequencies, and GPS coordinates. Each tool's output is stored in a dedicated named graph per directory, so that the provenance of every metadata value is always traceable to its source extractor. Running the three tools in parallel maximizes coverage: any surface properties the others may not capture, or cases where the same property is extracted by multiple tools with differing values, are preserved for cross-validation [3].

Post-extraction integrity verification. Once metadata extraction is complete, SHA-256 values of all source files are newly recomputed and compared against the Integrity baseline. Any discrepancy is flagged and logged, certifying that no source file was altered during processing.

Named graph registry. A final named graph formalizes the dependencies between all named graphs produced across directories and extraction tools. It functions as a centralized index that facilitates identification of graphs relevant to specific queries and supports targeted maintenance operations, preserving modularity while providing a navigable view of the full knowledge base.

3.2. Phase 2: Validation and Rule-Based Enrichment

Phase 2 operates exclusively within the semantic domain on the knowledge base produced by Phase 1. It comprises two sequential sub-processes: systematic validation of the graph's logical consistency, followed by rule-based enrichment that makes implicit knowledge explicit. All triples generated in this phase are stored in a dedicated named graph (`updated_relations`), keeping them separate from the Phase 1 baseline and ensuring complete reversibility.

⁷<https://docs.python.org/3/library/os.html>

⁸<https://tika.apache.org/>

⁹<https://exif.tools/>

Validation is conducted through a suite of 51 SPARQL queries, formulated as competency questions covering entity distributions and label coverage, structural and hierarchical integrity, metadata extraction coverage, and hash compliance and duplication. The set was derived directly from the entities and relations defined by the BoDi/RiC-O data model, each query targeting a specific structural or referential invariant that Phase 1 output is expected to satisfy; it is not claimed to be formally complete, but designed to cover the main risk categories in born-digital ingestion identified in Section 2.

Rule-based enrichment builds on the outputs of the informational queries to generate new triples through six operations that surface knowledge already latent in the data:

- (1) *Hash-based duplicate detection*: files sharing identical SHA-256 values across input directories are linked to each other surfacing duplication patterns and potential backup trajectories across media.
- (2) *Cross-tool metadata alignment*: metadata type labels produced by the three extraction tools (os library, Apache Tika, and ExifTool) were reviewed by a digital archivist with expertise in digital preservation, who identified 96 semantically equivalent pairs by triangulating lexical similarity of field names, cross-tool agreement of extracted values on the same files, and consistency with each tool's technical documentation. The pairs of metadata types are formalized as owl:sameAs relations and loaded into the knowledge graph.
- (3) *Semantic clustering*: metadata types are grouped into nine functional categories covering file system attributes, document content, image, audio, video, email, executable, archive, and security metadata, broadening the navigational granularity of the knowledge base.
- (4) *Document type assignment*: Major media types (MIME) are mapped to human-readable labels and assigned to each file's archival entity, improving the readability and filterability of the collection.
- (5) *Title formalization*: file and folder names are modeled as dedicated title entities according to the RiC-O model, enabling richer title metadata.
- (6) *Date formalization*: temporal metadata is normalized and modeled as dedicated date entities, distinguishing dates pertaining to the intellectual content of a record from dates pertaining to its physical digital carrier, and encoded in both machine-readable and natural-language representations.

3.3. Phase 3: Semantic Enrichment via Specialized Knowledge and Generative Models

Phase 3 extends the knowledge base beyond what can be inferred from the archive itself, introducing knowledge from two external sources: a specialist bibliographic expertise and a locally executed generative language model.

Literary work mapping. A domain expert conducts a bibliographic survey of the author's works, identifying individual titles and their eventual groupings into cycles, series, or trilogies — categories illustrative of literary production and not an exhaustive or fixed taxonomy: for archives of a different nature, this stage can be adapted by substituting the relevant grouping unit while retaining the same part-of modeling pattern. These are modeled as work entities following the LRMoo ontology [6], with part-of relations encoding the hierarchy between cycles and their constituent works. Individual archival records and record sets are then linked to the relevant work entity; hierarchical propa-

gation ensures that a record associated with a specific work is also implicitly associated with any cycle or series containing it. This integration enables content-aware querying across the knowledge base. All generated triples are stored in a dedicated named graph. In the Evangelisti case study, the bibliographic survey and mapping of 30 works into 8 cycles and trilogies was carried out by a single domain expert over approximately three working days, a cost that scales with the size and complexity of the author's bibliography rather than with archive size.

Automatic description generation. For each file in the archive, the system automatically generates a concise natural-language description by combining deterministic retrieval from the knowledge graph with a locally executed LLM. For each archival instantiation, a SPARQL query retrieves all associated technical metadata, organized by semantic priority: identifying properties first, followed by structural and descriptive fields. This structured context is passed to Llama 3.2 3B [19], executed locally via Ollama, with a prompt instructing the model to write a brief, factual, 2–3 sentence description based only on the supplied metadata, without speculation, and with units and dates normalized (the full prompt template is available in the code repository).

The system shares some characteristics with retrieval-augmented generation approaches, while relying on deterministic graph retrieval rather than vector similarity. It differs from conventional KGQA systems by producing narrative output from structured input, rather than structured tuples from narrative queries. Given the sensitive nature of these archival materials, local execution is a requirement rather than a preference: no file content or metadata is transmitted to external infrastructures, and the use of open-source model weights ensures full auditability and reproducibility.

An illustrative example from the case study shows the system's interpretation capability. Given the following technical metadata for an RTF document:

Content-Type: application/rtf		Pages: 5		Words: 1739
CreateDate: 2008:02:15 12:00:00		ModifyDate: 2008:07:29 04:43:00		
RevisionNumber: 33		TotalEditTime: 6.0 hours		
LastModifiedBy: Valerio Evangelisti		st_size: 100735		

The system generates: *“Rich Text Format (RTF) document, 5 pages with 1,739 words. Created by Valerio Evangelisti in February 2008, revised through 33 versions with 6 hours total editing time, last modified by Valerio Evangelisti in July 2008. Titled ‘3’, likely a chapter or section of a larger work. File size 98.4 KB.”*

The output demonstrates unit conversion (bytes to KB), date normalization, and contextual inference from structured metadata. Generation parameters (temperature = 0.3, num_predict = 150, top_p = 0.9) favor descriptive consistency over lexical variety. Each generated description is annotated with full provenance — generating activity, model identity, supervising agent, and timestamp — and flagged for potential human review. Descriptions are inserted incrementally with checkpoint persistence, enabling resumption after interruption, and stored in a dedicated named graph.

Both processes are independently configurable: the scope of the work mapping and the subset of records targeted for description generation can be adjusted to the specific archive and access policy. In this way, the two processes can also be made interdependent — as in the Evangelisti case study, where description generation is restricted to the records made visible by the work mapping step, focusing computational resources on the materials that will be effectively accessible to users.

3.4. Phase 4: Chain of Custody Reconstruction

Phase 4 reconstructs and formalizes the physical chain of custody connecting the original storage media to the working copies used for processing. Documenting this history is archivistically essential: each transfer, copy, or migration operation may have left traces in the files themselves — through system-specific metadata, format conversions, or file system timestamps — and the awareness of who handled the materials, under what conditions, and through which technological environments is necessary to interpret anomalies and assess the reliability of the archive as received. In the absence of prior inventories — a frequent condition in personal archives — the phase begins with a manual census of all identified storage supports, recording medium type, provenance, current location, and conservation conditions. This census provides the empirical grounding for all subsequent modeling.

Each storage medium and its physical location are represented as dedicated entities using the PREMIS vocabulary, extended by BoDi to connect them to the corresponding archival instantiations in the knowledge graph. Derivation relationships between successive instantiations — documenting copy, migration, or transfer operations from one medium to another — are modeled explicitly, with each relationship temporally anchored to the date of the operation and linked to the agents involved.

The resulting named graph transforms the custodial history from background knowledge held by archivists into queryable, machine-readable provenance — making it possible to connect every file in the archive to the full sequence of hands, media, and environments it passed through before reaching the researcher.

3.5. Phase 5: Selective Redaction for Publication

Phase 5 prepares the knowledge base for dissemination by replacing sensitive label strings with a standardized placeholder rather than removing entities, preserving the graph's relational structure while concealing sensitive content; a full backup precedes any operation, ensuring reversibility. Sensitive data in born-digital personal archives is distributed not only in file contents but also in metadata fields — creator names, last-author values, file paths — requiring field-level, not just record-level, screening.

Redaction follows differentiated, configurable logic per entity type. Archival records are redacted by default and remain visible only if they satisfy one of three criteria agreed with rights holders (in the Evangelisti case, with the Associazione Valerio Evangelisti – Il Sol dell'Avvenire): explicit linkage to a literary work (`rico:isRelatedTo` to an `lrmoo:F1_Work`), inclusion in an authorization list, or hierarchical containment (`rico:isOrWasIncludedIn`) in an authorized `rico:RecordSet`. Matching is implemented as a SPARQL classification against these properties; entities queued for redaction have their label and title replaced via SPARQL DELETE/INSERT with the placeholder “*Redacted information*”, and every entity is marked with a boolean `bodi:redactedInformation` flag, providing a queryable audit trail. Folders follow the inverse logic (visible unless blacklisted). For visible entities, metadata fields referring to third parties are additionally screened and redacted field-by-field. Defining these criteria with the archive's custodian was an iterative, one-time policy decision independent of archive size. Post-redaction consistency checks verify that there is no mismatch between entities and their instantiations.

4. Results

4.1. Dataset and Infrastructure

The pipeline was applied to the born-digital partition of the Valerio Evangelisti (1952–2022) Archive — an Italian science-fiction and historical-fiction writer — including materials from three storage media: a copy of the main Windows PC hard disk, an external hard disk (Western Digital My Book 2TB), and the extracted contents of 93 floppy disks, spanning approximately 1984–2022 and totaling 2.1 TB.

Phase 1, in particular, was conducted on a Dell PowerEdge R7525 server (2× AMD EPYC 7313 16-core CPUs, 256 GB RAM) running Ubuntu 24.04.3 LTS, using Python 3.12.3 with `rdflib` 7.1.4, `SPARQLWrapper` 2.0.0, and `Apache Tika` 3.2.1. The pipeline generated 61,154,322 RDF triples across 13 named graphs, stored in a `Blazegraph` journal of 18.33 GB. `Blazegraph` was deployed with default configuration settings and without query optimization; the reported performance figures should therefore be interpreted as a reproducible baseline rather than an optimized upper bound. Table 2 shows the distribution of archival entities across storage media.

Table 2. Distribution of archival entities across storage media.

Storage medium	Files	Folders
Main hard disk	56,171	9,624
External hard disk	19,810	1,394
Floppy disks	2,230	117
Total	78,211	11,135

4.2. Knowledge Graph Quality

The knowledge graph represents 78,211 files and 11,135 folders, each carrying one fixity and one location entity. Technical metadata extraction yielded 5,574,418 metadata entities, distributed across the three tools as follows: `Apache Tika` 2,221,401 values, `ExifTool` 2,191,519 values, `os library` 1,161,498 values. The tools exhibit markedly different and complementary type coverage. The `os library` produces a stable set of 13 metadata types per directory, capturing folder-level attributes unavailable to the other tools. `Apache Tika` yields 960 distinct types on the main hard disk, 867 on the external hard disk, and 244 on the floppy corpus, reflecting progressively narrower format diversity. `ExifTool` surfaces up to 106,211 distinct types on the main hard disk, reflecting the exceptional density of multimedia content. No single extractor would have captured the full descriptive depth represented in the knowledge base alone. Post-extraction integrity verification confirmed that all SHA-256 values were identical before and after extraction, providing evidence that the source materials remained unaltered during processing.

The 51-query `SPARQL` validation suite was executed against the complete knowledge base, completing in 403 seconds with 100% execution success (0 errors, 0 failures). The validation suite detected no structural violations across any functional group: no hierarchical inconsistencies, no orphan entities, no metadata values without provenance, no hash format violations, no multiple hash values per file, no invalid file paths. 349 files

remain classified as `application/unknown` or `application/octet-stream` — a known limitation of MIME-based classification for corrupted or legacy binary formats, consistent with the format obsolescence documented in the floppy disks corpus. Query execution times range from under 5 ms for simple entity counts to 44 seconds for full provenance coverage scans over metadata entities, demonstrating acceptable SPARQL performance on a dataset of this scale without query optimization.

4.3. Semantic Enrichment

Expert-curated cross-tool alignment produced 96 equivalence pairs covering properties from file size and timestamps to camera model, audio sample rate, PDF version, font family, and GPS coordinates, enabling unified cross-tool querying. Hash-based duplicate detection identified 9,080 hash groups shared across files, involving 35,577 file instances in total; the most replicated hash (1,816 occurrences) corresponds to the universal empty-file value, surfacing a systematic pattern of placeholder or migration artifacts that would be entirely invisible in a traditional finding aid. Structural analysis revealed a maximum nesting depth of 18 folder levels on the main hard disk, with the densest concentration of files between levels 9 and 12. MIME type classification identified 60 distinct types across 57,015 classified files; images and documents together account for over 86% of the corpus, with RTF as the dominant document format (9,319 files), reflecting a sustained use of legacy word-processing environments. The most frequent types are reported in Table 3.

Table 3. Most frequent media types in the Valerio Evangelisti Archive.

MIME Type	Occurrences
image/jpeg	17,778
text/rtf	9,319
text/plain	6,204
image/png	4,933
image/svg+xml	3,462
Other (55+ distinct types)	15,319

Literary work mapping linked 6,073 archival entities to 30 works organized into 8 narrative cycles and trilogies, enabling content-aware querying across the knowledge base. The combination of work linkage and temporal metadata formalization made it possible to correlate archival activity with the author’s editorial output. For example, the biennium 2005–2006 emerges as the peak of computational activity in the archive, coinciding with the publication of two novels and a short story collection — a correlation that the knowledge graph surfaces automatically from metadata alone, without access to file contents.

The full transfer history of each storage medium from its original physical location to the working copies used for processing. Three derivation sequences are recorded — one for the main hard disk, one for the external hard disk, and one for the floppy disk corpus. The resulting named graph allows to trace, for any file in the archive, the sequence of hands, media, and environments it passed through before reaching the processing infrastructure.

4.4. Description Generation Quality

The automatic description system produced 5,548 natural-language technical descriptions in 3 hours and 46 minutes on a consumer laptop (Apple M1 Pro, 16 GB RAM) running Llama 3.2 3B locally via Ollama. As a preliminary quality assessment, a domain expert reviewed a sample of 100 descriptions covering the five most frequent MIME types, evaluating each against its source metadata on three criteria: factual correctness of all stated properties, absence of hallucinated content, and appropriate unit conversion and date normalization. All 100 descriptions passed all three criteria. This evaluation confirms the basic factual adequacy of the generation system; a more systematic validation protocol — addressing sample size, single-evaluator design, and binary criteria — is planned as outlined in Section 5. Manual production of equivalent descriptions is estimated at 5–10 minutes per file, placing the equivalent manual effort between 58 and 116 person-working-days.

4.5. Knowledge Base Deployment

The resulting knowledge base is deployed on a ResearchSpace [20] instance dedicated to the Valerio Evangelisti Archive, providing a browser-based interface for SPARQL querying and visual exploration of the archive, built on top of the knowledge base produced by the pipeline presented here and described separately in [21]. Navigation is organized around a file system metaphor, with every visualization generated by live SPARQL queries executed directly against the underlying knowledge graph. Given the sensitive nature of the materials, access is currently restricted pending final review by the Associazione Valerio Evangelisti – Il Sol dell’Avvenire, custodian of the archive and of the rights of its contents.

5. Discussion

5.1. A Shift in Archival Epistemology

The pipeline and BoDi together instantiate a shift in how archival description of born-digital materials can be approached: rather than requiring an archivist to inspect and document each item individually, the pipeline extracts, formalizes, and connects the descriptive information already embedded in the digital objects, producing a queryable knowledge graph that provides a rich, structured baseline for targeted curation and expert intervention.

This changes the archival workflow rather than merely accelerating it: producing technical descriptions of thousands of files individually is operationally impossible, so the archivist is freed to focus on interpretation and contextualization — bibliographic references, access policy, editorial judgment — while the bottleneck shifts from producing descriptions to verifying them, and from comprehensive coverage to targeted intervention.

Scale itself becomes an interpretive resource rather than merely a technical challenge: the analyses reported in Section 4 — from cross-media duplicate detection to the correlation of archival activity with editorial output — are the kind of insight that scalable reading [22] affords, moving fluidly between macro-level patterns and targeted close

reading, with the knowledge graph functioning as a navigation instrument rather than a static inventory.

This granularity is a structural consequence of the KG format, not merely a byproduct of automation. Preservation-oriented formats (e.g., OAIS packages, PREMIS-only records) safeguard bit-level integrity but remain opaque to cross-entity querying; relational approaches such as PAD (Section 2) would need a proliferation of join tables to capture the same cross-cutting, many-to-many relations that the graph model represents natively. No manually curated finding aid reaches a comparable level of detail — a scale of curation that contemporary born-digital archives render impracticable. A formal, comparative evaluation of this added value against a traditional finding aid has not yet been conducted and is identified as future work below.

5.2. *Limitations and Future Work*

The most structural limitation affects Phase 5. The current architecture encodes access policies through static whitelists and blacklists, effectively reproducing the binary logic of paper-based access control within a digital environment. While operationally sufficient, this approach does not fully exploit the expressiveness of the knowledge graph. A graph-native access architecture based on differentiated named-graph layers and role-based visibility policies would provide flexible and maintainable solution, allowing access conditions to be updated independently from the processing pipeline itself. More importantly, such an approach would fully exploit the relational nature of the graph, allowing researchers with different authorization levels to navigate the same knowledge base at different degrees of resolution [23].

The pipeline has been validated on the Valerio Evangelisti Archive, to our knowledge the largest known born-digital personal archive preserved by an Italian cultural heritage institution; although validation relies mainly on a single case study, the archive's exceptional heterogeneity in formats, typologies, and organizational structure provides a demanding testbed. Phases 1, 2, and 4 are fully archive-agnostic, while Phases 3 and 5 require adaptation to the target collection, while the underlying modeling pattern and the rest of the pipeline remain structurally independent from the Evangelisti case. Computationally, Phases 1 and 2 scale linearly with archive size, while Phase 3 (description generation) and SPARQL query execution — ranging from 5 ms to 44 s on the current dataset without optimization — are the principal bottlenecks for substantially larger corpora, motivating planned parallelization, graph partitioning, and query caching. Broader empirical validation is ongoing: beyond tests on smaller sample archives, the pipeline is currently being applied, at a preliminary stage, to a second, structurally different archive — the unallocated but migrated files of the Gabriel García Márquez archive — expected to provide further evidence of the pipeline's behavior outside the Evangelisti Archive conditions.

Additional limitations concern metadata enrichment and semantic classification. The literary work mapping currently depends on domain expertise and prior bibliographic reconstruction, limiting transferability to archives lacking comparable scholarly resources. MIME type classification also remains intrinsically limited by the semantic ambiguity of certain media types; content-level disambiguation combining OCR and layout analysis is therefore envisaged as a complementary layer of interpretation.

The knowledge base is not yet publicly accessible, pending final review by the Associazione Valerio Evangelisti – Il Sol dell'Avvenire, the custodian of the archive.

The preliminary evaluation of automatically generated descriptions discussed in Section 4 likewise requires substantial extension. The planned protocol will assess a stratified sample of 300–500 descriptions across MIME types and metadata-density levels, independently reviewed by an archivist (factual accuracy, disciplinary adequacy) and a non-specialist user (accessibility, clarity). Evaluation will move from a binary pass/fail model to a Likert-scale framework covering factual accuracy, completeness, and linguistic quality, complemented by automatic checks of formally verifiable properties (date formats, unit conversions).

6. Conclusion

The five-phase pipeline presented in this paper shows the practical viability of large-scale semantic description for born-digital personal archives. Applied to the Valerio Evangelisti Archive, the system transformed a corpus whose manual processing would have required years of sustained archival work into a machine-queryable knowledge environment.

The significance of the contribution, however, lies not only in automation or scale, but in the underlying shift discussed in Section 5: from document-centric description toward a relational, computationally explorable model. At the same time, the work highlights the extent to which born-digital archival processing remains an unresolved methodological and infrastructural problem. Questions of access control, semantic classification, provenance modeling, evaluation methodology, and computational scalability require further experimentation across archives of different provenance and institutional context. The present study should therefore be understood not as a definitive solution, but as a scalable operational framework and a proof of methodological feasibility for knowledge-graph-based archival description.

More broadly, born-digital archives are not simply digital equivalents of paper fonds requiring new preservation strategies, but fundamentally different documentary environments whose complexity can only be adequately represented through relational and machine-processable models.

Author Contributions

Lucia Giagnolini designed and developed the pipeline, conducted the case study, and wrote the manuscript. Paolo Bonora supervised the work and contributed to the design of the pipeline.

Declaration on Generative AI

During the preparation of this work, the authors used Claude (Anthropic) in order to perform language editing and restructuring. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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Distributed Knowledge-Driven Data Exchange and Validation Framework for Networked Smart Buildings

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Abstract. Building Automation and Management Systems offer centralised control and monitoring by integrating smart devices into the environment. Together, these provide building owners and stakeholders with data about energy consumption and other relevant characteristics which can, among others, be used for decision making and the automation of actions. However, due to the proliferation of different protocols and ecosystems, it often remains challenging to seamlessly aggregate and analyse the data streams provided by the connected devices. To overcome these problems, we introduce a distributed, knowledge-driven data exchange and validation framework, built upon *Semantic Web* technologies, that a) facilitates the transparent flow of data between smart devices, b) validates these data and generates reports on detected anomalous behaviour in real-time, and c) aggregates and displays the data and any derived insight on dashboards and other interactive web-accessible interfaces. Our framework consists of two core components: the *Knowledge Engine*, which acts as an information broker and translation layer between device-specific languages and *RDF*, and the *Knowledge Validator*, which continuously monitors these *RDF* streams and reports on deviations from nominal behaviour. We evaluate our framework via requirement-based testing and demonstrate its functionality in a real-world case study in the smart buildings domain, showing that it can facilitate seamless communication between devices of different make and model and that it can help building owners recognise abnormal device behaviour.

Keywords. Smart Buildings, Internet of Things, Interoperability, Semantic Web Technologies, Anomaly Detection, Knowledge Graphs, Distributed

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1. Introduction

Building Automation and Management Systems facilitate centralised control and live monitoring capabilities by integrating smart sensors and other *Internet of Things* (IoT) devices into living environments. Common examples include lighting control systems, solar panels, electric vehicle charging stations, and HVAC systems [1,2]. By making their operational status accessible via available networks, IoT devices provide building owners and stakeholders with valuable insight about their working condition and energy usage, as well as many other relevant characteristics. These data can be used, amongst others things, to improve overall efficiency and comfort, for anomaly detection and predictive maintenance, and for strategic decision making and the automation of actions [3]. The fragmentation of standards and ecosystems, however, has made device integration and interoperability challenging and data aggregation and analysis non-trivial [3,4]. This work attempts to resolve these challenges by proposing a data exchange and validation framework for IoT environments based on *Semantic Web* technologies.

Integrating smart devices into their properties can help building owners reach sustainability goals and reduce energy costs. This has made the transformation to smart building an attractive investment, stimulating the adoption of building automation systems worldwide [2], whereas the development of mature standards and technologies for IoT ecosystems has enabled device manufacturers to incorporate *smart* capabilities into their products. This has lead to a multitude of available smart devices on the market, offering ample choice to building owners, yet has also contributed to the ongoing fragmentation of the IoT landscape, with manufactures adopting different, often closed and frequently incompatible, standards and ecosystems for their products [5,6].

The proliferation of different standards and ecosystems poses challenges to building owners and stakeholders, who now have to contend with device integration and interoperability issues, as well as with further complications downstream, in particular tasks requiring data aggregation such as forecasting, anomaly detection, and predictive maintenance [4,5]. Other challenges include reduced system life span and higher long-term costs, due to the retirement of closed standards, vendor lock-in, and the complexity involved with maintaining heterogeneous ecosystems [3]. Mitigating interoperability-related issues is therefore a key focus in current IoT research and industry.

Arguably the most successful efforts that prioritise interoperability are *Matter* [7] and *OPC UA* [8], which cater to the consumer market and industry, respectively, offering a uniform interface to products from different vendors. While both are build around open standards and protocols, however, *Matter* still requires built-in support from factory, whereas, in the case of *OPC UA*, system operators must explicitly expose the device properties they are interested in, using self-defined property paths. Instead, we argue for the adoption of semantic data, backed by domain ontologies, to standardise properties, annotations, and structure, facilitating effortless, transparent communication between devices via shared conceptualisations [9], as well as seamless data aggregation and processing as is inherent to graph data [10].

Combining *Semantic Web* and IoT technologies is an idea that goes back more than a decade. This idea has been formalised by the W3C as the *Web of*

*Things*², and is being pursued by various academia, industry, and standardization bodies [11]. Numerous contributions have been made since, many of which address interoperability and data aggregation [12]. The methods underlying these contributions can be broadly grouped into two approaches, performing semantic enrichment either 1) client side, enriching incoming data and leaving the data flow in the network untouched (e.g., [13]), and 2) via a single, centralised service, translating and, in some cases, aligning data streams, and forwarding these to relevant services (e.g., [14]). Instead, our framework employs a distributed topology, by enriching data streams at their origin and using peer-to-peer data exchange and reasoning. Doing so makes our framework more robust—no single point of failure—provides a form of access control, and reduces concern related to data sensitivity.

This work proposes a distributed, knowledge-driven data exchange and validation framework for building automation and management systems that is built upon Semantic Web technologies. The framework consists of two core components: the *Knowledge Engine*, which acts as information broker and translation layer that relays structured graph data between knowledge producers and knowledge consumers, offering reasoning on the fly, and the *Knowledge Validator*, which continuously monitors the data streams and reports on deviations from the devices' nominal behaviour in real time and with human-readable explanations. By embracing open standards and protocols such as RDF and HTTP, and by adopting industry-endorsed vocabularies such as SAREF and SHACL, our framework aims to alleviate device interoperability issues and to improve data interpretability and utility, while simultaneously emphasizing the reuse, sharing, and sustainability of building knowledge, information, and data.

We evaluate our framework using requirement-based testing, by eliciting relevant requirements via a combination of domain and task analysis, as well as from scenarios which are drafted together with building owners. The requirements are verified through testing and by demonstration in a controlled and real-world setting, the latter of which involves an ongoing case study in The Netherlands as part of the EU-funded *HEDGE-IoT* project. All software and resources developed during this research are open source and freely available online.

To summarise, the main contributions of this work are **a)** a distributed data exchange and validation framework based on Semantic Web technologies, that **i)** facilitates the transparent flow of data between smart devices, **ii)** validates these data and generates reports on anomalous behaviour on the fly, and **iii)** aggregates and displays the data and any derived insight on dashboards and other interactive web-accessible interfaces, and **b)** a demonstration in a real-world case study in the smart buildings domain.

2. Related Work

This section will provide background and related work of concepts that are important to our research, specifically IoT, the integration with *Semantic Web* technologies, and graph anomaly detection.

²See <https://www.w3.org/WoT/>

2.1. Internet of Things

The *Internet of Things* is a system of networked sensors and smart devices to monitor and control an environment, providing owners with various data and measurements that can be used for decision making, process automation, and analytics [15]. Communication between devices can occur via numerous different standards, amongst which are ZigBee, KNX, and ModBus, on top of which data is exchanged in various (often vendor-specific) formats. Alternatives that emphasise interoperability exist (e.g., Matter and OPC UA) but require built-in support or lack formal semantics.

IoT systems can be divided into several layers [16]. The physical devices typically form the lowest layer, whereas the application or business layer usually form the top. The intermediate layers are responsible for all communication and for data collection and aggregation. Our framework concentrates on several of these layers, in particular on the network, data abstraction, and application layers. Herein we differ from most other IoT-related studies, which predominantly focus on the device layer [5].

A common message architecture in IoT systems is the *publish-subscribe* pattern (pub-sub). Publishers tag their messages and broadcast them indiscriminately to the network, whereas subscribers listen on the same network for messages that match the tags they are interested in [17]. The pub-sub pattern supports anonymous, asynchronous, and many-to-many communication, which makes it well suited for IoT environments. The same architecture forms the basis of our framework, which consists of knowledge producers (publishers) and knowledge consumers (subscribers).

Early work on the integration of Semantic Web technologies into IoT by [14] proposes the *Semantic Gateway* as a centralised communication hub between local devices and cloud services. Upon reception, raw device data are enriched with semantic annotations and forwarded to the services, and vice versa. Similarly, in [13], the authors propose to semantically enrich the data streams in OPC UA ecosystems via client-side translation modules, by defining mapping tables and using these to convert the native OPC UA format to semantic data. A different approach is proposed in [18], which only translates the raw data to its semantic counterpart on request, as in the case when alignment between sensor readings is needed. All other communication remains untouched. Several other studies have explored the integration of formal semantics in IoT ecosystem (e.g., [19,20,21]) but these tend to focus more on ontology-driven retrieval and validation and are therefore less relevant to this work.

Similar ideas to resolve interoperability issues have been explored in specific contexts. The work by [22] considers the smart cities domain, addressing the information flow from smart sensors to citizens. The adoption of ontologies and open standards are seen as key components for interoperability, with the *Open-Messaging Interface* and *Open-Data Format* as prime contenders. The same standards are proposed in [23], which investigates their effectiveness for smart vehicles, and [24], which demonstrate their use in an IoT system for patient monitoring devices in the health domain.

2.2. Graph Anomaly Detection

Anomaly detection involves the detection of deviations from an established pattern [25]. These deviations are called outliers when both pattern and deviations are part of the same dataset; when, instead, the goal is to detect deviations in *new* observations, as is the case with streaming data, the task is more often called novelty detection [26]. Our framework incorporates novelty detection techniques as part of the knowledge validation component.

A distinction can be made between three different types of anomalies [25,27]: 1) point anomalies, which exceed expected bounds (e.g., temperature spike) 2) contextual anomalies, which are within bounds yet unexpected given a context (e.g., heating on high during the summer) and 3) collective anomalies, which are within bounds yet unexpected with respect to similar data sources (e.g., high temperature reading for just one of several sensors in the same building). In our framework, we consider bounds belonging to continuous (e.g., sensor reading) and discrete (e.g., sensor location) features.

Graph anomaly detection extends the notion of anomaly detection to graph-shaped data, by not only looking for anomalies in numerical features, as is often the case with traditional anomaly detection methods, but by also detecting deviations in structure and, in some cases, semantics [28,29]. Moreover, neighbourhood information can be leveraged to serve as context for more informed anomaly detection.

Anomaly detection in semantic IoT environments is still in its infancy [30]. Instead, recent studies on graph-based approaches primarily target traditional IoT ecosystems, and typically involve the use of *Graph Neural Networks* (GNN) or other *Deep Learning* techniques to detect anomalies in solitary data streams or on the network level [31,32]. Moreover, semantics and raw data (as opposed to structure) are largely left unused. Studies into pattern recognition and data validation on graph data in general do exist, but, to the best of our knowledge, we are among the first that apply such methods in semantic IoT environments.

Notable research is done by [33], who learn device behaviour and detect deviations thereof. Hereto, the authors use graph representation learning with the help of GNNs, which they feed both the data as well as several characteristics of the network. In [34], the authors likewise use GNNs to leverage the spatial context of the network. A *LSTM*-based architecture is then used to model temporal dependencies. The work in [35] emphasises the importance of these dependencies, using graph transformers to find anomalies in multivariate time series data in the IoT domain. For a more-complete overview of the other deep anomaly detection methods (which differ too far from our work to mention here) we refer to one of the excellent review papers [30,32].

3. Framework Overview

An architecture diagram of our framework is shown in Figure 1. Depicted are several example devices, each connected to a dedicated *Knowledge Engine* instance via a so-called smart adaptor. Adaptors are vendor specific and take care of the

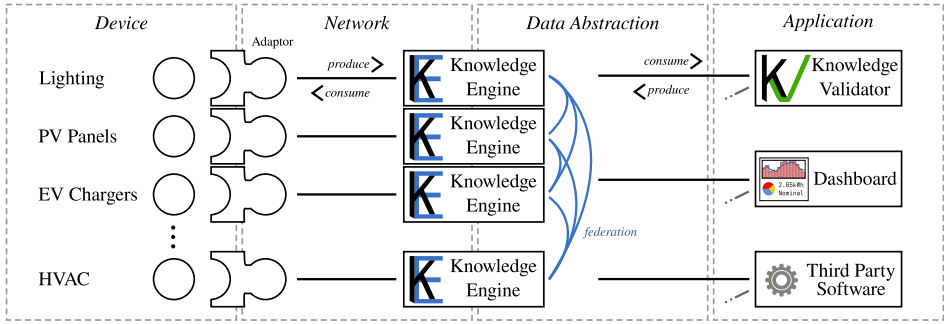


Figure 1. Framework architecture with example devices and applications and the layers they reside in. Some arrows have been omitted for clarity.

translation from and to graph data. The *Knowledge Engine*, if requested, reasons over these data and relays them to interested consumers, either directly or federated via other *Knowledge Engine* instances. Consumers include the same or other devices as well as dashboards and other applications. One such application is the *Knowledge Validator*, which monitors all data streams from the devices it is subscribed to and which uses the same network to publish validation reports. All communication from the adaptors onwards uses HTTP.

Together with the deployed *Knowledge Engine* instances, the registered knowledge producers and knowledge consumers constitute the nodes in a semantic IoT network. These nodes can either exclusively produce or consume knowledge, or they can take on both roles at the same time. Nodes that produce knowledge are treated as dynamic knowledge bases in which each statement is associated with a specific timestamp.

Each published message is assumed to be a self-contained knowledge graph in RDF format. The graph's structure and semantics are defined by a producer's initial message, whereas subsequent messages from the same producer and with the same identifier are viewed as updates to this graph. The stream of messages is expected to be ordered sequentially.

3.1. Knowledge Engine

The *Knowledge Engine*³ is a semantic gateway [14] that facilitates the exchange of graph data within a network of knowledge nodes. Examples include small components such as sensors or switches, as well as larger assets, data sources, and services such as HVAC systems, databases, and APIs. Each node in the network is commonly paired with a dedicated *Knowledge Engine* instance, which handles all communication from and to this node. This means that, in a typical IoT network, there are as many deployed *Knowledge Engine* instances as there are registered knowledge nodes. Imposing this one-to-one relationship guarantees that data ownership remains with the device (cf. a single, shared triple store) mitigating concerns about data sensitivity and access control.

³Code and documentation is available at <https://github.com/TNO/knowledge-engine>

Listing 1: An example graph pattern for a smart temperature sensor. Parameters are preceded by a question mark. Note that prefix declarations are omitted.

?sensor	rdf:type	saref:Sensor ;
	ex:isContainedIn	:Building_1 .
?measurement	saref:madeBy	?sensor ;
	saref:isMeasuredIn	?unit ;
	saref:hasValue	?temperature ;
	saref:hasTimestamp	?timestamp .

Federation is used to support data exchange between multiple *Knowledge Engine* instances. For this purpose, the various instances can communicate peer-to-peer with each other, leveraging the distributed topology of IoT networks without a single point of failure. To accommodate the dynamic nature of typical IoT environments, in which nodes join and leave on a frequent basis and without warning, the *Knowledge Engine* maintains appropriate mechanisms to automatically add and remove nodes and pub/sub registrations. This enables existing subscribers to receive messages from newly added sensor node automatically.

Knowledge producers must register their data model at their associated *Knowledge Engine* instance before they can publish their messages to the network. Similarly, knowledge consumers must register their interests before they can receive these messages. In either case, this registration is achieved via an HTTP operation with a parametrised graph pattern as payload. The parameters serve as named variables that can take on different values with each message, similar to the bindings in the answer to a SPARQL query. Listing 1 shows an example of a parametrised graph pattern for the group of smart temperature sensors in a specific building (:Building_1).

Adaptors are placed between *Knowledge Engine* instances and their associated knowledge nodes to facilitate the translation from vendor-specific standards and data models to graph data, and vice versa. In this particular setting, the graph data are modelled using the Smart Appliances REference⁴ vocabulary (SAREF); an industry-endorsed reference model for the smart appliances domain. The *Knowledge Engine* itself is data model agnostic, however, making swapping or combining vocabularies trivial.

In addition to their role as information broker, the *Knowledge Engine* also offers reasoning on the fly, exploiting semantic annotations and domain knowledge to infer implicit knowledge and to support distributed query answering. A typical use case concerns class hierarchies, for example to obtain messages from all HVAC devices irrespective of their specific type.

⁴Specification available at <https://w3id.org/saref/>

3.2. Knowledge Validator

Anomaly detection and predictive maintenance are two popular use cases for smart environments. [1,3,4]. The *Knowledge Validator*⁵ caters to this need by continuously monitoring the data streams for faulty or otherwise abnormal behaviour that could indicate possible (near-future) anomalies. For this purpose, the validator employs continuous online learning to capture and update patterns of nominal behaviour, deviations of which are reported together with possible explanations.

Patterns are tied to a device (or a group of identical devices) and are modelled as conjunctions of clauses $\phi = c_i \wedge c_j \wedge \dots \wedge c_k$, with $k \geq 1$, where each clause is a weighted binary predicate $c^w = p(f_i, f_j)$ that represents the relationship p between probability distributions f_i and f_j . Weight w represents the relevance of the clause to the pattern. Upon receiving a message from an unknown device, a new pattern is generated by converting the message's structure into the aforementioned form, distributing weights equally amongst its clauses, and by replacing all entities and literals with discrete or continuous distributions. Background information, including class and datatype annotations, is leveraged to determine appropriate distributions.

Existing patterns are updated after the successful validation of a new message. Updating involves adding new values to their respective distributions and forgetting past values (after a set decay period). Similarly, new clauses are included and existing clauses removed upon their weights exceeding or falling under a set threshold. The weights are adjusted by the (non-) appearance of the associated clauses in each new message. Together, this allows for patterns that adapt to gradual natural changes in behaviour (e.g., different seasons). Patterns can also branch off to accommodate significant natural shifts, maintaining different patterns per situation (e.g., weekends and non-weekends).

Incoming messages are validated against their associated patterns (if available) by evaluating their semantics, structure, and data. Semantic validation includes consistency checks for all observed (data) types and relations, whereas the structural validation looks for missing and unexpected relations. Data validation is accomplished using prediction intervals (point anomaly) and hypothesis testing (contextual and collective anomalies) with certain pre-set thresholds corresponding to nominal, suspicious, and critical assessments. The same tests are also used to monitor stream health, detecting delayed or absent messages. Messages with anomalies are skipped when updating patterns.

Validation reports are made up after each validation round and are published to the network in graph format, using the *W3C* endorsed industry standard *SHApe Constraint Language* (SHACL)⁶. Depending on the number of detected anomalies, the reports contain zero (no anomalies) or more validation results, together with the offending graph element, the severity, and human-readable explanation. Knowledge consumers that are subscribed to these reports can then dis-

⁵Code and documentation is available at <https://gitlab.com/wxwilcke/gladoss>

Listing 2: An example validation report of a faulty reading from a temperature sensor in Turtle syntax⁷. Note that prefix declarations and datatype annotations are omitted.

:report	rdf:type	sh:ValidationReport.
	dct:date	"2026-09-15T12:12:45";
	dct:subject	ex:observation_k;
	dct:conformsTo	"https://www.w3.org/TR/shacl/";
	dct:hasPart	:results_i;
	sh:conforms	"false".
:result_i	rdf:type	sh:ValidationResult;
	sh:focusNode	ex:measurement_i;
	sh:resultPath	saref:hasValue;
	sh:value	"47.09";
	rdfs:label	"Critical Value Violation"@en;
	sh:resultMessage	"Observed value not within 99% prediction interval. Expected value in (19.2, 24.6]."@en;
	sh:resultSeverity	:severity_i.
:severity_i	rdf:type	sh:Severity;
	rdfs:label	"CRITICAL"@en;
	rdfs:comment	"Critical Anomaly"@en.

play them and, if needed, act accordingly. Listing 2 shows an example validation report about a critical point anomaly caused by a faulty temperature sensor.

4. Evaluation

We evaluate the proposed framework with the help of requirement-based testing. First, we describe the requirement elicitation process and the requirements obtained through it. Next, to assess the requirements, we employ both demonstration-based and test-based verification. For the demonstration, we simulate a smart environment with several knowledge nodes. The same setup is also used to test each task requirement programmatically. Deployment scripts and unit tests are available⁷ for reproducibility. A second, larger-scale demonstration involves an ongoing case study in a real-world environment.

4.1. Requirement Elicitation

A combination of domain and task analysis, and scenarios were employed to elicit requirements [36]. The domain analysis involved a literature review of the challenges and (yet unrealised) opportunities pertaining building automation and management in IoT. A condensed version of this analysis is included in the in-

⁶Specification available at <https://www.w3.org/TR/shacl/>

⁷Available at <https://github.com/TNO/knowledge-engine/tree/master/examples/data-exchange-and-validation>

Table 1. List of domain requirements

#	Description	Source
RQ-1	Standardization of data formats and models	[38, pp. 60]
RQ-2	Standardization of communication protocols	[38, pp. 60]
RQ-3	Effortless interoperability and integration	[38, pp. 65]
RQ-4	Full (meta-) data accessibility and transparency	[38, pp. 62]
RQ-5	Real-time stream monitoring and management	[38, pp. 63]
RQ-6	Automated, real-time data validation	[38, pp. 62]
RQ-7	Real-time, low-latency anomaly detection	[38, pp. 62]
RQ-8	Predictive maintenance and analytics	[38, pp. 67]

roduction and related work sections of this paper. The task analysis and scenarios were formalised using UML diagrams, narratives, and use case descriptions which were iteratively developed together with stakeholders. This process has been documented across several technical documents [37,38,39].

The elicitation activity resulted in a set of broad domain requirements (Tab. 1) and two sets of more narrow task requirements, one for the *Knowledge Engine* (Tab. 2) and one for the *Knowledge Validator* (Tab. 3). The domain requirements serve as context for the task requirements: a task requirement is met if, and only if, it can be verified under the conditions defined by the domain requirements.

It is important to note that the domain requirements represent an idealistic set of conditions: rather than being hard requirements they serve as goals to strive towards. We therefore consider a domain requirement met if our framework significantly improves upon the current state within the building automation community.

4.2. Simulation

A minimal working Docker environment has been created to verify the requirements in a controlled setting. This simulated environment (Fig. 2) consists of two smart buildings with two temperature sensors each, one reporting the temperature in degrees Celsius and the other in degrees Fahrenheit. The environment also includes a simple building management system, which provides information about the building a temperature sensor resides in, and a unit-of-measure converter service, which converts measurements in degrees Fahrenheit to degrees Celsius. Finally, a single *Knowledge Validator* instance is running in the environment to monitor and validate the data sent by the four sensors, as well as a rudimentary dashboard that listens for validation reports and prints these to the standard output.

Each of the eight knowledge nodes (four sensors, one database, one service, and two applications) is paired with its own *Knowledge Engine* instance. While not strictly necessary (one instance can support multiple nodes) this provides a more robust setup in which communication issues or downtime of one or more *Knowledge Engine* instances does not impair the entire network. Communication between these instances is achieved through federation, enabling them to relay requests and data between knowledge nodes. Domain knowledge (Listing 3) is

Table 2. List of task requirements for the *Knowledge Engine*

#	Description
RQ.KE-1	Process node registration requests
RQ.KE-2	Process publication and subscription requests
RQ.KE-3	Accept data from registered publishers
RQ.KE-4	Send data to registered subscribers
RQ.KE-5	Forward data from registered publishers to registered subscribers

provided to enable automated type inferencing via RDFS rules. For this demonstration, this includes ontological background information about sensor class hierarchy (`x rdfs:subClassOf y`) as well as the RDFS-9 entailment rule⁸.

The four temperature sensors publish their readings periodically, using their vendor’s native unit of measure. These readings are received by the associated *Knowledge Engine* instance and translated to graph data by the smart adaptors. The resulting graphs are forwarded to interested knowledge nodes. One of these nodes is the unit-of-measure converter, which, triggered by temperature reading in degrees Fahrenheit, automatically converts these to degrees Celsius. The *Knowledge Validator*, which is configured to monitor temperature readings, will now only receive measurements in this unit, enabling it to learn patterns that generalise over all temperature sensors. Since the validator is only interested in readings from a specific building (see Listing 1) its associated *Knowledge Engine* instance requests information from the building management system about the locations of the sensors. This information is integrated into the original graph and relayed to the validator, which, upon detection of an anomaly, publishes a validation report. The reports are received and then displayed by the dashboard. Faulty sensor readings are injected periodically.

⁸Specification available at <https://www.w3.org/TR/rdf11-mt>

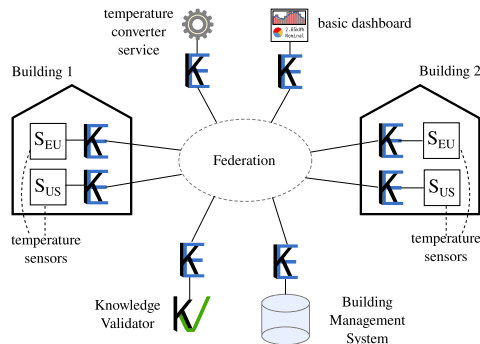


Figure 2. Network architecture of the simulated environment used for requirement verification. Shown are eight knowledge nodes, each with their own *Knowledge Engine* instance, across two buildings.

Table 3. List of task requirements for the *Knowledge Validator*

#	Description
RQ.KV-1	Real-time monitoring of data streams
RQ.KV-2	Learn patterns of nominal behaviour
RQ.KV-3	Update patterns of nominal behaviour
RQ.KV-4	Detect deviations from nominal behaviour
RQ.KV-5	Report on detected deviations
RQ.KV-6	Provide explanations for detected deviations

4.2.1. Verification by Demonstration

The demonstration illustrates several key features of our framework. Firstly, data streams from different knowledge nodes can, if needed, be integrated on the fly. This ability made it possible for the *Knowledge Validator* to receive temperature readings together with the buildings they reside in. These data come from different nodes—the readings from the sensors and the location from the building management system—making integration of data and structure non trivial. Another key feature are the reasoning capabilities, which, in the demonstration, are used to infer that two different type of temperature sensors belong to the same group of devices. A final key feature is the ability to integrate real-time data processing services, as was demonstrated with the unit-of-measure converter service.

To verify the task requirements we apply a *proof by negation*: if any of the requirements were not met, the demonstration would not have been successful. Since the demonstration *did* succeed, we infer that all task requirements (RQ.KE.1-5, RQ.KV.1-6) must have been met. How these requirements align with the domain requirement is addressed in the discussion (Sc. 5).

4.2.2. Verification by Testing

Unit tests were written to verify the requirements programmatically. These unit tests, one per task requirement, check whether a specific action has happened and whether the number of occurrences of that action is as expected. Checking the unit tests against the simulation showed that all 11 tests passed successfully. This implies that all task requirements were met in the controlled setting.

The verification by unit tests suggests that the framework meets all task requirements: the *Knowledge Engine* successfully registered the various knowledge nodes (RQ.KE.1) and their publication and subscription requests (RQ.KE.2). Data from these nodes was also successfully accepted by the *Knowledge Engine* instances (RQ.KE.3) and relayed to other nodes (RQ.KE.4/5). Similarly, the *Knowledge Validator* was able to monitor the data streams in real time (RQ.KV.1), learn and update patterns from them (RQ.KV.2/3), and detect and report on deviations from these patterns (RQ.KV.4/5). Human-readable explanations were provided in these reports (RQ.KV.6). How the task requirements align with the domain requirements is addressed in the discussion (Sc. 5).

Listing 3: Domain knowledge used in the demonstrator, defining two facts and one entailment rule (RDFS-9).

```
ex:EUTemperatureSensor rdfs:subClassOf ex:Sensor .
ex:USTemperatureSensor rdfs:subClassOf ex:Sensor .

(?z rdf:type ?y) :- (?x rdfs:subClassOf ?y), (?z rdf:type ?x) .
```

4.3. Use Case

*HEDGE-IoT*⁹ is an ongoing EU project that aims to revolutionise energy systems by deploying IoT components across different levels of the energy infrastructure. As contributing partner in this project, we are collaborating with the *Electricity Campus*¹⁰ at Arnheims Buiten in a pilot study in The Netherlands to improve energy grid efficiency and resilience. The electricity campus is equipped with its own local energy grid, independent from the national grid, making it an ideal location for innovative energy research.

Energy flows on the local grid are controlled by a grid energy management system which distributes the power amongst several distinct buildings. These buildings are gradually being made more energy efficient and intelligent over the course of the project, by integrating various smart devices including lighting control, solar panels, electric vehicle charging stations, and HVAC systems. Each building is equipped with its own independent building and energy management systems. These systems can communicate with the grid-wide system for more intelligent power distribution.

The current arrangement utilises the *Knowledge Engine* to manage the exchange of knowledge, information, and data between many of the aforementioned smart devices, with more being added on a regular basis. Other communication flows that are managed by the *Knowledge Engine* include those from and to sub-metering devices and the national grid. The *Knowledge Validator* monitors these data flows and provided real-time anomaly detection and predictive maintenance on the local campus grid. Several web-based dashboards offer live feeds, as well as summaries and visualizations to building owners and stakeholders.

5. Discussion

Interoperability issues and other downstream problems pose genuine challenges to building automation, hampering sustainability goals and reducing overall efficiency. These issues and challenges have long been known by the smart buildings community and are core goals of the *HEDGE-IoT* project. Our framework has been specifically designed to address the complications experienced by this

⁹See <https://hedgeiot.eu>

¹⁰See <https://arnhembuiten.nl/electricitycampus>

community, and thus aligns well with the needs and expectations of its members. Moreover, since our framework is data-model agnostic, only requiring the data to be encoded as *RDF*, applying it to other domains and use cases is trivial.

By setting up a simulated environment we were able to successfully demonstrate our framework and show that it works as expected in a controlled setting. This was further substantiated by the running of unit tests against this environment, all of which passed. Taken together, the two verification procedures indicate that all task requirements (RQ.KE.1–5, RQ.KV.1–6) were met. Moreover, the requirements were met under the conditions set by the domain requirements.

That the task requirements satisfy the domain requirements can be motivated as follows. Firstly, all task requirements were met using open, well supported standards and protocols (RQ.1/2) which, together with the inferencing capabilities, provide improved interoperability and integration (RQ.3). By hiding protocol and data format fragmentation behind adaptors and by providing location-agnostic data exchange via federation, the framework provides improved data accessibility and transparency (RQ.4). Similarly, the demonstration showed real-time, low-latency data validation through edge-based anomaly detection (RQ.6/7). While outside the scope of the simulation, the validation reports and accompanying human-readable explanations can be used by operators to assess device status and to schedule maintenance actions (RQ.8). Finally, the adoption of open web standards makes it trivial to monitor and manage data flows in real time (RQ.5).

It can be argued that the domain requirements have not been met to their fullest extent. Indeed, conditions such as effortless interoperability and full data transparency are difficult to measure, and therefore also difficult to verify. However, the domain requirements represent an ideal—a goal to strive towards as a community. With this in mind, we believe that our work moved the goal post significantly closer to this ideal situation.

The use case shows that our framework also satisfies the requirements outside a controlled environment and that it can readily be incorporated into existing IoT environments with little effort. Preliminary results suggest the *Knowledge Engine* can indeed facilitate interoperability within a network of many different smart devices, control systems, and applications, and that it is robust towards the dynamic and distributed nature of such a system. Likewise, the findings suggest that the *Knowledge Validator* can successfully learn graph patterns and recognise deviations from them in real time, and that the validation reports can be used by operators and building owners for more-informed decision making. A proper evaluation of these components in this setting will occur in the near future.

Since our framework depends on adaptors to translate the data to and from graph format, is it necessary to setup or, if unavailable, create appropriate adaptors. This can be seen as moving the interoperability issues from the devices (or aggregators) to the adaptors. However, by placing the adaptors directly after the devices, our framework ensures that all further data flows and downstream applications are freed of these issues, facilitating seamless integration, utility, and interpretability down the line.

6. Conclusion

This work proposed a distributed, knowledge-driven data exchange and validation framework, built upon Semantic Web technologies, that aims to overcome interoperability (-related) issues in building automation and management systems. Our framework was born from the needs of building owners and stakeholders in the smart buildings domain, who are looking for means to make their properties more intelligent and efficient. To evaluate our framework we employed requirement-based testing, by first eliciting requirements via domain and task analysis, as well as from scenarios which were drafted together with building owners. The requirements were verified against our framework via a demonstration and unit tests in a controlled and real-world environment. By collaborating in an ongoing pilot study in The Netherlands, combining research with practical application and tangible results, we strive to position our framework as an open, sustainable, and extendable addition to the IoT landscape.

There are several avenues for future work. One important direction involves scalability: while small graph patterns and a limited number of nodes perform reasonably well, the performance starts to noticeably drop when graph patterns and the network become larger. Using parallel computing or computation offloading might alleviate this problem. Another direction is the incorporation of more-sophisticated security and access control. The current setup assumes complete trust between actors within the network, yet this is often rare in practice. A solution that combines end-to-end encryption with public/private keys and ontology-based access control [40] is being considered. A final direction of future work is a thorough evaluation of the *Knowledge Validator*, by casting the anomaly detection task as a multi-class classification problem and by assessing its performance using appropriate metrics and across various datasets. Preparations for this are in the works.

Declaration on Generative AI

Generative AI was not used during any step in the creation of this manuscript nor in any of contributions presented in it.

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DQV4AI: Representing AI Quality Dimensions for ISO and EU AI Act Using the Data Quality Vocabulary

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Abstract. As the use of AI proliferates rapidly, its development and deployment across all stages requires understanding and documentation regarding aspects such as its accuracy, fairness, and transparency—collectively called ‘quality characteristics’. This has led to standardisation at forums such as ISO, and creation of obligations within regulations such as the EU’s AI Act. To support this developing landscape, we assess the sufficiency of the Data Quality Vocabulary (DQV), which is a W3C Working Note, for representing AI quality dimensions. We find that while the DQV can be useful, some concepts require changes to the definition to enable use beyond datasets, for AI models and systems. Based on our findings, we propose updates to DQV concepts to allow a more general quality representation and provide taxonomies based on ISO/IEC DIS 25059 standard on AI quality models and the EU AI Act to represent key AI quality dimensions and examples of their use. We conclude with a discussion on the future of AI quality documentation, the role of Semantic Web standards, and facilitating interoperability across value chains and data spaces.

Keywords. AI quality, Data Quality Vocabulary (DQV), ISO/IEC 25059, EU AI Act, Semantic Web

1. Introduction

As Artificial Intelligence (AI) continues to advance at an unprecedented pace and is being rapidly deployed across high-stake domains, delivering *high-quality* AI systems are crucial to ensure safe and trustworthy use of AI. Further, documented AI incidents, in repos-

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itories such as AI, Algorithmic and Automation Incidents and Controversies (AIAAIC) repository of AI incidents², show that poor quality, in different aspects such as accuracy and robustness, is one of the root causes of AI harms. Therefore, realising high-quality AI systems, that provide value to stakeholders while avoiding potential harms, requires understanding, measuring, and documenting quality attributes across the AI development and deployment lifecycle stages. This need has led to standardisation activities within the International Organization of Standardisation (ISO), particularly through its technical committee for AI standardisations, JTC 1/SC 42³. In regard to AI quality attributes, building upon the long-standing 25000 SQuaRE (Systems and software Quality Requirements and Evaluation) series of standards, ISO/IEC 25059 [26] was published in 2023, defining a quality model for AI systems. However, following the updates to the 25000 series, it is expected to be replaced by ISO/IEC DIS⁴ 25059 [27] in the near future.

In parallel, emerging regulatory frameworks, such as the European Union's (EU) AI Act [1], impose obligations for evaluation and documentation of AI system quality. Particularly the AI Act requires evaluation and documentation of quality of datasets, models, and AI systems. *Transparency* (Art. 13), *accuracy* (Art. 15), and *robustness* (Art. 15) are among the quality attributes for AI systems required by the Act to be assessed and documented through measures such as benchmarking, testing, and participating in AI regulatory sandboxes [19].

Given the need to log, trace, and update AI quality information as the system evolves and to exchange and integrate quality-related information across the AI lifecycle, e.g. drawing on training data quality for model assessments, interoperable, machine-readable specifications of AI quality information are not only beneficial but essential. Prior research on AI documentation, including [9,39,20,19,28], argue that Semantic Web-based approaches are promising in delivering auditability, interoperability, scalability, and enable automation for regulatory compliance tasks.

Within the existing standardised vocabularies, Data Quality Vocabulary (DQV) [7] is a well-established (meta)data model for data quality developed by the W3C Data on the Web Best Practices (DWBP) Working Group. In the decade since the publication of the latest version of DQV in 2016, the technological landscape has been changed substantially, with AI (and particularly large language models) moving to the forefront. While DQV has been reused in modelling quality characteristics of entities other than data, e.g. in [19], there is a key semantic issue in directly reusing DQV for AI systems and models as DQV concepts and properties are explicitly defined for quality of *data or data distributions*. Further, DQV is an extension of DCAT V1 (published in 2014) and has not been updated as per subsequent version of DCAT, namely V2 (2020) and V3 (2024). While DQV provides a well-established framework for dataset quality assessment, its concepts are coupled with dataset-oriented semantics derived from DCAT V1. In this paper, we identify and analyse the semantic limitations of directly applying DQV to AI systems, particularly regarding contextual and task-dependent quality evaluation (see subsection 4.1). To address this limitation, this work generalises DQV by introducing a more flexible abstraction based on `dcate:Resource`, enabling the modelling and evaluation of diverse AI artefacts beyond datasets. Since DQV is a widely-used and well-established framework, we avoid introducing a new ontology under a separate names-

²<https://www.aiaaic.org/aiaaic-repository>

³<https://www.iso.org/committee/6794475.html>

⁴Draft International Standard

pace. Instead, following the approach taken by DCAT, we propose our recommendations to extend the vocabulary to support AI-specific quality dimensions, contextual evaluation scenarios, and semantically interoperable quality metadata while remaining compatible with existing Semantic Web standards and linked data practices.

As no established open models or taxonomies for AI quality dimensions exist, we develop taxonomies of AI quality attributes based on our proposed DQV extension using two primary sources: the ISO/IEC DIS 25029 and the EU AI Act. It is worth noting that since this draft version of the standard (DIS) has been published on the ISO portal, we use it as source for developing the taxonomy, while acknowledging that it is subject to changes. The paper further contributes by aligning the proposed extension with existing AI semantic modelling efforts and demonstrating how Semantic Web technologies can support explainable, interoperable, and reusable AI quality assessment frameworks.

Therefore, the contributions of this paper are:

- DQV4AI, which is described as a set of recommendations to update DQV (based on change in DCAT v1 to v3 with examples), supporting the representation and assessment of quality for AI systems and AI models.
- Taxonomies of AI quality dimensions based on ISO/IEC DIS 25959 standard and requirements of the EU AI Act.

The remainder of this paper is structured as follows: section 2 provides an overview of the existing semantic models for data, software, and AI quality as well as models for quality based on the AI Act. Section 3 presents the ontology engineering methodology used. Section 4 presents DQV4AI, the proposed extension to the DQV, alongside the developed taxonomies for AI quality based on ISO/IEC DIS 25059:2025 and the AI Act. Section 5 illustrates the applicability of the proposed model through three proof-of-concept examples. Section 6 discusses the benefits and section 7 concludes the paper.

2. Related Work

2.1. Standards for Data Quality

A number of established standards and frameworks address data quality across different domains. ISO 8000⁵ provides a comprehensive foundation by defining semantic, syntactic, and pragmatic aspects of data quality, along with associated dimensions and management practices. In the geospatial domain, ISO 19157⁶ focuses specifically on evaluating and reporting the quality of geographic data, introducing a set of dimensions tailored to spatial datasets. ISO 25012⁷, part of the System and Software Quality Requirements and Evaluation (SQuaRE) standards, offers a widely adopted general data quality model for structured data, covering a broad range of dimensions and serving as a key reference in this work due to its extensive scope. Complementing these standards, W3C's Data Quality Vocabulary (DQV) [6] and related best practices support the publication and assessment of data quality on the web, particularly in Linked Data contexts. Another approach

⁵<https://www.iso.org/standard/50798.html>

⁶<https://www.iso.org/standard/32575.html>

⁷<https://www.iso.org/standard/35736.html>

proposed for Linked Data is the daQ ontology [14], which inspired the design of DQV's metric representation and provides a structured model for representing quality metadata through categories, dimensions, and metrics, enabling machine-readable quality assessment and comparison across datasets. In this space, prior work further extends quality modelling by defining multiple dimensions across categories, which inform mappings and interoperability across heterogeneous data quality frameworks.

2.2. Semantic Models for Software Quality

Several semantic models exist for representing software quality in general. Bajnaid et al. [10] developed a Software Quality Assurance Ontology mainly based on the ISO/IEC 9126:2001 standard for product quality in software engineering. The Software Quality Ontology [31] is constructed on elements of the ISO/IEC 25010:2011 standard as well as various object-oriented metrics, and consequently applied in a software quality assessment. Similarly, Blas et al. [12] published an ontology for the documentation of a software quality scheme based on ISO 25010. The SQuAP ontology [13] combines the modelling of software quality, architecture and processes in software engineering, based on ISO standards (ISO/IEC 25010:2011, ISO/IEC 12207:2008 and ISO/IEC 42010:2011 respectively), with a focus on software quality in banking systems.

2.3. Semantic Models for AI Quality

There is a body of work on semantic resources that enable represent quality of AI models and AI systems across several dimensions. The *Semantic Machine Learning Model Ontology (SML)* [29] allows the description and storage of various characteristics of ML models. Most relevant to for quality, this includes the *SML:Evaluation* class, alongside a *SML:Score* for a *SML:Metric* in a *SML:Category*. *ML-Schema* [34] provides a wider scope of representation, offering concepts for ML algorithms, datasets and experiments, including *EvaluationProcedure*, *EvaluationMeasure*, *EvaluationSpecification* and *ModelCharacteristic* for modelling quality. The *Intelligence Task Ontology and Knowledge Graph (ITO)* [11] is a resource for tracking AI tasks, benchmark results, and performance results, and as such suitable to representing quality in AI. The *Model Card Report Ontology (MCRO)* [8], as the name suggests, is a semantic representation of the commonly used model card approach [30] to the documentation of AI models. Like its non-semantic predecessor, it includes concepts such as *Metrics*, *Evaluation Data* and *Quantitative Analyses*, enabling the modelling of quality aspects across a range of criteria. *NIST's Machine Learning Lifecycle Ontology (MLLO)* [15] claims to include information on ML models' "performance and uncertainties", which would reasonably pertain to quality aspects. However, since this aspect of the ontology is not explained in detail in the paper and the MLLO GitHub repository site⁸ cannot be reached at the time of writing, the extent to which these dimensions can be modelled is uncertain. Finally, *FIDES* [16], an ontology and framework for accountability of statistical ML systems, includes the concepts of *EvaluationSpecification*, *ModelEvaluation*, *EvaluationMeasure* and *EvaluationProcedure* alongside other information about the system (such as *Model*, *Data*, *Algorithm*, etc.), with the goal of providing details for reproducibility, replicability and accountability.

⁸<https://oagi.atlassian.net/wiki/x/DgDhKAE>

While hence there are some semantic resources that allow a general representation of AI quality without pre-determining quality dimensions, none of them align with the relevant ISO standards (ISO/IEC 25059:2023 and ISO/IEC TS 25058:2024). MLLO asserts to have leverage ISO/IEC 22989:2022 alongside other ML standards, but this standard is limited to the standardization of key ML model concepts.

Apart from abstract representation of quality in AI, several semantic models specialize in modelling specific quality aspects. For explainability, XMLPO [40] focuses on the representation of explainable AI (XAI) models, such as the XAI metrics used for the evaluation. For fairness, the Fairness Metrics Ontology (FMO) [17] provides concepts for *ML model evaluation* and corresponding *evaluation metrics*, where *evaluation metrics* are derived from some *fairness metric* which in turn measures some *fairness notion*. Doc-BiasO [38] can be regarded as to some degree complementary to FMO: while FMO represents the metrics used to measure fairness, Doc-BiasO models the terminology of and relationship between potential biases (which are the likely outcomes of a fairness evaluation) in ML, including their measures.

2.4. Semantic Models for Quality Based on the EU AI Act

A few semantic models are directly based on the EU AI Act, while covering some aspects of quality in AI: The *AI Risk Ontology (AIRO)* [20] represents information of AI systems classified as high-risk under the AI Act, derived from the requirements of both the AI Act and ISO 31000. This includes the concept of *AI quality*, without however introducing further subcategories enabling the detailed representation of quality evaluation. It is also not aligned with ISO/IEC 25059:2023. *AI Cards* [19] extends AIRO to introduce a human- and machine-readable format to share information required by the AI Act, regarding technical specifications, context of use, and risk management. For modelling quality characteristics that are explicitly required by Art. 15 of the AI Act, including accuracy, robustness, and cybersecurity, DQV was reused. The *Fundamental Rights Impact Assessments (FRIA) Ontology* [36] enables creating and managing FRIA, required by Article 27 of the AI Act. As such, while being aligned with the AI Act and allowing the representation of a narrow aspect of quality of an AI system, it is not suited to represent AI quality evaluation more generally. Finally, *AI Data Governance Ontology (AIDGO)* [3] is designed to map the AI Act requirements in regard to data governance (Art. 10) to the requirements ISO/IEC 5259 Data quality for analytics and machine learning (ML). While it addresses data quality concepts, it does not reuse DQV and is primary focused on alignment of requirements.

3. Ontology Development Methodology

The development of the quality taxonomies followed the Linked Open Terms (LOT) methodology [32] with the following providing an overview of the steps:

Requirements Specification: The requirements concerning the representation of the quality of AI systems have been extracted from the EU AI Act and the relevant standards on AI quality, namely ISO/IEC DIS 25059.

Implementation: In order to ensure interoperability and alignment with existing common practices in data quality representation, the implementation have reused DQV since

it is a well-established vocabulary for representing *data* quality characteristics, with minor changes proposed to the definitions to enable its use beyond datasets and for AI models and systems. Adopting a modular approach, we use separate namespaces for the taxonomies, which are `/dqv4ai/iso-25059/` for quality characteristics derived from ISO/IEC DIS 25059 and `/dqv4ai/eu-aiact/` for the quality aspects extracted from the AI Act. This allows the alignment of dimensions and categories in DQV with the characteristics and sub-characteristics of the product quality, quality-in-use, and AI service quality models established in ISO/IEC DIS 25059:2025, as well as with quality aspects required by the AI Act.

Evaluation: The evaluation of the proposed changes to DQV and the developed taxonomies are conducted by modelling examples use cases (see [section 5](#)). Furthermore, the quality is ensured through the application of the Ontology Pitfall Scanner (OOPS!) [33] and the Ontology Pitfall Scanner for the FAIR Principles (FOOPS!) [18].

Documentation & Publication: The HTML documentation was created using the Respec template commonly used in W3C documents⁹. The taxonomy of AI quality characteristics from ISO/IEC DIS 25059:2025¹⁰ and the AI quality taxonomy from the EU AI Act¹¹ have been published online under the CC BY 4.0 license.

Maintenance: For stability and ongoing maintenance, we have contributed our work as a proposal to the W3C Data Privacy Vocabulary Community Group (DPVCG) for inclusion in the next version of the Data Privacy Vocabulary (DPV). In the future, we aim to submit our recommendation to the rechartered W3C Dataset Exchange Working Group¹² by proposing an update of DQV to support a more general quality representation that also encompasses AI models and AI systems.

4. Overview of DQV4AI

4.1. Limitations of DQV for representing AI Quality

A key challenge in directly reusing DQV for AI systems and models lies in the fact that its concepts and properties are semantically defined around the evaluation of *datasets*. Although the overall DQV approach is adaptable to AI quality assessment, thanks to its minimal ontological commitment and the absence of enforced domain restrictions in the accompanying DQV RDF specification [7], several core concepts, including `dqv:QualityMeasurement`, `dqv:QualityMeasurementDataset`, `dqv:QualityAnnotation`, `dqv:QualityCertificate`, and `dqv:UserQualityFeedback`, as well as properties such as `dqv:computedOn`, `dqv:hasQualityMeasurement`, `dqv:hasQualityMetadata`, and `dqv:hasQualityAnnotation`, explicitly reference datasets or distributions in the W3C Working Note formal definitions. Consequently, even concepts without direct dataset-oriented descriptions become semantically constrained through their relationships with these classes and properties.

A further limitation arises from the close alignment of DQV with the DCAT vocabulary, particularly the reliance on `dcat:Dataset` and `dcat:Distribution`. This coupling re-

⁹<https://www.w3.org/respec/>

¹⁰<https://w3id.org/dqv4ai/iso-25059>

¹¹<https://w3id.org/dqv4ai/eu-aiact>

¹²<https://www.w3.org/2026/04/dx-wg-charter.html>

stricts the semantic applicability of DQV for modelling AI systems, AI models, and their operational characteristics, which differ conceptually from static or versioned datasets.

Another challenge concerns the contextual nature of AI quality evaluation. In AI systems, quality is highly task- and context-dependent; for example, a model may perform poorly in reasoning tasks while achieving high accuracy in classification tasks. DQV's quality measurement approach is primarily designed for data quality evaluation, and therefore offers limited support for representing quality assessments in the context of AI. While DQV provides a strong foundation for representing general quality metadata, measurements, and annotations, its direct applicability to AI systems remains limited for task-specific and domain-specific quality assessment scenarios. In practice, AI quality evaluation often requires additional semantic resources and vocabularies to represent concepts derived from regulatory and standardisation frameworks, such as ISO standards and the EU AI Act. Therefore, complementary vocabularies, including the AI quality taxonomies proposed in this paper, are necessary to capture AI-specific quality dimensions and compliance requirements. Furthermore, the current DQV representation of quality measurements requires additional investigation to determine whether it can adequately support AI-oriented evaluation processes, including verification procedures, continuous monitoring, and Human-in-the-Loop (HITL) assessments, which are essential components of trustworthy and accountable AI systems.

4.2. Extending DQV for AI Quality

As mentioned earlier, DQV provides a well-established framework for representing quality metadata, measurements, annotations, and metrics, making it a strong foundation for interoperability and compatibility with existing Semantic Web standards. However, its current design is closely aligned with dataset-centric concepts derived from DCAT V1, particularly `dc:Dataset` and `dc:Distribution`.

To address this limitation, we propose generalising the evaluation target by replacing direct dependencies on `dc:Dataset` and `dc:Distribution` with the more abstract class `dc:Resource`. This approach enables the representation and quality assessment of a broader range of entities, including AI systems, AI models, agents, and other AI-related artefacts defined within specific use cases. This direction is also consistent with the DCAT specification itself, particularly since version 2 [5], which explicitly states that:

When describing a resource which is not a `dc:Dataset` or `dc:DataService`, it is recommended to create a suitable sub-class of `dc:Resource`, or use `dc:Resource` with the `dc:terms:type` property to indicate the specific type.

Thus, it becomes sufficient to model the target of evaluation as an instance or subclass of `dc:Resource`, depending on the specific AI artefact whose quality is being assessed. This approach is already reflected in existing AI semantic modelling efforts such as AICat [21], where concepts including `airo:AIModel` and `airo:AISystem` are defined as subclasses of `dc:Resource`. Such an approach provides a semantically flexible foundation for extending DQV beyond datasets towards the quality assessment of AI systems and models.

Figure 1 shows an overview of DQV4AI as DQV extension for AI quality assessment, with modifications and extensions to DQV concepts and properties highlighted in

Table 2. Proposed changes to expand DQV properties to AI systems and models. **Bold** indicates added text and ~~skos:broader~~ indicates removed text.

DQV property	Proposed changes
dqv:computedOn	Refers to the resource (e.g., a dataset, a linkset, a graph, a set of triples, an AI model, an AI system) on which the quality measurement is performed. In the DQV context, this property is generally expected to be used in statements in which objects are instances of deat:Dataset or deat:Distribution dcad:Resource .
dqv:hasQualityMeasurement	Refers to the performed quality measurements. Quality measurements can be performed to any kind of resource (e.g., a dataset, a linkset, a graph, a set of triples, an AI model, an AI system). However, in the DQV context, this property is generally expected to be used in statements in which subjects are instances of deat:Dataset or deat:Distribution dcad:Resource .
dqv:hasQualityMetadata	Refers to a grouping of quality information such as certificates, policies, measurements and annotations as a named graph. Quality information represented in such a grouping can pertain to any kind of resource (e.g., a dataset, a linkset, a graph, a set of triples, an AI model, an AI system). However, in the DQV context, this property is generally expected to be used in statements in which subjects are instances of deat:Dataset or deat:Distribution dcad:Resource .
dqv:hasQualityAnnotation	Refers to a quality annotation. Quality annotation can be applied to any kind of resource, e.g., a dataset, a linkset, a graph, a set of triples, an AI model, an AI system . However, in the DQV context, this property is generally expected to be used in statements in which subjects are instances of deat:Dataset or deat:Distribution dcad:Resource .

4.3. Taxonomies of AI Quality Characteristics

4.3.1. Taxonomy of AI Quality based on ISO/IEC DIS 25059:2025

ISO/IEC DIS 25059:2025 provides the quality models for AI systems and services. By introducing AI-specific quality characteristics, such as “robustness” and “environmental sustainability”, it extends three conventional SQuaRE quality standards, namely ISO/IEC 25010:2023 [24], which describes the product quality models for ICT and software products, ISO/IEC 25019:2023 [25] on quality of information systems and IT products when they are used in a specific context, and ISO/IEC TS 25011:2017 [22] on IT services qualities.

106 concepts from the ISO/IEC DIS 25059:2025 were manually extracted. These concepts are classified into 3 top-level categories (dqv:Category) defined by the standard itself: Product Quality, Quality In Use, and AI Service Quality. Following the structure of quality models provided in the aforementioned conventional SQuaRE, ISO/IEC DIS 25059:2025 introduces quality characteristics, which are further divided to sub-characteristics, which represent a unique aspect of the quality characteristic they are associated with. We consider both characteristics and sub-characteristics as dqv:Dimension, with skos:broader relation for specifying the relations between a quality characteristic and its sub-characteristics. For example, both Security (characteristic) and Confidentiality are defined as type of dqv:Dimension, with Confidentiality having a skos:broader relationship with Security (see Figure 2). Using this modelling approach resulted in 103 dqv:Dimension for AI quality characteristics. It is worth noting that ISO/IEC DIS 25059:2025 defines the AI quality model

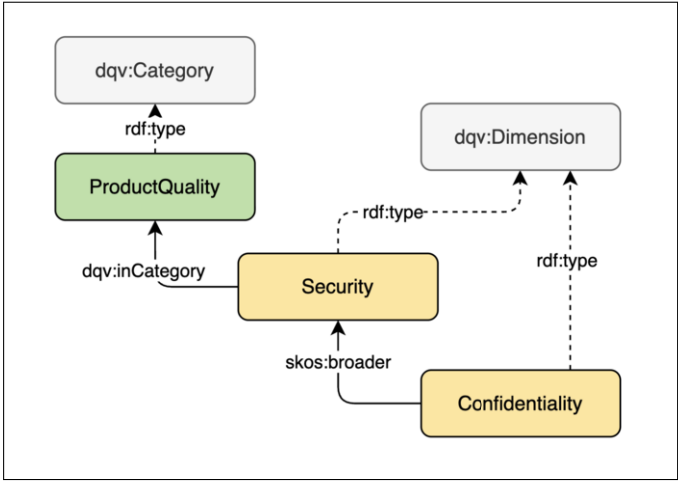


Figure 2. An example of modelling the relation between characteristics and sub-characteristics in the ISO/IEC DIS 25059:2025 taxonomy (dqv4ai/iso-25059) using DQV

and therefore does not prescribe any concrete metrics, given that these metrics might be technology- or context-specific. Such metrics can be identified from some of existing standards such as ISO/IEC TS 4213:2022 on assessment of machine learning classification performance and SO/IEC TS 42119-2:2025 on AI testing. Despite these efforts, currently, there is a lack of universally accepted metrics for some of key AI quality attributes such as fairness and trustworthiness and AI metrology is not matured yet. As the AI metrology field progress, the taxonomy of ISO/IEC DIS 25059:2025 AI qualities can be extended to include metrics. The taxonomy is made available online under the dqv4ai/iso-25059 namespace at <https://w3id.org/dqv4ai/iso-25059>.

4.3.2. Taxonomy of AI Quality based on EU AI Act

From the text of the EU AI Act, 15 quality characteristics (dqv:Dimension) and 4 metrics (dqv:Metric) were *manually* extracted in relation to data, model, and AI systems qualities. Using a bottom-up approach, we categorised these concepts to 3 high-level dqv:Category, namely System Quality Dimensions, Model Quality Dimensions, Data Quality Dimensions. Figure 3 shows the dimensions and metrics identified under each category from the EU AI Act. In the implementation of the taxonomy, dimensions and metrics are linked to AI Act articles there are identified from using dct:Source. The taxonomy is available online at <https://w3id.org/dqv4ai/eu-aiact> under the CC BY 4.0 licences, allowing free reuse and further enhancements as new guidelines are published by authorities including the EU AI Office.

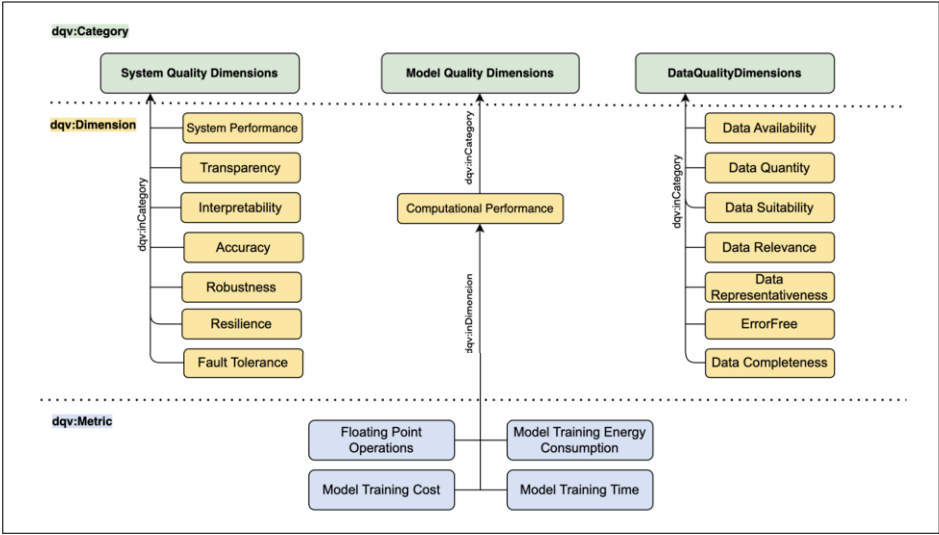


Figure 3. Quality dimensions and metrics identified from the EU AI Act taxonomy (/dqv4ai/eu-aiact). The green boxes represent `dqv:Category`, the yellow boxes are `dqv:Dimension`, and blue boxes are `dqv:Metric`.

5. Usage Examples

In this section, we provide examples to illustrate how we envisage the use of DQV for AI quality. Table 3 lists the prefixes and namespaces used in this section.

Table 3. Prefixes and namespaces used in the example section.

Prefix	Namespace
dqv	http://www.w3.org/ns/dqv#
ex	http://example.org/dqvexamples#
it6	http://data.europa.eu/it6/
prov	http://www.w3.org/ns/prov#
skos	http://www.w3.org/2004/02/skos/core#
eu-aiact	https://w3id.org/dqv4ai/eu-aiact#
iso-25059	https://w3id.org/dqv4ai/iso-25059#

5.1. Example 1: Expressing a Quality Assessment with Quality Metrics

The first example shows how a resource and its related quality assessments can be expressed in a machine-readable format. In the example, we consider an AI model (`ex:myAIModel`), as a `dcat:Resource`, to which quality measurements can be linked as per our DQV4AI proposed recommendations. According to DCAT usage note, for resources other than dataset or data service, either a sub-class of `dcat:Resource` should be created or the `dcterms:type` property should be used to indicate the resource type. We used `it6:MachineLearningModel` from the MLDCAT-AP¹³—an extension

¹³<https://semiceu.github.io/MLDCAT-AP/releases/2.0.0/>

of DCAT for machine learning models— to specify the resource type. The model has two quality measurements attached to it: one checking whether the model's endpoint is reachable, and one recording its factual accuracy score on a held-out benchmark. The quality dimensions (ex:operationalReliability, ex:factualAccuracy) and their associated metrics are defined inline. Each dimension belongs to a broader quality category.

```

ex:myAIModel
  a dcat:Resource ;
  a it6:MachineLearningModel ;
  dqv:hasQualityMeasurement ex:measurement1, ex:measurement2 .

#### - Quality measurements attached to the model deployment
ex:measurement1
  a dqv:QualityMeasurement ;
  dqv:computedOn          ex:myAIModel ;
  dqv:isMeasurementOf      ex:endpointAvailabilityMetric ;
  dqv:value                "true"^^xsd:boolean .

ex:measurement2
  a dqv:QualityMeasurement ;
  dqv:computedOn          ex:myAIModel ;
  dqv:isMeasurementOf      ex:factualAccuracyMetric ;
  dqv:value                "0.82"^^xsd:double .

#### - Quality metrics
ex:endpointAvailabilityMetric
  a dqv:Metric ;
  skos:definition          "Checks whether the model's inference API endpoint
↪ is reachable and returns a successful response."@en ;
  dqv:expectedDataType      xsd:boolean ;
  dqv:inDimension           iso-25059:Availability .

ex:factualAccuracyMetric
  a dqv:Metric ;
  skos:definition          "Ratio of factually correct statements in model
↪ outputs over a curated factual benchmark test set."@en ;
  dqv:expectedDataType      xsd:double ;
  dqv:inDimension           eu-aiact:Accuracy .

```

5.2. Example 2: Documenting the Provenance of the Quality Metadata

The second example aims to illustrate how the provenance of quality metadata can be modelled. It specifically shows how to provide specifications for the quality measurements of ex:myAIModel that were generated by a specific evaluation service (ex:myEvalService) during a timed activity.

```

ex:myAIModel
  dqv:hasQualityMetadata ex:myQualityMetadata .

ex:myQualityMetadata

```

```

a dqv:QualityMetadata ;
prov:wasAttributedTo ex:myEvalService ;
prov:generatedAtTime "2024-11-10T14:00:00Z"^^xsd:dateTime ;
prov:wasGeneratedBy ex:myEvalRun .

#### - The evaluation activity
ex:myEvalRun
  a prov:Activity ;
  rdfs:label "Evaluation run for myAIModel"^^xsd:string ;
  prov:wasAssociatedWith ex:myEvalService ;
  prov:used ex:myAIModel ;
  prov:generated ex:myQualityMetadata ;
  prov:startedAtTime "2024-11-10T12:00:00Z"^^xsd:dateTime ;
  prov:endedAtTime "2024-11-10T14:00:00Z"^^xsd:dateTime .

#### - The automated evaluation service
ex:myEvalService
  a prov:SoftwareAgent ;
  rdfs:label "AI Model Evaluation Service v3.1"^^xsd:string .

```

5.3. Example 3: Documenting the Provenance of a Single Quality Measurement

The third example shows how provenance of a quality measurement can be documented using DQV. In the example, the hallucination-rate measurement for `ex:myAIModel` is attributed directly to the evaluation agent and activity.

```

ex:myAIModel
  dqv:hasQualityMeasurement ex:hallucinationMeasurement .

ex:hallucinationMeasurement
  a dqv:QualityMeasurement ;
  dqv:computedOn ex:myAIModel ;
  dqv:isMeasurementOf ex:hallucinationRateMetric ;
  dqv:value "0.07"^^xsd:double ;
  prov:wasAttributedTo ex:myEvalService ;
  prov:generatedAtTime "2024-11-10T13:45:00Z"^^xsd:dateTime ;
  prov:wasGeneratedBy ex:myEvalRun .

#### - Metric definition
ex:hallucinationRateMetric
  a dqv:Metric ;
  skos:definition "Proportion of model responses containing at least
    ↪ one factual claim that cannot be grounded in the reference
    ↪ corpus."@en ;
  dqv:expectedDataType xsd:double ;
  dqv:inDimension ex:factualAccuracy .

```

6. Discussion of Benefits

Documenting quality characteristics is a key aspect of many best practices on data, model, and AI documentation approaches. For example, Model Cards [30] include a sec-

tion on *Metrics*, which requires reporting relevant metrics for the AI model, including its performance measures; and AI Factsheets comprise multiple sections for different quality attributes such as *Performance and Reliability* and *Safety*, including aspects such as *Explainability* and *Fairness*. When the quality information is provided in conventional text-based documentation, its generation, maintenance, assessment, comparison, and exchange can become significant hurdles, considering the volume of information and the need for continuous updates. Therefore, a primary benefit of our proposal is to facilitate representation and exchange of quality information related to different entities, from datasets to AI systems, in a standards-based, interoperable, and consistent manner, enabling interoperability across the AI value chain and data spaces.

Moreover, compliance with regulatory frameworks such as the AI Act requires assessment and documentation of quality characteristics, such as accuracy and robustness. Using DQV for modelling quality in the context of AI Act's technical documentation has been suggested by [19], however the limitations of DQV in modelling AI quality characteristics are not addressed. Our proposed extension, together with the additional taxonomies for quality characteristics based on ISO and the AI Act itself, provides a semantic foundation for representing AI quality information and allows automated self- and third-party compliance checking and auditing through querying and reasoning.

A limitation of this work is that it has not yet been evaluated in real-world settings, and further investigation is required to assess whether DQV4AI and the proposed taxonomies are sufficient for representing AI-specific evaluations and assessments required by standards and regulatory frameworks.

Moving forward, the Semantic Web community is well positioned to play a key role in providing technical approaches and best practices for AI quality documentation. The increasing uptake of structured, machine-readable documentation formats, such as Open DataSheets [37], Croissant [4], and AI Cards [19], to name a few, suggests a convergence towards an interoperable ecosystem for data, model, and AI documentation. The proposed DQV extension contributes to this ecosystem by providing an revised version of DQV as a standards-based foundation for expressing AI quality, one that can be extended with domain- and task- specific taxonomies, as we shown in this paper by developing taxonomies based on an ISO standard and the AI Act, and used by automated compliance and auditing tools.

7. Conclusions & Future Work

In this paper, we provided a detailed proposal to update DQV in order to make it suitable for modelling AI models and systems quality. The proposal includes: (1) replacing `dcat:Dataset` with `dcat:Resource` to align it with the latest version of DCAT (V3) and expanding the scope beyond data to AI models and systems; (2) changes to definitions of 5 DQV classes; and (3) changes to definitions of 4 DQV properties. This proposal is going to be presented to the Dataset Exchange Working Group, as it reconvenes, to be included in the forthcoming version of the vocabulary.

We further provide taxonomies of AI quality dimensions and metrics based on ISO/IEC 25059 and the EU AI Act. Adopting a modular design, the taxonomies are published under separate namespaces under the CC BY 4.0 licence, allowing flexibility in choosing quality dimensions and metrics and enabling mappings between different sets

of quality attributes. The taxonomies collectively comprise 128 concepts, of which only 4 are metrics. This reflects the immaturity of AI metrology, as acknowledged by the AI Act (Recital (74) and Art. 15(2)), which calls for development of further standards and benchmarks. Moreover, since AI quality is often technology- or context-dependent, it is neither expected nor appropriate for general AI quality standards and regulations to prescribe exhaustive metrics. Our work provides basis for developing domain-specific or AI technology-specific taxonomies can be developed based on the state-of-art, for example the metrology for AI in medicine [35] or the ISO/IEC technical report (TR) on machine learning classification performance [23].

In our future work, we aim to expand the EU AI Act taxonomy by analysing its related guidelines and codes of practice, including the General-Purpose AI Code of Practice [2] and anticipated guidelines by the EU AI Office. We further aim to update the ISO taxonomy when the new version of ISO/IEC 250259 is published as an *International Standard*.

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Declaration on Use of Generative AI

During the preparation of this work, some of the authors used ChatGPT and Claude in order to: grammar and spelling check, paraphrase and reword. After using these tools, these authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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Blue Sky Ideas – Visionary Perspectives

From Sequential Language to Structural Intelligence: Toward Relationship-Aware AI

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Abstract. Large language models have transformed artificial intelligence by showing how much can be learned from language at scale. Their success has made language appear to be the dominant substrate of intelligence. This Blue Sky paper argues that this view is incomplete. We focus on one specific gap: what may be missing when relationships and structure are learned primarily through language.

We argue that LLMs face two language bottlenecks. First, they are token-sequence systems: even with attention and rich internal representations, they process and generate language through sequences of tokens. Second, their training data is human language, which is the sequential expression of cognition rather than the full relational process from which cognition emerges. Neuroscience gives reason to take this distinction seriously: the connectome and network neuroscience describe the brain in terms of elements, connections, and graph-like organization, while recent work argues that language is primarily a tool for communication rather than thought itself.

This suggests that LLMs learn from linguistic traces of intelligence, but not necessarily from the full relational processes that produce those traces. We therefore argue for relationship-aware AI as an open research direction: not a solved architecture, but a search for ways to combine linguistic fluency with explicit awareness of relationships and structure. The central distinction is between inference-time use of relationships and training-time learning from relationships. Many current systems add structure after a language model has already been trained, through retrieval, graph augmentation, tools, memory, or verification layers. These approaches are important, but the deeper Blue Sky challenge is to make relationships part of representation learning itself, so that structural awareness is learned by the model rather than supplied only after training.

Keywords. large language models, relationship-aware AI, structural intelligence, relational inductive biases, language and cognition, representation learning

1. Introduction

Large language models have moved artificial intelligence into the center of scientific, industrial, and public attention. Their fluency is striking. They summarize, translate, ex-

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plain, generate code, write arguments, and increasingly interact with tools, images, documents, and structured data. Because of this fluency, it is tempting to treat language as the dominant substrate of intelligence. If a system can communicate well, perhaps it understands. If it can answer in language, perhaps it can reason in the same way humans do.

This paper argues for a more careful view. The question is not whether LLMs are powerful. They clearly are. Nor is the question whether they will continue to improve. They almost certainly will. Better architectures, larger contexts, multimodal inputs, tool use, memory, planning, and reinforcement learning may reduce many current limitations. The deeper question is different: if LLMs scale intelligence through language, what kind of intelligence are they scaling?

A central premise of this paper is that this question concerns the training stage, not only the inference stage. At inference time, a trained LLM can be connected to external structure: a knowledge graph, a database, a retrieval system, a tool, a memory module, or a verifier. These mechanisms are especially useful for supplying current, domain-specific relationships needed for a particular decision. The deeper question is whether relationships and structure should also participate directly in training, so that models learn not only relational patterns but also that different relationships carry different evidential responsibilities. Some may be inferred probabilistically, while others—such as provenance, authorization, dependency, temporal validity, or scientific evidence—must remain explicit and testable when a decision depends on them.

This paper does not claim to explain all limitations of LLMs. Language bottlenecks are only one part of a broader landscape. For example, LeCun argues that autonomous intelligent systems require predictive world models, intrinsic motivation, hierarchical planning, and mechanisms for learning representations from observation rather than relying only on language-like prediction [1]. Related work such as I-JEPA reflects this broader movement toward systems that learn internal models of the world from non-linguistic signals [2]. Multimodal and vision-language models already broaden representation learning beyond text, but non-linguistic inputs do not by themselves guarantee that relationships are represented with explicit type, provenance, or evidential status. Our focus is narrower: this paper examines what may be missing when intelligence is learned primarily through language, especially relationships and structure.

We argue that LLMs face two language bottlenecks. The first lies not simply in their token-sequence format, but in the training objective defined over it. Transformer-based models represent and process sequences of tokens [3]. Standard next-token prediction can learn complex relational patterns, yet it rewards likely linguistic continuations without inherently requiring relational properties, provenance, or evidential status to be preserved. The bottleneck is therefore not what the architecture can represent, but what the learning objective requires it to represent reliably.

The second bottleneck is in the training signal itself. LLMs are trained largely on human language, but human language is a compressed expression of cognition: it communicates the results of cognitive processes without necessarily preserving the relational organization that produced them. Written and spoken language carry traces of human thought, but they are not identical to the processes that produced them. A sentence may express the result of perception, memory, spatial reasoning, social understanding, unconscious recombination, or structural insight, but it does not necessarily preserve the full relational organization behind that result. The two bottlenecks therefore compound one

another: relational organization may already be compressed in the training signal, while the learning objective does not inherently require the remaining relational structure or its evidential status to be preserved.

This distinction is not only a modern neuroscience concern. Einstein, reflecting on his own scientific thinking, emphasized that words and language did not appear to play a primary role in the early stages of thought. Instead, he described his thinking as relying on signs and more or less clear images that could be combined, with conventional words or symbols appearing only later [4]. For the present argument, the important point is not biographical detail, but the distinction between internal reasoning and its later expression through language.

Neuroscience gives reason to take this distinction seriously. The connectome perspective describes the brain through neural elements and their interconnections. Sporns, Tononi, and Kötter proposed the human connectome as a structural description of the brain in terms of elements and connections, emphasizing that understanding brain function requires understanding not only parts, but also their interconnections [5]. Network neuroscience extends this view by studying structural and functional brain systems using graph-theoretic concepts such as hubs, modularity, hierarchy, clustering, and small-world organization [6]. At the same time, Fedorenko, Piantadosi, and Gibson argue that language is primarily a tool for communication rather than thought itself [7]. This sharpens the distinction between cognition and its linguistic expression.

If language is a communication channel, then LLMs are trained on linguistic traces of intelligence, not directly on the full relational processes that produce those traces. Some relationships are explicitly expressed and supported by identifiable sources, while others remain implicit, distributed across context, or are inferred during reasoning. Both explicit and inferred relationships may be useful, but they play different evidential roles. In high-consequence settings, an inferred relationship cannot substitute for one whose provenance and validity must be traced to supporting evidence.

This paper therefore argues for relationship-aware AI as a broader research direction: systems that combine linguistic fluency with explicit awareness of relationships, structure, and evidential status. Existing approaches point toward this direction, but they do not yet define its final form. The goal is not to replace LLMs with graphs, nor to claim that any current graph method solves the problem, but to develop structural intelligence: learned representations shaped by relationships, so that later inference can test whether generated claims are structurally supported and whether the relationships on which they depend are appropriate evidence.

The rest of this paper develops this argument in three steps. First, it describes the two language bottlenecks and why they matter for relational learning. Second, it discusses relationship-aware AI as an emerging but incomplete direction, distinguishing inference-time structure from training-time relational learning. Third, it frames the unresolved problem of representation and structural selection through the idea of a Poincaré-like bridge.

2. The Two Language Bottlenecks

LLMs are token-based systems: documents, questions, and answers are represented and generated as sequences of discrete units [3]. Transformer-based models can learn com-

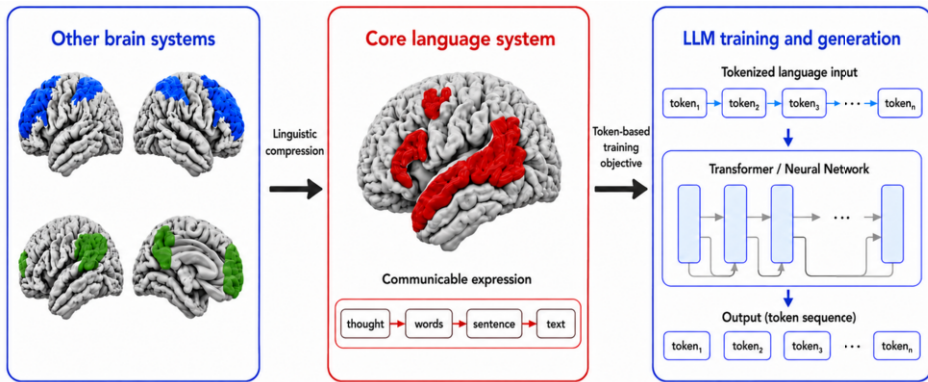


Figure 1. Linguistic compression and token-based training. Human cognition involves multiple brain systems, but communicable thought is compressed into words, sentences, and text before becoming training data. LLMs then learn from and generate token sequences. One bottleneck lies in the compression of cognitive processes into linguistic traces. The other lies in a token-based training objective that does not inherently require relational properties, provenance, or evidential status to be learned and preserved reliably. The key issue is not only what structure can be supplied during inference, but what relational regularities and evidential distinctions are learned and preserved during training.

plex dependencies, abstractions, and latent relational patterns, but standard next-token prediction does not reliably induce relational patterns that satisfy structural constraints or generalize across equivalent formulations. The reversal curse, for example, shows that models trained on a relation in one direction may fail to recover its inverse, even when the underlying fact has been learned [8]. The same objective also does not inherently require provenance or evidential status to be preserved. The bottleneck therefore concerns both the fidelity with which relational structure is learned and the status attached to the relationships that are produced.

The second bottleneck lies in the training signal itself. LLMs are trained largely on human language, but language is a compressed expression of cognition: it communicates the results of cognitive processes without necessarily preserving the relational organization that produced them. This is different from saying that a model lacks enough context during inference. The deeper issue is what the model learned from in the first place. A sentence may express the result of perception, memory, spatial reasoning, social understanding, or structural insight without preserving the full set of relationships involved in producing that result.

Although modern neural networks were historically inspired by biological neurons, they are engineering models rather than faithful models of the brain. Neuroscience nevertheless provides a useful way to frame the relational issue. The connectome perspective describes the brain through neural elements and their interconnections. Sporns, Tononi, and Kötter introduced the human connectome as a structural description of the human brain, emphasizing a connection matrix of neural elements and the importance of interconnections for understanding brain function [5]. In this formulation, the connectome is not merely a list of parts, but a structural map of relations among parts.

Network neuroscience develops this idea further. Bullmore and Sporns review how structural and functional brain systems can be studied as complex networks, using graph-theoretic concepts such as small-world topology, hubs, and modularity [6]. The point for AI is not that the brain is literally a graph, or that future AI should copy the brain.

The point is more modest but important: biological intelligence is deeply relational. It depends not only on units, but on patterns of connection among units.

Anokhin's cognitome hypothesis extends this relational perspective by proposing cognitive functional elements organized within a neural hypernetwork [9]. The proposal is notable here because the relevant units are considered together with their functional relations and organization, rather than only as isolated neural elements.

Language has a different form. It converts complex cognitive processes into a communicable stream of words and sentences. Two studies from the Fedorenko lab approach this distinction from complementary directions. One argues that language is primarily a tool for communication rather than thought itself [7]. The other distinguishes formal linguistic competence from functional linguistic competence in large language models [10]. Together, these perspectives suggest that success at linguistic form does not by itself establish that the relationships expressed in language are grounded, current, or supported by traceable evidence.

LLMs therefore learn primarily from linguistic traces of intelligence rather than from the full relational processes that produced them. Some relationships are explicitly stated in language, including causes, dependencies, hierarchies, ownership, similarity, analogy, and sequence. Others may be compressed, implied, distributed across context, nonverbal, structural, or never verbalized at all. The issue is not simply whether LLMs can compute relational patterns; it is whether those patterns are learned reliably and represented in a form that preserves type, provenance, and the distinction between observed evidence and inferred completion. A social network is not reducible to descriptions of people. A supply chain is not reducible to a set of documents. A molecule is not reducible to a list of atoms. A scientific field is not reducible to isolated abstracts. In each case, meaning depends on relationships.

The implication is that LLMs may be trained on linguistic traces of intelligence rather than the full relational processes behind them. Adding relational evidence at inference time can improve grounding and verification, but it does not fully address relational organization that was absent, compressed, or learned unreliably during training. This gap motivates relationship-aware AI as a broader research direction toward systems that learn from relationships as well as language.

3. Toward Relationship-Aware AI

Because relational organization may be compressed, omitted, or learned unreliably from language, linguistic fluency alone is insufficient for some forms of reasoning and verification. Future AI systems therefore need ways to make relationships available more directly and to preserve the distinctions that determine how those relationships may be used.

Several existing directions already provide partial forms of relationship awareness. LLMs may learn relations from language and structured data, while knowledge graphs, GraphRAG, and related methods make those relations more explicit [11,12,13,14,15]. The challenge is not only whether a relationship is available, but whether it is used for prediction, as a constraint, or as evidence. Evidence substitution occurs when a plausible inference replaces the explicit evidence required for a valid decision. Relationship-aware systems must therefore distinguish learned structure from relations that need to remain explicit and traceable at inference time.

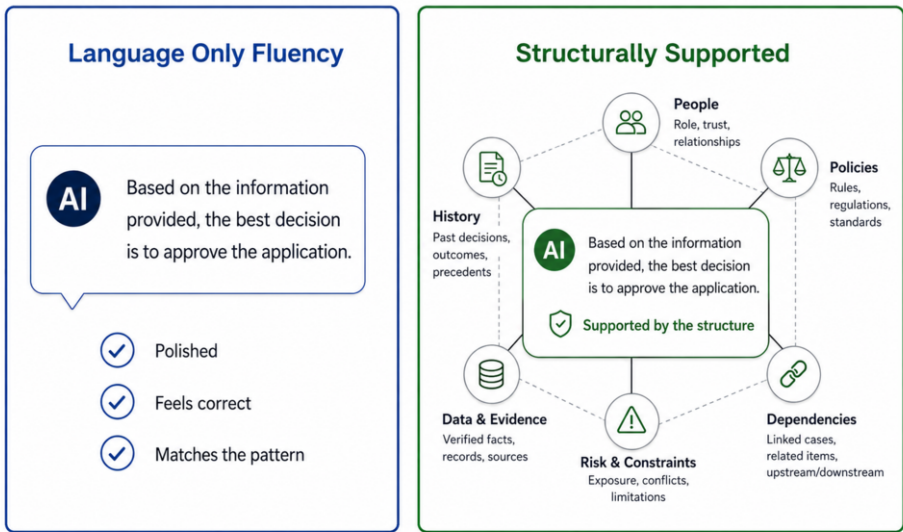


Figure 2. Language-only fluency versus structural support. A language-only system may produce a polished and plausible answer while missing relationships beneath the surface. A relationship-aware system connects the answer to structural evidence, including history, data, policies, dependencies, people, and constraints. Inference-time structure can support, constrain, or verify an answer during or after generation; the broader research challenge is to make relational structure part of what models learn. The point is not only that an answer sounds correct, but that it is grounded and verifiable through relationships. The distinction is especially important when a relationship is not merely useful for prediction, but required as evidence for a valid decision.

In the common pretrain-then-deploy pattern, training shapes the model’s representations, while inference uses those representations together with current data, graphs, or tools. Inference-time structure can improve grounding and verification, but it does not change how an already-trained model learned to represent relationships. The deeper challenge is to make relational structure part of training through data, objectives, or architecture [16].

Neuroscience-facing models provide one example of training beyond text. Brain-graph learning represents neural systems through explicit relational structure [17], while BrainLM learns from large-scale neural recordings and brain-graph foundation models combine neural activity with graph organization [18,19]. These approaches show that relational and temporal signals can participate directly in training, although they do not resolve how decision-critical evidence can be kept explicit and traceable.

World-model and multimodal research provide another route beyond language-only learning [20]. PointWorld learns three-dimensional state–action dynamics from visual observations and robot actions [21]. Theory of Space instead evaluates whether foundation models can construct and revise spatial beliefs through active exploration, and reports continuing failures in maintaining coherent beliefs under changing observations [22]. Together, these works show both the promise of non-linguistic relational learning and its current limitations. Such learned structure does not by itself provide explicit, traceable evidence.

Taken together, these directions show that parts of relationship-aware intelligence are already emerging. What remains unresolved is how models can learn relational struc-

ture reliably while preserving the distinction between prediction and evidence. The open problem is not simply to attach graphs to LLMs or improve retrieval, but to connect linguistic fluency with learned structural competence while keeping decision-critical evidence explicit and traceable.

4. The Poincaré-Like Bridge

Henri Poincaré was one of the rare scientists whose work crossed mathematics and physics, including geometry, topology, dynamical systems, and celestial mechanics. He is relevant here not because of any single theorem, but because he repeatedly connected areas that had been treated as separate and recognized common structure beneath different forms. This intellectual ability motivates the idea of a Poincaré-like bridge for AI: a way to connect linguistic fluency with relational organization so that models can perform structural selection.

Poincaré's work on topology provides a historical example. Problems of continuity, connectedness, and global structure were not naturally expressed by the dominant methods of the time, and progress required new ways to represent structural properties [23]. The relevance for AI is not that topology supplies the missing method, but that a problem may remain difficult until it is expressed in the right form.

Poincaré also described mathematical invention as conscious preparation followed by unconscious recombination, illumination, and later verification [24]. He associated the selection of fruitful combinations with a sense of mathematical beauty and harmony. Recast for the present argument, the relevant point is that many combinations are possible, but only some preserve the structure of the problem and survive verification.

This distinction matters for AI because LLMs are already powerful combinatorial systems. They can generate answers, analogies, hypotheses, and plausible connections from language. But Poincaré's harder problem remains: which combinations are structurally meaningful? Structural intelligence requires learned structural competence: more than producing possible connections, a model must recognize which relationships belong to the organization of the problem.

The deeper analogy is that the problem may not yet have found its proper form. Existing approaches provide pieces of relationship-aware intelligence, but the representation that unifies linguistic fluency with structural selection remains unresolved.

In this paper, the missing form concerns training: how relational structure can shape what models learn before inference begins. The current evidence required to justify a consequential decision must remain explicit and auditable, but the model must learn the structural competence needed to recognize when such evidence is required.

In this sense, AI is missing its Poincaré-like bridge: a way to move from linguistic fluency to structural selection. The challenge is not only to generate possible connections, but to learn which relationships belong to the organization of the problem and when their support must remain explicit.

5. Conclusion

Large language models have transformed what AI can do with language. Their fluency shows how much intelligence can be reconstructed from linguistic traces, but it does not

establish that language preserves the full relational organization behind thought, knowledge, and action. Some relationships are compressed or omitted in linguistic expression, while others may be learned unreliably even when they are present in the training signal.

This motivates relationship-aware AI as a research direction rather than a single architecture. The deeper challenge is to make relational structure part of training so that models develop structural competence: the ability to learn relational regularities, recognize constraints, distinguish plausible from structurally meaningful connections, and recognize when explicit evidence is required. Structure may enter training through data, objectives, or architecture, while the current evidence required to justify a consequential decision must remain explicit and auditable at inference time.

The distinction between prediction and evidence is therefore central. A model may infer a relationship correctly, but a plausible inference cannot replace the traceable support required for a consequential decision. This failure is evidence substitution. Trustworthy AI must combine learned structural competence with external evidence that can be inspected, updated, and verified.

AI is therefore still missing its Poincaré-like bridge: a way to move from linguistic fluency to structural selection. The open question is not only how models can generate possible connections, but how they can learn which relationships are structurally meaningful and when those relationships require explicit support. The task is to test whether new architectures, objectives, relational data, or combinations of existing methods can improve structural selection while preserving the distinction between prediction and evidence.

Declaration on Generative AI

During the preparation of this work, the author used ChatGPT (OpenAI), Claude (Anthropic), and Fable (Anthropic), under active author direction, to support literature and source discovery, language editing, rephrasing, and structural refinement. The central research question, arguments, conceptual framework, interpretations, and conclusions were developed by the author. The author guided the use of these tools through specific instructions, independently verified relevant sources and factual claims, and critically reviewed and revised all incorporated outputs. The author takes full responsibility for the publication's content.

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From Fact to Feeling: Narrative Knowledge Graphs for Human-AI Interaction

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Abstract.

Knowledge Graphs (KGs) and Large Language Models (LLMs) each capture a different dimension of human knowledge, and neither captures both. KGs excel at representing factual, structured information, yet remain blind to the perspectival and narrative texture through which humans construct and communicate meaning. LLMs absorb this texture implicitly from language, yet do so without explicit semantic structure, leaving their interpretive operations opaque, difficult to revise, and impossible to hold accountable. The next major challenge for Semantic Web research is developing formal infrastructures for the knowledge that falls between these two: perspectival, narrative, and metaphorically organized. We propose Narrative Knowledge Graphs (NKGs) as such an infrastructure. Grounded in computational hermeneutics, NKGs extend standard RDF foundations along three axes: perspective-annotated named graphs anchoring assertions to situated standpoints; narrative ontologies representing how the same events yield different accounts from different positions; and metaphorical frames, grounded in frame semantics, making explicit which conceptual structure a community or individual projects onto a domain. Together, these axes define a semantic layer designed to mediate between human experience and LLM-generated meaning. We position NKGs as a new fundamental layer for human-AI interaction, aimed at making machine-supported interpretation more explicit, revisable, and faithful to human meaning-making.

Keywords. Narrative Knowledge Graphs, Hermeneutics, Large Language Models

1. Introduction

In Ted Chiang's short story *The Truth of Fact, the Truth of Feeling* [1], a narrator living in a near-future where a wearable technology called Remem can retrieve any past experience with perfect fidelity reflects on two radically different kinds of truth: the truth of what actually happened (verifiable, recordable, retrievable) and the truth of what something meant, how it was experienced, the narrative one builds around it. The story's dystopian undercurrent is not the technology itself but what it reveals: that human mem-

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ory was never a recording device. It was always a narrative engine, selecting, recomposing, and meaning-making in ways that perfect recall quietly destroys. The consequence of this objectivized memory is a shift in epistemic authority: lived recollection becomes subordinate to the external record. What is lost is the narrative and affective labor through which experiences become memorable, communicable, and engaging. Chiang's provocation maps directly onto a distinction cognitive science has long maintained between *semantic* memory that is general, factual, decontextualized knowledge, and *episodic* memory, which is autobiographical, situated, narratively organized experience [2].

In this paper, we argue that this tension is the foundational problem of knowledge representation in the age of Large Language Models (LLMs). The knowledge infrastructures we have built so far are technologies aiming for factual truth: they could excel at storing what happened, who did what, and which entity belongs to which category. However, they are structurally blind to the perspectival, narrative, metaphorical texture through which humans actually construct and communicate meaning (see Fig. 1). On the other hand, LLMs, trained on the surface of human language, absorb this episodic aspect implicitly and uncontrollably². We argue that the next major challenge for Semantic Web research is to develop semantic infrastructures for perspectival, narrative, and metaphorical knowledge that are more truthful to human experience and interaction with both other humans and machines. Metaphor is a central instance of this narrative layer: when a patient frames their illness as a *battle* and a physician frames the same condition as a *dysfunction* to be managed, they are not using decorative language; they are operating within structurally different narrative worlds, with different implications for agency, outcome, and meaning.

This paper proposes a vision for closing this gap: Narrative Knowledge Graphs (NKGs), designed to represent knowledge as it is actually experienced and communicated: perspectival, situated, metaphorical, and narratively organized; and to integrate this layer formally into the human-LLM interaction loop. We ground this proposal in the emerging framework of computational hermeneutics [3], a framework that brings philosophical theories of interpretation to bear on the evaluation and design of AI systems. We argue that the missing infrastructure in current human-AI interaction is precisely a layer capable of attending to both the semantic and episodic nature of knowledge. The vision is organized around three interconnected contributions: a theoretical framework for perspectival knowledge representation rooted in hermeneutic theory; an architectural proposal for the human-LLM-NKG triangle as the basic unit of semantically grounded interaction; and a long-term roadmap toward community adoption, illustrated through motivating use cases in the cultural heritage domain, the healthcare domain and as an answer to the problem of benchmark contamination in LLM evaluation.

2. The road so far

Recent theoretical work has reframed LLMs as fundamentally interpretive systems: generators of contextually appropriate responses shaped by the statistical regularities of hu-

²Alignment techniques may partially reduce the variability of model outputs by encouraging more stable, normatively preferred responses. However, this reduction in variability can also come at the cost of under-representing marginal, ambiguous, culturally specific, or idiosyncratic ways of narrating experience. In this sense, alignment can improve consistency while further narrowing the expressive range through which episodic meaning is represented.

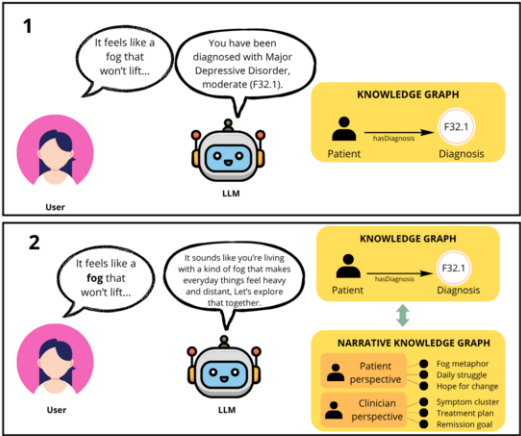


Figure 1. The standard human-LLM-KG triangle (top) versus the NKG-mediated triangle (bottom). In 1, the patient’s fog metaphor and illness narrative leave no trace in the interaction. In 2, the NKG maintains a perspectival semantic layer: patient and clinician perspectives coexist as linked subgraphs, neither collapsed into the other, conditioning the LLM’s response and updated through each exchange.

man meaning-making [3]. This framing, which the authors describe through the lens of computational hermeneutics, positions LLMs as performing something structurally analogous to what Hans-Georg Gadamer called “the fusion of horizons”: the encounter between an interpreter’s existing framework and a new text or situation, producing meaning that neither party “had” beforehand [4]. This is a powerful observation, but it raises an immediate question: if LLMs are interpretive systems, whose interpretations are they performing?

The answer, in current architectures, is unsatisfying: an average of all the interpretations present in the pretraining corpus, shaped in ways that are opaque to both users and developers. The LLM has no explicit representation of the perspective it is adopting, the community whose assumptions it is deploying, or the narrative frame through which it is organizing information [5]. Meaning happens inside the model, and it leaves no trace.

Knowledge Graphs (KGs) have been proposed as a corrective: an explicit, structured, human-readable layer of factual grounding that can anchor LLM outputs [6]. But standard KGs, as we argued above, are designed for factual truth. They can tell you that a patient was diagnosed with condition X on date Y, but not what that diagnosis meant to the patient, which metaphors they used to make sense of it (and possibly act on the basis of those metaphorical frames [7,8]), how their understanding evolved over time, or how their perspective on their own illness diverges from their physician’s. Recent works have conceptualized different strategies for representing perspectives with KGs, including the polyvocality [9] and the perspectivization approach [10,11]. Both approaches, however, do not fully account for the richness of the act of interpretation: the ambiguities, emotional framings, narrative negotiations, and socially situated meanings through which knowledge is created and recorded. The semantic layer that could hold this kind of knowledge does not yet exist in a structured and especially widely adopted form.

NKGs emerge at this point of tension. Recent work on structured representations for narratives shows that the Semantic Web community already has many of the building blocks for representing events, agents, locations, and some inter-event relations, but that the more demanding dimensions of narration—perspective, goal, outcome, conflict, and evolving interpretation—remain only partially modeled and unevenly standardized [12]. In this sense, the problem is not that narrative has been ignored, but that it has largely been treated as an object of study, as in the representation of storytelling, rather as a fundamental mode through which knowledge is organized. [13]. Our position is that narrative should therefore serve as an additional layer for representation overall: its representation is to be considered a primary epistemic layer in its own right.

We are still missing a semantic infrastructure that can represent *how* what happened is organized into a meaningful account for someone, from somewhere, under specific social and interpretive conditions [12]. NKGs, as we envision them here would add a formal layer on top of KGs for perspectival and interpretive knowledge that can mediate between human experience and LLM-generated meaning.

3. Narrative Knowledge Graphs

3.1. *Hermeneutics, perspective, and narrative*

Hermeneutics, the philosophical tradition concerned with the theory and practice of interpretation, offers a rich conceptual vocabulary for NKGs. From a hermeneutical point of view, meaning is always constructed by an interpreter, from a particular standpoint, in dialogue with a text, an artifact, or a cultural object. In our proposal, we draw on three central concepts from this tradition: situatedness, perspectival plurality and ambiguity.

Situatedness names the condition that all interpretation happens from a position, that is, a specific historical, cultural, embodied standpoint that shapes what can be understood and how. There is no view from nowhere. A knowledge representation system that does not encode situatedness implicitly claims to offer one, and is epistemically dishonest.

Perspectival plurality names the condition that multiple valid, mutually incompatible interpretations of the same phenomenon can coexist. Differently from relativism (not all perspectives are equally well-evidenced or equally coherent), we consider it a recognition that the space of valid interpretations is not a singleton. Cultural knowledge about phenomena such as historical events, aesthetic objects, illness experiences, and social practices is necessarily plural, and representations that collapse this plurality into a single ground truth are systematically distorting.

Ambiguity names the irreducible openness of meaning that persists even within a given situated perspective. Ambiguity is a structural feature of natural language and symbolic representation that cannot be reduced simply to a failure of specification or a problem to be resolved through additional data. The same expression or narrative can remain underdetermined with respect to its precise interpretation, allowing for multiple coherent but non-equivalent readings. Ambiguity is a condition that enables interpretive flexibility: it allows meaning to remain responsive to new contexts, evolving knowledge, and shifting interpretive goals.

While these three concepts are analytically distinct and manifest differently across domains, they share a deeper commonality: they all point to the fact that meaning is not

exhausted by formalization. Following Paul Ricoeur's account of narrative as a fundamental mode of human understanding, we thus argue that **narrative** provides a unifying principle across these dimensions [14]. Narrative, in this sense, is the primary operation through which temporal human experience becomes intelligible as identity, causality, and change over time. This process is intrinsically interpretive: it binds together discordant elements of experience into concordant wholes while never fully eliminating tension, ambiguity, or reinterpretation across time.

Metaphor plays a constitutive role in this architecture of meaning, functioning as a cognitive operation that enables the re-description of reality by projecting structure across domains. As later developed in cognitive linguistics, metaphor allows abstract and institutional domains of knowledge to be grounded in embodied, experiential schemata, enabling narrative comprehension of phenomena that would otherwise remain conceptually inaccessible [15]. Consider how the conceptualization of ILLNESS as a BATTLE, pervasive in discourse, imports a whole structure of meaning: an enemy to be defeated, a fighter who may win or lose, a war that can be won or surrendered. This framing shapes what counts as a good outcome, what emotional responses are legible, and whose agency is foregrounded. A patient who resists this metaphor and understands their condition instead as a JOURNEY is organizing the same clinical situation into a fundamentally different narrative, with different implications for treatment, identity, and meaning.

In this view, NKGs are designed to represent knowledge along all three of these dimensions: structured enough to be formally manipulable, rich enough to hold perspectival and ambiguous content, and explicitly connected to the individuals and communities from whose standpoint the knowledge is valid.

3.2. Operationalizing NKGs

The three hermeneutic dimensions explained above are directly operationalizable. A NKG is built on standard RDF/OWL foundations, extended along three axes that correspond to them: perspective-annotated named graphs operationalize situatedness by anchoring every assertion to a standpoint; narrative ontologies operationalize the temporal and sequential dimension of situated experience by allowing the same events to yield different narratively organized accounts; and metaphorical frames, grounded in frame semantics, operationalize ambiguity by making explicit which conceptual structure a community projects onto a domain, thus which aspects of meaning remain open to reframing.

Perspective as a first-class dimension. Every assertion in a NKG is anchored to a perspective node encoding the community, epistemic tradition, temporal and cultural context, and evidential basis of the claim. Following RDF versioning frameworks [16], we repurpose change-tracking from the temporal to the perspectival dimension: a NKG answers the question: *from whose standpoint is this true?* instead of *when is this true?*. RDF 1.2 triple terms now enable statement-level perspective annotation natively, where prior approaches required reification or named graphs with their respective limitations [17,18]. Existing provenance infrastructures such as PROV-O [19] and nanopublications [20] provide compatible building blocks for perspective anchoring; NKGs extend their scope from provenance documentation to active perspectival conditioning within a generative interaction loop.

Narrative ontologies. NKGs represent narrative sequences. Drawing on existing narrative ontology work [21,22] and extending it with perspectival annotation (e.g., sim-

ply with provenance explicitation), the same sequence of events yields different narratives depending on the standpoint from which it is told.

Metaphorical framing. Building on FrameNet-based semantic resources for metaphor [23,24], metaphorical frames are represented as named graph connectors linking experiential source domains to abstract target concepts. A patient’s JOURNEY metaphor and a physician’s MANAGEMENT metaphor over the same chronic illness become formally visible and queryable divergences. Within frame semantics, each metaphorical mapping activates a distinct frame structure with its own role slots, entailments, and inferential consequences, making divergent framings formally distinguishable [23].

3.3. Including LLMs: the Human-LLM-NKG Triangle

The three NKG axes are active resources in a triadic interaction loop. A human user interacts with an LLM mediated by a NKG: the human carries a situated perspective; the NKG maintains the persistent narrative subgraph built incrementally through interaction; the metaphorical frame mappings, stored in the NKG, provide the basis for evaluating whether LLM outputs are coherent with the user’s active framing. The interaction proceeds as a loop. When the user submits a query, the NKG supplies whatever perspectival and narrative context has accumulated, conditioning the LLM’s response against a structured background. The LLM’s output is evaluated for perspectival appropriateness alongside factual accuracy, then written back to the NKG as a new annotated event. When the user revises a framing, the subgraph is updated with the previous version retained as an auditable layer. Over time, personal subgraphs can be aggregated, with appropriate privacy controls, into community-level NKGs representing the perspectival landscape of any domain.

4. Use cases

Here we present three use cases of NKGs as a response to open problems in the Semantic Web.

4.1. Documenting Perspectives in Cultural Heritage

Up to now, the application of perspective-conditioned representation using KGs has seen the most success in the cultural heritage domain. A constitutive factor of the meaning of heritage artifacts are the stories about the creation, the origin, the transfer of property and the reasons for current possession by a certain cultural institution. Lately, significant effort has been made to recontextualize ethnographic collections in order to highlight the colonial history of unlawfully appropriated works [25].

Several modeling efforts have since developed schemas for representing, storing, and querying perspectival metadata about heritage objects, including requirements for adequate representation, multilayered documentation frameworks, and provenance-tracking models based on named graphs [26,27,28]. NKG perspective annotations and narrative subgraphs can be layered on top of existing CIDOC-CRM [29] models and extensions. While such advancements have pushed forward the literature on contextualized knowledge representation, they still tend to treat perspective primarily as meta-

data attached to otherwise stable factual structures. What remains underrepresented is the narrative dimension through which cultural heritage is actually experienced and interpreted. Heritage objects do not only possess provenance histories; they accumulate layers of memory, trauma, symbolism, and contested meaning that differ across communities and evolve over time. An ethnographic artifact may simultaneously function as a scientific specimen within a museum classification system, as a sacred ancestral object within an Indigenous community, and as evidence of colonial violence within postcolonial discourse. These are different narrative worlds that organize the object's meaning through distinct metaphors, temporalities, and moral frameworks. Current polyvocal and provenance-aware KG approaches can document the coexistence of such perspectives, but they generally lack mechanisms to formally represent how these perspectives are narratively constructed, how they change historically, or how metaphorical framings shape interpretation itself. NKGs thus enable heritage knowledge to be represented not only as collections of contextualized facts, but as interconnected, situated narratives whose meaning emerges through perspective-conditioned interpretation.

4.2. *Empowering the Patient in Healthcare*

The health domain also illustrates the need for NKGs. A patient interacting with a LLM-based health assistant brings a rich narrative: symptoms interpreted through personal conceptual frameworks, an illness story organized around metaphors (the condition as enemy, companion, or teacher), and a lay understanding of medical concepts that may diverge substantially from clinical usage [30]. Current health KGs and clinical ontologies represent diagnoses, procedures, and anatomical relations with precision, yet are structurally silent on everything that makes an illness experience meaningful to the person living it [31]. The LLM, drawing on medical literature and clinical guidelines, has no mechanism for tracking where its assumptions diverge from the patient's, and no way to flag when expert-community metaphors conflict with the patient's own meaning-making, and therefore their empowerment. The gap is primarily one of communication [30]. In a NKG-mediated interaction, the patient's illness narrative is structured as a perspective-annotated subgraph (Fig. 1): symptoms are nodes, causal beliefs are edges, metaphorical frameworks are encoded as frame mappings, and narrative progression is a temporally ordered sequence. This subgraph is linked to, but not collapsed into, the clinical knowledge layer representing the physician's perspective. The LLM operates against this structured background; its outputs are evaluated for whether they respect the patient's narrative framework, deploy metaphors coherent with their framing, and flag divergences rather than silently resolving them. Concretely, a patient's journey-framed belief that their condition is improving and a physician's contrary clinical assessment coexist in the NKG as unasserted quoted triples, each attributed to its source. A query for the patient's perspective returns their narrative; a query across perspectives surfaces the divergence.

4.3. *Mitigating the Data Leakage Problem*

The data leakage problem in LLM evaluation, i.e., the systematic contamination of benchmarks by pretraining data, is, from the NKG perspective, a symptom of the deeper epistemological problem: benchmarks built on factual ground truths can be passed by memorization, not understanding, because they do not test perspectival competence.

NKG-based evaluation replaces fixed ground truths with perspectively valid answers: the correct response to a query is not a single fact but an answer that is appropriate given the perspective specification embedded in the query. A model cannot pass this kind of evaluation by memorization, because the space of (situation $\times$ perspective $\times$ narrative frame) combinations is generative and unbounded: novel evaluation instances can be created on demand that have never appeared in any pretraining corpus. Moreover, provenance tracking in the NKG provides a formal mechanism for detecting contamination when it does occur: if a model's response matches a NKG assertion at the level of both content and perspective-encoding, this is evidence of memorization. Perspective-annotated provenance chains make the structure of knowledge explicit enough to distinguish genuine understanding from pattern matching. The stakes of this shift extend beyond technicalities, as benchmarks built on web-scale pretraining data reproduce the perspectival distribution of that data: communities, illness experiences, and cultural frameworks underrepresented in the corpus are underrepresented in evaluation. A model certified as accurate on such benchmarks has been tested primarily against dominant perspectives. NKG-based evaluation, by grounding correctness in explicit perspective specifications, makes the perspectival coverage of a benchmark a measurable and improvable property.

5. Long-Term Vision

The long-term vision of this paper is the articulation and adoption of a new representational layer for human-AI interaction: NKGs. In this sense, the ambition of NKGs is to reorganize the relation between formal knowledge representation, human interpretation, and machine generation. From this perspective, solving the problem of narration is not one challenge among others, but a leverage point through which many other smaller open problems become tractable. A semantic infrastructure capable of representing narrative organization, perspectival plurality, and ambiguity would simultaneously improve explainability, provenance tracking, human-AI alignment, user modeling, benchmark design, and cross-community communication. We aim to use NKGs so that perspective becomes natively embedded in KGs of all domains rather than exceptional, isolated perspectives for specific use cases. What today appears as polyvocal metadata, user context, prompt conditioning, or post-hoc explanation could, in a mature NKG ecosystem, become part of a single integrated representational logic.

For this vision to become viable, NKGs must move beyond one-off prototypes into a shared research program, developing reusable design patterns for perspective annotation, and ontologies of narratives and metaphorical phenomena across all domains. In the longer horizon, we envision the human-LLM-NKG triangle as a basic pattern for neurosymbolic systems. In such systems, the LLM would no longer function as an opaque reservoir of implicitly absorbed perspectives, but as a generative interface operating over an explicit, revisable, and inspectable narrative substrate. We thus propose NKGs as one step toward that future: a future in which formal knowledge representation becomes more faithful not only to the structure of the world, but to the ways human beings inhabit, interpret, and narrate it.

Declaration on Generative AI During the preparation of this work, the authors used ChatGPT in order to: Grammar and spelling check. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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Can a Knowledge Graph Fall Down?

Embodied Knowledge Graphs for Humanoid Robots

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Abstract. Knowledge graphs (KGs) are powerful tools for representing, integrating, and querying knowledge. Humanoid robots expose a limitation of current world models: they can predict future states, but they rarely make explicit the semantic commitments needed to decide whether an object is reachable, a surface is supportive, a human is too close, or an action is unsafe. This paper argues that the next long-term challenge for Semantic Web and Knowledge Graph research is to build *Embodied Knowledge Graphs*: dynamic, multimodal, uncertainty-aware semantic world models whose statements are grounded in geometry, perception, proprioception, affordances, physical constraints, and predicted action consequences. Such graphs would not merely answer “what is true?” but also make explicit which commitments are grounded enough to act upon, which remain risky, and which require further perception. This is not a proposal for adding a knowledge graph to a robot, but for redefining what a knowledge graph must be when its statements can trigger physical action. Using humanoid semantic world reconstruction as a motivating stress test, the paper outlines a research agenda around joint geometric-semantic reconstruction, structural abstraction, affordance transfer, provenance, and self-maintenance. The proposal is intentionally high-risk: it asks whether knowledge graphs can make embodied action safer, more recoverable, and more auditable than systems based only on latent world models or scene graphs.

Keywords. embodied knowledge graphs, humanoid robotic, semantic world models

1. The provocation

The Semantic Web was born from a promise that machines could act more intelligently if data were given explicit meaning, shared identifiers, and machine-processable relations. Two decades later, KGs support search, recommendation [1], data integration [2], question answering [3], scientific discovery [4], and retrieval-augmented language models

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[5]. Yet most KGs still represent facts without representing the physical conditions under which those facts become actionable. A triple encodes a relation, but not the physical conditions under which that relation may fail. An ontology can describe constraints, but it does not encode the physical dynamics through which those constraints are violated.

Humanoid robots make this limitation impossible to ignore. A humanoid acting in a home, factory, or hospital must reason across language, geometry, objects, body posture, balance, human proximity, affordances, and safety. A statement such as `:knife :hasPart :blade1` and `:blade1 :hasQuality :sharp` is useful, but not sufficient. The robot also needs to know where the sharp blade of the knife is, which grasp regions are safe, whether the object is reachable without overextending the body, whether the human is in the handover region, and whether the next action could transform a safe scene into a dangerous one. In such settings, the central semantic question is no longer only “what does this symbol denote?” It becomes: “what actions or decisions does this statement allow the robot to take?”

This paper argues for an ambitious scope: the Semantic Web community should treat embodied robots as a forcing function for the next generation of KGs and force us to rethink them. The goal is not to attach a KG to a robot as an external database, but to make it an active component of the robot’s world model that informs perception, planning, safety checking, and replanning. An *Embodied Knowledge Graph* (EKG) is a dynamic, multimodal, uncertainty-aware, action conditioned graph whose statements are grounded in perception, geometry, proprioception, temporal change, provenance, and expected consequences of action. This paper is not a proposal for adding a KG to a robot; it is a proposal for redefining what a knowledge graph must be when its statements can cause physical action.

The question in the title is therefore not a joke. A knowledge graph cannot fall down, but a robot can fall because it trusts the wrong assertion at the wrong moment. If knowledge graphs are to become the semantic substrate of embodied AI, their representational commitments must include the possibility of physical failure. More broadly, this agenda sits within a long-standing and still active debate on embodied, situated, and grounded intelligence in AI. Recent syntheses argue that robust intelligence cannot be understood as detached symbol manipulation alone, but must be studied through the coupling of perception, action, morphology, and environment [6,7]. EKGs are not a return to disembodied AI. They ask what an embodied agent must represent explicitly in order to act safely, justify its choices, and recover from error.

Why this problem matters for the Semantic Web. Humanoid robots are not the application domain of this paper so much as the stress test that exposes what current KGs still cannot represent: grounded, risky, time-sensitive, action-relevant commitments. This stress test raises four long-term challenges for the Semantic Web community: (i) when is a statement grounded enough to act upon; (ii) how can a graph record the sensorimotor evidence that makes an assertion usable; (iii) how can queries express what may become true, risky, or impossible after an action; and (iv) how can a graph repair its own uncertainty by triggering perception, probing, replanning, or human clarification? The risk is that explicit semantics may not scale to the speed, uncertainty, and continuous change of embodied action. The gain is an accountable semantic layer through which embodied AI can explain why it acted, why it replanned, and why it sometimes chose not to act.

2. Why current graphs and World Models are not enough?

The state of the art is impressive, but fragmented. Semantic Web standards provide mature foundations for graph data, ontologies, constraints, and provenance, including RDF [8], OWL [9], SHACL [10], and PROV-O [11]. KG research has developed rich methods for schema modelling, entity linking, reasoning, completion, and learned graph representations [12]. Commonsense resources such as ConceptNet², CSKG [13], and ATOMIC 2020 [14] provide structured background knowledge about object properties, uses, events, and expectations.

Robotics has developed a related but largely separate line of work. Semantic maps have long supported robot task planning [15], and KnowRob showed how symbolic knowledge processing can support robot cognition and task reasoning [16]. More recent work on 3D dynamic scene graphs, including Kimera and SceneGraphFusion, moves toward spatial world models that integrate geometry, semantics, objects, places, rooms, and dynamic agents, and that can be built incrementally from RGB-D streams [17,18]. Affordance benchmarks such as 3D AffordanceNet further show that functional object understanding requires 3D physical grounding, not only image labels [19]. These lines of work are essential precursors, but they address different parts of the problem: KnowRob focuses on symbolic robot knowledge, Kimera and SceneGraphFusion focus on scene structure grounded in geometry and semantics, whereas EKGs focus on explicit commitments for action, including grounding, uncertainty, provenance, and conditions for safe use.

Recent work has also started to connect knowledge graphs more directly with embodied perception, manipulation, and planning. Large-scale knowledge graphs have been used to enrich robotic perception with semantic priors that improve the interpretation of observed scenes and objects [20]. Other work has shown that behaviours and geometric constraints for manipulation can be represented in graph form, making control structure more explicit and interpretable [21]. Recent work by some of the authors has shown that external commonsense knowledge graphs can help robots resolve ambiguity and decompose tasks more safely when planning from natural language instructions [22]. These directions are important precursors, but in all of them the graph still functions primarily as a prior, an annotation layer, or a planning aid. It is not yet the persistent semantic world model that an EKG is meant to be: one that makes uncertainty explicit and represents commitments for action. In short, KnowRob emphasizes symbolic robot knowledge, Kimera and SceneGraphFusion emphasize scene structure grounded in geometry and semantics, whereas EKGs emphasize explicit commitments for action, with grounding, uncertainty, provenance, and conditions for safe use.

At the same time, robot learning is scaling through large cross-embodiment datasets, diffusion policies, vision-language-action models, and learned world models for prediction and control [23,24,25,26,27,28]. Yet these systems do not by themselves expose which semantic commitments make an action safe, feasible, or risky. This is precisely the gap that Embodied Knowledge Graphs are meant to address.

The gap is not a missing component, but a missing contract between KGs and world models. KGs generally excel at explicit meaning, interoperability, and reasoning, but they often lack dense grounding in sensorimotor reality. Robot world models excel at

²<https://conceptnet.io/>

perception-action coupling, but they often lack explicit, reusable, inspectable semantics. Current world models can predict future states, but they do not naturally expose the semantic structure needed to act on those predictions: object parts, affordance provenance, and safety constraints whose violation may create physical or social risk. Nor do they represent these commitments in a form that can be queried, audited, and revised.

Scene graphs, robot knowledge bases, and learned world models each address an important part of the problem, but none takes explicit commitments for action as its primary representational object. The central challenge is therefore to define a representation that integrates semantic commitments, spatial grounding, evidential provenance, and action relevance, and that can serve as the explicit, queryable layer of a semantic world model. The claim is not that robot control should be rewritten in RDF. The claim is that the commitments that guide robot action should be represented in explicit graph form whenever this makes them easier to query, check, revise, and use for safer behaviour. Recent work by the authors of this paper has already begun to explore this space by using external commonsense knowledge graphs to improve ambiguity resolution and risk-aware robotic task decomposition. However, the graph still functions mainly as a planning aid rather than as a persistent, embodied semantic world model [22].

3. The Embodied Knowledge Graph

An EKG is a knowledge graph in which the basic unit of meaning is not simply a triple, but a structured claim that combines semantic content with spatial grounding, temporal validity, uncertainty, provenance, and action relevance. Its nodes include objects, object parts, surfaces, containers, tools, humans, robot body regions, free-space regions, contact patches, affordance regions, skills, policies, goals, tasks, and safety constraints. Its edges include classical semantic relations such as *is-a*, *part-of*, and *used-for*; spatial relations grounded in measurable geometry such as *on*, *inside*, *near*, and *occludes*; functional relations such as *support*, *containment*, *graspability*, and *safe handover*; and predictive relations such as *expected reachability*, *balance risk*, and *expected action effect*.

A further distinction is important here: affordances should not be collapsed into signifiers. An affordance is a physically grounded possibility for action that depends on the relation between a body, an object, and a context; a signifier is a perceptual, geometric, or conventional cue that indicates that such a possibility may be available. For an EKG, the graph should therefore represent both actionable possibilities and the evidence that makes them perceptually or socially legible in a given situation. The robot itself must also be represented as a special class of embodied claims: not only as an acting subject, but as an object of semantic description whose body regions, reachability limits, balance envelope, load state, visibility cones, and operating constraints determine what the graph may safely license as actionable.

The important move is not to invent more predicates. It is to change what a predicate is allowed to mean. An embodied assertion can be represented as $c = (s, p, o, \tau, G, U, P, A, R, M)$, where s, p, o denote the semantic claim, τ temporal validity, G geometric grounding, U uncertainty, P provenance, A the usable action context, R the risk condition, and M the maintenance action if the claim is not grounded enough. In the running scenario Figure 1 where an older adult asks a humanoid for a glass of water, the claim `glass1 canBeHandedTo human1` is meaningful only if it is tied to a specific

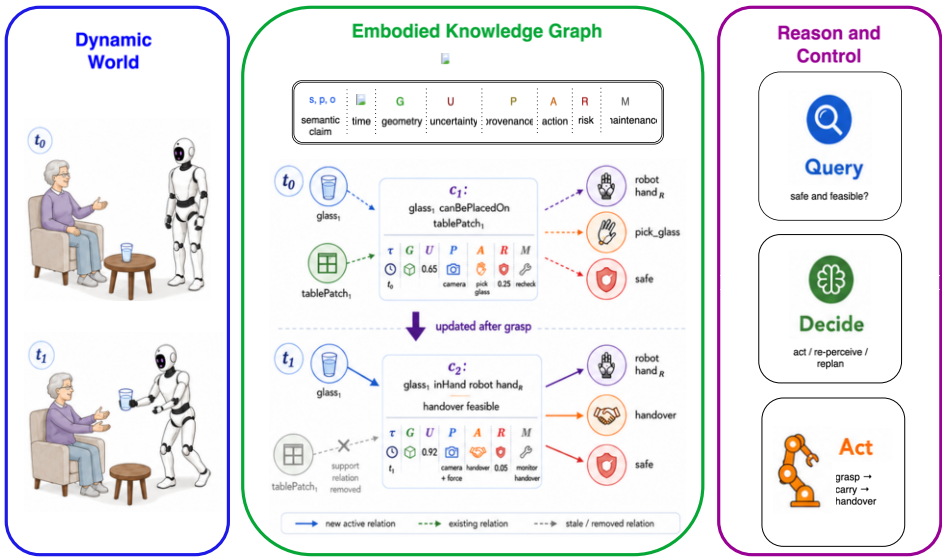


Figure 1. Embodied Knowledge Graph framework. The physical scene changes from t_0 to t_1 , and the graph updates its embodied assertions accordingly, linking semantic claims with geometry, uncertainty, provenance, action, risk, and maintenance to support safe reasoning and control.

glass pose, a handover region, the robot’s balance margin, recent sensor evidence, and a repair policy if that evidence is outdated. This changes the role of query answering. A graph query asks which facts match a pattern. An embodied graph query asks which commitments are safe enough to act upon. Example queries include: Which surfaces can support this object with sufficient confidence? Which grasp regions are reachable without destabilizing the humanoid? Which facts would have to be checked before the next skill is executed? Which semantic constraints are relevant to the next manipulation phase?

Such claims should support pseudo-queries for action feasibility and repair, for example whether $\text{place}(\text{glass1}, s1)$ is admissible or requires lookAgain . The EKG does not directly control the robot’s motors or joint forces. Its role is to expose the commitments that guide control, such as feasible objects, admissible grasp regions, active safety constraints, unresolved risks, and missing evidence. Not every sensorimotor fact needs to be materialized declaratively. The EKG should expose only those commitments whose explicit form changes planning, safety checking, repair, transfer, or post hoc accountability.

4. Grounding Semantic World Models

Humanoid semantic perception offers a concrete stress test for this vision. Consider the running scenario in Figure 1: an older adult asks a humanoid robot for a glass of water. The graph must know whether glass1 is visible and reachable, whether tablePatch3 is a stable support surface, whether human1 is already in the handover corridor, and whether the robot can reach without losing balance.

The *first* challenge is joint geometric and semantic reconstruction. Sensor setups that combine RGB-D observations from multiple views, including cameras mounted on the

wrist or placed externally, help address the partial observability of any single camera: external views provide global context, while wrist or egocentric views capture close object interactions under occlusion and large viewpoint changes [29]. Classical volumetric fusion based on signed distance functions [30] and newer radiance-based representations such as 3D Gaussian Splatting [31] show that dense geometric and appearance reconstruction is feasible. The Semantic Web challenge starts one step later: *how do surfaces, free space, visibility, timestamps, and human proximity become graph statements with evidence, confidence, and consequences for action?*

The knowledge represented in an EKG cannot come from a single source. It must combine live perception, self-observation and proprioception, prior structured knowledge, embodied experience, and external input. Some commitments are learned from sensorimotor interaction; others are hand-authored, extracted from external resources, or transferred from prior executions and grounded templates. Action traces, failures, recovery episodes, and human corrections turn embodiment into a cumulative source of semantic evidence. Knowledge acquisition is therefore not a preprocessing step, but a continuous problem of grounding, updating, revising, and deciding what may safely transfer across situations, tasks, and bodies.

The *second* challenge is structural abstraction. Dense point clouds and neural volumes are too expensive to serve directly as long-horizon planning bases or real-time collision models. For an EKG, primitive fitting and abstraction at the level of object parts, including decompositions on superquadrics, provide an intermediate layer between raw perception and symbolic knowledge. They transform pixels and point clouds into object parts, surface patches, contact regions, and geometry relevant to affordances, which can then be represented in graph form.

The *third* challenge is to generalize functional knowledge across objects that differ in shape, size, pose, material, and affordance structure. A semantic world model is not useful if every new mug, drill, chair, or bottle requires relearning affordances from scratch. Canonical correspondence methods such as NOCS [32] suggest a path forward because they turn generalization into a structured grounding problem. The system aligns a newly observed shape with a canonical template and transfers grasping zones, support regions, and action vectors through geometric homology rather than local pixel similarity. For EKG research, affordance transfer should become a first-class semantic operation, not an external preprocessing step.

The *fourth* challenge is scene graph compilation as operational semantics. Instance segmentation, affordance prediction, topological relation extraction, and commonsense enrichment must be fused into a hierarchical 3D scene graph whose nodes represent environments, regions, objects, parts, and whose edges capture support, containment, proximity, reachability, and risk dependencies. Commonsense remains useful only as defeasible hypothesis. “A glass can contain water” matters only if this glass is open, not already occupied, not partially hidden, and not currently in a human’s hand.

Finally, the world-state graph must operate as a dynamic query basis for prediction. It has to update when objects move, when humans intervene, when occlusions appear, and when predicted effects fail. In the running scenario, it should answer questions such as: *if the robot reaches now, does the balance margin become unsafe? if the glass is lifted, what support relations disappear? if the evidence is outdated, should the next graph action be to look again, replan, ask, or refuse?*

This pipeline is more than a robotics module. It is a challenge for semantic technologies. The key question is whether a graph can represent the world at a level that a body can safely use. It should be evaluated against strong baselines, including latent world models and scene graphs. If explicit semantics does not improve safety, recovery from outdated beliefs, failure explanation, or uncertainty reduction, then embodiment has not justified its symbolic cost.

5. Five Foolish Principles

The following principles are deliberately overstated. Their purpose is to break the assumption that knowledge graphs are primarily stores of descriptions. These principles are not a claim that all control should become symbolic; they are hypotheses about which commitments should become explicit above the reactive control layer in a hybrid neuro-symbolic system.

- **No actionable fact without sensorimotor grounding:** Every embodied assertion should carry the cost of making it usable for action: perceptual, geometric, temporal, causal, or social. A graph system should rank assertions by this grounding debt and recommend actions that reduce it.
- **From truth conditions to action conditions:** For embodied action, the key question is not only whether a statement is true, but whether it is valid, grounded, and safe for a particular action under current conditions.
- **Relations should say what they permit:** Relations such as support and containment should describe not only what is true, but also what physical conditions must hold for the relation to guide action.
- **A query tests whether a claim is grounded enough for action:** In embodied graphs, uncertainty should sometimes trigger epistemic action: look from a new viewpoint, probe, ask a human, or choose a safer alternative.
- **A self-healing graph must be allowed to move:** An embodied graph should not only detect outdated or inconsistent assertions, but also generate new evidence through movement, probing, or reversible action.

6. Research agenda for the Semantic Web Community

The EKG vision defines a long-term agenda for the Semantic Web community, not only for robotics. Its central questions are semantic: what must be represented explicitly, what must be queried before action, what must remain revisable under uncertainty, and what may or may not transfer across bodies and situations. Some knowledge, such as object categories, task schemas, and affordance templates, may be shared across robots; other commitments, such as reachability, occlusion, balance, and local risk, remain grounded in a particular body and scene.

1. Embodied Ontology Design: Semantic Web ontologies usually model categories, properties, events, and relations. Embodied ontologies must also represent regions, contact patches, uncertainty, affordance distributions, body configurations, skill phases, failure modes, and safe operating boundaries. This requires modelling hybrid entities that are both symbolic and geometric: a handle is an object part, a grasp region, and a role in

an action; the area around a human is a spatial region, a social constraint, and a safety limit. The challenge for the Semantic Web community is therefore to design ontological patterns in which such entities and relations can be represented explicitly, queried, validated, and revised. Embodied ontology design should make commitments for actions visible: what a relation means, what evidence supports it, when it remains valid, what risks it carries, and what must be done when the graph is not grounded enough to act.

2. *Query languages for action and counterfactuals*: The most important embodied queries are counterfactual: what will become true if the robot executes this skill, which facts must remain invariant, and which constraints would be violated? SPARQL and graph query languages are excellent for graph pattern matching, but they do not directly express action effects over learned world models, physical constraints, and risk thresholds. A useful first step would be a small operational fragment rather than a complete language: declarative constraint patterns such as a `SupportRelationShape`, together with an action-query operator such as `ASK_ACTION`, returning whether an action is admissible, which constraints are violated, which evidence is missing, and which repair action is required. This would already support experimental evaluation of recovery from outdated beliefs, safe refusal, and failure localization.

3. *From data provenance to action traceability*: PROV-O and related vocabularies provide a foundation for representing sources, activities, and agents. Embodied provenance should extend this foundation to action. A graph statement should record not only who produced it, but also which camera pose, segmentation mask, action attempt, or human correction supports it. This is essential for failure analysis. If the robot drops an object, the system should help determine whether the cause was poor geometry, outdated knowledge, an incorrect affordance, or policy execution.

4. *Representing uncertainty as part of meaning*: Uncertainty is not a scalar decoration. It is structured. The robot may be uncertain about object identity but certain about occupied space; uncertain about material but certain about weight; uncertain about a human's intention but certain about proximity. EKG research should treat uncertainty not as a single confidence score, but as a structured semantic element: it has a type, a source, a temporal validity, and a different meaning depending on the action under consideration. This could connect Semantic Web formalisms with probabilistic programming, neurosymbolic learning, space planning, and risk estimates.

5. *Graph foundation models for grounded symbolic reasoning*: Graph foundation models are often imagined as large learned models operating over graph structured data. For embodied graphs, the challenge is more nuanced: learned models must generalize from graph structure without losing the explicit entities, relations, constraints, provenance, and risks that make the graph useful for action. They should predict missing affordances, likely state changes, and relevant constraints, while still exposing reasons, uncertainty, and counterexamples. The graph should not disappear into a latent vector, but remain inspectable enough to support accountability and repair.

6. *Benchmarks that measure whether semantics improve action*: The community needs benchmarks where descriptive semantics alone is insufficient, such as fragile object handling, tool handover, navigation in dynamic human spaces, and recovery from failed grasps. The relevant baselines should be strong, including latent world models without explicit semantic layers, dynamic scene graphs with only perceptual semantics, and KGs attached to perception outputs after the fact. Evaluation should measure not only task completion, but also unsafe action avoidance, safe refusal, recovery from out-

dated beliefs, failure localization, latency, and the usefulness of explanations for diagnosing why an action was executed, delayed, replanned, or refused. **A graph that answers many questions but still lets the robot collide has not earned the right to exist.**

7. A ten-year horizon

The path to adoption can be imagined in four phases. It begins with richer semantic scene graphs, moves to world models that support control, then to embodied graphs that can answer counterfactual and action-related queries, and ends with semantic systems that can repair outdated beliefs and explain why an action was taken, delayed, revised, or refused. This trajectory is risky because it asks robotics, Semantic Web research, machine learning, and cognitive systems to converge on the same formal object. Its potential gain is that semantic technologies would take on a concrete role in embodied AI as explicit structures that connect perception, prediction, action, safety, and explanation.

8. Risks and failure modes

This vision can fail in several ways. One risk is trying to force continuous and uncertain contact dynamics into logical forms that are too rigid, so explicit semantic commitments must coexist with learned continuous models. Another risk is making explicit what should remain reactive; the EKG should constrain, justify, and repair action, not replace fast control. A further risk is ontological inflation, namely adding predicates, affordances, and provenance fields without changing robot behaviour or improving failure explanation. Speed also matters, since a slow graph cannot guide humanoid action, and fast control must therefore remain distinct from slower semantic reasoning even if the two stay connected. External knowledge and suggestions from language models must remain defeasible, must carry provenance, and must be checked against current perception when the risk of action is high. Finally, benchmarks should measure real effects on robot behaviour, including safer actions, fewer unsafe attempts, better recovery, and less uncertainty before execution.

9. Conclusion

This paper has argued for an ambitious claim: knowledge graphs should grow a body. For humanoid robots, semantic structures cannot remain labels placed on top of perception; they must help determine what is reachable, safe, useful, fragile, occupied, and worth checking again. The graph of the future will not only know that a glass is a container. It will know whether this glass, here and now, under current uncertainty, can be grasped and handed over without harming someone. If this agenda fails, the failure will still be informative: it will show where explicit semantics stops helping embodied AI act more safely, recover more effectively, and explain its failures more clearly.

Declaration on Generative AI

During the preparation of this paper, the authors used ChatGPT to support grammar and spelling revision. The authors subsequently reviewed and edited the text and take full responsibility for the final content of the publication.

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